

Exhibit CC-122

IN THE UNITED STATES DISTRICT COURT
FOR THE EASTERN DISTRICT OF TEXAS
MARSHALL DIVISION

NETLIST, INC., (CAUSE NO. 2:21-CV-463-JRG
)
Plaintiff, ()
vs. ()
SAMSUNG ELECTRONICS CO., LTD., ()
et al., () MARSHALL, TEXAS
() APRIL 17, 2023
Defendants. () 8:30 A.M.

VOLUME 2

TRIAL ON THE MERITS

BEFORE THE HONORABLE RODNEY GILSTRAP
UNITED STATES CHIEF DISTRICT JUDGE
and a jury

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1 THE COURT: Be seated, please.

2 Are the parties prepared to read into is the record those
3 items from the list of pre-admitted exhibits used by last
4 Friday's portion of the trial?

5 MS. TRUELOVE: Yes, Your Honor.

6 THE COURT: Please proceed.

7 MS. TRUELOVE: The following JTX exhibits were used
8 Friday during trial. That would be: 002, 003, 004, 005, 006,
9 0021, 0024, 0029, 0043.

10 The following PX numbers were also used: 0446, 0464,
11 0586, 0621, 1663, 1756, 1778, and 1816.

12 And my understanding is Defendants don't have anything to
13 tender to the Court.

14 THE COURT: All right. Do Defendants have any
15 objection to the rendition offered by Plaintiff?

16 MS. DEGNAN: No, Your Honor.

17 THE COURT: And can you confirm Defendants don't
18 have additional items from the list of pre-admitted exhibits
19 to put into the record?

20 MS. DEGNAN: That's correct.

21 THE COURT: All right. Mr. Milton, if you'll return
22 to the witness stand, please, sir. I remind you, you remain
23 under oath.

24 Is there anything, Counsel, I need to hear from you on
25 before I bring in the jury?

1 MR. SHEASBY: Nothing from Plaintiff, Your Honor.

2 THE COURT: Okay. And you are in the middle of the
3 cross-examination, Mr. McKeon. You may go to the podium and
4 prepare.

5 MR. McKEON: Thank you, Your Honor.

6 THE COURT: Let's bring in the jury, please.

7 (Whereupon, the jury entered the courtroom.)

8 THE COURT: Welcome back, ladies and gentlemen.
9 Please have a seat.

10 We will continue with the testimony of Mr. Scott Milton.
11 We recessed on Friday evening with cross-examination by the
12 Defendants underway, and we'll continue with Defendants'
13 cross-examination of Mr. Milton at this time.

14 Mr. McKeon, you may proceed when you're ready.

15 MR. McKEON: Thank you, Your Honor.

16 SCOTT MILTON, PREVIOUSLY SWORN,
17 testified further under oath as follows:

18 CROSS EXAMINATION continued

19 BY MR. McKEON:

20 Q. Good morning, Mr. Milton.

21 A. Good morning, Mr. McKeon.

22 Q. I want to start where we left off and we are talking
23 about patent claims. And just to reset, you understand that
24 you had testified that you cannot comment on claims or claim
25 language. Isn't that right?

1 A. That is a true statement.

2 Q. And you're also not qualified to assess claims, including
3 for the '918 and '054 Patents. Correct?

4 A. That's correct.

5 Q. So you're not here to answer questions on what Netlist
6 claims the patent claims cover. Isn't that right?

7 A. That's correct.

8 Q. You are not here to offer opinions on whether Samsung's
9 patents infringe the claims. Isn't that right?

10 A. That's correct.

11 Q. Now, you recall on Friday, you testified about the Patent
12 Office procedures and you seemed to know a lot about what
13 happens in the Patent Office. Do you recall that testimony?

14 A. I do, sir.

15 Q. And you're aware that when you file a patent application
16 in the Patent Office, the inventors, whether sole inventor or
17 joint inventors, are required to sign a declaration swearing
18 to a number of different things. Do you recall that?

19 A. I do, sir.

20 Q. Okay.

21 MR. McKEON: So let's pull up admitted exhibit PX
22 1816, please.

23 Q. (BY MR. McKEON) And this is an exhibit, sir. Do you
24 recognize this exhibit?

25 A. Yes, I do, sir.

1 Q. Okay. And this is the file history for the -- one of the
2 patent applications relating to '918 and '054. Isn't that
3 right?

4 A. That is correct.

5 Q. Now, I want to go to page 82 of this document.

6 MR. McKEON: And if we can just get it a little
7 larger.

8 Q. (BY MR. McKEON) And you recognize this page, sir?

9 A. Yeah. It looks like what Mr. Chen signed.

10 Q. Yes. And this is the -- the declaration, this is an
11 example of a declaration that we just talked about. Do you
12 see that? In the top there, it says Declaration?

13 A. I see that.

14 Q. And here you can see the title of the invention,
15 Non-volatile Memory Module. This is in the chain of the
16 application relating to the '918 and '054. Isn't that right?

17 A. I see that.

18 MR. McKEON: And if we go to page 83.

19 Q. (BY MR. McKEON) And we see your signature there. Do you
20 see that?

21 A. I do see that, sir.

22 Q. Okay. And then let me go back to page 82, and we see
23 here the things that you told the Patent Office when you
24 signed that. And, of course, if you look at the bottom here,
25 you were making these statements under, you know, potential

1 imprisonment if you weren't being honest and truthful. Do you
2 see that?

3 A. I do see that.

4 Q. And then let's go to the statement in the middle here.
5 You see here it says, "I have reviewed and understand the
6 contents of the above identified application, including the
7 claims and any amendment filed herewith and identified above."
8 Did I read that correctly?

9 A. You did, sir.

10 Q. So you told the Patent Office, when you filed the patent
11 application, you signed this declaration, you said, I
12 understand the claims, I know all about claims. But you just
13 told this jury that you're not qualified and you can't assess
14 claims. Isn't that right?

15 A. That's not entirely correct, sir.

16 Q. Well --

17 A. Understanding and being a lawyer and stating with legal
18 authority about the claims is much different than
19 understanding them as an inventor.

20 THE COURT: And, Mr. Milton, he didn't ask you to
21 explain why you didn't agree.

22 THE WITNESS: Okay. I'm sorry, Your Honor.

23 THE COURT: Follow the scope of the question. And
24 as I've told you, Mr. Sheasby's going to get a chance --

25 THE WITNESS: Understood. I apologize.

1 THE COURT: -- to ask follow-up.

2 THE WITNESS: Understood.

3 THE COURT: Go ahead, Counsel.

4 Q. (BY MR. McKEON) You're not here as a lawyer. Isn't that
5 right, sir?

6 A. I'm not here as a lawyer.

7 Q. You told the jury you're here as an inventor on the
8 patents. Is that right?

9 A. I did, sir.

10 Q. And you are here as a representative of the company.
11 Isn't that right?

12 A. That's true.

13 Q. And you told the jury you here not qualified to assess
14 patent claims. Isn't that right?

15 A. Not legally. Correct.

16 Q. Right. And you signed this document in the Patent
17 Office, you didn't sign it as a lawyer. Right?

18 A. That's correct.

19 Q. You signed it as an inventor. Right?

20 A. That is true.

21 Q. And you told the Patent Office that you understood the
22 claims, not legally but you understood them as an inventor.
23 Isn't that right?

24 A. That is correct.

25 Q. All right. Let's move on, if we can.

1 MR. McKEON: You can take that down.

2 Q. (BY MR. McKEON) Now, you did not write the '918 Patent
3 application or its claims. Correct?

4 A. I did not write the application, that is correct.

5 Q. And you did not write the '054 Patent specification or
6 its claims. Correct?

7 A. That is correct, I did not write those.

8 Q. You didn't draft the figures for either of those patents.
9 Isn't that right?

10 A. That is true.

11 Q. And you didn't conduct any searching of prior art before
12 you filed the patent applications. Correct?

13 A. That is true as well.

14 Q. And, in fact, you did not even review the patent
15 application for the '054 Patent before it was filed with the
16 Patent Office. Isn't that right?

17 A. The final application? No, I did not.

18 Q. Okay. Can you -- sir, you were deposed in this case. Do
19 you recall that?

20 A. I do, sir.

21 Q. And when you were deposed, you took an oath to tell the
22 truth just like you did here in court. Isn't that right?

23 A. That is true.

24 Q. If you can get your binder, you should have it in front
25 of you. It's a binder that says Cross Examination, Scott

1 Milton. Do you see that?

2 No, that's the one that Mr. Sheasby handed you. That's
3 the one, right. If you would go to the second tab in there,
4 it's your deposition transcript. And I'd like you to go to
5 page 81. Do you have that deposition transcript, sir?

6 A. I believe I do. I believe I do.

7 Q. Okay. And if you open -- I want you to find page 81.
8 And I want you to read -- when you get there, let me know.

9 A. Right here at the top where it's -- line 2 is, Did you
10 write any of the '054 Patent specification?

11 Q. No. I want you to read it to yourself. Okay, sir?

12 A. No, no. I'm sorry, Mr. McKeon. I just want to make sure
13 I'm on the right page.

14 Q. Okay. 81, do you see that?

15 A. Yes, sir.

16 Q. And lines 12 to 17, I want you to read that to yourself.

17 A. Okay. Yes, sir.

18 Q. Let me ask the question again. You did not even review
19 the patent application for the '054 Patent before it was filed
20 with the Patent Office. Correct?

21 A. I -- I stated that. I did not read the final
22 application.

23 MR. McKEON: Your Honor, may I publish this, please?

24 THE COURT: He hasn't testified inconsistent with
25 his deposition, so you can't publish it.

1 Q. (BY MR. McKEON) Now -- okay. You didn't -- you say you
2 didn't do the final one. Is that right?

3 A. That is correct.

4 Q. Now -- so the lawyers, for all you know, they could have
5 claimed -- wrote claims that didn't reflect your actual
6 invention. You just don't know. Isn't that right? Because
7 you didn't review it.

8 A. Well, I had reviewed the claims previously.

9 Q. But you didn't review it before it was filed. Isn't that
10 right?

11 A. Not right before. That is correct.

12 Q. Now, on Friday, do you recall that you testified that
13 your invention in these patents is intelligent on-module power
14 management? Do you recall that testimony?

15 A. I recall parts of it, yes.

16 Q. It's okay. Let's put it up. If I can get transcript at
17 195 through 11, and we see here you tell the -- answering a
18 question, and you explain to the members of the jury, So what
19 we're calling our product, our invention in this case, is
20 intelligent on-module power management. Do you see that?

21 A. I do, sir.

22 Q. Okay. And here you're trying -- you're explaining to the
23 jury about what your invention is all about. Is that fair?

24 A. Yeah, that is fair.

25 Q. Now, sir, would it surprise you that the word

1 'intelligent' doesn't appear anywhere in your patent?

2 A. No, it does not.

3 Q. Okay. And, you know, if this was your invention as you
4 described it to the jury, do you think the word -- the very
5 word, sort of the first word that describes the invention,
6 that would be in your patent. Right?

7 A. I guess not necessarily.

8 Q. Okay. So -- but you understand, though, that the patent
9 itself doesn't use the phrase 'intelligent on-module power
10 management.' Correct?

11 A. That is a correct statement.

12 MR. McKEON: Now, if we can go to -- bring up PX
13 464, please.

14 Q. (BY MR. McKEON) And do you recognize this document? You
15 testified about it on Friday.

16 A. That is true. That is correct.

17 Q. This was a document that was presented to Samsung in
18 April 2015 when Netlist was proposing a partnership with
19 Samsung. Is that right?

20 A. I'm actually not sure who proposed what to whom, but that
21 is -- it was presented then, yes.

22 Q. You testified to the jury about this document on Friday.
23 Do you recall that?

24 A. Yes.

25 Q. And this is a document that says Netlist. It was

1 Netlist's presentation to Samsung. Isn't that right?

2 A. Oh, yes. This is a Netlist presentation. That is
3 correct.

4 Q. And it says, Samsung/Netlist partnership opportunity.
5 Isn't that right?

6 A. It does, sir.

7 Q. And Netlist was proposing a partnership with Samsung.
8 Correct?

9 A. I honestly don't know who proposed the opportunity first.
10 This presentation was our response to that.

11 Q. It was a response to what?

12 A. Again, we prepared a presentation to give to Samsung. I
13 don't know who approached who first. I do know that this
14 document we presented to Samsung.

15 Q. Okay. And, of course, you wanted -- of course, you
16 clearly wanted to have a partnership with Samsung because
17 Samsung is a reputable company with a great name. Isn't that
18 right?

19 A. Yes, sir.

20 Q. And if we go to page 43 of this document, you testified
21 about this page. Do you recall that?

22 A. Yes, sir.

23 Q. And you were specifically -- if you look here, it says
24 here -- you were providing testimony about the '870 Patent.
25 It's on the slide here. Do you see that?

1 A. Yes, sir.

2 Q. And the right-hand columns, it lists several patents that
3 Netlist says are related to hybrid memory products. Do you
4 see that?

5 A. I do, sir.

6 Q. And the hybrid memory products, as we all know from your
7 testimony on Friday, are the combinations of the DRAM and the
8 flash. Is that right?

9 A. That is part of the patents, yes.

10 Q. And then listed here is the '833 and the '831. These are
11 hybrid patents. Isn't that right?

12 A. They are listed under that column, sir.

13 Q. So you're telling the -- in this presentation, you are
14 communicating to Samsung that these two patents are the
15 combination patents where we have on the module flash memory
16 chips and DRAM chips. Isn't that right?

17 A. Okay, sir.

18 Q. Is that correct?

19 A. Correct.

20 Q. And, of course, the '831 and the '833 Patents are sort of
21 the great and the great-great -- I'm not sure I got all the
22 greats, but the grandparents of '918 and the '054. Is that
23 right?

24 A. That is a true statement.

25 Q. Now, there is no indication on the slide that -- that

1 these two patents are related to LRDIMM or RDIMM. Is that
2 right?

3 A. On this slide, they're under hybrid. That is correct.

4 Q. And no indication they're related to UDIMM or SODIMM.
5 Isn't that right?

6 A. Those words do not show on this slide, uh-huh.

7 MR. McKEON: If we go to page 19.

8 Q. (BY MR. McKEON) This is another slide you presented --
9 that was presented in the presentation. Is that right?

10 A. That is correct.

11 Q. And, by the way, you weren't at this presentation. Isn't
12 that right?

13 A. That is correct. I was not.

14 Q. And do you see here we have the Netlist IPs identified,
15 and Netlist IP, that stands for intellectual property. Isn't
16 that right?

17 A. Intellectual -- yes, uh-huh.

18 Q. And that, of course, is patents, is what we're talking
19 about in this courtroom. Right?

20 A. Yes, sir.

21 Q. And here you have Netlist IP, then you have the DRAM plus
22 the NAND, and that's the flash memory chips. Isn't that
23 right?

24 A. That is correct.

25 Q. So what you're communicating in this slide is that, hey,

1 we're a great company, Netlist, we have great IP, and the IP
2 is the DRAM plus the flash. Isn't that right?

3 A. That -- that is part of it.

4 Q. Okay. Now I want to go, if I can get the elmo, and I
5 want to put up a slide that you presented on Friday, if we
6 can. A second here, sir, if I can get this the right way.
7 Okay.

8 Do you recall this slide you presented, sir, from your
9 testimony?

10 A. Yes, sir.

11 Q. It's 2.37. Do you see that?

12 A. 2.37. Oh, yes, yes, sir.

13 Q. Okay. Now, this is a slide that you were talking about
14 what's called the restriction requirement in the Patent
15 Office. Do you recall that testimony?

16 A. I do, sir.

17 Q. And you highlighted these certain parts of the
18 restriction requirement, the document from the patent
19 examiner. Isn't that right?

20 A. That's correct.

21 Q. Okay. But in this document, sir, you didn't
22 highlight -- well, let's make sure we're on the same page.
23 The Roman two there says, claims 21 to 36 drawn to a power
24 module. Do you see that?

25 A. I do.

1 Q. And what you just told the jury was, oh, well, see, this
2 is a separate invention, the power module. I think that's
3 what you said to the jury. Isn't that right?

4 A. And I believe that's what this document from the PTO
5 tells Netlist, yes.

6 Q. Okay. But you didn't highlight this part of it. Right?
7 Because it says for volatile and non-volatile memory. Right?
8 So it's talking about --

9 A. Sure.

10 Q. It's talking a the combo. Right? That's the -- the
11 hybrid. Correct?

12 A. Well, it depends on how you read it. I mean, it's for
13 volatile and non-volatile. It doesn't mean that they have to
14 both be there.

15 Q. Well, sir, you understand that this application, this
16 became the '833 Patent. Do you understand that?

17 A. I do.

18 Q. All right. And the '833 Patent you just confirmed for
19 the jury was the combo. Right?

20 A. Yeah. That patent was, absolutely.

21 Q. So the patent examiner said, hey, you have a separate
22 invention here, and then you went and continued to prosecute
23 it and you got a patent for the combo. Right?

24 A. Well, my understanding is, is that the -- the words down
25 there it has separate utility means that it could apply to

1 other things.

2 Q. Sir, this went on to become the '833 Patent. Isn't that
3 right, sir?

4 A. That is correct.

5 Q. Whether it's other things or not, what you went on to
6 patent was the hybrid combination of a module with flash
7 memory chips and DRAM chips. Correct?

8 A. In the '833, absolutely.

9 Q. Thank you.

10 MR. McKEON: If I can get back. Thank you.

11 Q. (BY MR. McKEON) All right. Now, of course, you're
12 listed as an inventor on the '918 and the '054 Patents, but
13 you're not an inventor on the '339. Right?

14 A. That is a correct statement.

15 Q. Because you were not an inventor, you wouldn't know about
16 it. Is that right?

17 A. That's not correct.

18 Q. Okay. Well, let's -- let me ask you another question.
19 Because you're not an inventor, you had to talk to somebody
20 about the '339 Patent in order to testify about it. Right?

21 A. That -- that is correct.

22 Q. Okay. Let's get your deposition, if I can --

23 A. Uh-huh.

24 Q. -- and I want you to read -- go to the same binder, sir,
25 at 97, page 14 to 98 -- sorry, 97, line 14, to 98:13. And are

1 you there?

2 A. Yes, sir.

3 Q. Just read that to yourself.

4 A. Okay. I've read down to line 25.

5 Q. Okay. And did you go to 97:14 to 98:13?

6 A. Oh, I'm sorry, sir.

7 Q. Okay.

8 A. Yes, sir.

9 Q. And because you're not an inventor -- let me ask this
10 question again. Because you're not an inventor with respect
11 to the '339 Patent, you wouldn't know about it. Is that
12 right, sir?

13 A. I know about the concepts. I don't know -- I wasn't part
14 of drafting the claims or reading the application. That is
15 correct. But the concepts I know about.

16 Q. All right. Now, that same is true with the '060 and
17 '160. Right? You're not an inventor and you happened to talk
18 to someone else to find out what the patents were all about.
19 Isn't that right?

20 A. Yeah, that's absolutely correct.

21 Q. The same is true for '339. Correct?

22 A. That is correct, the '339.

23 Q. Now, sir, earlier you presented the '918 Patent, and part
24 of what you said was your invention on Friday was the idea of
25 taking this power management and taking and sort of from the

1 mother board and putting it on the actual DIMM module. Isn't
2 that right?

3 A. That is correct.

4 Q. Right. And you said to the jury that this was a big deal
5 kind of -- kind of thing, is what you told the jury. Isn't
6 that right?

7 A. I did.

8 Q. Okay.

9 MR. McKEON: Now, I want to pull up DTX 28.

10 Q. (BY MR McKEON) Now, this is a document, sir, that's from
11 an industry organization and a bunch of people show up and
12 they talk about technology. You familiar with that?

13 A. I believe this is a JEDEC document, sir.

14 Q. Well, yes. And you understand that -- that Mr. Martinez
15 was attending this meeting.

16 MR. McKEON: If we can go to page 2, if we can show
17 Mr. Martinez. Highlight Mr. Martinez down here.

18 Q. (BY MR. McKEON) Do you see here he's from Netlist?
19 Right?

20 A. That is correct.

21 Q. So he was at this meeting. It's just a record of who was
22 there. Right?

23 A. I believe -- could you scroll up real quick?

24 Q. Sure.

25 MR. McKEON: If we can scroll up.

1 THE WITNESS: The meeting minutes, yes.

2 Q. (BY MR. McKEON) Yes.

3 A. Yes, sir.

4 Q. And Micron, Qualcomm, Renesas, Samsung was there. A
5 whole lot of industry people show up at this meeting. Isn't
6 that right?

7 A. That is correct.

8 MR. McKEON: And if I can go to page 4 of this
9 document, and about the bottom there, highlight this.

10 Q. (BY MR. McKEON) This is one of the things that was
11 discussed at the meeting was DDR5 VR on DIMM requirement for
12 SODIMM and UDIMM. Do you see that?

13 A. I do.

14 Q. And SODIMM and UDIMM are accused products in this case.
15 Isn't that right?

16 A. That is correct.

17 Q. And the DDR5 VR on DIMM means -- this means putting the
18 voltage regulator on the DIMM module. Isn't that right?

19 A. That is correct.

20 Q. And that's what you claim you invented in the '918
21 Patent. Isn't that right, sir?

22 A. That is correct.

23 MR. McKEON: And if I could go to the page 5, which
24 is the next page.

25 Q. (BY MR. McKEON) And Netlist at this meeting commented on

1 this proposal and whether, sir, this is a good thing or a bad
2 thing. Do you understand that?

3 A. What's a good thing or a bad thing?

4 Q. Well, let's highlight what Netlist said because you
5 understand this is a record of what Netlist said at the -- at
6 this meeting with the industry folks. Do you see that? We've
7 highlighted it?

8 A. I do, sir.

9 Q. Let's read it together. "VR on commodity DIMM has no
10 technical benefit and is an additional cost. VR could
11 generate additional heat on the DIMM. There is no reason to
12 require it on the DIMM." Do you see that?

13 A. I do.

14 Q. "Netlist was present but did not add further
15 comments." Do you see that?

16 A. I do.

17 Q. So here you're saying to the industry and the people that
18 were there talking about this stuff, this is a bad idea, don't
19 do it. Isn't that right?

20 A. At the time of this meeting, that was correct.

21 Q. Okay. And, in fact, if you go to the next page, page 6,
22 and you see Samsung was at the meeting, too.

23 MR. McKEON: Blow that up.

24 Q. (BY MR. McKEON) And you see Samsung says, while this
25 proposal shows advantages in the terms of pin count, et

1 cetera, also it shows limited advantages in terms of DRAM chip
2 design and test and will not provide any further benefit for
3 tighter DRAM voltage tolerances. Do you see that?

4 A. I -- I see it.

5 Q. And it goes on to say, VR on client aggravates complexity
6 of the module design, increases BOM and test costs, increases
7 power consumption and can result in thermal throttling and
8 increased EMI due to absence of ground pins. Do you see that?

9 A. I do.

10 Q. Okay. So Samsung, they don't agree in this courtroom,
11 but actually at this meeting they agreed. They both said,
12 hey, this VR on DIMM is a bad idea. Isn't that right?

13 A. At the time of this meeting, that is correct.

14 Q. And if we go to page 8, there's another discussion with
15 the DDR5 SODIMM pin assignments on PMIC on client. Do you see
16 that?

17 A. I do.

18 Q. That's the same issue--right?--putting the PMIC on the --
19 on the module. Isn't that right?

20 A. That is correct.

21 Q. And if we go to the -- down a little bit, again Netlist
22 repeats its criticism. It says, VR on the DIMM has no
23 technical benefit and is an additional cost. VR could
24 generate heat on the DIMM. There's no reason to require it on
25 the DIMM. Do you see that?

1 A. I do.

2 Q. So Netlist is saying, hey, this is a bad idea to the
3 industry, we shouldn't do this. Isn't that right?

4 A. Yeah. At the time for commodity DIMMs, that's what we
5 believed.

6 Q. It doesn't say anything about the commodity DIMMs here,
7 sir. Correct?

8 A. Oh. Well, it's implied. UDIMMs and SODIMMs are
9 commodity DIMMs.

10 Q. Sir, this phrase here, when you were at the meeting, the
11 record reflects Netlist's observation that the DIMMs shouldn't
12 be -- that VR should not be on the DIMM. It doesn't say
13 anything about commodity. Isn't that right? Right here in
14 this record.

15 A. In that record, no, it's -- yeah, okay.

16 Q. Thank you.

17 MR. McKEON: You can pull that down.

18 Q. (BY MR. McKEON) Now, you testified on Friday about the
19 '060 and '160 Patents. You recall that testimony. Right?

20 A. Yes, sir.

21 Q. I want to pull up a slide. This is PDX 2.63, a slide
22 that you presented to the members of the jury. Do you recall
23 this, sir?

24 A. I do, sir.

25 Q. This is a slide that, you know, describing the -- the

1 invention -- the inventions of these two patents. Right?

2 A. Yes, sir.

3 Q. And you described the invention here as relating to
4 unique structures within the array dies called TSVs, is what
5 you told the jury. Do you recall that?

6 A. That is one of the structures, yes.

7 Q. And TSV is short for through-silicon via. Right?

8 A. That is correct.

9 Q. Meaning that they have connections, the via goes -- goes
10 through the dies as they go up. Isn't that right?

11 A. It goes -- yes, it goes through the -- through the dies.

12 Q. But Netlist did not invent TSVs. Correct?

13 A. That is correct.

14 Q. And Netlist was certainly not the first company to use
15 stacked chips. Isn't that right?

16 A. That is correct.

17 Q. That was before -- before Netlist. Isn't that right?

18 A. Sure, uh-huh, yes, sir.

19 Q. And Netlist didn't invent stacking chips with TSVs going
20 through them. Correct?

21 A. That is correct.

22 Q. And, in fact, if we could go -- take -- pull up the '060
23 Patent, and if we can go to -- let's see here. The
24 patent -- and also column 1, line 30 to 33.

25 This is figures 1 and 1B from the patent, and it actually

1 shows the stacking of the DRAM with the TSVs going through it.

2 Isn't that right?

3 A. That is correct.

4 Q. And that's prior art, and the inventors told the Patent
5 Office this is prior art, this is not what we invented. Isn't
6 that right?

7 A. That's absolutely correct.

8 MR. McKEON: Let's pull up -- if you can pull that
9 down, FTX 2.65.

10 Q. (BY MR. McKEON) This is another slide you presented on
11 Friday. Do you recall that?

12 A. Yes, sir.

13 Q. Okay. And described -- you had these labels on the side,
14 data port and bypass. Do you see that?

15 A. I do, sir.

16 Q. Now -- and the labels, by the way, the labels are not in
17 the patent like this. Isn't that right?

18 A. I'm sorry. The labels are not in the patent?

19 Q. Yeah. It says data port -- this particular image on the
20 right that you've created here, that's not in the patent
21 itself. Isn't that right?

22 A. It's -- well, I think it's like pulled out and additional
23 annotation added.

24 Q. Okay. Well, let me ask you this question, sir.

25 A. Yeah.

1 Q. In the labels on the right, you labeled something and
2 called it bypass. Do you see that?

3 A. Yes.

4 Q. Let's get your testimony from Friday at transcript 223,
5 10 to 16. And this is your testimony from Friday, and you say
6 here, when they're not connected, they have what's called a
7 bypass. Do you see that?

8 A. Yes, sir.

9 Q. And you testified that -- yeah, so let's go back and look
10 at your testimony.

11 MR. McKEON: Let's go back to the slide, if I can.

12 PDX 2.65. Okay.

13 Q. (BY MR. McKEON) Now, sir, did you know that the word
14 'bypass' does not appear anywhere in the '060 and '160
15 Patents?

16 A. That could be true.

17 Q. And, in fact, the patent describes this as a
18 through-hole. Do you see that? You have it highlighted here
19 on the left.

20 A. Uh-huh, yes, sir.

21 Q. The patent doesn't use the word 'bypass'. Correct?

22 A. Okay. I agree.

23 Q. You agree with that. Yes. And let me ask you this
24 question, sir. Did the word 'bypass', was that your idea to
25 put that word on this slide?

1 A. I honestly don't remember whose idea it was.

2 Q. And if we can get your testimony up from Friday as well,
3 225 at 8 to 11.

4 THE COURT: Counsel, approach the bench, please.

5 MR. McKEON: Yes.

6 (The following was had outside the hearing of the
7 jury.)

8 THE COURT: Let me just remind both of you, and if
9 you will when it's convenient, share it with your co-counsel,
10 I've told the jury they will not have the transcript of the
11 testimony --

12 MR. McKEON: Okay.

13 THE COURT: -- to consider during deliberations.

14 MR. McKEON: Okay.

15 THE COURT: So if you want to talk about what you
16 said Friday, but let's try to avoid the word 'transcript' per
17 se.

18 MR. McKEON: Got it.

19 MR. SHEASBY: I also want to flag that last question
20 was an attempt to elicit the privilege, whose idea was it to
21 bypass. I didn't raise an objection, but he should not ask --
22 obviously those slides were prepared with counsel, and he
23 should not be asking any questions --

24 THE COURT: Well, you didn't object so we are going
25 to move on.

1 MR. SHEASBY: Thank you.

2 (The following was had in the presence and hearing
3 of the jury.)

4 THE COURT: All right. Let's proceed, please.

5 Q. (BY MR. McKEON) Now, sir, with respect to the word
6 'bypass', are you aware that this word, it doesn't appear in
7 your patents, but it's a word that the jury's going to see
8 later in this trial? It's a word that's actually used in
9 Samsung's confidential documents. Did you know that?

10 A. Actually I did not know that.

11 Q. You did not know that. And it's a word that's not in
12 your patent, and this is your slide that puts that word on the
13 slide. Isn't that right?

14 A. That is true.

15 Q. And the word didn't come from you. Isn't that right,
16 sir?

17 A. I don't believe so. I honestly don't remember.

18 Q. All right.

19 MR. McKEON: Now we can take that down.

20 Q. (BY MR. McKEON) Now, of course, you're the VP of
21 engineering at Netlist. You testified about that earlier.
22 Isn't that right?

23 A. That is true.

24 Q. And you've been in the memory business a long time.
25 Isn't that right, sir?

1 A. That is correct, sir.

2 Q. And the jury has already heard a lot in this trial about
3 DRAM and all about it now. Right?

4 A. As much as they can probably take.

5 Q. They've heard a lot. Right? And, of course, we know
6 that stands for, I think it's been said in this courtroom,
7 dynamic random access memory. Isn't that right?

8 A. That is correct.

9 Q. Now, let me ask you a question. Well, DRAM, if I have
10 DRAM in my hand, that DRAM, you agree, contains sort of
11 billions -- these days the technology is so incredible, it
12 contains billions of memory cells. Isn't that right?

13 A. Absolutely, yes.

14 Q. And a memory cell, it holds a 1 or a 0. Isn't that
15 right?

16 A. Yes.

17 Q. And that's the data. That's your photographs, emails,
18 your pictures is the 1 or the 0. Isn't that right?

19 A. Uh-huh. That's how it's encoded.

20 Q. And, of course, a memory cell comprises a transistor
21 that's wired to a capacitor. Isn't that right?

22 A. That's the very, very basic structure, yes.

23 Q. Okay. But let's put a --

24 MR. McKEON: If I can get the elmo.

25 Q. (BY MR. McKEON) And what I put on here is a basic, sort

1 of textbook version of what a sort of memory cell looks like.

2 Isn't that right?

3 A. That is correct.

4 Q. And it's got this -- this transistor we talked about,
5 it's got that capacitor which stores, and they're all kind of
6 connected by wires together. Isn't that right?

7 A. Well, they're -- they're connected by -- by copper, but
8 yeah.

9 Q. Copper. Right? It's got to be metal because the
10 electrons are flowing through that. Right?

11 A. That's correct.

12 Q. And, of course, you agree that a DRAM has circuits.
13 Correct?

14 A. All electronic components have circuits.

15 Q. Right. And that includes a DRAM. That has circuits.
16 Isn't that right?

17 A. All components, yeah.

18 Q. And, of course, sometimes the word 'integrated circuit'
19 is used in the industry as well. Isn't that right?

20 A. That is a true statement.

21 Q. And sometimes people refer to that as an IC. Isn't that
22 right?

23 A. That is correct.

24 Q. And sometimes DRAM is used in the -- in the same way that
25 DRAM IC is used. They are used interchangeably. Isn't that

1 right?

2 A. That's possible.

3 Q. Okay.

4 MR. McKEON: Let's get, if we can, the JTX 62.

5 Q. (BY MR. McKEON) And you recognize this as the annual
6 report, or 10-K, that Netlist filed? Isn't that right?

7 A. Where's the date?

8 MR. McKEON: Blow it up a little.

9 Q. (BY MR. McKEON) It's fiscal year ending January 2022.
10 Do you see that?

11 A. Very good, yes, sir.

12 Q. This is a Netlist document that they filed with the
13 Securities & Exchange Commission, that they have to do -- all
14 public companies have to do it every year. Isn't that right?

15 A. That is a true statement.

16 Q. In fact, if we go to page 22 of the document and blow up
17 the signature, Mr. Chung Hong, who goes by Mr. Chuck Hong. Is
18 that right?

19 A. That is true.

20 Q. And he is the president and CEO of the company. Isn't
21 that right?

22 A. That is true.

23 Q. And he is, of course, verifying that everything in this
24 document is accurate. Isn't that right?

25 A. He is.

1 Q. Okay.

2 MR. McKEON: And if we go to page 14 of the
3 document, and we can blow up the middle here.

4 Q. (BY MR. McKEON) And just to confirm what we were talking
5 about here --

6 MR. McKEON: If we can blow up net sales and then
7 the DRAM IC or DRAM.

8 Q. (BY MR. McKEON) In this document in various places,
9 including here, it says -- it uses interchangeably the word
10 'DRAM IC' and DRAM. Isn't that right?

11 A. For this document, yes.

12 Q. And reflecting the notion that, of course, everybody in
13 the industry knows that a DRAM has circuits in it. Right?

14 A. A DRAM has circuits in it.

15 Q. All right.

16 MR. McKEON: You can put that down.

17 Q. (BY MR. McKEON) Now, during your direct testimony, you
18 provided testimony about some emails and presentations between
19 Samsung and Netlist. Do you recall that?

20 A. There were several of those, I --

21 Q. Okay.

22 A. Yeah.

23 Q. Let me get grounded here so we can talk about it.

24 MR. McKEON: If we can pull up one of your
25 demonstratives, PDX 1.23.

1 Q. (BY MR. McKEON) Or, actually, this is a demonstrative, I
2 believe, that was used in opening. Do you recall this
3 demonstrative, sir?

4 A. Yes, sir.

5 Q. Okay. And this is PX 586. Do you see that document
6 referenced?

7 A. I do.

8 MR. McKEON: Okay. Let's pull that up.

9 Q. (BY MR. McKEON) And this is PX 586. And do you
10 recall --

11 MR. McKEON: If we can blow up so Mr. Milton can see
12 here.

13 Q. (BY MR. McKEON) This is a document you also discussed in
14 your direct testimony. Do you recall that?

15 A. That is true.

16 Q. And it's an email chain. If we can start sort of at the
17 bottom, it starts off with an email from Mr. Neal Knuth at
18 Samsung's Semiconductor to -- to Netlist. Do you see that?

19 A. I do.

20 Q. And Steven Yu is -- at the end he's identified -- he's a
21 Netlist employee. Right?

22 A. He is, yes.

23 Q. Is he with the company still?

24 A. He is.

25 Q. And Neal Knuth, Mr. Neal Knuth, is in California in the

1 same general area as Netlist. Isn't that right?

2 A. Yeah. I believe he's in northern California.

3 Q. And if we can go up further in the document, it says,
4 Check requesting a meeting. Do you see that?

5 A. I do.

6 Q. Let's go down. If I go down again, we can establish
7 that. So Mr. Knuth is requesting a meeting with Netlist.
8 Isn't that right?

9 A. That is correct.

10 Q. And they want to discuss the VRoD, voltage regulator on
11 DIMM. Isn't that right?

12 A. Yes, sir.

13 Q. Okay. And if we go up further, it says -- response from
14 Mr. Yu says, yeah, our team will be available on October 2nd.
15 Do you see that?

16 A. I do.

17 Q. Now, you understand that -- well, first of all, we've
18 already established, of course, that the Samsung and the
19 Netlist, these companies were -- had a joint development
20 agreement at this time. Right?

21 A. That is correct.

22 Q. Right. And this email is dated -- can we see the date on
23 it? It's 2019, and the joint development agreement was from
24 2015 to 2020. Isn't that right?

25 A. That is correct.

1 Q. And -- and you weren't suggesting to the jury there was
2 something sort of unusual about a party to a joint development
3 agreement reaching out to the other party to request to set up
4 a meeting to discuss sort of technical issues that are being
5 discussed in the industry. There's nothing unusual about
6 that. Right, sir?

7 A. Oh, not at all. In fact, I believe I testified it was
8 fairly common.

9 Q. Right. And these kind of things happen particularly with
10 partners where you are in a joint -- under a joint agreement,
11 working together. It's not unusual to say, hey, I want to
12 show up and have a meeting. Isn't that right?

13 A. Yeah, that is correct.

14 Q. Okay. Now, on Friday you said -- you said, well, I
15 don't -- the language you used was, you know, in fact, let me
16 read it. I don't have my readers today so I'm having a little
17 trouble reading my notes. You know, in fact, we were meeting
18 with Samsung fairly frequently. So I assume that this meeting
19 did occur. Do you recall that testimony?

20 A. I do recall that.

21 Q. Okay. And you told -- you assume it occurred --
22 Thank you.

23 But, sir, you don't know, you don't really know if that
24 meeting happened. Right?

25 A. That is a correct statement. I wasn't at this meeting,

1 so I can't -- and we couldn't find the notes. So I can't say
2 that.

3 Q. Okay. I'm going to, if I may --

4 MR. McKEON: May I hand Mr. Milton a document?

5 THE COURT: Is this some pre-admitted exhibit or --

6 MR. McKEON: It's not, Your Honor. It's a document
7 to establish the occurrence of the meeting or not.

8 THE COURT: Why don't you disclose it to opposing
9 counsel. And if there's a problem, I'll hear about it. If
10 not, we'll move forward.

11 MR. SHEASBY: May I approach, Your Honor?

12 THE COURT: Approach the bench.

13 (The following was had outside the hearing of the
14 jury.)

15 MR. SHEASBY: This is a screenshot. It was never
16 disclosed to us in discovery or at any point in time. And so
17 there's nothing -- it's not in evidence. It can't be used
18 to -- it's not even authentic. We have no idea whether --
19 where it came from, whether it was altered for the purposes of
20 the deposition. It refers to people who are not even on this
21 email chain. None of this can be used.

22 THE COURT: This looks like a surprise out of left
23 field, Mr. McKeon.

24 MR. McKEON: Your Honor, this -- we had the witness
25 on Friday. I didn't know the witness would be talking about

1 this stuff on Friday, but he's not even been to any of these
2 meetings. So we what we did is we obtained a record showing
3 this is a picture of a phone --

4 THE COURT: When did you get this?

5 MR. McKEON: We got it over the weekend.

6 THE COURT: Why is Mr. Sheasby hearing about it in
7 open court for the first time?

8 MR. McKEON: Well, Your Honor, the witness confirmed
9 his testimony that he didn't know about this, whether this
10 meeting occurred and --

11 MR. SHEASBY: That's not impeachment.

12 MR. McKEON: Well, I'm going to -- I'm going to show
13 him this document so he can confirm that --

14 MR. SHEASBY: Look, look, we don't even know if this
15 document --

16 THE COURT: We're not going to use this.

17 (The following was had in the presence and hearing
18 of the jury.)

19 THE COURT: Let's proceed.

20 MR. McKEON: Thank you, Your Honor.

21 Q. (BY MR. McKEON) All right.

22 MR. McKEON: Now, if we can pull up 1756, PX 1756.

23 Q. (BY MR. McKEON) This is another document, sir, that you
24 discussed on -- on Friday. Do you recall that?

25 A. Yes, sir.

1 Q. Okay. And you see your page 1 of this email, there's an
2 attachment. Right? And if you look at the email, it's from
3 Mr. Kim at PriceWaterhouseCoopers to Ms. Gail Sasaki at
4 Netlist. Do you see that?

5 A. I do, sir.

6 Q. And there's an attachment. Do you see that? It says
7 here, Attachments?

8 A. Yes, sir.

9 Q. And we go down -- if we go down to the document, what
10 this is, this is an email --

11 MR. McKEON: Go up one page. Yeah.

12 Q. (BY MR. McKEON) The initial email is from Samsung, and
13 Samsung is sending this attachment to Mr. Kim at
14 PriceWaterhouseCoopers, and they say, I'm sending our
15 company's response to the questions. Do you see that?

16 A. I do.

17 Q. Okay. And you on Friday testified about some of these
18 answers to the questions. Do you recall that?

19 A. Yes, sir.

20 Q. Okay. And if you go up, just to confirm that
21 PriceWaterhouseCoopers then takes that email and sends it to
22 Ms. Sasaki at Netlist. Is that right?

23 A. Yes, sir.

24 Q. Okay. And it says here, Samsung prepared the attached
25 additional information for submission to the authorities, and

1 I told them to wait until we have provided them with the green
2 light to proceed. Do you see that?

3 A. I do, sir.

4 Q. Okay. So trying -- saying, hey, the government asked for
5 this, and we gave them -- we have proposed answers, but you
6 know what, Netlist, we want to check with you to make sure
7 that you're okay with us going forward with these answers.
8 Correct?

9 A. That is correct.

10 Q. Okay.

11 MR. McKEON: And if we can go to the attachments so
12 it's at page 4. And if we can highlight Question 2.

13 Q. (BY MR. McKEON) Now, if you recall a discussion relating
14 to the bottom paragraph, it says, Accordingly -- this is a
15 question that's being answered, and the question is, Can you
16 tell us why Samsung Electronics signed the joint development
17 and license agreement? Do you see that?

18 A. I do, sir.

19 Q. And the bottom clause, Accordingly, Samsung came to
20 resolve the risk of patent infringement for the period before
21 and after the JDLA and used their patents without entering
22 into a separate royalty agreement or payment. Do you see
23 that?

24 A. Yes, sir.

25 Q. Okay. But you also see the top which wasn't discussed,

1 and it says, The main reason for entering into the agreement
2 with Netlist was that we wanted to resolve the risk of Netlist
3 filing a patent infringement suit under the circumstances of,
4 among others, their assertion of infringement by Samsung with
5 respect to the patents.

6 Do you see that?

7 A. I do, sir.

8 Q. So what the answer is saying is we don't want to do
9 litigation, we don't want to be in court, we want to have
10 cooperation, so we did the JDLA. So we don't want to be in
11 court, we want to work, move forward under a JDLA with
12 Netlist. Isn't that really what that's saying?

13 A. That's what it's trying to say.

14 Q. Okay. I mean, you weren't suggesting to the jury by
15 showing this to them that -- that Samsung somehow knew it was
16 infringing the patents and therefore it had to take a license.
17 This doesn't say that, does it, sir?

18 A. I mean, it says what it says. They wanted to resolve the
19 risk of patent infringement.

20 Q. All right. Let's go to page 7.

21 A. Uh-huh.

22 Q. And this is an answer -- Question 7 you hadn't shown the
23 jury. It's another question, Can you briefly tell us which
24 patents held by Netlist before and after the signing of the
25 JDLA have been mainly used by Samsung? And it goes on, you

1 know, what products Samsung produced and sold using the
2 technology. Do you see that?

3 A. I do.

4 Q. And the answer is, The patent issue was resolved by
5 signing the cross license. Right? Perpetual cross license.
6 So Netlist's patent list and company's use of the technologies
7 weren't even checked. Right?

8 A. They weren't -- well, we -- we sent them a list of our
9 patents and what products they applied to.

10 Q. Well --

11 A. So --

12 Q. Focus on my question, sir.

13 A. Uh-huh.

14 Q. This document, the -- the answer to this question is
15 indicating that Samsung hadn't even looked at it because they
16 had a cross license. So we didn't need to review whether we
17 were using the patent or not. Isn't that right? Isn't that
18 what the answer says?

19 A. Well, I mean, I don't know what Samsung did or did not
20 do. All I know is that we had already sent them stuff. So it
21 says here -- I'm sorry -- the technologies were not checked.
22 They -- they wouldn't have to. We gave them the information.

23 Q. Well, sir, this is the time period where Samsung licensed
24 your patents. Right?

25 A. Yeah, this is before that was -- that ended, yes.

1 Q. Right. And we're going to get to the -- the
2 communications on that. But you understand that the
3 communication here, at least from Samsung's perspective, was
4 --

5 A. Uh-huh.

6 Q. -- we didn't need to check because we took a license. Is
7 that fair?

8 A. Yeah. All the information was provided, so sure.

9 Q. Let's go to another document, PX 1663. And this is
10 another document you discussed on Friday. Do you recall that?

11 A. Yes, sir.

12 Q. And we have, again, an email from Samsung to Ms. Gail
13 Sasaki at Netlist. Isn't that right?

14 A. That is correct.

15 Q. And like the document we just saw, again the date of this
16 document is August 2019, the JDLA, the companies were still
17 working together under the JDLA. Correct?

18 A. That is correct.

19 Q. And you say, again, the request by Samsung, Please send
20 us Netlist's written consent with signature on submission to
21 NTS by Monday next week. Do you see that?

22 A. I do.

23 Q. And, again, here Samsung is requesting, hey, Netlist,
24 what do you think of this? Let us know if we can move
25 forward. Isn't that right?

1 A. Yes, sir.

2 Q. Okay. And then if we go to the page 7 here, there's a
3 reference to Company --

4 MR. McKEON: If you go to 7 of the document. There
5 we go. Go down.

6 Q. (BY MR. McKEON) There's a reference to Company N. Do
7 you see that? Our company did not check whether Company N --
8 do you see that Company N?

9 A. I do.

10 Q. And that's Netlist. Isn't that right?

11 A. That is correct.

12 Q. And you were here for opening arguments in this case,
13 weren't you? Opening statements?

14 A. I'm sorry. Complete your question, sir.

15 Q. Yeah. You were here for opening statements?

16 A. I was, sir.

17 Q. Okay. And do you recall Netlist's counsel telling the
18 jury that, well, sometimes, especially in federal courts, we
19 are allowed to pull back the curtain and see what Samsung
20 actually thinks. He's talking about discovery, and he goes,
21 you know what, in discovery we get to pull back the curtain
22 and -- and see what -- really what's going on. Do you recall
23 that, him saying that to the jury?

24 A. I do, sir.

25 Q. And he said, This is an example of it. And he cites the

1 PX 1756. Did you know that?

2 A. I don't recall the exact --

3 Q. I can tell you he cited to PX 1756 which was the
4 document -- not this email, but the email prior to that. Do
5 you recall that, 1756 we just looked at?

6 A. Okay.

7 Q. Okay. And --

8 THE COURT: Counsel, approach the bench, please.

9 (The following was had outside the hearing of the
10 jury.)

11 THE COURT: Let me renew my discussions with you
12 here at this juncture. It seems like we're barreling toward
13 who's at fault for the termination of the JDIA, and we're not
14 going there.

15 MR. McKEON: No. I have no intention of going
16 there. I was going to say that -- my reason of this cross is
17 that these documents were communicated to Netlist, they were
18 not hiding anything, they weren't like -- he suggested in
19 opening we were hiding them. I'm not touching that.

20 THE COURT: All right.

21 MR. McKEON: I'm not touching that.

22 THE COURT: I just want to make sure we're all
23 clear.

24 MR. McKEON: I'm not touching that.

25 MR. SHEASBY: Your Honor, he talked about their

1 interactions with this perpetual license and talked about
2 that, and I would like a curative instruction emphasizing the
3 agreement is terminated and there are no rights on a
4 going-forward basis.

5 THE COURT: You can address it on redirect. The
6 JDLA is in evidence. You can make it clear that it's not
7 forever and ever, no matter what.

8 MR. SHEASBY: Thank you, Your Honor.

9 THE COURT: We can address it later.

10 MR. McKEON: Thank you, Your Honor. Thank you.

11 THE COURT: All right.

12 (The following was had in the presence and hearing
13 of the jury.)

14 THE COURT: All right. Let's proceed.

15 MR. McKEON: Thank you, Your Honor.

16 Q. (BY MR. McKEON) All right. So just so we're clear that
17 -- and also another thing that was said in opening was, you
18 know, the code word, Company N, was a secret code word at
19 Samsung. Do you recall Mr. Sheasby, my colleague over here,
20 saying that in opening?

21 A. Yeah. I don't remember if he said secret or not, but, I
22 mean, that's common we have code words for company, yeah.

23 Q. But what you do agree with me, sir, is that these last
24 two emails--right?--these last two emails weren't secret, sir?
25 Right? And there wasn't no opening the curtain. Samsung sent

1 the attachment to Netlist saying, please, Netlist, let us know
2 what you think, we want to move forward with this document,
3 please review it. Isn't that right?

4 A. From the emails, yes.

5 Q. The curtain was wide open--right?--before any litigation.
6 Samsung was just getting confirmation from Netlist. That's
7 fair, isn't it?

8 A. For -- for these documents? Yes.

9 Q. And the code word, you know, the N code word, wasn't much
10 of a secret. Right? We're telling Netlist about it. Isn't
11 that fair?

12 A. Yeah, that is true.

13 Q. All right. Now, I'd like to put up another document you
14 discussed on Friday, PX 446. And this is a document, an
15 email, you discussed with counsel. Do you recall that?

16 A. Yes, sir.

17 Q. Okay. And it's an email from -- from Jibum Kim to -- to
18 Samsung. Jibum Kim is a Netlist employee that actually lives
19 and works in Korea. Isn't that right?

20 A. I actually don't know where he lives.

21 Q. Okay. But he sent this email to Samsung. And you had
22 discussions on Friday, and I want to talk to you about -- a
23 little about the content. There's a chart, an example, and
24 that was patents and products they cover. Do you see that?

25 A. Yes, sir.

1 Q. Okay. And the point here is that this was sent to
2 Samsung. And at this time period, of course, this is when
3 Samsung was licensed to all of the patents. Isn't that right?

4 A. For 2016, that is correct.

5 Q. Okay. And if we can go to page 2 of the document, and
6 you recall discussing this with the members of the jury on
7 Friday?

8 A. Yes, sir.

9 Q. Okay. And you see you reference to '006, but you also
10 reference to '833 and the '831. Do you see that?

11 A. Yes, sir.

12 Q. Okay. And what it indicates here on the right is that
13 the '831, the covered product is an SSD. Do you see that?

14 A. I do, sir.

15 Q. That stands for solid state drive. Correct?

16 A. That is correct.

17 Q. And they're not accused products in this case. Isn't
18 that right, sir?

19 A. In this case? No.

20 Q. And the '831, sir, is not identified as relating to DDR4,
21 RDIMM, or LRDIMM. Is that right?

22 A. That is correct.

23 Q. The NV DIMM product that's identified in the slide here,
24 that's the hybrid product we've been talking about in this
25 case. Isn't that right?

1 A. That is correct.

2 Q. And '833 is identified as corresponding to the hybrid
3 product. Correct?

4 A. That is correct.

5 Q. All right. Now, sir, in the opening, Mr. Sheasby made
6 the statement that: And the infringement in this case does
7 not relate to any of the behavior from before the termination
8 of the relationship between the parties. The infringement in
9 this case is about Netlist's behavior after the agreement is
10 terminated.

11 Do you recall that statement from counsel in opening?

12 A. I don't believe he said Netlist's behavior after
13 termination.

14 Q. That's what was recorded in the transcript, but I think
15 that was a mistake. Right? He said, infringement in this
16 case is about Samsung's behavior.

17 A. That's what -- that's what I believe, sir.

18 Q. All right. So we're all on the same page in this
19 courtroom that, in light of the statement, the infringement in
20 this case does not relate to the behavior before the
21 termination. Correct?

22 A. Correct.

23 Q. And that was in July of 2020. Correct?

24 A. Yes.

25 Q. Okay. And the emails we just talked about and the

1 documents that were sent to Samsung, including this one, that
2 was during a period of time when there was a license and it
3 was before the termination. Isn't that right?

4 A. Yes, that is correct.

5 Q. So those documents don't have any relevance in this case
6 at all right now. Isn't that right, sir?

7 MR. SHEASBY: Your Honor, I object. That calls for
8 a legal conclusion. They've already been admitted as
9 relevant.

10 THE COURT: Sustained.

11 Q. (BY MR. McKEON) All right. Those documents, sir, don't
12 bear on the question of Samsung's infringement in light of
13 counsel's opening. Isn't that right?

14 MR. SHEASBY: Your Honor, objection, calls for a
15 legal conclusion.

16 THE COURT: This witness is not here to say what is
17 and isn't infringement.

18 MR. McKEON: Okay.

19 THE COURT: Sustained.

20 MR. McKEON: Thank you, Your Honor.

21 Q. (BY MR. McKEON) The documents, you agree with me they
22 are before the termination of the agreement while Samsung was
23 still licensed. Isn't that right?

24 A. That is correct.

25 Q. All right. Well, thank you, Mr. Milton. I appreciate

1 your time.

2 MR. McKEON: Pass the witness, Your Honor.

3 THE COURT: Is there redirect from the Plaintiff?

4 MR. SHEASBY: Yes, Your Honor.

5 THE COURT: Let's proceed.

6 MR. SHEASBY: Good morning, ladies and gentlemen of
7 the jury.

8 REDIRECT EXAMINATION

9 BY MR. SHEASBY:

10 Q. Good morning, Mr. Milton.

11 A. Good morning, Mr. Sheasby.

12 MR. SHEASBY: Mr. Huynh, can we have PDX 14?

13 Q. (BY MR. SHEASBY) I want to first talk about the patented
14 technology relating to on-module power management. This is PX
15 621. Who was presented with this document?

16 A. This was presented to Dr. J.B. Lee.

17 Q. That's a Samsung employee?

18 A. Samsung Doctor J.B. Lee.

19 Q. He is now the president of Samsung. Is that correct?

20 A. That is correct.

21 Q. And you were describing your technology as on-module
22 power distribution. Is that correct?

23 A. Intelligent on-module power distribution. Correct.

24 Q. The phrase 'intelligent on-module power management
25 distribution' was not a phrase that was -- was the concept of

1 intelligent on-module power distribution created for this
2 lawsuit?

3 A. No, sir.

4 Q. And I want to talk about Mr. -- my brother, opposing
5 counsel, made these statements about --

6 THE COURT: Counsel, don't tell the jury what you
7 want to talk about. Ask the witness a question.

8 Q. (BY MR. SHEASBY) But you were here when opposing counsel
9 was making reference to the presence or absence of the word
10 'intelligent' in the specification of the patents. Is that
11 correct?

12 A. That is correct.

13 Q. I want to show you a slide. Can you describe how this
14 technique, which is from your patent at column 23, lines 1
15 through 27, adds intelligence to on-module power management?

16 MR. McKEON: Objection, leading, Your Honor.

17 THE COURT: Restate the question, please.

18 Q. (BY MR. SHEASBY) Can you -- can you describe
19 whether -- is there any relationship between the concept of
20 intelligent on-module power management and this passage at 23,
21 1 through 27, of your patent?

22 A. Is there? Yes, there is.

23 Q. Can you describe it?

24 A. So what we're doing is adding -- if we start at the top
25 there with the underlined -- the logic element and FPGA, what

1 that is is a controller that we combine with the regulators
2 that provide the voltage to provide additional control for
3 being able to detect failures that might occur and take other
4 action in the event. That's the intelligent or control
5 function.

6 Q. And I noticed that makes reference to the presence of
7 flash in the controller. Do you see that?

8 A. I do see that, sir.

9 Q. Does Samsung's DDR5, as you've examined it, use flash
10 within the controller?

11 A. It does, sir.

12 Q. And I want to go back one slide. Does the patent
13 specification provide any information or designs in which
14 flash is not present on the memory subsystem?

15 A. Yes. That's present in -- in the two figures that you
16 have as well as there's some discussion of those figures in
17 the specification.

18 Q. So this is your patent, and it's figures 15A and 15B, and
19 what is the memory subsystem comprised of?

20 A. So in this case it's the -- what we call the RCD, that's
21 the box 1160, the DRAM in 1180, and then the control function
22 which is 1070.

23 Q. And in that control function, there can be some flash.

24 A. That is correct, sir.

25 Q. The -- counsel -- do you remember when counsel made

1 reference to the fact that NV doesn't have a relationship to
2 DDR5?

3 A. Yes, sir.

4 Q. And hybrid doesn't have a relationship to DDR5?

5 A. That's correct.

6 Q. Do you have -- from a technology standpoint, is there a
7 relationship between Samsung's DDR5 and NV and hybrid?

8 A. I believe that that is true, yes.

9 Q. Why?

10 A. So for two reasons. One is, is that the -- the PMIC that
11 we've been talking about, which is the -- which includes the
12 power management, does have flash in it. That flash is used
13 to save information about the state of the system so it could
14 be debugged in the event of failure, which is something that
15 the -- we actually also did on the original product we had.

16 And the second thing is, is that this power management
17 functionality was taken from, you know, as been stated, a
18 hybrid product. So by putting that on the standard modules,
19 we've essentially made those hybrid products. They're
20 incorporating things that we use from hybrid into now today
21 standard products.

22 Q. And by standard -- do you mean Samsung's RDIMM, UDIMM,
23 SODIMM?

24 A. I'm sorry. Yes, sir.

25 MR. SHEASBY: Your Honor, may I approach briefly?

1 THE COURT: Approach the bench.

2 (The following was had outside the hearing of the
3 jury.)

4 MR. SHEASBY: Counsel showed a passage from the
5 JEDEC proceedings talking about the fact that DDR5 was not
6 useful for SODIMMs and RDIMMs.

7 There's a subsequent presentation that's been disclosed
8 and produced in the case. Because of exhibit limits, we could
9 not produce it. We could not designate it.

10 I'd like permission to publish it to the jury. It's
11 obviously not hearsay; it's been disclosed to them. And I'm
12 removing all reference to JEDEC, just to rebut what counsel
13 had said about the value of --

14 THE COURT: Let me show it to Mr. McKeon.

15 MR. McKEON: I mean, you know, it's produced. It's
16 not on the exhibit list.

17 MR. SHEASBY: It's on the exhibit list. It's just
18 not on the narrowed list.

19 THE COURT: And you've got the references to JEDEC
20 redacted?

21 MR. SHEASBY: All redacted, Your Honor.

22 THE COURT: All right.

23 MR. SHEASBY: Thank you, Your Honor.

24 THE COURT: Without objection, it's granted.

25 (The following was had in the presence and hearing

1 of the jury.)

2 MR. SHEASBY: Can we turn to 2.87, Mr. Huynh?

3 Q. (BY MR. SHEASBY) Counsel for the Defendant represented
4 to the jury that there was a belief that on-module power
5 management was not ultimately going to be valuable for
6 commodity DIMMs. Do you recollect that?

7 A. Yes, sir.

8 Q. Do you recollect you kept saying -- he showed you
9 exchanges, and you kept saying, at the time, at the time?

10 A. That is correct, sir.

11 Q. Was there a subsequent consensus reached as to the
12 importance of on-module power management for Samsung's DDR5
13 modules?

14 A. Yes, there was.

15 Q. So I'm showing you this document. Can you describe what
16 the data showed and what the determination was about the
17 importance?

18 A. Yes. So for additional information -- so at the time of
19 that document or at the time of the meeting minutes stating
20 that it was not necessary or not needed for those classes of
21 memory, that's what the industry believed at the time. And,
22 in fact, we were -- we were on board with that.

23 But subsequent to that, folks from the industry did
24 additional testing to see, you know, if, in fact, it would
25 provide benefit for those products, and what came back was,

1 yes, the benefit outweighed the cost and extra power or the
2 heat, as it was discussed, for the product and, in fact, is a
3 good idea because of the fact that, as it says here, for the
4 SODIMMs and the UDIMMs, which are two of the module types,
5 without the voltage regulator, you can't meet the targeted
6 tolerance, which is, again, that precise amount of power that
7 needs to be provided to the various components.

8 Without putting that functionality on the DIMM, you could
9 not hit that precise tolerance.

10 Q. And what was the ultimate conclusion as to whether the
11 on-module power management was required?

12 A. It is in the designs today. So the modules that
13 are -- the R --

14 Q. Let me ask you -- let me withdraw the question.

15 A. I'm sorry.

16 Q. This says at the conclusion was that if you didn't put
17 on-module power management, what would be the impact?

18 A. I'm sorry, Mr. Sheasby. One more time?

19 Q. Sure. This talks about why the supply voltage tolerance
20 increases DRAM size, reduces yields, and increases power. Do
21 you see that?

22 A. Okay.

23 Q. And what does on-module power management do in terms of
24 tolerances?

25 A. Okay. So it ensures that we can keep within the tighter

1 tolerance which means that, you know, we can have the -- we
2 can use the components in a way that, you know, the DRAMs
3 are -- are smaller and they have higher yield and use less
4 power.

5 Q. Was there any barrier that you're aware of to Samsung
6 showing you this document and examining you about it?

7 A. Was there any barrier?

8 Q. Exactly.

9 A. Samsung could have showed me this document? Absolutely.

10 Q. Did they show it to you --

11 A. They did not.

12 Q. Tell the ladies and gentlemen of the jury.

13 A. No, they did not.

14 Q. Now, I want to turn and talk briefly about PX -- there
15 was discussion about DRAM and DRAM ICs. Do you remember that?

16 A. I do, sir.

17 MR. SHEASBY: Let's turn to PX 2.70. PDX 2.70.

18 Q. (BY MR. SHEASBY) Counsel was asking about DRAM and DRAM
19 ICs. Is that correct?

20 A. That is correct.

21 Q. What is the left-hand side?

22 A. So the left-hand side is what we're calling a DRAM
23 circuit.

24 Q. And what is the right-hand side?

25 A. The right-hand side is the -- the array dies with TSVs

1 and data ports.

2 Q. And counsel was showing you a passage from Netlist's
3 annual report where it discusses DRAM and DRAM IC. Do you
4 remember that?

5 A. I do, sir.

6 Q. Has Netlist been able to market or sell array dies yet?

7 A. Not yet, sir.

8 Q. Would the 10-K have any discussion of the array die
9 design with TSVs and data ports?

10 A. Umm --

11 Q. The annual report. Let me re-ask the question. Would
12 your annual report that counsel showed the jury have any
13 discussion of array dies in it?

14 A. I don't believe so, sir.

15 Q. And why is that?

16 A. Because it's not something that we were selling.

17 MR. SHEASBY: Let's turn to page 65.

18 Q. (BY MR. SHEASBY) On the left-hand side, it talks about
19 through-holes or vias. Do you see that?

20 A. I do, sir.

21 Q. And the right-hand side uses the phrase 'bypass'?

22 A. Yes, sir.

23 Q. Is bypass a common or uncommon phrase in the industry to
24 describe through-holes?

25 MR. McKEON: Objection. Leading, Your Honor.

1 THE COURT: Sustained.

2 THE WITNESS: Can I --

3 THE COURT: No.

4 MR. SHEASBY: I'll re-ask the question.

5 THE WITNESS: Okay.

6 Q. (By MR. SHEASBY) What is the relationship between
7 bypass -- is there a relationship between bypass and
8 through-hole vias in common parlance in the industry?

9 A. Well, in this particular case, what we're -- what we're
10 showing is that those vias are going through the stack. And
11 if there's no -- if there is what we call a keep-out around
12 that via, it's effectively bypassed.

13 So we talk about in our document those keep-out areas.
14 And if it doesn't hook up to the data port, it means there's
15 no electrical communication; whereas, above that, if there is
16 a data port present, then we will have that electrical
17 communication.

18 MR. SHEASBY: Your Honor, may I approach one more
19 time?

20 THE COURT: You may.

21 (The following was had outside the hearing of the
22 jury.)

23 MR. SHEASBY:

24 THE COURT: What is it, Mr. Sheasby?

25 MR. SHEASBY: I intend with your permission to

1 elicit the testimony that Samsung was required to provide this
2 information to us.

3 THE COURT: What information?

4 MR. SHEASBY: This is the emails that they showed
5 and suggested that they were doing something generous or open
6 with us by sharing with us and, in fact, they were required to
7 provide it to us. I'm not going to elicit why they were
8 required to. Just that they were required to provide it to
9 us.

10 MR. McKEON: What's the requirement? I'm not aware
11 --

12 THE COURT: You're talking about discovery
13 obligations?

14 MR. SHEASBY: No. This is the KTT form that counsel
15 was examining Mr. Milton about and suggesting that they were
16 doing -- they are being open and up front by providing this
17 information. And in reality, they were required to.

18 MR. McKEON: There's no evidence of that at all.
19 There's no evidence that there was any kind of requirement at
20 all.

21 MR. SHEASBY: Yes. Mr. Milton's going to put that
22 evidence in.

23 MR. McKEON: We object to that. That's not in the
24 record at all that there's a requirement.

25 MR. SHEASBY: Yes, Your Honor. It's the testimony

1 that -- that we've repeatedly provided under oath. I didn't
2 ask them to say -- to use this as them being up front and
3 open.

4 THE COURT: I don't know how this witness has
5 personal knowledge of what is and isn't required to
6 be produced.

7 MR. SHEASBY: This wasn't produced. This was
8 required to be sent by the Korean Tax Authority so we could
9 approve it.

10 THE COURT: I don't understand how this witness has
11 personal knowledge of what's required through the Korean Tax
12 Authority or not.

13 MR. SHEASBY: He does. They were very well aware of
14 these documents when they were sent because he was a member of
15 the JDLA team.

16 MR. McKEON: Your Honor, I object to that.

17 THE COURT: I'm going to sustain the objection.

18 MR. SHEASBY: Thank you, Your Honor.

19 THE COURT: Let's move on.

20 (The following was had in the presence and hearing
21 of the jury.)

22 MR. SHEASBY: Let's go to PX 2.33.

23 Q. (BY MR. SHEASBY) You were here in the instructions given
24 by Judge Gilstrap when he confirmed that the JDLA was
25 terminated as of July 2020. Correct?

1 A. That is correct.

2 Q. What did Samsung candidly report of its views of
3 Netlist's technology before this lawsuit?

4 A. As underlined here, it's -- they stated that we had
5 unique and proprietary know-how.

6 MR. SHEASBY: I pass the witness, Your Honor.

7 THE COURT: Is there additional cross examination?

8 MR. McKEON: Your Honor, just have a couple of
9 questions.

10 THE COURT: Proceed.

11 RE CROSS EXAMINATION

12 BY MR. McKEON:

13 Q. Now, Mr. Milton, you said some testimony about what's on
14 Samsung's PMIC. Do you recall that testimony you just gave?

15 A. Yes, sir.

16 Q. Because you haven't examined Samsung's internal technical
17 documents. Isn't that right?

18 A. That's correct.

19 Q. And you haven't examined the source code. Correct?
20 Samsung, any source code from Samsung? Isn't that right?

21 A. No. What we did is we tested it.

22 Q. Sir, you haven't examined any Samsung internal source
23 code. Correct?

24 A. Oh. That is a true statement.

25 Q. Samsung's circuits schematics that are internal to

1 Samsung, you haven't reviewed those. Right?

2 A. That is correct.

3 Q. Now, you understand that there was a document --

4 MR. McKEON: If I can get it up, PDX 2.89. If I can
5 have, Mr. Sheasby, your slide to present that?

6 Q. (BY MR. McKEON) This was a document that Mr. Sheasby
7 presented to you just now. Do you recall that?

8 A. Yes, sir.

9 Q. Okay. And the tolerance here, do you see the tolerances
10 here, the plus six to minus 3. Do you see that?

11 A. I do, sir.

12 Q. So that's a total of nine percent window. Isn't that
13 right?

14 A. Yes, sir.

15 Q. And in the DDR5 tolerances, it's a 10 percent window.
16 Isn't that right?

17 A. In the DDR5 tolerances, it's a 10 percent?

18 Q. In the technical documents --

19 A. I believe it's the plus 6 minus 3 is what was agreed upon
20 for -- for DDR5.

21 Q. In the -- if you look at the documents related to DDR5
22 that the industry agreed on, you understand that it's plus or
23 minus -- it's 1 to 5. Isn't that right? Plus or minus 5.

24 A. I actually believe that it's what it's stated here.

25 Q. In the actual -- for DDR4, sir, plus or minus 5, do you

1 recall that in the --

2 A. Oh, for DDR4?

3 Q. Yes.

4 A. Oh, for DDR4, yes.

5 Q. Plus or minus 5. Isn't that right? And that's a total
6 of 10 is the tolerance. Is that right?

7 A. That's correct.

8 Q. Okay.

9 MR. McKEON: No further questions, Your Honor.

10 Thank you. I pass the witness.

11 THE COURT: Anything further, Mr. Sheasby?

12 MR. SHEASBY: No, Your Honor.

13 THE COURT: All right. You may step down, Mr.
14 Milton.

15 THE WITNESS: Thank you, Your Honor.

16 THE COURT: You're welcome.

17 MR. SHEASBY: Your Honor, may we clear binders?

18 THE COURT: We're fine.

19 Plaintiff, call your next witness.

20 MR. SHEASBY: Your Honor, Plaintiffs call Dr. Bill
21 Mangione-Smith.

22 THE COURT: All right. Doctor Mangione-Smith, if
23 you'll come forward and be sworn by our Courtroom Deputy.

24 And, Counsel, if you have binders to distribute for this
25 witness, let's do that, too.

1 MR. MANZIN-MONNIN: The binders are up there.

2 THE COURT: They're already there. Okay.

3 (Whereupon, the oath was administered by the Clerk.)

4 THE COURT: Please come around, sir. Have a seat on
5 the witness stand.

6 MR. SHEASBY: Your Honor, may I approach?

7 THE COURT: You may approach.

8 MR. MANZIN-MONNIN: Your Honor, may I approach?

9 THE COURT: Hand it to the Court Security Officer.
10 He'll hand it to the witness.

11 MR. MANZIN-MONNIN: Thank you.

12 THE COURT: Let's get everybody in their places and
13 have a seat, please.

14 All right, counsel. You may proceed with direct
15 examination of this witness.

16 MR. MANZIN-MONNIN: May it please the Court.

17 WILLIAM HENRY MANGIONE-SMITH, Ph.D., SWORN,
18 testified under oath as follows:

19 DIRECT EXAMINATION

20 BY MR. MANZIN-MONNIN:

21 Q. Good morning. Could you introduce yourself to the jury?

22 A. Good morning. My name is William Henry Mangione-Smith.

23 I live with my wife who I've been married to for 36 years, and
24 we have two adult children.

25 Q. And what are you going to be testifying regarding today?

1 A. Well, I'm going to be testifying about two sets of
2 patents, the intelligent on-module power management patents
3 and the load-reduced patents.

4 MR. MANZIN-MONNIN: Can we bring the slide down?
5 Mr. Huynh, can you make sure we have the latest version of the
6 slides? There's been a few changes this morning. Can you
7 check that? We'll continue.

8 Q. (BY MR. MANZIN-MONNIN) And can you describe your
9 professional background?

10 A. I'd be happy to.

11 Q. We'll do without slides for the moment.

12 A. Sure.

13 Q. Thank you.

14 A. I got my doctorate at the University of Michigan, and I
15 worked at a number of companies, including Motorola, where I
16 was involved in testing and designing memory controllers as
17 well as my first patent was from Motorola, and it had to deal
18 with the interface to a memory.

19 Then I worked for 10 years at UCLA where I was a
20 professor in electrical engineering.

21 After that, I went to work for a company called
22 Intellectual Ventures for about four years. And since then,
23 I've been running my own technical consulting company.

24 Q. Have you used the continuation process to find under
25 United States patent law when filing for your own patents?

1 A. Yes. I am a named inventor on over 120 patents, most of
2 which belong to Intellectual Ventures. And the continuation
3 process is something that they use regularly as well as most
4 companies that I'm aware of.

5 Q. And can you explain how the continuation process works?

6 A. Sure. So basically what happens, as I understand it, I'm
7 not an attorney, is an inventor or a company will file a
8 patent application. And the Patent Office will come back and
9 say, we're applying restrictions, which means, we think
10 there's two sets of related -- of unrelated inventions,
11 there's two sets of inventions that should be continued
12 independently.

13 And so a company will say, okay, we'll immediately
14 continue with set A, and we'll follow up with set B as a
15 continuation.

16 Q. And you were here for opening statements. Correct?

17 A. Yes, sir.

18 MR. MANZIN-MONNIN: Could we bring up PDX 4.5,
19 please?

20 Q. (BY MR. MANZIN-MONNIN) And you recall that Samsung
21 showed this slide to the ladies and gentlemen of the jury.
22 Correct?

23 A. Yes.

24 Q. And I wrote this down. He said, They filed six
25 continuations, ladies and gentlemen. They kept it pending

1 year after year after year.

2 Now, does Samsung use the continuation process?

3 A. Sure.

4 Q. Could you show us some examples?

5 A. I'd be happy to. I no longer appear to have control of
6 the slides.

7 Here is the '509 Patent from Samsung, and you'll see that
8 I've highlighted that this was a number of different
9 continuations. And, in fact, from when they first filed their
10 application until that '509 Patent issued, it covered 16
11 years.

12 Q. Any other examples?

13 A. Indeed. If we flip forward.

14 MR. MANZIN-MONNIN: Could we have the next slide,
15 please, Mr. Huynh? We don't have control. I don't know if
16 there's a clicker that's --

17 THE WITNESS: All right. It seems like maybe I have
18 control now.

19 This is the '894 Patent. You'll see it also is a
20 continuation of a set of continuations. And, in fact, since
21 the original application was filed for this patent until the
22 '894 issued, it spanned 14 years.

23 Here's the '700 Patent, which, again, was a continuation.
24 And from the time the original patent was filed until that
25 patent issued, it spanned 13 years.

1 Q. (BY MR. MANZIN-MONNIN) So in your experience
2 continuations are common practice.

3 A. They are very common, yes.

4 MR. MANZIN-MONNIN: Let's skip to slide 12, please.

5 Q. (BY MR. MANZIN-MONNIN) Can you tell us about
6 Intellectual Ventures?

7 A. Intellectual Ventures is a company that creates ideas and
8 patents them. That's how I got my 120 patents. It starts
9 up -- it then attempts to find companies that are interested
10 in starting up new companies using that patented technology.

11 Q. What companies have you worked with as a consultant?

12 A. There are a number. Best known would be Hewlett Packard.
13 In addition, there a number of start-up companies that were
14 developing technology that I had expertise in, and I consulted
15 with them.

16 Q. What do you do when you are asked to consider taking on
17 an investigation of a patent?

18 A. When somebody asks me to consider a particular patent, I
19 look at the claims and I decide whether I think it's new,
20 novel, potentially important. If it's in my area of
21 expertise, and if all of those things are true, and I think
22 the technology is exciting and interesting, I'll agree to work
23 on the matter and offer my opinions about the technology in
24 the claims.

25 MR. MANZIN-MONNIN: Your Honor, we offer Doctor

1 Mangione-Smith as an expert on memory modules, including on
2 the subject matter of the '918, '054, and '339 Patents.

3 THE COURT: Is there objection?

4 MR. McKEON: No objection, Your Honor.

5 THE COURT: Without objection, the Court will
6 recognize this witness as an expert in those designated
7 fields.

8 Please continue, Counsel.

9 Q. (BY MR. MANZIN-MONNIN) How did you approach your task in
10 this case?

11 A. Well, I used my standard approach--if you would advance
12 to the next slide, please--which involves beginning by --

13 MR. MANZIN-MONNIN: Slide 13, please, Mr. Huynh.

14 THE WITNESS: I employed my standard approach which
15 involves looking at the -- beginning by looking at the patents
16 and the entire history of the patents and the patent claims.

17 Then I looked at confidential Samsung documents that were
18 made available to me to review.

19 I looked at top secret Samsung source code and considered
20 that.

21 And then I evaluated the testimony of Samsung's corporate
22 representatives and engineers. And only after looking at all
23 that information and digesting it, did I reach opinions
24 regarding the patent claims in this matter.

25 Q. (BY MR. MANZIN-MONNIN) And what is a patent claim?

1 A. A patent claim is the invention. So the patent claims
2 define the bounds of the invention.

3 Q. And how do you assess the meaning of the claims?

4 A. You assess the meanings of the claim from the point of
5 view of a person of ordinary skill in the art--that's the term
6 that's used--at the time of invention. So if somebody was
7 able to -- was technically skilled enough to work on solving
8 the problems the patent was trying to solve at that time, what
9 would they know the patent terms to mean.

10 But, additionally, in some cases, the Court has construed
11 certain terms and said, when you see this word or this phrase,
12 you need to apply this meaning, this definition.

13 Q. And how is infringement determined?

14 A. So infringement is determined, as I understand it, and
15 again I'm not an attorney, in two ways. One is called literal
16 infringement. If I look at the patent claims and it says
17 there is this thing that must be present, and I find it
18 literally present in the accused products, well, then it's
19 literally infringed.

20 But there's another concept which is called the doctrine
21 of equivalents, this has been explained to me, which means
22 that if the thing isn't literally there but there is something
23 that performs substantially the same function in substantially
24 the same way to achieve substantially the same result, then
25 they can infringe under the doctrine of equivalents.

1 Q. About how many hours have you spent working on this case?

2 A. Oh, I'm not sure, but certainly more than a hundred, I
3 would say.

4 Q. And did you prepare reports in this case that provide the
5 details of your analysis?

6 A. Yes. I prepared reports that total over a thousand
7 pages.

8 Q. Are you being compensated for your time working on this
9 case?

10 A. I am. I'm being compensated at my ordinary hourly rate
11 just for my time. It has -- my compensation doesn't change
12 based on who prevails in this matter or exactly what I testify
13 to.

14 Q. So let's turn --

15 THE COURT: Let me interrupt just a minute. Go back
16 to the prior slide, please. Just so the jury's not confused,
17 this is the Eastern District of Texas, not the Western
18 District of Texas that's on that seal.

19 MR. MANZIN-MONNIN: I apologize, Your Honor.

20 THE COURT: All right. Let's go forward.

21 MR. MANZIN-MONNIN: Can we have slide 18, please?

22 Q. (BY MR. MANZIN-MONNIN) Let's turn to the technology at
23 issue in the '918 and '054 Patents. What components in a
24 computer system will you be discussing?

25 A. I'm mostly going to be discussing memory modules, which

1 on this slide you'll see on the right-hand side. Memory
2 modules are where the data is stored that a computer is using.
3 So that might be -- well, it might be your email or images or
4 whatever.

5 That data is processed by a processor. That's the brains
6 of the computer where software runs.

7 And in between them is something called a memory
8 controller. We'll hear a fair amount -- I'll talk about a
9 memory controller.

10 And all of these components reside on a motherboard.

11 Q. And what is a memory module?

12 A. Here's an example of a memory module. This is called a
13 dual in-line memory module. What that means is that there are
14 electrical contacts, edge connectors. I'm showing one side of
15 the memory module, but it's dual. So there's pins on the
16 other side as well.

17 And also there are DRAM memory chips.

18 And, again, there's a circuit board that all of these
19 components are mounted on, and it plugs into a socket that we
20 saw previously on a mother board.

21 Q. Are memory modules important for system performance?

22 A. They're very important. I would say, in general, nobody
23 ever has enough memory. We always want more; we always want
24 it to run faster.

25 Q. Are there challenges to designing fast memory modules?

1 A. There are significant challenges. Most relevant here is
2 that if you take a circuit and you make it run faster, it's
3 going to use more power.

4 Now, back in the days when I first started in the
5 industry, that wasn't very important, the power and the
6 temperature that -- the heat that came off as a result of it,
7 but it's been very important for the last several decades.

8 Now we're in a situation where there's too much heat in
9 the system, it uses too much power, and it's a fundamental
10 engineering problem that we always have to deal with.

11 Q. Are there any other challenges related to delivering
12 precise voltages?

13 A. There are significant challenges. We've talked about --
14 Mr. Milton was talking about the tolerances involved. If you
15 need to provide a voltage at a precise -- precise voltage
16 within a certain tolerance, that's a tough engineering design
17 problem.

18 Q. Now, how do the '918 and '054 Patents address these
19 challenges with power management?

20 A. Well, they address the challenges by introducing a
21 concept -- here's a -- here's a DIMM, and I've highlighted
22 that it provides on-module power management. And it uses this
23 on-module power management to provide reliable and precise
24 power to memory, and it does it in what I would characterize
25 as an intelligent way.

1 If it sees that there's a problem with the voltage, it's
2 too high or it's too low, it adjusts to that situation.

3 Q. When Samsung put out the accused products, did they tout
4 the power benefits of including power management in the memory
5 module?

6 A. They did. They talked about DDR5 compared to DDR4, that
7 one generation of memory compared to the previous generation,
8 and they said that by using an on-DIMM PMIC, power management
9 IC, they were able to achieve a 30 percent improvement in
10 power efficiency.

11 Q. And that word there, PMIC, is that just one part of
12 Samsung's on-module power management?

13 A. That's just one part of their power management which is
14 on-module, but just one part of the DIMMs.

15 Q. Do the '918 and '054 Patents describe what types of DIMMs
16 the invention can be applied to?

17 A. Yes. They give a number of examples, in particular what
18 are called small outline DIMMs or SODIMMs, unbuffered DIMMs or
19 UDIMMs, and registered DIMMs, RDIMMs.

20 Q. Now, what types of DIMMs are accused of infringement in
21 this case?

22 A. Well, I highlighted those specific examples because those
23 are exactly the kind of DIMMs that are accused. They are
24 among the accused products from Samsung--SODIMMS, UDIMMs, and
25 RDIMMs.

1 Q. Did Samsung face challenges with power management when
2 developing the DDR5 memory modules at issue in this case?

3 A. They did, just as everybody in the industry did. In
4 fact, they acknowledge that going from DDR4 to DDR5 in order
5 to achieve the doubling in performance, one of the things they
6 needed to do was to move the voltage regulation as close to
7 the DRAMs as possible.

8 Q. And what did Samsung do after it realized that it needed
9 to solve the power problem that existed with DDR5?

10 A. Well, as was previously discussed by Mr. Milton, they
11 sent an email to Netlist saying, hey, can we have a technical
12 conversation with you to discuss exactly how to put the PMIC
13 on the -- on the DDR5 DIMMs.

14 MR. MANZIN-MONNIN: And, Mr. Huynh, let's jump to
15 slide 29.

16 Q. (BY MR. MANZIN-MONNIN) Let's look at Samsung's accused
17 products. Can you tell me about them?

18 A. Sure. So these are the accused products, the entire
19 DIMMs. There is the RDIMM at the top, which RDIMMs are
20 typically used in high-end server boxes by like Amazon and --
21 and eBay, for example.

22 There are UDIMMs in the middle. These are typically used
23 on desktop computers.

24 And then there are SODIMMs at the bottom, which are
25 small. They're typically used in laptops and tablets.

1 Q. And does your analysis apply to all of these products?

2 A. I analyzed all of these products for the claims I'm going
3 to be talking about next. There are a set of claims that only
4 apply to RDIMMs, and I will point that out. But the majority
5 of the claims that I'm going to be talking about apply to all
6 three types of accused products.

7 Q. So there was no material difference between these
8 products in terms of infringement?

9 A. That's correct.

10 Q. Do each of Samsung's products include on-module power
11 management?

12 A. Yes, they do.

13 Q. And we saw earlier what is the name for the part
14 of -- what is one -- excuse me. What does Samsung call one
15 part of the on-module power management?

16 A. So they call it a power management IC, or I refer to it
17 as a PMIC and so do they, as you've been hearing. This is the
18 cover sheet from something called a data sheet that Samsung
19 produced. A data sheet is like a user's manual, somebody is
20 going to use this part, Samsung is telling them exactly how it
21 operates and how they should use it.

22 MR. MANZIN-MONNIN: Let's turn to page 54 of JTX 11.

23 Q. (BY MR. MANZIN-MONNIN) Is Samsung's on-module power
24 management applicable to all types of DIMMs, including
25 NVDIMMs?

1 A. Well, they say that it works with DDR5 RDIMMs, LRDIMMs,
2 and NVDIMMs.

3 Q. How does that compare to what the patent says about which
4 DIMMs the invention can be used with?

5 A. That's exactly what the patent had said previously that I
6 highlighted.

7 Q. Did you do a patent claim analysis?

8 A. Yes, I did.

9 Q. And can you tell us about claim 16 of the '918 Patent?

10 A. So here's claim 16 of the '918, and a claim analysis is
11 just where I actually look at the claims and compare it to
12 exactly how the products operate, the accused products.

13 Q. And so what is your opinion on element 16 -- let me take
14 a step back. Can you break these up into elements for us to
15 make it a little easier?

16 A. Yes, sir.

17 Q. And what is your opinion on element 16A?

18 A. Well, I'll point at the top right, it says DDR5 SODIMMs,
19 UDIMMs, and RDIMMs. So this is one of those patent claims
20 that apply to all of the accused products. And element A
21 says, a memory module.

22 Q. And are the accused products memory modules?

23 A. Yes, they are. And here's a picture of one of the
24 accused products, and I've copied a portion from the data
25 sheet for JTX 60, 61, and 9 for the SODIMMs, UDIMMs, and RDIMM

1 products.

2 Q. And what is your opinion on element 16B?

3 A. So 16B begins by reciting a printed circuit board, PCB.
4 And a printed circuit board is just that green sort of
5 substrate that the board that all of these components are
6 mounted on.

7 And then it says that it has an interface. And it can go
8 into a slot and communicate with a host system. Well, the
9 interface are those pins along the bottom, and we saw examples
10 of slots that they can plug into.

11 Q. How do the host system and memory module communicate?

12 A. They communicate over the edge connector, those pins,
13 those gold sort of fingers along the edge, and they
14 communicate by sending power.

15 I've highlighted what's called VIN_bulk from JTX 60. So
16 they communicate with power that comes over a set of those
17 pins. And they communicate by sending data bus, data signals,
18 along another set of pins, and then command and control lines
19 over another set of pins.

20 Q. So that covers all the words of 16B. Let's turn to the
21 next element.

22 A. Sure. And I'm going to put a checkmark, a green mark,
23 indicating that we've covered all of that element as well.

24 Q. And so what is element C?

25 A. Element C begins by talking about a first, second, and

1 third buck converter. And from JTX 11, it shows -- this is
2 from one of the data sheets of the PMIC. It shows a buck
3 converter, which I've highlighted in yellow here.

4 Q. And what is the next requirement for this element?

5 A. Well, next, I'm going to point out a converter circuit,
6 which is in green in the lower left-hand corner, that's
7 labeled LDO 2. LDO stands for low drop-out.

8 And the next element is that it receives a pre-regulated
9 input voltage. I've highlighted the inputs to the chip which
10 are -- with is -- which are pre-regulated input voltage.

11 And then the last element is that the three buck
12 converters and the converter circuit produce four regulated
13 output voltages. And I've highlighted where those output
14 voltages are produced by the PMIC.

15 Q. And so those blocks you have yellow and green there, each
16 of those blocks receives the pre-regulated input voltage and
17 produces a regulated output voltage?

18 A. Yes, sir.

19 Q. Why do you say that the input voltage is pre-regulated?

20 A. Well, the users manual the data sheet tells me it's
21 pre-regulated. Here's an excerpt from JTX 12, and it says
22 that during power on, during operation, the host--the
23 computer--needs to provide a stable supply voltage, VIN_BULK.
24 And VIN_BULK has to be between four-and-a-quarter volts and
25 five-and-a-half volts.

1 Q. You identified the LDO as the converter circuit?

2 A. Yes, sir.

3 MR. MANZIN-MONNIN: If we can flip to the next
4 slide.

5 Q. (BY MR. MANZIN-MONNIN) Do you remember this slide from
6 Samsung's opening?

7 A. Yes, I do.

8 Q. And do you agree with Samsung that an LDO is not a
9 converter circuit?

10 A. No. I believe they're wrong.

11 Q. Have you seen testimony from Samsung's engineers that
12 supports your opinion?

13 A. I certainly have. Here's testimony from Mr. Park. He is
14 the engineer, he is the technical leader that Samsung has
15 indicated is their representative. He's their corporate
16 represent on DDR5 matters. And he agreed that an LDO reduces
17 an input voltage to a lower output voltage.

18 Q. I don't see the word 'converter' there. Did he also
19 testify about whether this LDO is a converter?

20 A. He did. He said that the LDO converts VIN_BULK to a
21 regulated output.

22 Q. Now, is Mr. Park a low-level engineer?

23 A. No. He was asked about this. He said that he is the
24 person most knowledgeable about PMICs at Samsung.

25 Q. And in addition to the Samsung engineers, have you seen

1 testimony from anyone else who agrees with your opinions?

2 A. Yes. There is a third party called Renesas who makes
3 PMIC chips, and they sell them to Samsung. They are used in
4 some of the accused products.

5 Mr. Lo was their representative. He is a design
6 engineer, and he agreed that an LDO is a voltage converter,
7 And I put a figure for JTX 30 on the right-hand side which is
8 the data sheet for Renesas's PMIC.

9 Q. So PMIC is also used in Samsung's infringing products?

10 A. Yes, sir.

11 Q. If Samsung claims the LDO does not literally infringe, is
12 it equivalent?

13 A. Yes, I believe it is.

14 Q. And why is that?

15 A. It's equivalent because it does the same function which
16 is to convert; it does it the same way, which is to reduce the
17 input voltage; and it achieves the exact same result, which is
18 a regulated output voltage.

19 Q. So could you summarize for us, what is your conclusion
20 about element 16C of the '918 Patent under both literal
21 infringement and the doctrine of equivalents?

22 A. It's present under both forms of analysis.

23 Q. And let's turn to element 16D.

24 A. Sure. So 16D talks about the first, second, third, and
25 fourth regulated voltages having a specific voltage amplitude.

1 Now, voltage amplitude just means how high does the voltage
2 get. And JTX 9 specifies that, well, the amplitude of these
3 four different voltages is 1.1 volts or 1.8 volts or 1.0
4 volts.

5 Q. And let's turn to the next element.

6 What is element 16E?

7 A. 16E talks about a plurality of components that are
8 coupled to the PCB -- well, there is the PCB. There are
9 components on there and they don't fall off. They're stuck
10 to the board. And it includes a plurality of SDRAM devices,
11 which I have highlighted here in orange--this evidence is from
12 JTX 61--as well as other components.

13 Here I've highlighted on the right-hand side from JTX 60
14 something identified as SPD. That's the serial presence
15 detect. It's a small circuit that allows the host computer
16 to know how big is this DRAM, how fast is it. So it has
17 configuration information for the entire DIMM.

18 And you will see that those four output voltages are
19 connected to the SDRAM and the SPD, and, in fact, other
20 components on the DIMM.

21 Q. And so that brings us to the last element. What is 16F?

22 A. So 16F talks about a voltage monitoring circuit. And
23 going back to JTX 11, I've highlighted in purple the upper
24 left-hand portion which is labeled 'housekeeping'.

25 And in one element of the house -- one responsibility for

1 the housekeeping circuitry is to provide input over voltage
2 protection. If the voltage goes too high, bad things can
3 happen. Eventually the entire circuit will break and can
4 burn, begin to catch fire. So you really don't want that to
5 happen.

6 So there is this detection of the input voltage by the
7 housekeeping circuitry. It actively monitors the VIN_BULK
8 value, and it compares -- the VIN_BULK from the power rail,
9 and it compares it to a threshold. So it's programmed. It
10 says, by the way, if VIN_BULK gets above a certain level, then
11 we've got an overvoltage situation and you have to take
12 action.

13 Q. So what did you conclude about '918--excuse me--claim 16
14 of the '918 Patent?

15 A. Well, I also relied upon some testimony from Mr. Park who
16 was asked about this, and he said that the voltage monitoring
17 circuit produces a signal called VIN_BULK OVLO, which stands
18 for overvoltage lockout, which identified all of the elements
19 of claim -- of element F.

20 And since all the elements were present, I had to
21 conclude that claim 16 is infringed by all of the accused
22 products, both literally and under the doctrine of
23 equivalents.

24 Q. And what claim did you analyze next?

25 A. Next, I turned to claim 18. Claim 18 is called a

1 dependent claim, and it begins by saying basically, you
2 remember claim 16? We're going to add some additional
3 requirements.

4 So all of the evidence that I applied for 16 satisfies
5 element A of claim 18. And then it adds this additional
6 requirement where it talks about a controller and what the
7 controller needs to do.

8 So the controller that I've identified is called control
9 logic, and it's below the housekeeping circuitry. And I've
10 identified that in blue.

11 Q. And so with the last claim element you were discussing a
12 signal, and where is that signal you just described from the
13 housekeeping block provided?

14 A. So it's provided to the control logic, and I've
15 highlighted that in yellow. That's the OVLO signal.

16 Q. And does the control logic perform a write operation in
17 response to the signal?

18 A. The data sheets tell us that that's exactly what it does.
19 In response to that signal, it sets a bit in a particular
20 register, a piece of memory, and the exact location is called
21 0x04. Unfortunate engineering jargon. But that's a memory
22 location where a bit gets written to indicate that this
23 condition has happened.

24 Q. And what claim -- what was your conclusion about claim
25 18?

1 A. The accused products infringe the claim for the -- based
2 on the evidence I've shown.

3 Q. Let's turn to '918, claim 19. And what does this claim
4 recite?

5 A. So claim 19 says it depends from claim 18, and it adds
6 the requirement that the data be written to non-volatile
7 memory.

8 Q. So that 0x04 register you just pointed to, what type of
9 memory is it that the control logic is writing to?

10 A. It is non-volatile memory. And, again, that just means
11 if the power gets taken away completely, the information
12 that's stored in that memory is available once the power comes
13 back up.

14 And you'll notice that in JTX 11, it labels that block as
15 having NVM, non-volatile memory. And Mr. Park, the corporate
16 representative for DDR5, agreed that that is, in fact,
17 non-volatile memory.

18 Q. If Samsung claims the non-volatile memory on the power
19 management module does not literally infringe, is it
20 equivalent?

21 A. Yes, it is, in my opinion, because it is equivalent
22 because it does the same function. It provides a memory.
23 It does it in the same way, which is to store data in
24 non-volatile memory so that it can survive through a power
25 reset. And it provides the same result, which is it allows

1 the host computer to get access to that once the power comes
2 back.

3 Q. And so what did you conclude for '918, claim 19?

4 A. Based on my two analyses under literal infringement and
5 doctrine of equivalents, I found element B present and, thus,
6 the accused products infringe.

7 Q. And what claim did you analyze next?

8 A. Next, I turned to claim 1 of the '918. Now, as we'll
9 see, claim 1 only applies to the RDIMM products.

10 Q. And does the evidence you've already discussed for the
11 previous claims cover certain elements for claim 1?

12 A. It does, which is fortunate. It will help speed up some
13 of this process. On the left-hand side here, I've shown claim
14 16. On the right-hand side, I've shown claim 1.

15 Q. And so claim 16 is the one we just walked through all the
16 evidence for?

17 A. That's correct.

18 Q. And does -- let's start with element 1A on the right.
19 Are the DDR5 RDIMMs memory modules?

20 A. Yes. I showed that in claim 16, element A.

21 Q. Now let's turn to element 1B. Do the RDIMMs include
22 power data and address lines to communicate with the host
23 system?

24 A. Yes, via a PC board.

25 Q. And element 1C, do the accused products include the claim

1 buck converters and converter circuit?

2 A. They do.

3 Q. And do those circuits generate regulated voltages?

4 A. Yes, indeed.

5 Q. And element 1D on the right, are there components coupled
6 to the regulated voltages?

7 A. Yes.

8 Q. And that leaves '918, element 1E. Is there a circuit
9 between the edge connections and the SDRAM devices in the
10 RDIMM products?

11 A. There is. And so here I've broken up element F, and I'm
12 going to focus on evidence from JTX 9 which, again, relates to
13 the RDIMM products. And I have reproduced a figure from it.
14 The blue lines and the text below the figure are literally
15 present in the documents.

16 I've highlighted in yellow the RCD to call attention to
17 that. That's the circuit that's called a registered clock
18 device.

19 Q. And could you show us how the input and output voltages
20 connect through that circuit?

21 A. Sure. Let me correct myself. It's a registered clock
22 driver.

23 So they connect via the edge connectors at the bottom, as
24 I've shown here. And the SDRAMs at the top. The address and
25 control and clock lines come in along the edge connectors from

1 the host and then get registered and driven out to the SDRAM
2 devices over those blue lines.

3 Q. And could you show the input and output?

4 A. Sure. I've highlighted now in purple where the inputs
5 are coming into the RCD, and then the outputs in purple where
6 they're then driven to the individual SDRAM devices.

7 Q. And does that circuit receive regulated voltages?

8 A. Indeed, it does. This is evidence from JTX 26. Zooming
9 in on the RDIMM showing the RCD in yellow, that's the
10 controller circuit, again, and that it's driven -- it's
11 provided data from the PMIC. In fact, it's provided the
12 second regulated voltage and the fourth regulated voltage,
13 and as the evidence shows here, the VVDIO is below VDDQ.

14 Q. So what did you conclude about '918, claim 1?

15 A. All of the elements are present, so the accused RDIMM
16 products infringe this claim.

17 Q. Let's turn to claim 5. Have you shown evidence of
18 overvoltage monitoring?

19 A. Yes. Indeed, I have in my analysis of claim 16.

20 Q. And is that the only additional requirement of claim 5?

21 A. That is.

22 Q. And so what did you conclude about claim 5?

23 A. Claim 5 is infringed by the accused RDIMM products.

24 Q. Can you just show where that evidence was shown already
25 with claim 16? Thank you.

1 A. Yep.

2 Q. And so what claim did you analyze next?

3 A. Next, I turned to claim 13, which depends from claim 5.

4 Q. And are the first, second, and third buck converters
5 coupled to an input voltage from the host?

6 A. Yes. We saw evidence of that previously.

7 Q. So with '918, claim 16C, we've already seen that
8 evidence?

9 A. Yes, sir.

10 Q. Okay. So what did you conclude about '918, claim 13?

11 A. It's infringed by the accused RDIMM products.

12 Q. Which claim did you analyze next?

13 A. Next, I turned to claim 16 of the '054 Patent.

14 Q. And I'm hoping, again, did some of the evidence overlap?

15 A. Yes, quite a bit of it does. And, again, on the
16 left-hand side, I produced the '918 claim, and on the
17 right-hand side, I've shown the '054 claim.

18 Q. And can you walk us through how some of the evidence
19 we've already seen overlaps with claim 16 of the '054 Patent?

20 A. Yes. So we've already talked about there being a memory
21 module present. I've talked about the PCB having an interface
22 that connects to a slot connected to the host system. I've
23 talked about the first, second, and third buck converters,
24 which provide regulated voltage outputs, and I've talked about
25 the converter circuit as well.

1 Then I've talked about the plurality of components on the
2 PCB in the context of the '918 claim, and I've talked about
3 the controller on the PCB, the RCD. So what's left is this
4 last element which I have highlighted in yellow at the bottom.

5 Q. Could I just pause you there, Doctor, one moment?

6 A. Yes.

7 Q. So for element 1E of the '916 [sic], the controller there
8 I believe is the voltage monitor circuit.

9 A. Yes. Thank you.

10 Q. Have you given evidence so far of voltage monitoring
11 that's consistent with '054, claim 16, element E?

12 A. Yes, I have.

13 Q. Okay. And you have one element there at the bottom
14 highlighted in yellow. Can you tell us about that one?

15 A. Yes. So this reads, "the memory module transitions from
16 a first operable state to a second operable state."

17 Q. And what is a second operable state?

18 A. This is one of those situations where the Court has told
19 us what this term means in this context, and it means, a state
20 in which the memory module is operated after a transition.

21 Q. Do you remember during opening statements when Samsung's
22 attorney suggested there's just a single operable state and
23 otherwise the module is just off?

24 A. Yes, I recall that.

25 Q. Is that accurate?

1 A. No, it's not.

2 Q. And why is that?

3 A. It's not accurate because --

4 MR. MANZIN-MONNIN: I'm sorry. Can we take the
5 slides down, please? Slides down, please? There should be a
6 new version of that one. We'll come back to it.

7 Q. (BY MR. MANZIN-MONNIN) So --

8 THE COURT: Tell you what let's do. We need to take
9 a brief recess anyway. I'll let you work on that while
10 we're --

11 MR. MANZIN-MONNIN: Thank you, Your Honor.

12 THE COURT: -- on recess.

13 Ladies and gentlemen, if you will, simply close your
14 notebooks and leave them in the chairs. Take this opportunity
15 to stretch your legs and get a drink of water. We'll be back
16 and continue with Doctor Mangione-Smith's direct testimony
17 shortly.

18 Please follow all my instructions, including not to
19 discuss the case with each other, and we'll be back shortly.

20 The jury's excused for recess.

21 (Whereupon, the jury left the courtroom.)

22 THE COURT: All right. Let's get these slides fixed
23 over the break.

24 We stand in recess.

25 MR. MANZIN-MONNIN: Yes, Your Honor.

1 (Brief recess.)

2 THE COURT: Be seated, please.

3 MR. MANZIN-MONNIN: Excuse me, Your Honor. We need
4 to seal the courtroom, if it's possible, for some source code.
5 If now's a convenient time for the Court, we can do so.

6 THE COURT: Well, I'm not going to do it outside the
7 presence of the jury so they know what's going on.

8 MR. MANZIN-MONNIN: Yes, Your Honor.

9 THE COURT: You can ask for it as soon as we get
10 them back. Have we got all our slides corrected?

11 MR. MANZIN-MONNIN: Yes, Your Honor.

12 THE COURT: All right. Let's bring in the jury,
13 please.

14 (Whereupon, the jury entered the courtroom.)

15 THE COURT: Please be seated, ladies and gentlemen.
16 Counsel, you may continue with direct examination of the
17 witness.

18 MR. MANZIN-MONNIN: Your Honor, may we seal the
19 courtroom?

20 THE COURT: To protect confidential information?

21 MR. MANZIN-MONNIN: Yes, Your Honor.

22 THE COURT: All right. At counsel's request and to
23 protect confidential or proprietary information, I'll order
24 the courtroom sealed. If you are present in the courtroom and
25 not subject to the protective order that's been entered in

1 this case, then you should excuse yourselves and remain
2 outside the courtroom until it's reopened and unsealed.

3 (Courtroom sealed.)

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(Courtroom unsealed.)

MR. McKEON: Your Honor, may we have a sidebar?

THE COURT: Approach the bench, Counsel.

(The following was had outside the hearing of the jury.)

MR. McKEON: This eFlash thing was not in the expert report, and my understanding is that he never discloses eFlash

1 in that specific term. He never said that in connection with
2 this in his report.

3 MR. MANZIN-MONNIN: He said exactly that the logic
4 in the schematic is where the register that we cited is
5 stored, and that's precisely what we just showed.

6 MR. McKEON: Never -- that eFlash term was never
7 pointed out in the expert report. They never pointed that
8 specific logic --

9 THE COURT: All right. Well, we've done what we've
10 done. We're going to go forward. I will say this. There is
11 something off color about how all this was handled, and I'm
12 not sure who's the culprit here, but there is no way this
13 should have come down the way it came down.

14 MR. MANZIN-MONNIN: Understood, Your Honor.

15 THE COURT: Also, Counsel, I don't really like you
16 roaming around the courtroom from -- if you're at the podium,
17 you're at the podium. If you're not, you're at counsel table.
18 If you need to move somewhere else, you ask for leave before
19 you start wandering around and talking to people --

20 MR. MANZIN-MONNIN: Yes, sir.

21 THE COURT: -- even if we've got spectators coming
22 in and out during the unsealing process. Okay?

23 MR. MANZIN-MONNIN: Understood, Your Honor.

24 THE COURT: Let's proceed.

25 (The following was had in the presence and hearing

1 of the jury.)

2 THE COURT: Let's proceed.

3 MR. MANZIN-MONNIN: Could we flip back to PDX 4.91,
4 please, Mr. Huynh?

5 Q. (BY MR. MANZIN-MONNIN) And so before we left off, we
6 were covering '054, claim 16, and we've gotten to this last
7 element here.

8 Can you remind the ladies and gentlemen of the jury what
9 you have highlighted here at the bottom?

10 A. Yes. This is the additional element beyond that in the
11 '918 Patent, and it's the element that refers to 'operable
12 states'.

13 Q. And what is a second operable state?

14 A. A second operable state is a state in which the memory
15 module is operated after the transition.

16 Q. And so we had briefly reviewed the slide from opening
17 statements of Samsung's counsel, and I, again, will ask you,
18 is this an accurate depiction of how the products work?

19 A. It's accurate to the extent that the products can be
20 completely on and completely off, but that doesn't tell the
21 whole story.

22 Q. So let me ask a better question. Are there multiple
23 operable states that occur when the switch is on?

24 A. Yes.

25 Q. And can you show us what that looks like?

1 A. Sure. What I've illustrated here is that when the --
2 when, for example it goes into an overvoltage condition --
3 I'm sorry. Let me start with the first operable state.
4 Everything is fine, the DRAM is working, the host computer
5 can read and write data to the DRAM, it operates essentially
6 full capability.

7 In the second operable state, after an overvoltage
8 condition, for example, the DRAMs are turned off but other
9 elements in the system continue to operate, such as the SPD
10 that we talked about and temperature sensors and the RCD.

11 Q. In the second operable state, is there power on the DIMM?

12 A. There is. There is power provided at least to the RCD,
13 the SPD, the temperature sensor, and the PMIC.

14 Q. And have you seen any testimony about this?

15 A. Yes, I have. Mr. Park, again, Samsung's corporate
16 representative to talk about these subjects, was asked about
17 it, and he said that the LDOs continue to output voltage to
18 power the SPD and the temperature sensors even when you're in
19 an overvoltage condition.

20 Q. In the second operable state, are those components that
21 you've been discussing on the memory module active and
22 operating?

23 A. They are active and operating. And it's important that
24 they are; otherwise, there would be no way for the host
25 processor to turn the DIMM back on. The RCD needs to be

1 listening to commands from it.

2 Q. And have you seen testimony about that?

3 A. I have. Mr. Lo from Renesas was asked about this,
4 whether the LDOs continued to operate, and he said that the
5 components that are powered by them will still be on and
6 operating.

7 Q. And so that's the last element of claim 16. What did you
8 conclude about '054, claim 16?

9 A. All of the accused products, the SODIMM, UDIMM, and
10 RDIMM, infringe claim 16 of the '054.

11 Q. And what claim did you analyze next?

12 A. Next I turned to claim 17.

13 Q. And does the overvoltage monitoring already for '918,
14 claim 16F, relate to this element?

15 A. It does, because the first part of this element talks
16 about a monitoring circuit that detects when you go
17 overvoltage, and we've already talked about this.

18 Q. Does the housekeeping block that you've described also
19 monitor for an undervoltage?

20 A. It does. It says that -- JTX 12 says that it monitors
21 for good power status, which means reliable that it can use,
22 and, in fact, it recognizes when the input power supply is
23 failed or when VIN_BULK goes below a particular threshold.

24 Q. So what did you conclude about '054, claim 17?

25 A. It's infringed by the accused products.

1 Q. And can you just summarize your opinions on the '918 and
2 '054 Patent's infringement?

3 A. So '918 claims, 1, 5, 13, are infringed by the RDIMM
4 products, the accused RDIMM products. All of the other claims
5 are infringed by all of the accused products.

6 MR. MANZIN-MONNIN: And, Mr. Huynh, can you please
7 bring up slide 4.115?

8 Q. (BY MR. MANZIN-MONNIN) So let's talk a little more about
9 the benefits associated with these patents. How did speed
10 change between DDR4, the prior generation, and DDR5, the
11 products at issue here?

12 A. The speed doubled. It was a significant improvement.

13 Q. And how does that change in speed impact the power
14 requirements?

15 A. Well, as I mentioned before, in general, as circuits get
16 faster, they consume more power and give off more heat, and

17 Mr. Kim testified to exactly this fact. He agrees.

18 Q. Is Mr. Kim a low-level engineer?

19 A. No. He's Samsung's corporate representative to answer
20 questions on DDR5, and his title is senior engineer.

21 Q. Did the '918 and '054 Patent inventions address the
22 increasing power demand for DDR5?

23 A. Yes, they did.

24 Q. And what was the resulting increase in power efficiency?

25 A. As I mentioned previously, from JTX 24, Samsung saw an

1 increase of 30 percent in power efficiency.

2 Q. And do you agree with that?

3 A. Yeah, that seems right to me.

4 Q. How does that increase in power efficiency relate to
5 speed?

6 A. Well, I'm going to rely on a document from -- that was
7 developed by Intel and Samsung. This is PX 932. And they
8 were working together, I assume, to develop technology for
9 DDR5 to meet their performance goals, and what's shown here is
10 a slide that relates on the X axis the speed. And the speed
11 is measured in something called millions of transactions per
12 second. It just means a bigger number means faster speed, and
13 power on the left-hand side, on the vertical side. So the
14 yellow line that -- I've added the yellow highlighting to show
15 the relationship between speed and power.

16 Q. And can you show on that graph where the DDR5 max speed
17 was so you have it there? Can you describe what you have?

18 A. That's right. I've identified 4800 is the maximum speed
19 that DDR5 supports. And I've drawn this green line up to the
20 yellow line that relates power to speed, and I've used that
21 speed to determine what the power associated with it was, and
22 then I marked that on the left-hand side as the power at
23 maximum speed.

24 Q. And so what would be the impact if Samsung wasn't able to
25 use Netlist's inventions?

1 A. Well, Samsung said that they achieve a 30 percent power
2 efficiency benefit, so I would take away the 30 percent;
3 reduce the power they had available by 30 percent.

4 Q. And so what would be the maximum speed without the
5 inventions?

6 A. So without the invention, they would be capped at 3200
7 million transactions per second, or in essence they would
8 lose -- their speed would be reduced by 33 percent.

9 Q. Did Samsung have available alternatives to Netlist's
10 inventions?

11 A. They were not able to identify any viable alternatives.
12 This is from testimony from Mr. Park again.

13 Q. So let's turn now to the last family of patents you'll be
14 addressing. This is the '339 Patent.

15 Are there challenges to increasing capacity on a memory
16 module?

17 A. Yes, absolutely.

18 Q. Can you show us what that looks like?

19 A. Here's an example we've got. I'm going to start by
20 talking about ranks of memory. We've got one rank, and this
21 just means when the memory controller says write some data
22 into memory, it's going to also identify write it into a
23 particular address in rank 1 or write it into an address in
24 rank 2. We're just adding more memory chips to the DIMM.

25 Q. Excuse me, Doctor. How does adding a second rank impact

1 the load?

2 A. So it takes work for the memory controller to talk to a
3 rank, and we call that load. And when it talks to two ranks,
4 it's double the load that it needs to deal with.

5 Q. And what is the effect of doubling the load?

6 A. It's a significant -- it increases a significant
7 performance challenge in order to make the system work at
8 the speed you want to achieve.

9 Q. So more load, a lot harder for this thing to run
10 efficiently and run well and --

11 A. Exactly. Run fast.

12 Q. Are there other approaches to increasing memory capacity
13 in a computer system?

14 A. There are. So modern high-end systems have what are
15 called channels rather than just one DIMM. So you can plug
16 two DIMMs together and the memory controller can decide which
17 DIMM to address. So we plug one DIMM into one socket and
18 another DIMM into another socket, and the combination of all
19 of the DIMMs that are sort of connected together electrically
20 are called the memory channel.

21 Q. And if we go back to just one DIMM, what is the load here
22 on the memory controller?

23 A. Well, what I'm trying to illustrate here is that there
24 are four rows of memory chips so there are four ranks. So
25 just this one DIMM would have a rank of four--I'm sorry--a

1 load of four.

2 Q. And if we added a second DIMM into that channel, what
3 would the load be?

4 A. We doubled the load to eight.

5 Q. Can the memory controller handle that load at DDR4
6 speeds?

7 A. No, absolutely not.

8 Q. So -- and how does the '339 Patent address these issues?

9 A. So it addresses these issues by introducing something
10 called a buffer in order to provide isolation between the
11 memory controller and the individual SDRAMs.

12 Q. And so if we bring up figure 3A here, what is being
13 shown?

14 A. So this is from the '339 Patent. And on the left it's
15 discussing 3A, and on the right it's showing that there are
16 four DIMMs. And they're stacked on top of each other. And
17 there are four ranks on any individual DIMM. And I've
18 highlighted on the left that what they are creating is what
19 they refer to as a load-reduced memory module.

20 And there's one more addition. There is the data
21 transmission circuit, which I've highlighted in blue. And
22 what this means is that previously the wires from the system
23 memory controller would literally go up and be connected to
24 every single DRAM chip. In this case, that electrical
25 connection gets cut and the transmission circuit or the

1 buffers, those signals, and then forwards them as needed to
2 the DRAM.

3 Q. And so just focusing on just the front DIMM where you've
4 got those four ranks in orange, what is the load on the memory
5 controller after you add that buffer?

6 A. After you add that buffer, it's a load of one.

7 Q. So it's reduced from four all the way down to one by
8 adding the buffer?

9 A. Exactly.

10 Q. Can you show what this figure 3A looks like using an
11 illustration?

12 A. I created this illustration. On the right-hand side, it
13 shows four DIMMs in four sockets on one memory channel. There
14 is the system memory controller I've highlighted in green, the
15 DIMMs in purple, the SDRAM in orange, and these buffer
16 circuits in blue, and then, finally, the wires of the channel
17 itself in yellow.

18 Q. All right. So we've got the components matched up there.
19 Now let's flip back to the patent.

20 What does the patent tell us about how a write operation
21 occurs?

22 A. So it's very specific. When you use this invention, a
23 write operation is directed to one specific rank, so only one
24 rank, on one specific memory module.

25 Q. Now, are the data paths on the other DIMMs also enabled?

1 A. No. What happens is that the transmission circuits on
2 the remaining modules are disabled.

3 Q. And can we flip back to your diagram and take a look at
4 what that looks like? Sorry. Forward. Flip forward.

5 A. Yes. So -- well, ask.

6 Q. Yes. Can you show what figure 3A, what you just
7 discussed about one data buffer being turned on and the other
8 off, what are you showing there with your illustration?

9 A. I put a green checkmark next to the data buffers on the
10 first DIMM and a red X on the other three DIMMs because they
11 are turned off.

12 Q. So for a given data transmission, there's only a single
13 path for the data?

14 A. That's correct.

15 Q. Now, which Samsung products use the '339 inventions?

16 A. So that would be their DDR4 LRDIMM products.

17 Q. All right. Let's turn to the claims.

18 Can you tell us about '339, claim 1?

19 A. So '339, claim 1, element A, begins by talking about an
20 N-bit-wide memory module mountable in a memory socket. So
21 we've seen examples of these DIMMs several times and that they
22 get mounted into a socket.

23 Q. How does the DIMM communicate with the rest of the
24 computer system?

25 A. So the DIMM communicates with the rest of the computer

1 system over the signal lines that are carried over those edge
2 connectors. Now, data comes in, it's either -- in this case
3 it's either 64 bits or 72 bits, and that's why the patent
4 begins by saying an N-bit-wide. That just means there's some
5 number, it happens to be 64 or 72, so it communicates with the
6 memory controller via those pins on the edge connector.

7 Q. And did you draw those red lines into the figure?

8 A. No, I did not. They were there on the original figure.

9 Q. And that's JTX 28. How does the data communicate with
10 the memory module?

11 A. So the data communicates with the memory module over the
12 same set of pins.

13 Q. So it comes in through the edge connectors and into the
14 memory module?

15 A. Yes.

16 Q. So that's all the language in element A. What did you
17 conclude about that element?

18 A. So I found that element present in the accused products.

19 Q. And how about element 1B?

20 A. So 1B begins, again, by reciting a printed circuit
21 board and an edge connector and electrical contacts. Now,
22 I reproduced evidence from JTX 28, but we've already seen
23 examples of these DIMMs and edge connectors and printed
24 circuit boards.

25 Q. Are there DRAM coupled to the PCV?

1 A. There are. I've highlighted those in orange here.

2 Q. And in this picture you have here of the accused
3 products, how many ranks of DRAM are you showing?

4 A. So this image is showing two ranks, one on top and
5 one below.

6 Q. And are there sometimes two additional on the back?

7 A. Yes.

8 Q. So just like your illustration earlier, four ranks?

9 A. Exactly.

10 Q. Is there also a module controller coupled to the PCV?

11 A. There is. Let me highlight that in the middle. That's
12 the RCD that I had mentioned previously.

13 Q. And so that's all the language for 1B. What did you
14 conclude there?

15 A. That's present, so I put a checkmark next to it. And
16 then I moved on to 1C, which begins, "wherein the module
17 controller." So it's going to tell us more about the module
18 controller.

19 Q. So what happens when the memory controller, the memory
20 controller being off that DIMM component, when it wants to
21 perform a write, what happens?

22 A. Well, what it does is it sends address and control
23 signals to the DIMM so that it knows that it should get ready
24 for a write.

25 Q. And what does the module controller do after it receives

1 those input addressing control signals?

2 A. After it receives the input, it sends an output onto the
3 SDRAMs telling them, Hey, we're going to do a write; whatever
4 you need to do, get ready for it.

5 Q. And what do the address -- let me skip that.

6 So let's take a closer look at those address and control
7 signals. Let me pull up a couple of examples.

8 So you have here highlighted 'rank selection'. Can you
9 explain what that control signal does?

10 A. That's a control signal that is sent -- so the rank
11 select signals are sent to all of the ranks and cause one
12 rank to be active and the others to be turned off.

13 Q. And how about the text in green there, the highlighting
14 you have of Samsung's data sheet you have in green there?
15 What is that?

16 A. So JTX 28 tells us that there are address signals which
17 are used to indicate exactly where in the rank data should be
18 written.

19 Q. And do these address and control signals cause the memory
20 module to perform a write?

21 A. Yes, sir.

22 Q. And when does the memory module perform the write?

23 A. Well, it can only write the data after it receives the
24 data, and in this case I'm showing that, indeed, the 64 bits
25 of data are sent to the DIMM.

1 Q. What element did you look at next?

2 A. So next I looked at element D and F together, only
3 because they seem to go better together in an analysis, in
4 my opinion.

5 So I'm going to start by talking about D, which further
6 discusses how the module controller works.

7 Q. Why does the RCD send module controller signals to the
8 data buffers?

9 A. Well, it's already told the DRAM that they need to get
10 ready to do a write operation. Now it needs to tell the data
11 buffers that they need to get ready; otherwise, they wouldn't
12 know to be turned on. And it's -- the RCD sends that
13 information to the data buffers over something called the
14 BCOM. That's just a set of wires over which it passes
15 commands.

16 Q. So let's turn to those data paths in the buffers.

17 So Samsung showed this slide during its opening
18 statement, and they represented that it was your opinion that
19 this figure here shows a single path in the data buffers. Is
20 that green line there that's annotated on the figure, is that
21 a single data path?

22 A. Yes.

23 Q. Does that tell the full story of what's going on here?

24 A. No. As is often the case with complex systems like this,
25 there's another layer you can dig deeper into.

1 Q. Can you explain that for us?

2 A. Sure. If we zoom in on this figure a little bit more,
3 hopefully it becomes apparent that in the bottom right-hand
4 edge, it refers to a lower nibble and an upper nibble. A
5 nibble is just half of a byte.

6 And if you look on the right-hand side of that blow-up,
7 hopefully it's apparent that there's two sets of these
8 circuits stacked on top of each other. So there's actually
9 one circuit for the upper half byte and one for the lower half
10 byte.

11 Q. So how many bits of data travel or how many bytes of data
12 travel on the green path highlighted during Samsung's opening?

13 A. Four bits.

14 Q. And how many total bits travel through the buffer for a
15 given write transaction?

16 A. For a given write transaction, a byte is passed.

17 Q. So can you show us where the other four bits travel?

18 A. Sure. I've just reproduced that circuitry and I've shown
19 it here. So a half byte goes over the green path and a half
20 byte goes over the red path.

21 Q. And have you confirmed in the source code that's how it's
22 implemented in the accused products?

23 A. I did. Let me illustrate that a little bit. So I'm
24 showing it connected to the individual DRAM devices. We
25 haven't really talked much about what's -- how the SDRAM chips

1 themselves work, but each one reads or writes four bits at a
2 time, so half a byte. And JTX 28 shows this figure that I've
3 -- literally from their documentation.

4 Q. Now, I just want to make clear what's on the slide here,
5 because there's some exhibit stamps.

6 On the left, that's your illustration of how the product
7 works?

8 A. Yes, that's correct.

9 Q. And it's got some parts from the data sheets, but you've
10 arranged that one on the left?

11 A. That's right. I made that arrangement.

12 Q. But it matches precisely with Samsung's documentation up
13 on the right.

14 A. Yes.

15 Q. Okay. So let's take a closer look at the data buffers.

16 What is the first side of the buffer connected to?

17 A. So the first side of the data buffer is the side that's
18 connected to the memory controller through the socket and into
19 the host processor.

20 Q. And what is the second side connected to?

21 A. Well, the second side is the opposite side of the data
22 buffer that's actually connected to the DRAMs.

23 Q. Okay. So we've covered all of D. You said you are going
24 to do D and F together, and we'll jump to E after. Let's
25 move ahead to F. How does the buffer turn on the data paths

1 in the buffer?

2 A. It does that using control logic, and I've highlighted
3 the control logic here in purple. It's labeled 'transaction
4 control'.

5 Q. And what triggers the transaction control to enable the
6 data path?

7 A. Well, that would be the module control signals that the
8 RCD generated and passed down to the data buffers over the
9 BCOM bus, so when it receives a message saying get ready to do
10 a write, it gets ready to do a write.

11 Q. And have you seen testimony about this?

12 A. Yes, I have.

13 Q. Can you show us that testimony?

14 A. Sure. Mr. Davey from Renesas said that the transaction
15 control block is the brains of the circuit. It reacts to
16 write commands that come over the BCOM bus, for example.

17 Q. So what did you conclude about elements D and F of '339,
18 claim 1?

19 A. I found them both present in the accused products.

20 Q. And what element did you look at next?

21 A. That would be element E, which talks about the location
22 of the data buffers.

23 Q. Now, how does the position of the data signal lines
24 compare to the buffer?

25 A. Well, remember the data signal lines are the lines that

1 are coming in over the edge connector from the socket on the
2 motherboard. This image is from JTX 28, and it sort of shows
3 that there's a relationship, but if we look -- relationship in
4 terms of position. It shows that the signal lines go into
5 those data buffers. But if we actually look at the literal
6 product, you'll see that those data signal lines are arrayed
7 directly below each one of the buffers. There is, I would
8 say, an exact vertical correspondence.

9 Q. And you determined the location of those data lines
10 looking at JTX 28 which has an exact pin layout. Right?

11 A. Exactly. It tells me exactly which pins carries which
12 signals, and then which data buffer it goes to.

13 Q. What did you look at next?

14 A. So next I looked at the last element that would be
15 element G, which then goes on and talks furthermore about
16 how the data path is enabled.

17 Q. And is the data path enabled in accordance with the
18 latency parameter?

19 A. It is. That's essential to making this -- the data
20 buffers work at the targeted speed. Mr. Davey commented on
21 this. He said that after a latency delay, a cycle delay, a
22 phase delay--so there's all these carefully programmed
23 delays--the path is turned on.

24 Q. And you're showing -- it's a little faded in the back.
25 You're showing the back path is red and the front path is

1 green. Do these two paths get enabled at the same time?

2 A. No, they actually operate independently. They have their
3 own latency delays that are programmed in.

4 Q. The next part of the claim requires that the path be
5 enabled to drive data across the buffer. What does 'drive'
6 mean?

7 A. This is another situation where the Court has told us
8 what 'drive' means in this context, and it means enabling only
9 one of the data paths while the other possible data paths are
10 disabled.

11 Q. And is that how the system works? Is the first path
12 enabled while all the other possible paths are disabled?

13 A. That's exactly how it works.

14 Q. And can you show us some testimony supporting that?

15 A. Sure. Mr. Davey commented on this, and he said that the
16 paths are turned off by default. So all the paths are off,
17 and only when they receive a write command from the BCOM bus
18 and after the latency delay has occurred are they turned on.

19 Q. And can you show us what this looks like with your
20 illustration?

21 A. Sure. So here's another figure showing that there's a
22 checkmark next to the data buffers that are turned on and all
23 of the other data buffers are turned off.

24 Q. So just going back to your diagram, so when the first
25 four bits get driven through the buffer, the buffers on all

1 the other DIMMs are off. Is that right?

2 A. That's right. Only the buffer on DIMM 1 is on.

3 Q. And does this fork in the road approach actually work in
4 the products?

5 A. Yes. This is exactly how the product works.

6 Q. Now, let's look at how those paths precisely are turned
7 on and off. What is the triangle that you've made purple
8 there in the data path?

9 A. So it's labeled TX. That's for transmitter. And that
10 figure is used to represent what's called a tri-state buffer.
11 In electronics, a tri-state buffer is just a buffer that is
12 like a door. It's either opened or closed. The path is
13 turned on or it's turned off.

14 Q. And how do the transaction logic and the tri-state buffer
15 drive the data to the BCOMs?

16 A. Well, they do that by providing the information that was
17 received on the first side up to the second side.

18 Q. And have you seen testimony about that?

19 A. I have, yeah. Mr. Davey was asked about this, and it
20 does it in response to a message received over the BCOM bus,
21 and it does that by turning on one path for a particular write
22 transaction.

23 Q. And that's in response, as you said, to the BCOM module
24 control signals we've looked at?

25 A. Yes.

1 Q. Okay. So that's the last element of claim 1. What did
2 you conclude?

3 A. Since the evidence shows that all of the elements were
4 present, the accused DDR4 products practice this claim.

5 Q. What claim did you analyze next?

6 A. Next I turned to claim 8, which depends from claim 1, and
7 adds another limitation.

8 Q. Now, is the rank select signal we've looked at, the one
9 that made that one row orange, is it different for one rank
10 than the other ranks in the chip?

11 A. Yes. Each rank gets its own rank select signal.

12 Q. And is that what this claim signal is talking about?

13 A. Yes.

14 Q. So what did you conclude for '339, claim 8?

15 A. I found this element present and so the claim is
16 infringed.

17 Q. So that brings us to the final claim. Can you tell us
18 about claim 9?

19 A. Sure. Claim 9 depends from claim 8, and it further talks
20 about what's happening with the module data lines.

21 Q. And are the data lines from the buffers able to carry
22 data to a DRAM in each of the ranks in?

23 A. Yes --

24 Q. Let me rephrase. So in one write they might carry data
25 to the first rank and a different write to the second rank,

1 but they're capable of carrying data to either rank. Is that
2 right?

3 A. Yes, that's exactly how they operate.

4 Q. And so what did you conclude about '339, claim 9?

5 A. I found it present, and so -- I found the additional
6 limitation present, and so claim 9 is infringed.

7 Q. And can you summarize your opinion with regards to the
8 claims of the '339 Patent?

9 A. The accused products for the '339 infringe claims 1, 8,
10 and 9.

11 Q. Before we move to the benefits here, I just want to touch
12 back on one thing we spoke about. So you mentioned --

13 THE COURT: Counsel, there's no need to tell the
14 jury or the witness what you want to touch back on. You're
15 not here to testify.

16 MR. MANZIN-MONNIN: Yes, Your Honor.

17 THE COURT: If you want to testify, I'll swear you
18 in and put you on the witness stand. Your job is to ask
19 questions, not to make statements.

20 MR. MANZIN-MONNIN: Yes, Your Honor.

21 THE COURT: Understood?

22 MR. MANZIN-MONNIN: Understood, Your Honor.

23 THE COURT: All right. Let's proceed on that basis.

24 Q. (BY MR. MANZIN-MONNIN) So did Samsung -- how much speed
25 loss would Samsung have realized if they were not able to use

1 the '918 and '054 Patents in the DDR5 products we looked at
2 first?

3 A. That would be 33 percent speed reduction.

4 Q. And to confirm, the way they could have done that is
5 to move the power management off the module. Is that right?

6 A. That's right. They could have used the earlier
7 technology where it was not on-module.

8 Q. Okay. So now let's look at the benefits of the '339.

9 Has Samsung identified any alternative technology that
10 it could have used in place of the '339 inventions to achieve
11 DDR4 speeds in the LRDIMM accused products?

12 A. No, they have not.

13 Q. If Samsung was aware of an alternative, would you have
14 expected them to say so?

15 A. Yes. Typically that's what's happened, in my experience.

16 Q. And can you summarize the benefits of the '339 invention
17 that you've explained?

18 A. Sure. So here I've shown memory controller with four
19 channels. This is representative of, again, modern servers,
20 and each channel is populated with two DIMMs in this case.
21 Without the invention, they could only support one DIMM.
22 And so, as a result, the capacity with the '339 invention is
23 doubled.

24 Q. Would DDR4 LRDIMM customers purchase modules that are not
25 capable of running at DDR4 speeds with two or more modules per

1 channel?

2 A. No, I don't -- I can't imagine that.

3 Q. So let's turn to the last part of your analysis. Can you
4 describe the comparability analysis you performed?

5 A. Yes. I was asked to consider a licensing agreement that
6 involved a company called Rambus, consider the technology that
7 was related to that, and comment on whether I thought it was
8 comparable to the technology in these patents.

9 Q. And did you find that the agreement was comparable?

10 A. Yes, I did. What I found is that the Samsung technology
11 related to DDR memory systems and communicating to DDR memory
12 systems and improving the speed with which the communication
13 could happen and reducing the power that was required.

14 Q. And what did you conclude about the licensed Rambus
15 patents?

16 A. That the license was comparable. The technology in the
17 licensing agreement was comparable to the technology in this
18 case.

19 Q. Did the patents relate to similar issues to the asserted
20 patents?

21 A. Yes, very much.

22 MR. MANZIN-MONNIN: Nothing further. I pass the
23 witness.

24 THE COURT: All right. Ladies and gentlemen, we're
25 just a few minutes before noon, and I've been informed by the

1 Clerk's Office that your lunch is ready for you in the jury
2 room, so we're going to break at this point. We'll proceed
3 with the Defendants' cross-examination of Doctor
4 Mangione-Smith when you come back from lunch.

5 If you will, take your notebooks with you to the jury
6 room during the lunch break. Please remember to follow all my
7 instructions, including not to discuss the case with each
8 other. And we are at about seven minutes until noon. We'll
9 try to pick up in about 60 minutes from now.

10 With that, the jury's excused for lunch.

11 (Whereupon, the jury left the courtroom.)

12 THE COURT: Be seated, please.

13 I want to return with counsel briefly to the episode
14 with PX 1860 [sic].

15 Several months ago the Court entered a standing order in
16 this court regarding patent infringement cases, particularly
17 addressing the admission and pre-admission of exhibits. The
18 last paragraph of that standing order is very clear, in my
19 view, and it says, "Notwithstanding the foregoing and to
20 assuage the parties' concerns that something vital might have
21 been overlooked as a part of this practice, if a party
22 inadvertently omits a relevant document from its party exhibit
23 list, it may move to admit the omitted document into evidence
24 at trial until the close of evidence upon a showing of good
25 cause. However, no party shall seek the admission of more

1 than four omitted documents at trial, and no omitted document
2 shall be raised or mentioned in the presence of the venire
3 panel or the jury prior to its admission by the Court. Any
4 party objecting to the admission of the omitted document will
5 be heard by the Court outside the presence of the jury."

6 Both sides totally ignored my standing order. At no time
7 did the Plaintiff move to admit PX 1820 until I raised the
8 question with you after you'd already exposed it and shown it
9 to the jury. The objection was raised by the Defendant about
10 the scope of the expert's opinion, and I sent the jury out of
11 the courtroom. At no time did you direct any offer of good
12 cause to the Court; you merely told me that you took the time
13 to let the other side know you intended to do that. And then
14 you sprung it on the jury and the Court without any prior
15 notice. That does not comply with the standing order.

16 And, quite honestly, Defendants, you never objected to
17 what they did. You should have read the standing order, too.
18 And, Mr. McKeon, you should have said, They've never made any
19 showing of good cause, they've never offered this to the Court
20 before it was exposed or exhibited to the jury. You sat there
21 like a bump on a log and didn't raise the standing order at
22 all.

23 And, quite honestly, I'm very unhappy with the way that
24 went down on PX 1820. I feel like the Court was, if not taken
25 for granted, ignored intentionally. I feel like the standing

1 order makes it abundantly clear that there must be an offer
2 outside the presence of the jury. If there is an objection,
3 it's to be made outside the presence of the jury, and in no
4 event is the document to be shown -- until it's admitted by
5 the Court, it's not to be shown to the jury, and it's only to
6 be admitted upon a showing of good cause. I never saw any
7 good cause. And, lo and behold, I had to raise the issue of
8 whether this was being offered for admission, and that was way
9 after it had already been exposed to the jury.

10 Let me remind both sides of something that I have tried
11 to emulate throughout my time as a licensed attorney and as a
12 member of the judiciary. Lawsuits are a race for credibility,
13 first with the Court and then with the jury. And I will tell
14 you with regard to PX 1820, both sides, particularly the
15 Plaintiff, lost some credibility with the Court today.

16 And I do not expect that to occur again in this trial or
17 in the future. If there is any document that has not been
18 fully exposed to the light of day during the pretrial
19 practice--and heaven only knows, you-all took hours and hours
20 of argument over exhibits in front of Magistrate Judge Payne,
21 at least over a couple of days if not more than that--if there
22 is something that hasn't been exposed to the light of day
23 during the pretrial practice and you hope to have it admitted
24 during the trial pursuant to this standing order, you darn
25 well better comply with the order. And if you don't comply

1 with the order again, you're going to be sanctioned.

2 All right. We stand in recess for lunch.

3 (Lunch recess.)

4 End of part 8

5 .

6 .

7 Beginning of file 9.

8 THE COURT: Be seated, please.

9 Mr. McKeon, are you prepared to proceed with
10 cross-examination?

11 MR. McKEON: Yes, Your Honor.

12 THE COURT: All right. Let's bring in the jury,
13 please. Before you bring in the jury, I want to put one thing
14 on the record.

15 Previously there was an issue at some length outside of
16 the presence of the jury about PX 1820 as an item that
17 Plaintiff moved to be admitted under the Court's standing
18 order which was not addressed during the pretrial proceedings
19 in this case. The Court ordered it admitted as an exhibit. I
20 am rescinding that order. What is marked as PX 1820 is to be
21 considered a demonstrative but not an exhibit or part of the
22 record evidence in the case.

23 The testimony regarding stands, and I'm not striking
24 anything at this point or I don't -- and I don't intend at
25 this point to present any kind of curative instruction to the

1 jury. I will take up any related requests later in the trial
2 if and when they are presented, but that's my ruling as of
3 this time.

4 With that, let's bring in the jury.

5 MR. McKEON: Your Honor, may I stand at the podium.

6 THE COURT: Yes.

7 (Whereupon, the jury entered the courtroom.)

8 THE COURT: Welcome back from lunch, ladies and
9 gentlemen. Please have a seat. We'll proceed with the
10 cross-examination by the Samsung Defendants of Doctor
11 Mangione-Smith.

12 Mr. McKeon, you may proceed.

13 MR. McKEON: Thank you, Your Honor.

14 CROSS EXAMINATION

15 BY MR. McKEON:

16 Q. Good afternoon, Doctor Mangione-Smith. Nice to see you.
17 My name is Mike McKeon, and I represent Samsung.

18 A. Good afternoon, sir.

19 Q. Now, let's start with the '918 and '054 Patents. And
20 just to make it easier, they share the same specification and
21 the same figures and written description. Isn't that right?

22 A. Yes.

23 Q. And you gave an opinion that the incremental value of the
24 '918 and '054 Patents is moving the voltage regulation off the
25 mother board and onto the memory module. Correct?

1 A. I don't recall saying that. Could you refresh my memory?

2 I'd be happy to review.

3 Q. Well, you believe that there's an incremental value with
4 the invention of the '918 and '054 Patents. Right?

5 A. Sure, over the prior art, yes.

6 Q. Right. And that's the application of on-module voltage
7 regulation. Correct?

8 A. It's the whole claim.

9 Q. Right. So the incremental benefit to these patents is
10 having the voltage regulation on the module. Correct?

11 A. It's the whole claim is the benefit, yeah. That's part
12 of it.

13 Q. Okay. So -- okay. So you also -- and just so
14 we're -- it's easier for this discussion we're going to have,
15 when we talk about voltage regulation, does that mean the same
16 thing as power management to you?

17 A. No.

18 Q. There they're different things to you?

19 A. Yeah. It would depend on the context, they might be the
20 same, but in general they mean different things.

21 Q. Okay. And, of course, you know in this case Netlist is
22 asking for a lot of money based on the idea of taking the
23 voltage regulator and putting it on the module. You're aware
24 of that?

25 A. Based on -- their belief that claim 1 is infringed, yeah.

1 Q. Okay. Now, you agree, of course, that Netlist shouldn't
2 get credit for inventing on board voltage regulation if
3 someone else invented it first. You agree with that. Right?

4 A. Yes.

5 Q. And the work in this case, you never endeavored to figure
6 out what was new or unique about the '054 Patent. Correct?

7 A. Not beyond the claims themselves, that's correct.

8 Q. Well, let's make sure we've got the testimony again. So
9 I'm going to make you read your deposition. Okay? If you can
10 open the binder that you have, sir, and it should be the first
11 tab. And I want you to read page 59:10 to 25.

12 A. I'm sorry. You said page 59?

13 Q. Yeah, 59:10 to 25. When you read that, let me know.

14 A. I'm sorry. I turned to PDF page 59.

15 Q. It should be the deposition page.

16 A. Yeah.

17 THE COURT: The numbers are in the upper right-hand
18 corner.

19 THE WITNESS: Yes, sir. Thank you.

20 All right. I'm on page 59.

21 Q. (BY MR. McKEON) Lines 10 to 25.

22 A. Yes, sir, I see that.

23 Q. Okay. And, again, your work in this case, you never
24 endeavored to figure out what was new or unique about the '054
25 Patent. Correct?

1 A. Sure.

2 Q. And, likewise, you never tried to figure out what was new
3 or unique about the '918 Patent. Correct?

4 A. That's right.

5 Q. But you understand, of course, that some of the concepts,
6 including this concept of moving the voltage regulator onto
7 the module, that was -- that's something that was done before
8 these Netlist patents. Isn't that right?

9 A. No, I wouldn't agree with that.

10 Q. Okay. Well, let's take a look at it.

11 A. The best I recall.

12 Q. Let's take a look at it.

13 MR. McKEON: If we can get JTX 56 up, pre-admitted.

14 Q. (BY MR. McKEON) And you're familiar with the Harris
15 patent, sir?

16 A. Yes, sir.

17 Q. Okay. And this is a patent I bet you looked at in this
18 case. Isn't that right?

19 A. That's right.

20 Q. And you agree the patent came before the '918 and '054
21 inventions. Isn't that right?

22 A. Yes.

23 MR. McKEON: And let's go to figure 1A of this
24 patent.

25 Q. (BY MR. McKEON) And we see here in figure 1A, we see

1 something labeled 100A. Do you see that? It's up at the
2 right-hand corner.

3 A. Yes.

4 Q. Okay. And this is a -- what's labeled as 100-A is a dual
5 in-line memory module. Correct?

6 A. Yes.

7 Q. And you agree that Harris discloses putting a voltage
8 regulator on the memory module. Correct?

9 A. Harris does talk about that, yes.

10 Q. In fact, it's shown right here, element 102, voltage
11 regulator. Correct?

12 A. Yes.

13 Q. So the idea of putting a voltage regulator on the memory
14 module, that was done before these patents. Isn't that right?

15 A. Yes. The idea does talked about, absolutely.

16 Q. And you've seen the Amidi patent before. Correct?

17 A. Yes.

18 Q. Let's pull that up. It's JTX 054. And you recognize
19 this patent, sir?

20 A. I do.

21 Q. And you agree this is prior art to the '054 and '918
22 Patent. Right?

23 A. Yes.

24 Q. And that means, when we say prior art, it's kind of a
25 patent-ish term that means something that was done before the

1 patents-at-issue.

2 A. Yes. That's my understanding.

3 Q. Okay. And you agree that -- why don't we go to figure 5
4 here. And you agree that Amidi discloses putting the power
5 management block shown in figure 5 on the memory module.

6 Correct?

7 A. Yes.

8 Q. And, of course, you agree that the power management block
9 includes a voltage regulator. Correct?

10 A. Yeah, I believe so. From this figure, it certainly seems
11 to, yeah.

12 Q. And if we look at PDX 4.22, this is a slide you presented
13 earlier. Do you recall that?

14 A. Yes, sir.

15 Q. And you have this 30 percent efficiency number that was
16 represented in this document that's a Samsung document, a
17 public document. Correct?

18 A. That's correct.

19 Q. And this is -- this efficient number is associated with
20 putting the voltage regulator on the module. Correct?

21 A. Yes, sir.

22 Q. And that's been done before the patents in this case.
23 Correct?

24 A. It's been talked about, yeah.

25 Q. Thank you, sir.

1 Now, in your direct --

2 MR. McKEON: You can pull this down. Thank you.

3 Q. (BY MR. McKEON) In your direct testimony, you talked
4 about buck converters and converter circuits. Do you recall
5 that?

6 A. I do.

7 Q. Let's look at the claim at issue here.

8 MR. McKEON: Can we go to claim 1 of the patent, and
9 can we highlight the buck converters there, the first, second,
10 and third buck converters?

11 Q. (BY MR. McKEON) And there's three buck converters that
12 are required for the claim. Isn't that right?

13 A. Yes, sir.

14 Q. And there's also a converter circuit that's a separate
15 element of the claim. Isn't that right?

16 A. That's correct.

17 Q. And then we highlight converter circuits so we can all
18 make sure we are on the same page here. Okay. Right here.
19 There we go.

20 Okay. And you see that's where the converter circuit
21 element of the claim is. Right.

22 A. Yes, sir. That's the converter circuit.

23 Q. Now, you agree that a buck converter is the name of a
24 specific type of voltage converter that already existed before
25 the '918 and '054 Patents. Correct?

1 A. Sure.

2 Q. Okay. Let's go to figure 6 of Amidi. We can just take a
3 look at that.

4 MR. McKEON: And if we can get back the Amidi
5 reference, which is JTX 54.

6 MR. MANZIN-MONNIN: Your Honor, may we approach?

7 THE COURT: Approach the bench.

8 (The following was had outside the hearing of the
9 jury.)

10 THE COURT: Yes, sir.

11 MR. MANZIN-MONNIN: Yes. There's no obviousness
12 defense or anticipation defense in the case, and they're
13 walking through whether a particular claim elements were known
14 and specific prior art references that the Court has already
15 found could not be used to argue obvious or anticipation in
16 this case.

17 THE COURT: What's your response?

18 MR. McKEON: I'm just talking about the incremental
19 benefit, Your Honor, and that's what he testified to on
20 direct.

21 THE COURT: You're way past the incremental benefit.

22 MR. McKEON: Okay.

23 THE COURT: This looks like prior art invalidity
24 defense.

25 MR. McKEON: Okay. We'll move on.

1 THE COURT: Let's move on.

2 (The following was had in the presence and hearing
3 of the jury.)

4 THE COURT: All right. Let's proceed.

5 Q. (BY MR. McKEON) All right. Okay. Let's go --

6 MR. McKEON: We can pull this down, Mr. Sayers.

7 Thank you.

8 Q. (BY MR. McKEON) Now, in your direct testimony, you
9 talked about the load drop-out regulator. Do you recall that?

10 A. Yes, sir.

11 Q. And we've referred to that for short as an LDO. Is that
12 right?

13 A. That's what we do in the industry typically, yes.

14 Q. Okay. And in dispute in this case is whether an LDO is
15 satisfied by the requirement of a claim of a converter
16 circuit. Isn't that right?

17 A. Yes. That's one of the disputes as I understand it.

18 Q. Now, you agree that a buck converter is different than an
19 LDO. Right?

20 A. Yes, sir.

21 Q. And you believe that compared to an LDO, a buck converter
22 takes up more space because it requires the use of
23 comparatively bulky passive components such as inductors and
24 capacitors. Correct?

25 A. Yes, sir.

1 Q. And you also believe that a buck converter is more
2 complicated to implement, is more costly to implement, and is
3 more noisy than an LDO. Correct?

4 A. Yes, sir.

5 Q. In fact, you believe that buck converters and LDOs have
6 fundamentally different characteristics as well as operational
7 benefits and limitations. Correct?

8 A. That sounds correct, yes.

9 Q. All right. Let's put up claim 1 again, and I want to
10 talk to you about the converter circuit and I want to focus
11 you on the language of the claim.

12 MR. McKEON: Let's highlight that whole block,
13 please.

14 Q. (BY MR. McKEON) And the claim provides a converter
15 circuit configured to provide a fourth regulated voltage
16 having a fourth voltage amplitude. Do you agree with how
17 -- is that accurate as I read that, sir?

18 A. Yes, I believe you read it right.

19 Q. All right. And the Court didn't construe this particular
20 element, so the construction is the plain and ordinary meaning
21 of the phrase as understood by one of ordinary skill in the
22 art. Is that right?

23 A. Yes.

24 Q. Okay. And as part of that analysis, you look at the
25 specification to figure out how it was used by the inventors

1 in this context. Isn't that right?

2 A. Yes.

3 Q. Now, in your opinion, any circuit that converts a voltage
4 is a converter circuit. Correct?

5 A. Sure, yes.

6 Q. In fact, you believe that an LDO is a converter circuit.
7 Right?

8 A. Yes.

9 Q. And you've testified earlier in this proceeding and prior
10 to the trial, you said that you believe that a wire can be a
11 converter circuit. Isn't that right?

12 A. Yes, because it has resistance.

13 Q. Now, to find infringement, you did not look for any
14 structural element beyond the functional requirements that
15 something reduces the voltage. Correct?

16 A. I reviewed all the material available to me.

17 Q. Well, let me have you look at your deposition, sir. It's
18 the same document you looked at earlier. It's page 112:8 to
19 17.

20 A. Yes, I see that.

21 Q. Okay. Let me ask you again, you did not look for any
22 structural element beyond the functional requirement that
23 something reduced the voltage. Correct?

24 A. Yes, that was the structure I looked for.

25 Q. Now, as long as something converts voltage, you say that

1 it's a converter circuit. Is that right?

2 A. Yes.

3 Q. Now, you agree with me that the converter circuit, this
4 language requires structure in the patent. Right?

5 A. Yes. This is an apparatus claim, so --

6 Q. So just looking at function is not enough to find out
7 what infringes this claim element. Isn't that right?

8 A. From a legal point of view, I don't know if that's
9 accurate or not. I'm sorry.

10 Q. You agree with me that the claim element -- this is a
11 memory module comprising, so it's a physical thing. Right?

12 A. Yes.

13 Q. Okay. And a converter circuit is a physical thing.
14 Isn't that right?

15 A. Yes.

16 Q. And it has structure. Isn't that right?

17 A. Yes.

18 Q. And the claim requires structure in order to find -- in
19 order to find infringement, you have to find structure. Isn't
20 that right?

21 MR. MANZIN-MONNIN: Objection, Your Honor.

22 THE COURT: State your objection.

23 MR. MANZIN-MONNIN: He's asking for a legal
24 conclusion.

25 THE COURT: Overruled. It's an expert. We're

1 talking about structure.

2 Q. (BY MR. McKEON) Do you want me to repeat the question,
3 sir?

4 A. Yes, please.

5 Q. So you agree with me that the converter circuit language
6 of this claim, it requires some structure. Right?

7 A. Yes.

8 Q. And there's -- sometimes there's multiple ways to do a
9 particular function. Do you agree with that? Like if there's
10 a certain function you want to do, there's multiple ways to do
11 it.

12 Let me give you an example. If I want to put two pieces
13 of wood together, you can use a screw, you can use a nail, and
14 you can use glue. Right? And that would accomplish the
15 function of putting the two pieces of wood together. Do you
16 agree with that?

17 A. Yes.

18 Q. But you also would agree with me that a structure of a
19 nail, screw, and glue is very different. Isn't that right?

20 A. Yes.

21 Q. The structure makes a difference. Isn't that right?

22 A. I'm sorry. What was the question?

23 Q. Structure makes a difference. Isn't that right?

24 A. In that case, I don't know that it does. From whoever's
25 using it, maybe they don't care.

1 Q. Well, if you have glue, nail, and screw--right?--there's
2 different ways of accomplishing the function of putting those
3 two boards together. Isn't that right?

4 A. Sure.

5 Q. Okay. Now, you agree, sir, that the '918 --

6 MR. McKEON: You can bring down the claim.

7 Q. (BY MR. McKEON) You agree the '918 Patent does not
8 disclose an LDO. Correct?

9 A. It does not use the words LDO, yes.

10 Q. And it doesn't have the phrase load drop -- load drop-out
11 circuit. Isn't that right?

12 A. That's correct.

13 Q. Okay. Of course, we see the buck converters in the
14 patent. Right?

15 A. Yes.

16 Q. And if we pull up --

17 MR. McKEON: If I can pull up column 29, lines 28 to
18 32. And if we can highlight the sentence that says, the
19 sub-blocks.

20 Q. (BY MR. McKEON) If we look at this part of the
21 specification, sir, it says, the sub-blocks may comprise
22 various converter circuits such as buck converters, boost
23 converters, and buck-boost converter circuits for providing
24 various voltage values to the memory. Do you see that?

25 A. Yes, sir, I see that.

1 Q. And an LDO, when it talks about the converter circuit,
2 you don't see an LDO described here. Isn't that right?

3 A. The words LDO or load drop-out do not appear here.

4 Q. And if you go to figure 16 of the patent --

5 MR. McKEON: Figure 16, please.

6 Q. (BY MR. McKEON) And you've discussed this on your direct
7 testimony. Do you recall that?

8 A. Yes, sir.

9 Q. Okay. And this is the power module that's disclosed in
10 the patent. Isn't that right?

11 A. Yes, sir.

12 Q. Okay. And, of course, in this power module here, we see
13 four buck converters. Isn't that right?

14 A. Yes.

15 Q. And just so we're all on the same page, we've got what's
16 been designated as 1122, and then you got 1124, that actually
17 has got -- there's two in here. Right?

18 A. Yes, that's correct.

19 Q. And then you've got 1126. Right? And those are four
20 buck converters. Isn't that right?

21 A. Right. A buck, a dual buck, and a buck -- what's called
22 a buck-boost.

23 Q. Okay. And now in your deposition, you weren't really
24 able to identify in this figure which one of these
25 corresponded to a converter circuit. Isn't that right?

1 A. I'm sorry. Could you repeat the question?

2 Q. Well, you can't identify which one of these corresponds
3 to the fourth element in the claim, the converter circuit
4 element.

5 A. I did not identify one of these as the fourth converter
6 circuit, the converter circuit in the claim.

7 Q. And you agree with me that the buck converter labeled
8 1122 cannot be an LDO. Correct?

9 A. Yes. There's a difference between a buck and an LDO.

10 Q. And a person of skill in the art would understand that
11 you cannot use one or more LDOs as the buck converter labeled
12 1124. Correct?

13 A. Typically, yes, that's correct.

14 Q. And there's a buck-boost converter labeled 1126 and a
15 buck-boost converter labeled 1144 over here. Do you see that?

16 A. Yes, sir, I do.

17 Q. Okay. And a person of skill in the art would understand
18 you cannot use an LDO in place of the buck-boost converter.
19 Isn't that right?

20 A. Not typically. You wouldn't do that, yes.

21 Q. Because an LDO doesn't boost voltage at all. Isn't that
22 right?

23 A. That's right. Some of these bucks don't boost. But as
24 you mentioned, they also have other things that make them
25 different from LDOs.

1 Q. Okay. And a person of skill in the art would not be able
2 to reach a conclusion regarding whether you can or cannot
3 replace any one of those buck converters shown with an LDO.
4 Correct?

5 A. That's correct. It would depend on the particular
6 circumstances.

7 Q. All right. Okay. Now --

8 MR. McKEON: And we can pull this down.

9 Q. (BY MR. McKEON) Now, you analyzed a number of the
10 different documents related to the PMICs in this case. Isn't
11 that right?

12 A. Yes, sir.

13 Q. Okay. And you presented some of the data sheets in your
14 direct. Correct?

15 A. Yes, sir, I did.

16 Q. Okay. And PMIC stands for power management integrated
17 circuit. Right?

18 A. That's right.

19 Q. And in the accused products, you understand that the PMIC
20 chip is supplied from different vendors. Isn't that right?

21 A. Yes, sir. That's my recollection.

22 Q. Okay. For example, Renesas is a company that supplies a
23 chip to Samsung that it uses in its modules. Isn't that
24 right?

25 A. Yes, sir.

1 Q. And Texas Instruments is another vendor that Samsung
2 uses. Isn't that right?

3 A. That's my recollection, sir.

4 Q. And Monolithic Power System is another vendor that
5 Samsung uses. Correct?

6 A. I believe so, yeah.

7 Q. And, in fact, Samsung has its own chip manufacturing
8 business, and Samsung also gets that from -- from its business
9 that makes this chip. Isn't that right?

10 A. Yes, sir.

11 Q. Now, you reviewed all these data sheets. Isn't that
12 right, sir, in this case?

13 A. Yes, sir, all the ones that I cited and relied upon.

14 Q. Okay. And not one of the data sheets you reviewed
15 referred to an LDO as a converter circuit. Correct?

16 A. To be honest with you, I don't recall.

17 Q. Okay. We'll take a look at some of that.

18 MR. McKEON: Let's pull up JTX 30, please.

19 Q. (BY MR. McKEON) This is that Renesas data sheet. Do you
20 see that?

21 A. Yes, sir.

22 Q. And I believe you talked about this on your direct. Is
23 that right?

24 A. Yes.

25 Q. Okay. If we go down, we see the diagram, I think you

1 presented this to the jury on your direct. Is that right?

2 A. Yes.

3 Q. And that's the first -- that's on the first page of the
4 data sheet, this Renesas data sheet. Right?

5 A. Yes.

6 Q. So the chip vendor Renesas makes this chip, and with it,
7 it gives a data sheet so people can know about the chip, what
8 it does, and all that. Is that right?

9 A. That's right.

10 Q. Okay. And in the data sheet, it specifies two LDOs.
11 Right? One, two. These are in the upper right-hand corner,
12 and they're called LDOs. Isn't that right?

13 A. Yes, sir.

14 Q. Okay. And, of course, you have the opinion in this case
15 that these LDOs are converter circuit. Correct?

16 A. That has been my opinion, yes.

17 Q. But the Renesas document refers to these as LDO
18 regulators. Isn't that -- isn't that right?

19 A. As you were scrolling down, I saw that above.

20 Q. Let's go up a little here.

21 A. Yeah.

22 Q. Go up a little more? There we go. You see that, sir,
23 two regulators? It's referring to the two LDOs we marked here
24 in the figures. Isn't that right?

25 A. Yes, sir.

1 Q. Okay. And so, again, no reference in this document to
2 these LDOs being a converter circuit. Isn't that right?

3 A. I don't recall. We've looked -- there's no reference to
4 it here.

5 Q. But in this case you haven't cited anywhere in this
6 document that calls the LDO a converter circuit. Isn't that
7 right?

8 A. I believe that's true, yeah.

9 Q. Let's look at the -- at another one, JTX 11. And you
10 recall, sir, you provided some testimony on direct about this
11 document. Isn't that right?

12 A. Yes.

13 Q. Okay. And this is the Samsung PMIC. Isn't that right?

14 A. Yes, sir.

15 Q. Okay. And you go to page 69 of this document, we see a
16 designation here at the top, 3.4. It says, linear regulator
17 LDO. Do you see that?

18 A. Yes, I see that.

19 Q. Okay. And this is the part that you say -- this is
20 portion of the chip you say corresponds to the converter
21 circuit. Isn't that right?

22 A. Yes, although I think I only identified LDO 2
23 specifically. I don't know if this is 1 or 2 or they are
24 built the same.

25 Q. Okay. But this is the description of the LDO as we saw,

1 you know, for the particular chip here, the PMIC chip, this is
2 the LDO that's described in the document. Isn't that right?

3 A. Yes. That's correct.

4 Q. And, of course, it calls it a linear regulator. There's
5 no reference to this being a converter circuit. Isn't that
6 right?

7 A. Here it doesn't use the word 'converter'. I agree, yeah.

8 Q. And you haven't referenced anything in this document
9 prior -- previously in this case that says a linear regulator
10 LDO is a converter circuit. Isn't that right?

11 A. I believe that's true.

12 MR. McKEON: We can take that down.

13 Q. (BY MR. McKEON) Now, you understand that you have to be
14 consistent -- when we talk about whether there's infringement
15 of a -- of the patent claim, you've got to be consistent when
16 you're talking about whether a patent claim is invalid, the
17 claims that mean the same thing. Isn't that right?

18 A. If I'm following you, I agree you can't change your
19 opinions when looking at one issue versus the other.

20 Q. So if you say there's an infringement in element A means
21 it's a -- it's a cow, then you can't go on the invalidity side
22 and say, oh, element A is a chicken. Isn't that right, sir?

23 MR. MANZIN-MONNIN: Objection, Your Honor.

24 THE COURT: State your objection.

25 MR. MANZIN-MONNIN: There's no validity,

1 obviousness, or anticipation in this case. They haven't
2 raised that defense.

3 MR. McKEON: Your Honor, I'm about to -- may I
4 respond?

5 THE COURT: Please respond.

6 MR. McKEON: I'm about, Your Honor, to show an
7 inconsistency between what was stated in the validity portion
8 of the case versus the infringement portion of the case.

9 MR. MANZIN-MONNIN: May we approach, Your Honor?

10 THE COURT: I don't think we need to. I'm going to
11 sustain the objection.

12 MR. MANZIN-MONNIN: Thank you.

13 MR. McKEON: All right. Now, let's get the '918
14 Patent back up, if we can.

15 Q. (BY MR. McKEON) And we've seen this -- this front page
16 of the '918 before, and, of course, the title of the invention
17 is Flash-DRAM Hybrid Memory Module. Isn't that right?

18 A. Yes.

19 Q. And that's the same title for the '054 Patent. Correct?

20 A. Yes, that's correct.

21 Q. And here if you go down the first page, we see this
22 figure we've seen before in this trial where we have the --
23 the module has both NAND chips and DRAM chips. Isn't that
24 right?

25 A. I -- yes, I see that.

1 Q. Okay. And this is what was referred to as the hybrid
2 module. Do you recall that testimony?

3 A. Yes.

4 Q. Okay. And if we -- this description of the invention in
5 this patent, it's talking about when the power goes out and
6 the DRAM -- Right? We all know that requires power. Right?
7 The DRAM, you got to have power.

8 A. Yes.

9 Q. Yeah. So when the power goes out, what the patent
10 endeavors to do, the technology here, is to take the data
11 that's in the DRAM and back it up, move it over to the NAND
12 flash. Isn't that right?

13 A. That's -- that's one of the things that they disclose,
14 one of their -- one aspect of one of the inventions.

15 Q. Okay. So then so that you are backing up the DRAM data
16 when power is lost is an aspect of the invention. Do you
17 agree with that?

18 A. It's an aspect of some of the inventions.

19 Q. Now, if we go to figure 16, we see this -- we see this --
20 this module again. We've seen the power module, and you agree
21 with me, sir, that this supplies voltage to both flash and
22 DRAM. Isn't that right?

23 A. Yes, it does.

24 Q. Okay. And this element labeled 1140 here, that element
25 is the sort of the second back-up power supply in this module.

1 Isn't that right?

2 A. Yes, I believe so.

3 Q. And you agree that the '918 Patent says that the power
4 module can be part of the DIMM or separate from it. Isn't
5 that right?

6 A. I'm sorry. Could you repeat the question?

7 Q. Yeah. You agree that the '918 Patent says that the power
8 module can be part of the DIMM or separate from the DIMM.
9 Isn't that right?

10 A. Sure, yeah.

11 Q. So we were talking about the benefit of the patent of
12 moving the -- the power module, the voltage regulator, on the
13 DIMM, but the very same disclosure says you can actually have
14 that voltage regulator off the DIMM. Isn't that right?

15 A. That's right. You could use the prior art.

16 Q. All right.

17 MR. McKEON: We can take that down. And can I get
18 up PDX 4.29, please?

19 Q. (BY MR. McKEON) And this is a slide you presented
20 earlier with the accused products. Isn't that right?

21 A. Yes, sir.

22 Q. Now, just so we're clear on -- on what we're seeing here,
23 these accused products just have DRAM chips on them. Isn't
24 that right? In terms of memory, it's in chip form. It's only
25 in the DRAM. Isn't that right?

1 A. These chips have DRAM and non-volatile memory in the form
2 of flash.

3 Q. Well, we're going to talk about that. All right? But
4 there's no flash memory chip on any of these products. Isn't
5 that right?

6 A. That's correct.

7 Q. It's just DRAM chips. Isn't that right?

8 A. That's correct.

9 Q. And none of these modules have a -- have a situation
10 where you can back up the data and put it somewhere else.
11 That doesn't happen in these products. Isn't that right?

12 A. I can't agree with that.

13 Q. Okay. Well, let me ask you this question. There's
14 nothing in the modules about back-up power. Isn't that right?
15 Strike that.

16 There's nothing in these modules that perform the
17 function of backing up the power when the main power goes out.
18 There's nothing that happens in the products that does that.
19 Isn't that right?

20 A. I guess I'm not following your question.

21 Q. Well, let me just go to the issue that I think we're
22 going to end up where anyways. This is this idea that you
23 testified on direct that non-volatile memory where you have
24 hex 4 gets saved. Do you recall that?

25 A. Yes.

1 Q. Okay. And just so we're clear on this, you said that --
2 there was something you pointed to in the PMIC, it's called
3 eFlash. Do you remember testifying to that in direct?

4 A. Yes, sir.

5 Q. And that's embedded flash. Isn't that right?

6 A. That's right.

7 Q. That's not a separate flash chip. Correct?

8 A. That is right. It is not a separate chip.

9 Q. And it's used in the PMIC to hold register data.
10 Correct?

11 A. It's used in the chip to hold data that was stored in the
12 register. Sure.

13 Q. It does not back up the information in the DRAM chips.
14 Correct?

15 A. That's correct.

16 Q. And it cannot be used in the DDR5 products to store
17 programs, videos, images, emails, or anything like that.
18 Isn't that right?

19 A. Sure. It's not intended to by any means, yeah.

20 MR. McKEON: And we can pull that down.

21 Q. (BY MR. McKEON) Now, we just established, of course,
22 that the DRAM is -- you need that to run programs like Word
23 and Excel. Isn't that right?

24 A. Yes.

25 Q. Okay. And/or to browse the internet, you have to have a

1 DRAM to do that. Isn't that right?

2 A. Yes.

3 Q. And as we've established, when the power goes out, that
4 all gets wiped out. Isn't that right?

5 A. That's correct.

6 Q. Okay. So in a sense, without power DRAM is kind of
7 useless.

8 A. The DRAM chips themselves, I guess they certainly don't
9 operate when the power is off.

10 Q. Now, let's look up at the independent claim, if we can,
11 claim 16 of the '054 Patent. And let's go to the bottom. And
12 we're not going to focus on the rest of the claim. But first
13 operable state and second operable state, do you see that?

14 A. Yes, I do.

15 Q. And it requires -- the claim specifically requires these
16 first two -- these first states, first operable state and a
17 second operable state. Right?

18 A. That's right.

19 Q. And, of course, the claim's very specific about the
20 nature of the state. It has to be operable. Correct?

21 A. The states have to be operable, yeah.

22 Q. It may be obvious, but it doesn't say first non-operable
23 state and a second non-operable state. Right?

24 A. It does not.

25 Q. And it doesn't use the word 'off'. And a first operable

1 state and a second off state, it doesn't use that language,
2 either. Isn't that right?

3 A. That's correct.

4 MR. McKEON: And if we can get your slide 92. There
5 we go.

6 Q. (BY MR. McKEON) And, of course, the claim construction
7 we got from the Court, this is one of the examples where the
8 Court tells us what the term means. Isn't that right?

9 A. That's right.

10 Q. Okay. And we got to listen to the Judge when the Judge
11 says this is what it means. We all got to work from the same
12 definition. Isn't that right?

13 A. Yes, sir.

14 Q. And it says, state in which the memory module is operated
15 after transition. Do you see that?

16 A. Yes, I do.

17 Q. Okay. Now, in this case, there's no dispute between the
18 experts about there being a first operable state when the DRAM
19 memory is operating. Correct?

20 A. Not that I'm aware of, yeah.

21 Q. And so the dispute here is whether there's a second
22 operable state. That's the dispute the parties are having.
23 Isn't that right?

24 A. Yes.

25 Q. Now, the '054 Patent states that memory module can

1 operate in multiple states. Isn't that right? The
2 specification?

3 A. Yes, sir.

4 Q. Okay.

5 MR. McKEON: And let's go to column 25, line 54 to
6 67.

7 Q. (BY MR. McKEON) It says here a power module may be
8 supplied. Right? And it's talking about power to the module
9 here. Isn't that right? To the volatile memory system, which
10 is the DRAM. Isn't that right?

11 A. Yes, sir. That's what it says.

12 Q. And here it identifies a first state and a second power
13 supply. Do you see that?

14 A. Yes, I see that.

15 Q. And then you also have a second state. Do you see that?

16 A. Yes, sir.

17 Q. Okay. And then -- and certain embodiments, they talk
18 about even the third state. Do you see that?

19 A. Yes, I see that.

20 Q. And you agree, sir, that in the disclosure, all of these
21 states are talking about when you have power supplied to the
22 DRAM. Isn't that right?

23 A. I'm not sure. I'd have to review it a little further.

24 Q. Okay. Well, so you don't -- you're not aware one way or
25 the other the way the patent describes the first state, the

1 second state, and the third state. Is that fair?

2 A. No, I wouldn't say that. I mean, for example, in the
3 fourth line it's referring to the state of the memory system,
4 not of the memory chips.

5 Q. Okay. Sure. But -- and let's agree on this. You agree
6 with me that, at least with respect to the first state, second
7 state, and third state, there's always power to the memory
8 chips, the DRAM, in these three states.

9 A. Yeah, I believe that's the case.

10 Q. So there's no disclosure where there's an operable state
11 when you have no power to the DRAM. Isn't that right?

12 A. Oh, I'm sorry. There's no disclosed state here where
13 there is no power presented to the memory system.

14 Q. So stated a little differently, there's no state
15 disclosed, the first, second, or third state, where the memory
16 is off and there's no power to it. Isn't that right?

17 A. I wouldn't say that. I don't recall. It doesn't -- it's
18 talking about the power -- like -- most of the first state and
19 second state is talking about the memory system. The third
20 seems to talk about the volatile memory in particular. Well,
21 again, it's volatile memory subsystem.

22 Q. Okay. Well, let me ask you this question. The PMIC in
23 the accused DDR5, these products have overvoltage and
24 undervoltage protection. Isn't that right?

25 A. Yes, sir.

1 Q. And you testified about that on direct. Isn't that
2 right?

3 A. Yes, sir.

4 Q. Okay. And just so we're clear what that is, if the
5 voltage is too high for a period of time, the DRAM cells get
6 damaged. Isn't that right?

7 A. That's one of the things that could happen, yeah.

8 Q. And so overvoltage protection prevents that from
9 happening. Isn't that right?

10 A. Yes.

11 Q. And, likewise, if the voltage is too low, the DRAM cells
12 won't operate correctly. Isn't that right?

13 A. Yes, sir, or other key circuitry in the device.

14 Q. Okay. And the undervoltage protection is used to prevent
15 that from happening. Isn't that right?

16 A. Yes, sir. That's correct.

17 Q. Now, in response to an overvoltage --

18 MR. McKEON: And you can take this specification
19 down, Mr. Sayers. Thank you.

20 Q. (By MR. McKEON) In response to an overvoltage incident,
21 the PMIC and the Samsung accused DDR5 products disables the
22 buck converters powering the DRAMs. Correct?

23 A. Yes, sir, that's what happens.

24 Q. So, in other words, the buck converters turn off, so the
25 voltage supplied to the DRAM turns off. Isn't that right?

1 A. Yes, sir.

2 Q. And that's in the overvoltage condition. Correct?

3 A. That's right.

4 Q. And -- but, despite that, it's your opinion that
5 there's -- the module is still operable. Isn't that right?

6 A. Yes.

7 Q. No memory, but the module in your opinion is still
8 operable. Isn't that fair?

9 A. No, that's not fair.

10 Q. Well, let's -- let's -- let me ask you this question.
11 You indicated previously in this case that you have an iPhone.
12 Isn't that right?

13 A. Yes.

14 Q. And when using an iPhone, everybody knows this, of
15 course, you make phone calls and you can surf the net and
16 there's all sorts of things you can do on an iPhone. Isn't
17 that right?

18 A. Sure.

19 Q. And you agree that if I'm using a call -- if I'm making a
20 call to my iPhone -- and, of course, I don't have an iPhone,
21 you know, because I have to have the -- the other company's
22 phone, but when you're making a call, you -- you have an
23 operable iPhone. There's no dispute about that. Right?

24 A. It's able to do some operations. It may not be able to
25 do all operations.

1 Q. And then sometimes with these -- with these devices, you
2 get that dreaded depleted battery signal where it's like all
3 red and it's down, there's nothing left. The thing is gonzo.
4 You ever had that happen to you, sir?

5 A. No. I remember looking at a figure like that. There is
6 something like that, yeah.

7 Q. And you had in your deposition, we showed you that image.
8 Do you remember that?

9 A. Yeah. I just don't remember if it was from an iPhone.
10 But if not, there's something similar for an iPhone.

11 Q. But you agree, though, that when we had that depleted
12 battery, we had that -- that graphic that pops up, that
13 there's still some power supplying the screen to form that
14 graphic. So there's still something happening on the phone
15 when you have no power in the device. Isn't that right?

16 A. There is power to do that and other operations.

17 Q. And -- but you understand, though, of course, you can't
18 -- when you get that -- that happens to you, you can't make a
19 phone call, you can't surf the internet. There's all sort of
20 things you can't do with your device that you would normally
21 do when it's operable. Isn't that right?

22 A. Right. There's a difference between the two states.

23 Q. So you agree with me, though, when you get that -- get
24 that image on your phone, you agree with me that the phone's
25 inoperable.

1 A. The phone is operating. It's still doing many things.

2 Q. You think the phone's in an operable state?

3 A. The iPhone?

4 Q. When you have a depleted battery symbol?

5 A. Yeah. It's showing me the battery. And on top of that,
6 one of the things I want to do is I want to charge it, so it
7 has to be able to respond to having -- getting plugged in.

8 Q. Well, I'm not talking about charging -- sir, you're on
9 vacation, you're on a mountain, you just spent an hour
10 climbing way up, you're proud of yourself, and you get the
11 dreaded depleted battery symbol, you're not near power. At
12 this point, the iPhone is inoperable, isn't it?

13 A. It's inoperable for making phone calls. It's operable
14 for displaying the image on the screen and getting ready to
15 charge.

16 Q. So is it your testimony, sir, that if you're hiking with
17 your wife, you're going up the mountain, and you have the
18 depleted -- depleted symbol, the depleted battery symbol, and
19 you want to call somebody while you're up there and your wife
20 says, hey, let's call somebody, I want to call my -- my
21 children because I'm so proud of climbing to the top of the
22 mountain, is it your testimony you are going to say, well,
23 yeah, it's operable, let's go make this call?

24 A. No. It's in a state where you can't make a call.

25 Q. So it's not operable for doing the function it's supposed

1 to be doing. We can agree with that at least?

2 A. It's in an operable state. It's not operable to make a
3 phone call, sure.

4 Q. And doing all the other things everybody does every day
5 with these devices. Isn't that right?

6 A. Sure.

7 Q. All right. Let's switch gears a little and talk about
8 the '339 Patent.

9 MR. McKEON: And if we can get up JTX 2, please?

10 THE COURT: Can we slow down just a little bit, Mr.
11 McKeon.

12 MR. McKEON: Yes. Sorry, Your Honor.

13 Q. (BY MR. McKEON) And this is, of course, the '339 Patent
14 that you testified to on direct. You recall that?

15 A. Yes, I do.

16 Q. Okay. And there was three claims at issue here claims 1,
17 8, and 9. Do you recall that?

18 A. Yes.

19 Q. And just so we can get ourselves an agreement on this,
20 claim 1 is an independent claim, and claim 8 and 9 are
21 dependent claims. Isn't that right?

22 A. That's right.

23 Q. So if there's no infringement of claim 1, then you can't
24 have infringement of claim 8 and 9. Is that right?

25 A. I believe that's the case unless I guess if claim 1

1 is -- there's no infringement because it's found invalid, you
2 can still infringe the others, I believe.

3 Q. Okay. I'm talking about just infringement, not validity.
4 Just infringement. If you don't infringe claim 1, you can't
5 infringe the other ones. Correct?

6 A. That is my understanding, yes, sir.

7 MR. McKEON: Now, if we can go to the -- pull the
8 claim up, claim 1 of the patent. Let's get the whole claim.
9 See if we can get all of it up here. We got it here. Okay.

10 Q. (BY MR. McKEON) Now, I'm not going to go through
11 everything here, but I want to make sure we have agreement and
12 common understanding of what's required by the claim.

13 This claim, if we start at the top here, requires an N
14 bit wide memory module. Isn't that right?

15 A. Yes.

16 Q. So -- and then it has a bunch of elements. Right? A
17 printed circuit board, a double data rate, dynamic random
18 access memory. Right?

19 A. Yep.

20 Q. A module controller. Right?

21 A. Yes, sir.

22 Q. A plurality of byte-wise buffers. Do you see that?

23 A. Yeah, I do.

24 Q. And then it goes on and talks more about -- about the
25 buffers where you had the 'wherein' clause. Isn't that right?

1 A. That's right.

2 Q. So everything we just went through, that all has to be on
3 the module. Isn't that right?

4 A. Yes, sir.

5 Q. Right. So in order to prove infringement in this case,
6 you can't point to something that's not on the module. Isn't
7 that right?

8 A. I believe that's correct, yeah.

9 Q. Right. Because the claim says it's got to be on the
10 module. So if you're trying to prove infringement by pointing
11 at something that's off the module, that doesn't cut it.
12 Isn't that right?

13 A. Yeah, for the most part, yeah.

14 Q. All right. Now, one thing I want to talk to you about is
15 the limitation down here. We have the drive terms. We've
16 been talking about these drive terms. Do you recall that?

17 A. Yes, I do recall that.

18 Q. It talks about to actively drive a respective byte-wise
19 section of the N bit wide write data. Do you see that?

20 A. Yes, I do.

21 Q. And you are driving that essentially to a first side to a
22 second side of the buffer. Isn't that right?

23 A. Yes, that's correct.

24 Q. And all this is happening within the buffer that's on the
25 module. Correct?

1 A. Could I -- yes. That's right. It's all on the module.

2 Q. And there's something else required in the claim. It
3 talks about a byte-wise section. Do you see that? And a
4 byte-wise buffer, those two terms?

5 A. Yes, I see that.

6 Q. And byte-wise, so we're clear here, is eight bits. Isn't
7 that right?

8 A. No.

9 Q. Well, a byte is eight bits.

10 A. Yes.

11 Q. You agree with that. Right?

12 A. Yes.

13 Q. And you also agree that when a claim talks about
14 byte-wise, we're talking about moving data in a byte format.
15 Isn't that right?

16 A. Sure.

17 Q. Okay. And with respect to the --

18 MR. McKEON: If I can get back up the Court's
19 construction again, which is PDX 4.155.

20 Q. (BY MR. McKEON) Now, the Court construed the claim,
21 enabling only one of the data paths while the other possible
22 paths are disabled. Do you see that?

23 A. Yes, I see that.

24 Q. So there's some very important requirements of the claim,
25 and one of the first things is there has got to be multiple

1 data paths. Right?

2 A. That's my understanding, yeah.

3 Q. Because it's -- the word data paths is plural. So
4 enabling only one of the data paths. So you got to have at
5 least two of them, and only one will be enabled. Isn't that
6 right?

7 A. Sure.

8 Q. And when that happens, if you have two and one's enabled,
9 the second one will be disabled. Isn't that right?

10 A. Yeah, that's right, during the first time period -- the
11 start of the first time period, yep.

12 Q. And let's go with figure 5 of the patent so we can get a
13 little more grounding in what we're talking about. And this
14 is the figure 5 of the patent that you rely on, sir. Is that
15 right?

16 A. I'm familiar with it. I don't recall if I relied upon
17 it.

18 Q. All right. Well, the data -- so we all understand, the
19 data is coming here from the memory controller. Right?

20 MR. MANZIN-MONNIN: Objection.

21 THE COURT: State your objection.

22 MR. MANZIN-MONNIN: Comparing figures to claims is
23 just not relevant to --

24 THE COURT: What's your response?

25 MR. McKEON: I'm just trying to context, Your Honor,

1 for the -- the claim construction that the Court gave which is
2 one of ordinary skill in the art should be construing the
3 claim construction, and one of ordinary skill in the art would
4 look at the specification to figure out what this claim
5 element requires based on the plain and ordinary meaning.

6 THE COURT: Well, the problem is you're both right.
7 The only proper comparison for infringement is the language of
8 the claims as interpreted by the Court compared to the accused
9 products.

10 But, by the same token, that language from the claims
11 that has not been specifically construed is accorded its plain
12 and ordinary meaning as understood by one of ordinary skill in
13 the art.

14 That, unfortunately, is a problem we have in these kind
15 of cases on an ongoing basis. There are things that fairly
16 relate to one issue but can be used to address another issue
17 that really shouldn't be addressed that way.

18 I am going to allow you just another question or two on
19 context, Mr. McKeon, but then we're going to move on lest this
20 become an incorrect comparison. Okay?

21 MR. McKEON: Thank you, Your Honor.

22 THE COURT: All right.

23 Q. (BY MR. McKEON) And let me ask you a question, Doctor
24 Mangione-Smith, and I'm going to have a long question. So
25 I'll get to it. Okay?

1 If you go through here, the data comes up, comes over
2 where my finger is, and you go through here, you hit this
3 point here where you have a decision to make about whether to
4 go in the direction of A or the direction of B. Isn't that
5 right?

6 A. It's going to go to A and B.

7 Q. Well, this is sort of the fork in the road. Right? It
8 can go to A or B, depending on the state of these tri-state
9 buffers. Isn't that right?

10 A. No.

11 Q. Well, you understand the '504 and '506 are tri-state
12 buffers. Right?

13 A. Yes.

14 Q. And you understand that the data comes through path A and
15 to the other side, and through path B to the other side past
16 Y1, Y2. Isn't that right?

17 MR. MANZIN-MONNIN: Objection, Your Honor. He is
18 continuing to compare the claims and the figures.

19 THE COURT: Let's move on, Counsel.

20 MR. McKEON: All right. We'll move on. Thank you,
21 Your Honor.

22 Q. (BY MR. McKEON) Now, sir, you did an infringement report
23 in this case setting forth your opinions on how the Samsung
24 products infringe. Isn't that right?

25 A. Yes.

1 Q. And it's very important in these cases because we get an
2 opportunity to look at your theories and how you're applying
3 to the claims and why you believe there's infringement.

4 A. Yes, yeah.

5 Q. And isn't it the case, sir, that in your expert report
6 you gave an opinion that there is for the accused DDR4 LRDIMM
7 products, there is only a single data path --

8 MR. MANZIN-MONNIN: Objection. He shouldn't be
9 reading from the report. This is not impeachment. I don't
10 know what he's doing.

11 MR. McKEON: I'm just going through what the witness
12 said in the previous -- previously in this case, Your Honor,
13 about his opinion on this.

14 MR. MANZIN-MONNIN: You said, "In your expert
15 report," and began reading his expert report.

16 THE COURT: Yeah. I'm going to sustain that
17 objection. Ask the witness his question.

18 MR. McKEON: Yeah, sure.

19 THE COURT: If he testifies contrary to what's in
20 his report, then you can move to impeach.

21 MR. McKEON: Thank you, Your Honor.

22 Q. (BY MR. McKEON) Sir, you believe that there is in the
23 accused DDR4 LRDIMM products, there is only a single -- let me
24 ask it this way. You believe that in the accused DDR4 LRDIMM
25 products, you believe there is only a single byte-wise data

1 path in the byte-wise buffer. Correct?

2 A. Yes, for a particular write transaction. That's correct.

3 Q. Well, let me have you go to your report, sir.

4 It's -- that should be in your binder.

5 A. I have two.

6 Q. The one that we have, I believe, is the dark binder.

7 A. Ahh, I see it. Okay. Yes, sir.

8 Q. And if you go to page 208, paragraph 210. It's -- 210
9 starts at 207, and then it goes over to 208.

10 THE COURT: This is the opening report?

11 MR. McKEON: This is the opening report, Your Honor,
12 yes.

13 THE COURT: All right.

14 Q. (BY MR. McKEON) Are you there, sir?

15 A. Yes, I am now.

16 Q. Okay. And I'm going to direct your attention to page
17 208. Right above the quote you have of a document and right
18 above it you say something, and I want to direct your
19 attention to there. Okay?

20 A. Yes.

21 Q. Have you read that?

22 A. What I wrote or what they wrote or both?

23 Q. What you wrote.

24 A. Yes.

25 Q. Okay. So you believe that, in the accused DDR4 LRDIMMs,

1 they meet the limitation, this limitation you're talking
2 about, because there is only a single byte-wise data path in
3 the byte-wise buffer. Correct?

4 A. Yes, that's what it says, yep.

5 Q. And you also believe that Samsung actually admits this as
6 well because there is only one data path for a given
7 transmission in the accused DDR4 products. Isn't that right?

8 A. Yes.

9 Q. Okay. And this is an opinion you gave multiple times in
10 this case in your report. Isn't that right?

11 A. That's right.

12 Q. But you testified earlier that the claim construction of
13 the Court requires two write paths. Isn't that right, sir?

14 A. Two paths, yeah.

15 Q. Two write paths. Isn't that right?

16 A. I don't think it said write. But we were talking about
17 write paths so I wouldn't disagree with that, yeah.

18 Q. And, now, in your report -- and I'm going to blow up the
19 figure that was presented on direct. If I can go to paragraph
20 209, page 207, just focus on the figure that was presented to
21 you on direct testimony.

22 THE COURT: Put up whatever figure you want, ask him
23 whatever question you want, but let's don't specifically
24 reference his report unless you are going to refresh his
25 recollection or impeach his testimony. Okay?

1 MR. McKEON: Yes, Your Honor.

2 Q. (BY MR. McKEON) All right, sir. You had this figure you
3 talked about in your direct testimony with counsel. Do you
4 recall that?

5 A. Yes, I do.

6 Q. Okay. And this is a prior figure you prepared in this
7 case. Isn't that right?

8 A. This is a figure I prepared, yes.

9 Q. Okay. And you -- you agree that there is only one --
10 this, by the way, was your attempt to demonstrate that there
11 was infringement of this particular element of the claim we're
12 talking about, the drive path. Isn't that right?

13 MR. MANZIN-MONNIN: Objection, Your Honor. Just
14 again referring to the report.

15 MR. McKEON: I'm just trying -- they used this on
16 direct.

17 THE COURT: I'm going to overrule the objection, but
18 let's try to focus on specific questions and answers and not
19 the prior report --

20 MR. McKEON: Thank you, Your Honor.

21 THE COURT: -- unless we're in the context I
22 mentioned.

23 MR. McKEON: Thank you, Your Honor.

24 Q. (BY MR. McKEON) And what you're showing in this figure,
25 sir, is you have a green line--right?--that comes up and that

1 is the data path that you identify in this figure. Isn't that
2 right?

3 A. Yes. That's a data path, yep.

4 Q. And there's only one -- you have one green line, not two,
5 but just one that's shown in the figure. Isn't that right?

6 A. Yes.

7 Q. And give me a moment. And you agree that in the Samsung
8 accused products for the write operation, there's only one
9 byte-wise data path referred to as the write path. Correct?

10 A. Referred to as the write path where?

11 Q. Well, let me ask you again, sir. You believe in the
12 Samsung accused products, there's only one byte-wise data path
13 referred to as the write path. Correct?

14 A. This is one byte-wise write path that I'm showing here,
15 yes.

16 Q. Right. And you believe, sir, there is only one in the
17 Samsung accused products. Correct?

18 A. For a given transmission, yes.

19 Q. In fact, you have not really thought about whether there
20 was some other path that could be taken. Isn't that right?

21 A. No.

22 Q. Okay. Let me get your deposition. If we can go
23 to -- just read it. Pull it up. It's page 173 now, and at 14
24 to 24. Let me know when you're there.

25 A. Okay. I'm at 173. And which lines did you want me to

1 read?

2 Q. Sure. Lines 14 to 24.

3 A. Yes, sir, I see that.

4 Q. Okay. In fact, you have not really thought about whether
5 there was some other path that could be taken. Correct?

6 A. That's right. I think the write can only go to
7 the -- take the one path to get to the one write that it's
8 addressed to.

9 Q. All right. And you also agree, sir, that there is only
10 one feasible -- strike that.

11 You also agree it is only feasible to send write data
12 down the write path, not the read path. Correct?

13 A. Yes, I would agree with that.

14 Q. And you agree that sending write data down the read path
15 could result in damage. Correct?

16 A. Sure. I don't know how that would work. I agree.

17 Q. All right.

18 MR. McKEON: We can pull this down.

19 Q. (BY MR. McKEON) So let's get up a demonstrative you had,
20 PDX 137.

21 MR. McKEON: I'm sorry about that. It's 145. PDX
22 145.

23 Q. (BY MR. McKEON) Okay. This is a document you presented
24 or presentation you did in your direct testimony. Isn't that
25 right?

1 A. Yes, it is.

2 Q. Okay. And here you're saying -- you were telling the
3 jury that there's two paths. Isn't that right?

4 A. There are two paths for two different transmissions, one
5 on the upper half byte and one on the lower half byte.

6 Q. And you refer to them as nibble. The one on the left is
7 green, it's a nibble. And the one on the right is also a
8 nibble, and you have in here, I guess, as red. Isn't that
9 right?

10 A. Yeah, a nibble is a half of a byte, that's correct.

11 Q. Yes, sir. That's what I was going to get to. A nibble
12 is not eight bits. Isn't that right?

13 A. That's correct.

14 Q. And a byte, as you know, requires eight bits. Isn't that
15 right?

16 A. A byte is eight bits, yeah.

17 Q. Right. It is eight bits. So if you only have four bits,
18 you don't have a byte. Isn't that right?

19 A. That's right. You have half of a byte.

20 Q. Right. So if the claim requires a byte being eight bits,
21 pointing to just four bits doesn't meet the requirement.
22 Isn't that right?

23 A. The claim refers to a byte-wise data path, not a
24 byte-wide data path.

25 Q. Well, byte-wise, is your testimony because of the word

1 wise is used, that all of the sudden I can have less than
2 eight bits?

3 A. I take that to mean the width is measured in terms of
4 bytes. In other places, they use the term byte-wide, which
5 I'm more familiar with.

6 Q. Sir, have you ever provided that testimony in this case,
7 that opinion?

8 A. I -- my report is over 800 pages long. I haven't
9 memorized it.

10 Q. What we do know is that you're identifying two different
11 nibble paths with each one being just four bits. Isn't that
12 right?

13 A. Yes.

14 Q. Here in this instance, sir, it's like you took a banana
15 and you split it in half and you're calling it two bananas.
16 Isn't that what's going on here?

17 A. No. I'm calling them half a banana and half a banana,
18 exactly what they are.

19 Q. All right. Well, a half a banana and a half banana, if
20 you're really hungry for lunch, half a banana and a half a
21 banana, you'd be disappointed if someone just gave you a half
22 a banana. Right? You'd want the whole banana. Isn't that
23 right?

24 A. I -- I try not to eat bananas.

25 THE COURT: Does this have some relevance to a

1 patent case? Let's move on.

2 MR. McKEON: I'll move on.

3 Q. (BY MR. McKEON) All right. But we can agree so
4 there's -- so the record is clear on this, that if the claim
5 requires eight bits in the byte-wise buffer, you've only shown
6 the jury here, you've only shown the jury here, one path of
7 four bits and another separate path with four bits. Isn't
8 that right?

9 A. Yes. Yeah, I agree.

10 Q. All right.

11 MR. McKEON: We can pull that down.

12 Q. (BY MR. McKEON) Now, you provided testimony on your
13 direct about some technical benefits of the '339 Patent. Do
14 you recall that?

15 A. Yes, I do.

16 Q. Okay. And you agree that earlier generation of memory
17 modules earlier than DDR4, they are called DDR3. Isn't that
18 right?

19 A. Yes, that's correct.

20 Q. And DDR3 only had a single buffer. Isn't that right?

21 A. I'm not sure which buffer you're talking about. I
22 haven't really reviewed DDR3 in quite a long time.

23 Q. Well, is it your testimony that DDR3, this patent
24 requires a distributed buffer architecture. Isn't that right?
25 You've said that in the past. Isn't that right?

1 A. The claims I think cover distributed buffer architecture
2 as we understand it, yeah.

3 Q. And is it your testimony that DDR3 had that distributed
4 buffer architecture?

5 A. I don't think so, no.

6 Q. Okay. So there's a difference between DDR3 and DDR4.
7 Isn't that right?

8 A. Sure.

9 Q. All right. And the benefit that you testified to about
10 the '339 Patent is using this distributed buffer architecture.
11 Isn't that right?

12 A. Well, the benefit is it reduces the load and then allows
13 you to add more chips to a -- sorry, DIMMs to a channel.

14 Q. Right. But you reduce the load because you're having a
15 distributed buffer architecture. Isn't that right?

16 A. This particular one from the claims, yes.

17 Q. Okay. And -- but, of course, you understand that
18 distributed buffer architecture, that was not invented by
19 Netlist. That was done before. Isn't that right?

20 A. There -- so people have talked about having buffers
21 previously, but not in the way I've been talking about a
22 distributed buffer architecture in the claims.

23 Q. Okay.

24 MR. McKEON: Your Honor, may I approach?

25 THE COURT: Approach the bench, Counsel.

1 (The following was had outside the hearing of the
2 jury.)

3 MR. McKEON: Your Honor, I want to get into a prior
4 art reference. It's the Rajan reference that has a
5 distributed patent buffer. This is not been pre-admitted.
6 It's a -- an exhibit that's been produced in the case, but
7 it's not a pre-admitted exhibit. But I want to make the
8 demonstration that -- this witness provided testimony on this
9 previously. But I wanted to provide, show this to the jury
10 and ask questions about --

11 THE COURT: You want to show the jury a prior art
12 reference that's not a pre-admitted exhibit?

13 MR. McKEON: That's why I'm coming to sidebar. Can
14 I ask questions about this being known in the art and that he
15 previously provided testimony to that fact without pulling up
16 the document?

17 THE COURT: Is it in his report?

18 MR. MANZIN-MONNIN: He is not the expert opining in
19 response to 102 and 103. Doctor Brogioli is. He has no
20 opinions.

21 MR. McKEON: He gave in his deposition, Your
22 Honor -- we talked about this in his depo and we asked him.

23 MR. MANZIN-MONNIN: He opined --

24 THE COURT: Just one at a time now.

25 MR. McKEON: He provided testimony in his

1 deposition, and then here in trial he's saying how the
2 distributed architecture is a big benefit. And I just want to
3 make the point that the distributed architecture or the
4 buffers is well known in the art. I just want to make that
5 one point, and he's already looked at this and testified about
6 it.

7 I mean, if they're going to say that the value is from
8 that, I just want to make the point that that particular
9 feature was known.

10 THE COURT: Yeah. That's too far afield from what
11 his field of expertise is. I mean, he's an infringement
12 expert.

13 MR. McKEON: Right.

14 THE COURT: Not an invalidity expert, not an expert
15 on the state of the art.

16 MR. McKEON: He did provide testimony, Your Honor,
17 about the benefit, and that's what I'm getting at. I'm not
18 saying it's invalid. I'm not going to get into --

19 THE COURT: If he's addressed it in his reports, you
20 can go into it. If he got asked a question that's outside the
21 field of expertise in a deposition, that doesn't open the door
22 to it coming in before the jury.

23 MR. McKEON: Okay.

24 THE COURT: All right?

25 MR. McKEON: Let me just inquire with my colleague

1 on that one point, Your Honor? I think I'm going to move on,
2 but if I can just ask whether this was in the report or not.

3 THE COURT: If you want a minute to consult with
4 your co-counsel, you can have a minute.

5 MR. McKEON: Thank you, Your Honor.

6 MR. MANZIN-MONNIN: Do you want us to stay here?

7 THE COURT: If he comes back up, come back up with
8 him.

9 MR. MANZIN-MONNIN: Thank you, Your Honor.

10 (The following was had in the presence and hearing
11 of the jury.)

12 MR. McKEON: We're going to move on, Your Honor.

13 THE COURT: All right. Let's do that.

14 Q. (BY MR. McKEON) All right. Doctor Mangione-Smith, you
15 gave some testimony about the Rambus license. Do you recall
16 that?

17 A. Yes.

18 Q. Okay. And I believe your conclusion was the five patents
19 in this case are technically comparable to the entire Rambus
20 patent portfolio. Isn't that right?

21 A. I don't think that's exactly what I said. I thought the
22 technology covered by the portfolio is comparable to these
23 patents, yeah.

24 Q. Okay. And as I understand, you analyzed 12 Rambus
25 patents. Isn't that right?

1 A. I believe that's correct, yeah.

2 Q. But there's no dispute in this case that Rambus has
3 over -- it's 2,175 patents. They have a lot of patents.
4 Isn't that right?

5 A. They have a lot of patents, yes.

6 Q. You agree it's over 2,000. Isn't that right?

7 A. That's right.

8 Q. And you just analyzed 12 of them. Isn't that right?

9 A. That's right.

10 Q. And you also understand that Rambus has -- their patent
11 portfolio has many, many different technologies in addition to
12 DRAM patents. Isn't that right?

13 A. I would expect that they're almost all, if not all,
14 related to DRAM, but I haven't reviewed them all so I couldn't
15 say.

16 Q. Are you aware whether Rambus patents include baseband
17 processor systems?

18 A. Yes. And I believe it was in the context of
19 communicating with DRAM, but I'm not a hundred percent sure.

20 Q. And do you have an opinion whether Rambus includes
21 application processors, Rambus portfolio?

22 A. Yes, I would largely give the same answer.

23 Q. And in light-emitting diodes, they include light-emitting
24 diodes in that patent portfolio. Isn't that Right?

25 A. I don't know.

1 Q. But what we do know, sir, is that when making a
2 comparison of what we have comparable licenses here, you agree
3 that you looked at the five patents that are at issue in this
4 case, and you looked at a portfolio that includes over 2,000
5 patents, but you only examined 12 of the 2,000. Isn't that
6 right?

7 A. That's right.

8 Q. All right. And, sir, it's not your first patent case
9 that you've served as an expert witness. Is that right?

10 A. That's correct.

11 Q. And, of course, you're aware that you've served as an
12 expert witness over 50 different times in patent case. Did
13 you know that?

14 A. I believe I've had more than 50 -- I've been engaged in
15 50 different matters. I don't think I've testified or
16 provided evidence in all of them.

17 Q. Okay. And you only served as a -- a majority of these --
18 let me ask you this question. In a majority of those cases,
19 you served as an expert for the Plaintiff, not the Defendant.
20 Isn't that right?

21 A. Yes.

22 Q. You only served as a defendant three -- three cases of
23 the over 50. Isn't that right?

24 A. Yes.

25 Q. Thank you, Doctor Mangione-Smith.

1 MR. McKEON: I pass the witness, Your Honor.

2 THE COURT: All right. Is there redirect by the
3 Plaintiff?

4 MR. MANZIN-MONNIN: Yes, Your Honor. May we
5 approach?

6 THE COURT: Approach the bench.

7 (The following was had outside the hearing of the
8 jury.)

9 MR. MANZIN-MONNIN: They brought up eFlash directly
10 with Doctor Mangione-Smith, which is exactly what was at issue
11 in PX 1820. And in our opinion that opens the door for us to
12 have Doctor Mangione-Smith explain what that eFlash is. He
13 was asked about that on cross-examination.

14 MR. McKEON: I had no choice, Your Honor. He said
15 eFlash in his testimony; I have to cross-examine him on it. I
16 was just saying it's not a chip. I'm not going to leave that
17 alone. That's the first time we heard that from this witness.
18 We have to ask him something.

19 He never said eFlash in his expert report, not a single
20 time. The first time we heard it, here in this courtroom.
21 I'm just establishing -- I'm just established what it is.
22 That's all I asked.

23 THE COURT: I understand. That's why they're up
24 here asking for leave because it's not in his report.

25 MR. MANZIN-MONNIN: I'm sorry, Your Honor?

1 THE COURT: That's why I assume you're up here
2 asking for leave to go into it because it is not in his
3 report.

4 MR. McKEON: But it came in in direct testimony,
5 Your Honor. That's the only reason I asked about it.

6 THE COURT: I understand.

7 MR. MANZIN-MONNIN: The word 'flash' is in his
8 report. But the diagram, the description of the non-volatile
9 memory in the registry, is in the report. The Defendant chose
10 to make flash their theme of trial and use a special word for
11 a type of non-volatile memory.

12 But there's clear disclosure in the report that this
13 component has non-volatile memory. So assigning a special
14 word to it doesn't change the fact that the substance is
15 disclosed in the expert report very clearly.

16 THE COURT: Stay within the four corners of his
17 report. That's all I'm going to tell you. Okay?

18 MR. McKEON: Your Honor, can I ask you a question,
19 Your Honor, a clarification? If he goes into eFlash, which is
20 not in his report, he shouldn't be able to talk about eFlash.

21 THE COURT: Hold it down. We don't want the jury to
22 hear.

23 MR. McKEON: It's not in the expert report. That's
24 my point.

25 MR. MANZIN-MONNIN: It is.

1 MR. McKEON: I was just establishing that, what this
2 was, because it wasn't in the report. So I was just trying to
3 establish what it is.

4 MR. MANZIN-MONNIN: All eFlash means is embedded
5 non-volatile memory which is well disclosed in his report.
6 They are assigning a name to a specific eFlash is a brand of
7 non-volatile memory. They have chosen to make their case
8 about the word 'flash'. That doesn't mean the expert can't
9 talk about the flash that's non-volatile memory that's
10 discussed, Your Honor, without putting the brand name.

11 MR. McKEON: The first time we saw it was in trial
12 with respect to that document that you let in, previously let
13 in. They pulled it up and it said eFlash on it, and the
14 witness testified about it.

15 I was just saying eFlash is not a memory chip. Right?
16 It doesn't store emails. That's all I said.

17 THE COURT: If anybody wants to use the term eFlash,
18 they need to get leave before they do. You need to stay
19 within the four corners of his published report. All right?

20 MR. MANZIN-MONNIN: Can we show PX 1820 to the jury
21 and show the flash memory that's -- and refer to it as
22 non-volatile memory, not flash?

23 MR. McKEON: It has been stricken, Your Honor.

24 THE COURT: It's been stricken as an exhibit. It's
25 a demonstrative.

1 MR. MANZIN-MONNIN: Okay. And we have one other
2 question. So counsel put up JTX 11 at page 69. It's a
3 diagram of a LDO circuit.

4 And we'd ask for leave to be able to address what's
5 inside that circuit, the switch, which he cross-examined
6 Doctor Mangione-Smith about. It's not in his report, but this
7 particular figure in this particular document that he chose to
8 use in cross-examination is a switch, but we would like to
9 address the structure that was just shown. If he knows what
10 it is --

11 THE COURT: So you're saying that Defense counsel
12 brought up something outside his report for the first time on
13 cross-examination?

14 MR. MANZIN-MONNIN: The specific figure -- the
15 specific page of that exhibit is not identified as being a
16 switch in that diagram. But counsel cross-examined him on
17 that specific figure and asked him, is this LDO a converter
18 circuit? He made the same argument they've made since opening
19 statement, that it's not a converter circuit because it
20 doesn't include the structure necessary.

21 And all we want to point out is what is the structure in
22 the diagram, a switch, and move on.

23 MR. McKEON: Can I respond, Your Honor?

24 THE COURT: Yes.

25 MR. McKEON: I was cross-examining on a document

1 they presented in direct and a document that was in his
2 report. And I was simply asking him questions about the
3 document he presented in direct and the document he had in his
4 expert report, that it doesn't call it a converter circuit.
5 That's all I asked.

6 I certainly wasn't opening the door, I'm not going back
7 and doing stuff that was never in his report, that we've never
8 seen before, because now that's what they're trying to do,
9 they're trying to go deeper. And that's why they put up
10 circuit earlier today from that document. They now realize
11 they have a problem, and they want to go deeper now into that
12 chip.

13 THE COURT: All right.

14 MR. McKEON: And they haven't done it.

15 THE COURT: All right.

16 MR. MANZIN-MONNIN: One more moment, if I may.
17 The LDO is disclosed as having a switch. We just haven't
18 disclosed in our report from this specific figure that he
19 cross-examined our witness on. And all we're doing is
20 pointing out the same fact we already made.

21 THE COURT: If he put up a figure, you can put up a
22 figure. If you put up a figure, he can cross-examine on that
23 figure. Okay?

24 MR. MANZIN-MONNIN: Thank you.

25 THE COURT: Let's proceed.

1 (The following was had in the presence and hearing
2 of the jury.)

3 THE COURT: All right. Redirect examination by the
4 Plaintiff.

5 REDIRECT EXAMINATION

6 BY MR. MANZIN-MONNIN:

7 Q. Good afternoon, Doctor.

8 A. Good afternoon, sir.

9 Q. What do you consider when determining the value of a
10 patent?

11 A. Well, what I do is one of the things is I consider what
12 are called non-infringing alternatives, could somebody go down
13 another path.

14 Q. And as part of your analysis, do you try to figure out
15 which of the individual elements of the claim is unique?

16 A. No. I look at the claim.

17 Q. Counsel asked you about a reference named Harris. Do you
18 recall that?

19 A. Yes, I do.

20 Q. Now, does Harris teach the '918 and '054 inventions?

21 A. No.

22 Q. Is Samsung alleging that Harris teaches the inventions?

23 A. No.

24 Q. Has Harris ever been implemented in a product?

25 A. No.

1 Q. Would Harris work if it were implemented in a DDR5 DIMM?

2 MR. McKEON: Your Honor, objection.

3 THE COURT: I think we covered this enough. Let's
4 move on. You objected to Defendants' use of it as an improper
5 reference. You've addressed it enough. Let's move on.

6 MR. MANZIN-MONNIN: Thank you, Your Honor.

7 There was a second reference that made the same points.

8 THE COURT: Then don't go any further with the
9 second reference than you did with that one.

10 Q. (BY MR. MANZIN-MONNIN) You recall counsel asked you
11 about a second reference, Amidi?

12 A. Yes.

13 Q. Does Amidi teach the '918 and '054 inventions?

14 A. No.

15 Q. Is Samsung alleging that Amidi teaches the inventions?

16 A. No.

17 Q. Has Amidi ever been implemented in a product?

18 A. No.

19 MR. MANZIN-MONNIN: And could we bring up PDX 4.52?

20 Q. (BY MR. MANZIN-MONNIN) Now, counsel showed you this
21 figure here on the right and suggested that the LDO -- nowhere
22 in this document does it say it's a converter circuit. Do you
23 recall that?

24 A. Yes.

25 Q. And did we see testimony from Renesas' engineer on this

1 point?

2 A. Yes. He said that it was a converter.

3 MR. MANZIN-MONNIN: And let's bring up JTX 11 at
4 page 69. And if you could zoom in on the figure, please.

5 Q. (BY MR. MANZIN-MONNIN) And counsel asked you about this
6 figure here, which is the circuitry of the LDO. Correct?

7 A. Yes.

8 Q. And is there a -- excuse me. Is there a switch in the
9 circuitry for this LDO?

10 A. Yes.

11 MR. MANZIN-MONNIN: Could we have slide 1 of the
12 redirect module, please?

13 Q. (BY MR. MANZIN-MONNIN) Now, counsel for Samsung
14 suggested that the '918 and '054 Patents are about
15 non-volatile memory. I think they said that holds pictures
16 and emails and things like that. Do you recall that?

17 A. Yes.

18 Q. And what is the patent describing here?

19 A. The patent is describing that it writes data to
20 non-volatile storage, which is not part of the memory
21 subsystem.

22 Q. So in this embodiment of the patent, the non-volatile
23 memory is not on that memory module. Is that right?

24 A. That's right. And it's not even flash; it's a hard disk.

25 MR. MANZIN-MONNIN: Let's flip to the next slide,

1 please.

2 Q. (BY MR. MANZIN-MONNIN) And is there any non-volatile
3 memory shown in these embodiments?

4 A. No, other than the configuration data on the FPGA, which
5 we discussed previously, I believe.

6 Q. Okay. So there's some flash located in the FPGA that
7 holds configuration information?

8 MR. McKEON: Objection, leading, Your Honor.

9 THE COURT: Sustained.

10 Q. (BY MR. MANZIN-MONNIN) Is there a non-volatile memory
11 located in the block 1070, the FPGA?

12 A. Yes, but it's not used to store things like email or
13 images.

14 Q. And the primary memory here is the volatile -- what is
15 the primary memory in figure 15A?

16 A. That would be the DRAM.

17 MR. MANZIN-MONNIN: And could we bring up PTX 1820?
18 You know what? Let's bring up the slides from opening on that
19 document. One moment.

20 Your Honor, may I walk over to counsel table to get
21 something?

22 THE COURT: You may.

23 MR. MANZIN-MONNIN: Thank you.

24 I just want to -- hold one moment, please.

25 We need to seal the courtroom again for this. Sorry,

1 Your Honor. May we seal the courtroom?

2 THE COURT: All right. Based on counsel's request
3 to protect confidential information, I'll order the courtroom
4 sealed. Everyone present who's not subject to the protective
5 order that's been previously entered in this case should --

6 MR. MANZIN-MONNIN: Your Honor, we're going to skip
7 this. I apologize.

8 THE COURT: Have a seat, ladies and gentlemen.
9 King's X.

10 All right. We're not sealing the courtroom. Let's move
11 on, counsel.

12 MR. MANZIN-MONNIN: Let's have PDX 4.145, please.
13 It's not this one. Keep clicking forward. 145. Sorry.
14 That's not it. The two paths with the data buffer.

15 Q. (BY MR. MANZIN-MONNIN) Doctor Mangione-Smith, do you
16 recall being asked about one of your slides that shows a red
17 and a green path through the data buffer?

18 A. Yes.

19 Q. Is that a single path or two paths through the data
20 buffer?

21 A. Those are two paths which operate independently.

22 Q. This is the diagram you were describing?

23 A. Yes, sir.

24 Q. And these are two byte-wise data paths?

25 A. Yes.

1 Q. Thank you, Doctor Mangione-Smith.

2 MR. MANZIN-MONNIN: Nothing further. I pass the
3 witness.

4 THE COURT: All right. Is there additional cross?

5 MR. McKEON: Yes, there is, Your Honor.

6 THE COURT: All right. Let's proceed with
7 additional cross-examination by the Defendants.

8 MR. McKEON: Can we get up PDX 123, please?

9 RECROSS EXAMINATION

10 BY MR. McKEON:

11 Q. Sir, you just provided testimony about this slide. Do
12 you recall that?

13 A. Yes, I do recall that.

14 Q. Now, we've already established in this case that
15 everything that's required by the claim is on the memory
16 module. Isn't that right?

17 A. Yes.

18 Q. And what you're pointing to here, sir, that's a memory
19 controller that is off the memory module. Isn't that right?

20 A. Yes.

21 Q. And these little lines you got here on PDX 4.123, that
22 one here, you got one here, you got one here, you got one
23 here--you got four of them--that is all off of the memory
24 controller. Isn't that right?

25 A. That's right.

1 Q. And the buffer that the claim requires, you say that
2 buffer is located where you've got your little marks here,
3 here, here, and here. Isn't that right?

4 A. Sure. It's the blue boxes on the DIMMs.

5 Q. Right. And so to the extent you're trying to rely on
6 anything here in your testimony, that's irrelevant for the
7 purposes of infringement. Isn't that right? And when I see
8 'here', the memory controller and all the lines you got going
9 to the modules, that's off of the module. Isn't that right?

10 A. All of that is off of the module, yeah.

11 Q. It's not relevant to the question of infringement in this
12 case. Correct?

13 A. I think that's a legal conclusion. I'm not ready to -- I
14 don't have an answer to that.

15 Q. Okay. But you agree, though, everything that's required
16 in the claim, including the requirements of a buffer that
17 we're talking about in this case, that has to be on the
18 module. Isn't that right?

19 A. Yes, in order to infringe.

20 Q. And the drive limitation that we've been talking about
21 does not recite actively driving a half of a byte-wise section
22 of the write data along a half of a byte-wise data path.
23 Isn't that right?

24 A. That's right. That's not the language it uses.

25 Q. It doesn't say drive limitation, does not recite actively

1 driving a nibble-wise section of the write data along a
2 nibble-wise data path. Isn't that right, sir?

3 A. That's also correct.

4 MR. McKEON: I pass the witness, Your Honor.

5 THE COURT: All right. Any additional direct?

6 MR. MANZIN-MONNIN: Nothing further, Your Honor.

7 Thank you.

8 THE COURT: All right. You may step down, Doctor
9 Mangione-Smith.

10 THE WITNESS: Thank you, Your Honor.

11 THE COURT: You're welcome.

12 Ladies and gentlemen, I anticipate the next witness is
13 going to be lengthy, so we're going to take a short recess at
14 this point.

15 If you'll simply close your notebooks and leave them in
16 your chairs, follow all my instructions, including, as you
17 might expect me to remind you, don't discuss the case among
18 yourselves. And we'll be back shortly to continue with the
19 next Plaintiff's witness.

20 The jury's excused for recess.

21 (Whereupon, the jury left the courtroom.)

22 THE COURT: The Court stands in recess.

23 (Brief recess.)

24 THE COURT: Be seated, please.

25 Are you Doctor Brogioli?

1 THE WITNESS: I am, yes.

2 THE COURT: You don't need to be on the witness
3 stand. If you'll take a seat out there. When the jury's back
4 in the room, I'll ask the Plaintiff to call their next
5 witness, and then you can come in and be sworn.

6 THE WITNESS: Thank you, Your Honor.

7 THE COURT: Sure.

8 Let's go off the record for a minute.

9 (Discussion off the record.)

10 THE COURT: All right. Let's go back on the record.
11 Let's bring in the jury, please.

12 (Whereupon, the jury entered the courtroom.)

13 THE COURT: Welcome back from recess, ladies and
14 gentlemen. Please have a seat.

15 Plaintiff, call your next witness.

16 MR. SHEASBY: Your Honor, Plaintiffs call Doctor
17 Brogioli.

18 THE COURT: All right. Doctor Brogioli, if you'll
19 come forward and be sworn by the Courtroom Deputy, please.
20 And once you're sworn, please have a seat on the witness
21 stand.

22 (Whereupon, the oath was administered by the Clerk.)

23 THE COURT: Please have a seat, sir.

24 Mr. Burgess, you may go to the podium and prepare for
25 direct examination. The binders have been distributed already

1 for this witness?

2 MR. BURGESS: Yes, Your Honor.

3 THE COURT: Okay. Then you may proceed with direct
4 examination when you're ready.

5 MR. BURGESS: Thank you, Your Honor.

6 MICHAEL BROGIOLI, SWORN,
7 testified under oath as follows:

8 DIRECT EXAMINATION

9 BY MR. BURGESS:

10 Q. Good afternoon, Doctor Brogioli.

11 A. Good afternoon.

12 Q. Will you please introduce yourself to the jury?

13 A. My name is Michael Brogioli, and I am here to talk about
14 the Netlist '060 and '160 Patents.

15 Q. Are you going to also discuss some Samsung products?

16 A. Yes, I am.

17 Q. Which?

18 A. Samsung's HBM2, 2E, and HBM3 products.

19 Q. And did you prepare anything to help you explain your
20 testimony to the jury today?

21 A. I did. I prepared some slides. I think the first one's
22 up on the screen.

23 Q. Will you please describe your educational background?

24 A. Sure. I have a Bachelor's degree in electrical
25 engineering, a Master's degree in electrical and computer

1 engineering from Rice University here in Texas, and a Ph.D.
2 also in electrical and computer engineering from Rice
3 University as well.

4 Q. And how does your education relate to what you plan to
5 discuss today?

6 A. My background goes back to the '90s and working memory of
7 various kinds from smaller computers to big, high performance
8 computers, optimization of memory, sort of R&D configurable
9 memory systems, things like that.

10 Q. And once you finished your education, where did you work
11 professionally?

12 A. I worked at a number of places. Texas Instruments,
13 you've heard of this week, Intel, Free Scale Semiconductor
14 that was called Motorola.

15 And I've also held an Adjunct Professor of electrical and
16 computer engineering position at Rice since, I think, 2009,
17 and have in the last few years been doing some guest lectures
18 in the School of Engineering at University of Texas.

19 Q. And how does your professional work relate to your
20 testimony today?

21 A. So that work has also involved semiconductor design, work
22 with memory systems, high performance, IO things like that, as
23 well as working academia.

24 THE COURT: Let me ask both of you gentlemen to slow
25 down, both with the questions and the answers, please. All

1 right?

2 THE WITNESS: Thank you.

3 MR. BURGESS: Yes, Your Honor.

4 THE COURT: Thank you. Please proceed.

5 Q. (BY MR; bURGESS) Have you actually designed memory
6 systems yourself?

7 A. I have, yes.

8 Q. Can you describe that, please?

9 A. They were memory systems used in what's called digital
10 networking products. You have to get a lot of information in
11 and out in a certain way and really tight timing requirements.

12 Q. Have you written any publications that are related to
13 memory systems?

14 A. Yes. Probably some white papers or conference papers as
15 well as some book contributions or textbook contributions.

16 Q. Are you being compensated for your work in this case?

17 A. I am at my customary rate of \$700 per hour.

18 Q. And about how many hours would you estimate that you've
19 worked so far on the case?

20 A. Probably getting close to 400 or so, in that range.

21 Q. Which two Netlist patents are you going to talk about
22 today?

23 A. It will be the Netlist '060 and '160 Patents.

24 Q. And are you going to provide your opinion about whether
25 Samsung products infringe these two patents?

1 A. That's correct.

2 Q. And which Samsung products are those?

3 A. It's the HBM2, 2E, and HBM3 products.

4 Q. And what's the relationship between the '060 and the '160
5 Patent?

6 A. The two patents here are common specification or common
7 application you can see here on the slide.

8 Q. Can you provide an overview for the jury of how you went
9 about conducting your infringement analysis?

10 A. Sure. So I've listed about seven points here, but I
11 started by looking at the patents themselves and their
12 examination history.

13 I looked at the various Samsung schematics and source
14 code. Some of that was shown -- similar schematics and source
15 code types was shown by Doctor Mangione-Smith.

16 I looked at the Samsung witness testimony as well as
17 internal engineering documents, specifications, simulation
18 data, and so forth. Customer requirements.

19 Q. I'm sorry. I didn't mean to interrupt. Please continue.

20 A. I also looked at the Court's claim construction order,
21 some journal articles and conference papers we'll get into
22 later perhaps, and Samsung discovery responses.

23 Q. And did you ultimately prepare a written report that the
24 set out the details of your analysis and the conclusions that
25 you drew?

1 A. I did. I'm sure it's in here which is why they gave it
2 to me. It was probably 300 or so pages in length.

3 MR. BURGESS: Your Honor, I'd like to proffer Doctor
4 Brogioli as an expert in the field of memory systems.

5 THE COURT: Any objection?

6 MR. CORDELL: No objection, Your Honor.

7 MR. BURGESS: Thank you.

8 THE COURT: The Court will recognize this witness as
9 an expert in these fields.

10 Counsel, approach the bench quickly, please.

11 (The following was had outside the hearing of the
12 jury.)

13 THE COURT: Did I hear this witness refer to the
14 Defendants as Sam scum and then correct himself?

15 MR. BURGESS: I didn't catch that.

16 THE COURT: I'll be listening. If it happens again,
17 we're going to have a problem.

18 MR. BURGESS: Oh, no. If that came out, I'm sure it
19 was inadvertent.

20 THE COURT: I may have misheard it, but it didn't
21 strike me as being inaccurate from watching him.

22 MR. BURGESS: We would have no intention of ever
23 doing that.

24 THE COURT: I'm not talking about you. I'm talking
25 about him.

1 MR. BURGESS: I'll vouch for him, too.

2 (The following was had in the presence and hearing
3 of the jury.)

4 THE COURT: All right. Let's proceed.

5 Q. (BY MR. BURGESS) So which particular claims from the two
6 patents are you going to talk about today?

7 A. For the '060 Patent, it will be claims 1, 5, and 7. And
8 for the '160 Patent, it will be claim 5.

9 Q. And just so we don't bury the lead, what are your
10 ultimate conclusions in regard to whether or not the Samsung
11 HBM products infringe each of these four claims?

12 A. That the Samsung accused products do infringe each of
13 these claims.

14 Q. So you're here to talk about infringement. Do you expect
15 to hear Samsung make arguments about these two patents being
16 invalid at this trial?

17 A. No, I do not.

18 Q. Will you provide a little bit of technology background
19 for the jury and just at a high level describe what's the
20 technology that's in these two patents?

21 A. So the patents are talking about a stacked memories and a
22 way of going about doing that with specific groupings. And
23 we'll get into that today. But the way the patents
24 contemplate that allow for greater speed of the devices,
25 memory speed; you can build larger devices in terms of stuff

1 that they can store inside; and you can also get benefits in
2 power consumption.

3 Q. What is a stack of memory devices?

4 A. So if you look on the figure on the left here, this is
5 sort of the early days of how we stacked things. So you see
6 these big black rectangles that are literally stacked up, and
7 then you can see they're connected using these external
8 connections.

9 Q. And why did companies that manufacture memory modules
10 begin to stack chips like this?

11 A. So if you remember earlier, Mr. Milton held up one of the
12 memory modules. They're pretty small or relatively small. So
13 as in the industry we kept trying to put more and more memory
14 on it, eventually you run out of space on the little green
15 board. So we decided to start stacking it up, if you will.

16 Q. And so these -- these external connections that kind of
17 look like hairs coming off the sides of the memory modules,
18 were there any problems associated with those?

19 A. There were. As you can see in the picture, you can
20 really only connect them around the edge of the black
21 rectangle, and so you're limited in the number of, you know,
22 the amount of real estate you have.

23 Another problem is that, due to their length and them
24 being external, you can only run the signals over them so
25 fast. There's loading issues that come up with them and

1 things like that.

2 Q. And what's the problem with only being able to run memory
3 so fast? What's the problem with a slower memory system?

4 A. Sure. So, big picture, your computer is always wanting
5 more and more memory, and if the ability to get things in and
6 out of the memory to the CPU or something, for example, slows
7 down, everything starts to slow down. So it's generally been
8 a trend to just have higher and higher speed of memory.

9 Q. Do these external connections have an impact on the power
10 that was required to run the memory?

11 A. They did. You know, you have to use quite a bit of power
12 to run a system like this. And as others have talked about
13 this week, you want to get that power down, if you can, too.

14 Q. What's the problem with just turning up the power?

15 A. So it sounds good to just turn the knob up. But what
16 happens is, one, you're operating in an environment with more
17 than just the memory. So something else always needs power,
18 so there's a balance at the system level.

19 The other issue is that power turns into heat. And so if
20 we're talking about really small, you know, computer-type
21 chips, you've got to get rid of that heat. And then when
22 those are installed in servers and racks of servers, it really
23 starts to add up. You may damage the part and eventually melt
24 it.

25 Q. So did memory stacking technology evolve to address the

1 shortcomings?

2 A. It did, yes.

3 Q. So what do you show here that you've labeled Early
4 Internal Connection?

5 A. So this is what I would call an interim solution, so it's
6 still before we get to the patents. But what we started to do
7 was use each of these white rectangles or what are called an
8 array die. And on the bottom, those blue triangles we'll talk
9 more about today, but they're drivers. So they are sending
10 signals, if you will, up and down these vertical interconnects
11 through the array dies.

12 Those vertical interconnects are called through-silicon
13 vias, or TSVs. So now you're going up internally through
14 what's called an array die rather than externally connecting.

15 In addition, if you look at the red dots I've highlighted
16 here, each of those array dies has what's called a data port.
17 And we'll talk more about that later. But that allows them to
18 couple to these vertical interconnects we see now that are
19 internal.

20 Q. So this is an annotated figure that's actually from the
21 patent, isn't it?

22 A. That's correct.

23 Q. But it's not the invention?

24 A. No. It's still prior to the invention.

25 Q. So what's the loading problem with what you've called the

1 interim or early internal connection?

2 A. So you end up with a significant amount of loading going
3 on here, and that's what the patent is trying to avoid. And
4 the reason for that is for each one of those lines that's
5 going up with the blue driver on the bottom, you have each and
6 every one of the array dies connected to it. They all have a
7 data port connected. So there's more and more load getting
8 stuck on that line.

9 Q. Is the situation vis-a-vis loading better for the
10 structure on the right than for the structure on the left?

11 A. It's better in terms that it's getting in the right
12 direction, but there's still a problem.

13 Q. Does it get us where we need to be for today's market
14 demands for memory systems?

15 A. Not for today, no.

16 Q. So let's jump into your infringement analysis, and let me
17 ask you, did you review the operation and structure of all of
18 Samsung's HBM products?

19 A. Yes, that's correct.

20 Q. And are there three different products?

21 A. That's right.

22 Q. And what are they, please?

23 A. Again, it's the HBM2, HBM2E, and the HBM3.

24 Q. And are there any differences in the way those products
25 work or how they're structured that matter to your conclusions

1 regarding infringement?

2 A. Not as it relates to my conclusions on infringement.

3 Q. And does Samsung's expert, Doctor Robins, agree that
4 there aren't any differences between the products, the
5 different types of HBM memory, that matter for purposes of
6 determining whether there's infringement?

7 A. My understanding is he agrees.

8 Q. So let's look at a document that describes some of the
9 HBM memory that we're going to talk about today. This is JTX
10 14 at page 12. And before we dive into it, can you just tell
11 the jury what this document is, JTX 14?

12 A. Sure. So we're going to see a few of these today. These
13 are Samsung internal documents, and you'll see Samsung called
14 out there. And these are produced as part of discovery.
15 They're highly confidential documents.

16 Q. So these are the sorts of things we heard described as
17 top secret earlier?

18 A. That's right.

19 Q. And are we going to see different examples of this sort
20 of document throughout your testimony?

21 A. We will, yes.

22 Q. And do these documents contain actual evidence about the
23 structure and operation of Samsung's HBM products?

24 A. Yes, they do.

25 Q. So let's look at the -- the structure on the right from

1 JTX 14 at page 12, and using this structure as a reference,
2 can you tell the jury whether Samsung's HBM products have
3 stacks?

4 A. They do. So you can see the core dies I've highlighted
5 in blue or that are also called memory dies. That's a stack
6 of dies.

7 Q. So this -- this arrangement has -- are there eight core
8 dies?

9 A. That's correct.

10 Q. And so might we refer to this as an eight-high stack?

11 A. Sure.

12 Q. And what's actually in the core dies? Why are they
13 important?

14 A. Well, they are important because they have what are
15 memory cells, something we talked about storing your email,
16 for example. It ends up in bits that goes into those memory
17 cells.

18 Q. And what is the blue die at the bottom?

19 A. That is called a buffer die or a control die. So it's
20 separate type of die.

21 Q. Does it -- does it have a memory cells in it?

22 A. It does not.

23 Q. What's its purpose?

24 A. Its purpose is for controlling the system to send
25 information in and out of those core dies you see above it.

1 Q. And do all of Samsung's HBM products have this sort of
2 stack structure?

3 A. They do.

4 Q. So what's the first claim you're going to talk to the
5 jury about today?

6 A. It will be claim 1 of the '060 Patent.

7 Q. And is this the claim language on the screen?

8 A. It is.

9 Q. And I see that you've broken it up into, we'll call them,
10 limitations A through F?

11 A. That's right.

12 Q. Can you explain to the jury what you mean by the
13 annotations in the right-hand column that say either buffer
14 die or core die?

15 A. Sure. So each of these rows is a different limitation of
16 the claim. I think you've seen this format roughly used this
17 week. In the right-hand column, I've listed which portion of
18 these HBM devices pertain to that claim limitation.

19 So there's kind of two flavors. One's the buffer die we
20 saw or control die. The other's the core die and memory
21 arrays.

22 Q. And so leaving aside the first limitation for the moment,
23 the green limitations are the ones that relate to the buffer
24 die and the black limitations relate to the core die?

25 A. That's right.

1 Q. Is there any other significance to the green and the
2 black text?

3 A. There is. There's a lot packed in here. I apologize.
4 The green highlighted text of those limitations are ones
5 that Samsung concedes are met in these accused products. The
6 ones that I've left in black text have portions of them that
7 Samsung and Netlist disagree about.

8 Q. Let's start with the very first limitation, limitation A,
9 a memory package comprising. Are Samsung's HBMs memory
10 packages?

11 A. They are. Here is JTX 38 at 27.

12 MR. CORDELL: Your Honor, I hate to interrupt. If
13 we are going into this level of detail, we probably need to
14 seal the courtroom.

15 THE COURT: All right. I don't want to curtail your
16 examination unduly, so to protect confidential information,
17 I'll order the courtroom sealed.

18 I'll direct those present who are not subject to the
19 protective order to excuse themselves until the courtroom is
20 reopened and unsealed.

21 (Courtroom sealed.)

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(Courtroom unsealed.)

(Whereupon, the jury left the courtroom.)

THE COURT: Let me see Mr. Cordell and Mr. Sheasby
in chambers. Otherwise, we stand in recess.

(Brief recess.)

THE COURT: Be seated, please.

Do I gather you're prepared for cross examination?

MR. CORDELL: I am, Your Honor.

THE COURT: Then let's bring in the jury.

(Whereupon, the jury entered the courtroom.)

THE COURT: Welcome back, ladies and gentlemen.
Please have a seat. We will proceed with cross examination of
the witness by the Samsung Defendants.

Mr. Cordell, you may proceed.

MR. CORDELL: Thank you, Your Honor.

May it please the Court?

1 THE COURT: Please proceed.

2 CROSS EXAMINATION

3 BY MR. CORDELL:

4 Q. Good afternoon, Doctor Brogioli.

5 A. Good afternoon.

6 Q. It's been a couple of years. You doing okay?

7 A. Doing all right. Yourself?

8 Q. Not bad, thanks.

9 You're appearing here as an expert. Right?

10 A. That's correct.

11 Q. But you've never designed these HBM products that you
12 talked about this afternoon. Right?

13 A. HBM specifically, no.

14 Q. You've never designed a memory product that has TSVs at
15 all. Right?

16 A. That's correct.

17 Q. These vias that go up and down through the chips, you've
18 never designed a memory product that has those. Correct?

19 A. Correct.

20 Q. In fact, you've never designed any complete memory
21 product?

22 A. Portions of, but not a complete memory product.

23 Q. Now, you do a lot of expert witness work. Right?

24 A. A fair amount, yes.

25 Q. And you testified a little while ago that I think you

1 make \$700 an hour in this case. Is that right?

2 A. That's correct.

3 Q. And you've been paid, by my math, about \$280,000 for your
4 work.

5 A. That sounds about right.

6 Q. Well, is that too high or too low?

7 A. I was just keeping up with you on the math.

8 Q. And you treated both the '060 and the '160 Patents the
9 same. Right?

10 A. In terms of the specification, yes.

11 Q. Well, more than that. Your infringement analysis is the
12 same for both patents. Right?

13 A. As I recall, there's some differences for claim
14 limitations, but a lot of it's common, yes.

15 Q. Okay. And the big issue that you've testified about had
16 to do with a DRAM circuit. Did I understand you correctly?

17 A. One of the issues, yes.

18 Q. Well, and the DRAM circuit issue is in both the '060 and
19 the '160 Patents. Right?

20 A. As I recall, that's correct.

21 Q. So if this jury finds that Samsung uses DRAM circuits,
22 there's no infringement of these two patents. Correct?

23 A. I don't know that I performed an analysis on that, but
24 that sounds right.

25 Q. Okay. I'm trying to be as clear as I can. These claims

1 say don't use DRAM circuits. Right?

2 A. In that the addresses are different from them, yes.

3 Q. Well, and more than that. They say don't use DRAM
4 circuits. Correct? Got to do something different from a
5 DRAM circuit. Right?

6 A. Something different from the addresses, yes.

7 Q. Well, not different from the array dies; it says
8 different from a DRAM circuit. Right?

9 A. In the claim, you're asking?

10 Q. Yes.

11 A. Okay.

12 Q. As construed, the claims say do something different from
13 a DRAM circuit. Right?

14 A. Well, the construction, as I recall, was something
15 different from a DRAM circuit. I'm following you, yes.

16 Q. Okay. And that --

17 MR. CORDELL: Let me have the claim construction
18 just to make sure everybody is clear.

19 Q. (BY MR. CORDELL) Well, actually let's just keep going.
20 I'm sorry.

21 You gave us hundreds of pages of reports in this case.
22 Right?

23 A. That's correct.

24 Q. And the rules require you to write down all of your
25 opinions before you get on the stand to testify. Right?

1 A. That's my understanding, yes.

2 Q. And it would be unfair for you to hold back an opinion
3 and then express it on the stand for the first time. Right?

4 A. That's my understanding, yes.

5 Q. Now, you offered no opinion about whether my 16-year-old
6 is going to get an A in history this year. Right?

7 A. No, I didn't.

8 Q. You offered no opinion about whether it was smart for Kim
9 Mulkey, the basketball coach, to leave Baylor. Right?

10 A. I did not.

11 Q. You didn't offer any opinion about whether my dog at home
12 can hunt or not. Right?

13 A. No, sir.

14 Q. And, sir, you offered this jury no opinion about the
15 plain and ordinary meaning of the phrase 'DRAM circuit'.
16 Right?

17 A. I disagree.

18 Q. Well, in that binder we have your deposition, and I'd
19 like you to open it to page 129, your February 8, 2023,
20 deposition just a couple of months ago.

21 A. Okay.

22 Q. Do you have it?

23 A. The first tab. Is that right? The Brogioli depo tab,
24 I'm guessing.

25 Which page again?

1 Q. Page 129, lines 7 to 19.

2 You recall being deposed in this case. Right?

3 A. That's right. I think it was two months ago or so.

4 Q. And you get deposed a lot. Right?

5 A. A fair amount, sure.

6 Q. I'd like you to read the question and answer at line 7
7 through 19 just to yourself. Don't read it out loud.

8 A. Okay.

9 Okay.

10 Q. I'm going to ask you again, what is the plain and
11 ordinary meaning of the term 'DRAM circuit'?

12 A. Let's say it's a contested issue in this matter. It
13 depends on the context we're talking about.

14 MR. CORDELL: Your Honor, may I publish his answer?

15 THE COURT: You may publish 7 through 19.

16 MR. CORDELL: Thank you, Your Honor.

17 Q. (BY MR. CORDELL) So at your deposition, Doctor Brogioli,
18 you were asked, "What is the plain and ordinary meaning of the
19 term 'DRAM circuit'?"

20 And your answer was, "I don't recall discussing the plain
21 and ordinary meaning of that phrase. Let me go back to the
22 reference here."

23 You then fished around, and you say, "I'm trying to find
24 this section in my report here. It's on kind of page 25 or
25 so, starting around 6, and discussing how DRAM circuit, at

1 least in paragraph 67, that Rajan has this DRAM circuit and
2 that being different than -- or not usable in the Samsung
3 products. I think we talked about Rajan a bit earlier
4 today."

5 That was your answer. Right, sir.

6 A. For that question, yes.

7 Q. You didn't give us the plain and ordinary meaning of DRAM
8 circuit is a structure with a transistor and a capacitor.
9 Right?

10 A. I don't recall saying that, no.

11 Q. You didn't say, Oh, a DRAM circuit, that's the kind of
12 thing I teach my second year engineering students. Everybody
13 knows what that is. Right?

14 A. That may have been discussed in the depo. I don't
15 recall.

16 Q. I've been around DRAM for years; I can draw that circuit
17 in my sleep. Right?

18 A. In that response, no.

19 Q. And you're aware of the inventor of the '060 and '160
20 Patent, Doctor Hyun Lee?

21 A. Yes, I am.

22 Q. And you're aware he testified about what a DRAM circuit
23 is. Right?

24 A. I generally recall there was some testimony. I don't
25 recall the specifics.

1 Q. Well, do you recall that he said a DRAM circuit can just
2 be called a DRAM?

3 A. I don't recall his specific testimony there from memory.

4 Q. Well, let me see if I can help you.

5 A. Sure.

6 Q. In your binder there is the 11/29/22 deposition of
7 Mr. Hyun Lee. Dr. Hyun Lee.

8 A. Okay.

9 Q. Turn to page 67.

10 A. Okay.

11 Q. Starting at line 7 through -- I'm sorry. I said 67. 76.
12 Dyslexia. At line 7 through 11, read that to yourself.

13 A. Okay.

14 Q. So Doctor Lee said, "Well, you know, a circuit can be
15 called a DRAM." Right?

16 A. He's -- well, he's saying, "DRAM is an overly broad scope
17 or concept, and to simplify or simply put, a DRAM means a
18 dynamic random memory."

19 Q. And then he goes on to say, "So DRAM circuit -- a circuit
20 can be called a DRAM." Right?

21 A. He says that, yes.

22 Q. And he's the inventor of these patents. Right?

23 A. That's correct.

24 Q. He also said, you know, a die can be called a DRAM.
25 Right?

1 A. I'm going down the to line 15 or so, I think?

2 Q. Yes.

3 A. Yes. He says, Circuit can be called a DRAM; a die can be
4 called a DRAM, et cetera.

5 Q. Remind the jury what a die is. It's kind of an odd word.

6 A. The die, as I talked about earlier and I think others
7 did, is that piece that's cut out during the manufacturing
8 process.

9 Q. Well, so the way these things are made is they take a big
10 round disk of silicon. Right?

11 A. That's correct.

12 Q. And then using these really sophisticated machines, they
13 print the circuits on the silicon. Right?

14 A. That's fair, yes.

15 Q. Actually it's like an old-fashioned slide show where you
16 used to have to go watch somebody's vacation. Now they're
17 printing the circuits right on the face of the silicon.
18 Right?

19 A. Pretty much, yes.

20 Q. And they use photo resist to make sure that it sticks in
21 all the right places, but they end up creating thousands of
22 DRAM die on one round disk. Right?

23 A. That sounds right, yes.

24 Q. And then they saw them into pieces. When they saw them
25 into pieces, instead of dicing them up -- well, they dice them

1 up and they create die. Is that how the process works?

2 A. That's right.

3 Q. Okay. So a little piece of silicon from one of those big
4 wafers is what we call a die. Right?

5 A. Correct.

6 Q. And that die would have millions of DRAM circuits on it?

7 A. It could have millions of DRAM cells on it, sure.

8 Q. Well, sir, you seem to have changed my question. That
9 little piece of silicon would have millions of DRAM circuits
10 on it. Right?

11 A. I would say it has circuit components or circuitry on it.

12 Q. So you're resisting calling them DRAM circuits even
13 though it's DRAM die and the inventor calls them DRAM
14 circuits. Is that fair?

15 A. Well, you're describing a process of what sounds like
16 making DRAM versus the array dies I was talking about earlier.

17 Q. Okay. I'm just talking about in the real-world out
18 there, engineers, as they would pick up one of those die, they
19 would say, This is a DRAM die. You would agree with that.
20 Right?

21 A. I would say it would depend on the context you're asking
22 them.

23 Q. Okay. But if I just made that wafer with a thousand
24 little DRAM die on them and I cut them up, an engineer would
25 say, Yep, that's a good DRAM die. Right?

1 A. If it was a, you know, DDR3, or something like that,
2 DRAM and it passed the test, I think your answer was a good
3 DRAM die.

4 Q. Okay. And then there would be -- if that were true, if
5 it passed the test, it was a good product. Right? If it
6 passed the test, you would agree with me that there would be
7 millions of DRAM circuits on that DRAM die. Right?

8 A. Phrased differently in the context I talked about today,
9 there would be millions of DRAM cells on there.

10 Q. Again, sir, I'm asking you out in the real world. Before
11 you got involved in this case you would have looked at that
12 die and you would have said there are millions of DRAM
13 circuits on that die. Right?

14 A. Again, it would depend on the context, sir.

15 Q. So you can't answer my question.

16 A. As it's phrased, I can't.

17 Q. And yet you understand that a person of ordinary skill
18 back around, let's say 2011, would readily recognize the term
19 'DRAM memory die'. Correct?

20 A. I would say depending on the context, yes.

21 Q. That 99 out of a hundred engineers would know exactly
22 what you were talking about when you said 'DRAM die'.
23 Correct?

24 A. Absent of any context, they might have a notion of what
25 you're referring to.

1 Q. You know that 99 out of a hundred engineers back in 2011,
2 if you said, I've got a circuit here, they would know what you
3 were talking about. Right?

4 A. Not necessarily.

5 Q. You don't understand what a circuit is. Is that what
6 you're telling this jury?

7 A. No, that's not what my answer was.

8 Q. Okay. So when you teach class and you -- do you teach
9 circuits?

10 A. I have, yeah.

11 Q. What are you teaching now?

12 A. It's more graduate level Ph.D. candidate stuff.

13 Q. Like software?

14 A. Some software, but more hardware phasing.

15 Q. Okay. But you're not teaching any circuits classes now.
16 Right?

17 A. No.

18 Q. But you and I both took circuits as an undergrad. Right?

19 A. We did.

20 Q. And we knew exactly what people were talking about when
21 they said the word 'circuit'. Right?

22 A. Well, again, that's very context specific.

23 Q. And before you got involved in this case, you knew what a
24 DRAM circuit was. Right?

25 A. Again, it would depend on context.

1 Q. So you can't answer that.

2 A. Well, it would depend on the context what constitutes a
3 circuit or not.

4 Q. So the first time you ever encountered the phrase 'DRAM
5 circuit' is in this litigation. Is that what you're telling
6 us?

7 A. I don't know if that's true one way or the other.

8 Q. Well, and, in fact, when we asked you that in your
9 deposition, you couldn't tell us. Right?

10 A. Perhaps. I don't recall if that question was in there
11 or not. It sounds like it was.

12 Q. Well, before you became involved in this case, you
13 understood that a DRAM circuit is included pretty much any
14 time you see the word 'DRAM'. Right?

15 A. Again, same answer. It depends on the context you're
16 asking about asking the question.

17 Q. Well, in fact, you would need to think more about that
18 before you answered. Is that fair?

19 A. I guess depending on the context of the question, yes.

20 Q. So you showed the jury this slide. And I believe you
21 stood up and you said, A DRAM circuit, well, that's the thing
22 with these little legs coming off the side of it. Right?

23 A. That was one of the things I pointed out on this slide.

24 Q. Well, no, that was the thing. Right, sir? You said the
25 die or the pieces of silicon, those weren't what you referred

1 to; you referred to the little legs coming up. Do you
2 remember that?

3 A. I referred to those, yes.

4 Q. And those little legs are the connections to the DRAM
5 die. Right?

6 A. I would say they're part of the DRAM circuit in this
7 case.

8 Q. Well, it wasn't my question. My question was the little
9 legs are the connections to the DRAM die. Right?

10 A. No.

11 Q. Okay. You didn't tell this jury just a few minutes ago
12 that each of these little legs are what connects the DRAM die
13 to the support board underneath?

14 A. I don't recall using those words, no.

15 Q. So you dispute the idea that what you have shown in your
16 exhibit, this black rectangle here, you say that is not a DRAM
17 die. Is that it?

18 A. That in this context or my explanation it's a DRAM
19 circuit.

20 Q. Okay. So outside of this courtroom, if we were just back
21 in engineering school and you were teaching your students, you
22 would say that this black structure is a DRAM die and the
23 little wires connecting to it are the electrical connections.
24 Fair?

25 A. It would depend on the context you're asking me.

1 Q. Well --

2 A. Where I'm teaching it. Sorry.

3 Q. So in that context, can't you answer that question?

4 A. Can I hear the question one more time?

5 Q. Sure. You would tell them that this black structure,
6 that's a DRAM die and these little fine wires are the
7 connections to the DRAM die. Right?

8 A. I don't know that I would use that definition.

9 Q. And, in fact, these little connections, these are
10 microscopically separated from one another. Right?

11 A. They are certainly separated from one another, but
12 they're visible from the eye.

13 Q. You threw out a definition that made it sound like it was
14 10 millimeters, but this is on the order of microns. Right?

15 A. I didn't say 10 millimeters, no. That could be on the
16 order of microns, though.

17 Q. Okay. So it's your opinion, as you sit here, that if
18 somebody sees the phrase 'memory circuit', you've got to think
19 of these little wires coming off of the edges. Right?

20 A. In the context of these DRAM circuits, yes, that there's
21 external connections.

22 Q. So my question was really precise. When I see the word
23 'DRAM', I'm an engineer out there walking around in the world
24 in 2011, and I see a reference to 'DRAM circuit', it's your
25 opinion that I immediately think of these little wires coming

1 off the edges. Right?

2 MR. BURGESS: Objection, Your Honor. It's
3 criticizing the witness for not answering a different
4 question.

5 THE COURT: Overruled.

6 THE WITNESS: Again, it would depend on the context,
7 sir.

8 Q. (BY MR. CORDELL) You can't answer that question.

9 A. I believe I just did. It would depend on the context.

10 Q. Well, let's look at what Netlist said in its 10-K. Do
11 you recall that document?

12 A. I recall the document.

13 MR. CORDELL: So can I have DDX 116.6? And can I
14 have the paragraph under 'business overview'? There we go.
15 Blow up that first one, please. Thank you.

16 Q. (BY MR. CORDELL) Now, you understand what a 10-K is.
17 Right?

18 A. As I recall, I think it's a document filed with the SEC,
19 but perhaps you can clarify.

20 Q. Well, it's a document that a company has to file with the
21 SEC to talk to its stockholders. Right?

22 A. That's more or less my recollection. Thank you.

23 Q. And it's not nice to say something that's not accurate to
24 your stockholders. Right?

25 A. That's my understanding, yes.

1 Q. You get in trouble for that. Right?

2 A. We've seen that in the news recently.

3 Q. And do you see the sentence that begins right here, "We
4 were one of the first to bring NAND flash memory to the memory
5 channel with our NV vault non-volatile dual in-line memory
6 modules using software-intensive controllers and merging
7 dynamic random access memory integrated circuits DRAM IC or
8 DRAM and NAND flash."

9 Do you see that?

10 A. Yes, I do.

11 Q. So this is talking about that combination of NAND flash
12 and DRAM. Right?

13 A. I -- that's what it sounds like. I think may have been
14 discussed earlier today I didn't offer opinions on that,
15 though.

16 Q. You don't recognize that here, sir?

17 A. I do. I wasn't speaking about it earlier today is all
18 I'm saying.

19 Q. So you can't give your impression of this at all?

20 A. Oh, that's not what I said.

21 Q. Okay. Well, one thing we can agree on is that the
22 company as they're filing documents with the SEC called their
23 products dynamic random access memory integrated circuits.
24 Right?

25 A. I see that phrase here, yes.

1 Q. And then they say it's the same thing as DRAM ICs or
2 DRAM. Right?

3 A. Quotes DRAM ICs, in quotes DRAM, yes.

4 Q. So at least according to Netlist, the phrase DRAM IC
5 refers to a DRAM circuit. Right?

6 A. It looks like for this passage they're using them
7 interchangeably.

8 Q. And they also use DRAM and DRAM circuit interchangeably.
9 Correct?

10 A. It looks like that's what they're doing here, yes.

11 Q. Now, you have no reason to disagree with Netlist that
12 it's perfectly reasonable to call a DRAM chip a DRAM circuit.
13 Right?

14 A. It looks like in this document, yes.

15 Q. In other words, you don't have any reason to disagree
16 with them. Correct?

17 A. What they're saying here, no.

18 Q. Okay. So let's look at your paper.

19 MR. CORDELL: Can I have DDX 118? I'm sorry. I
20 think it's a book.

21 Q. (BY MR. CORDELL) You contributed a chapter to this book.
22 Right? "Software Engineering or Embedded Systems"?

23 A. That's right. There might have been a couple of
24 chapters, but I think at least one.

25 MR. CORDELL: Okay. Let's go to chapter 12.

1 Q. (BY MR. CORDELL) Now, you wrote this book before this
2 lawsuit. Right?

3 A. That's right.

4 Q. And it looks like the copyright date is like 2013. Is
5 that about right?

6 A. That looks right. I think it's on the bottom of the
7 page.

8 Q. There we go. And obviously you wrote it a little bit
9 before 2013. Fair?

10 A. At least that's what we tell people, but yes.

11 Q. Turning to chapter 12, page 344, so right here at the top
12 you talk about the underlying memory system, caches, DRAM, and
13 buses, et cetera. Right?

14 A. I see that.

15 Q. And there you said DRAM as if everybody's going to know
16 what you're talking about. Right?

17 A. I -- in this context, that looks like what I was doing,
18 yes.

19 Q. You didn't say DRAM with little wired external
20 connections. Right?

21 A. I don't see that I said that here.

22 Q. In fact, you wouldn't have cared how the DRAM was wired
23 for the purposes of this book. Right?

24 A. It's been a long time since I've looked at this, but I
25 don't think that would have necessarily been relevant in this

1 context one way or the other.

2 Q. You certainly didn't mean to suggest to a reader that you
3 were talking about the Rajan DRAM structure. Right?

4 A. In this context I don't think so, no.

5 Q. You just mentioned DRAM as a type of memory. Right?

6 A. It was described in the context of the components of a
7 memory system on the machine.

8 Q. And that was just like the way Doctor Lee described it.
9 Right? It's just a memory chip. Right?

10 A. Here it was in contrast to faster memory like caches or
11 faster memory like registers.

12 MR. CORDELL: Let's look at DDX 119.

13 Q. (BY MR. CORDELL) Do you recall this paper, Doctor
14 Brogioli?

15 A. I recall the title. It's been some time, but yes.

16 Q. Okay. And under table 1, once again, you start talking
17 about memory. Right?

18 A. That looks correct--memory systems.

19 Q. And you make reference to SRAM and DRAM modules. Right?

20 A. That's correct.

21 Q. You don't say anything about Rajan here, do you?

22 A. No, sir.

23 Q. You don't say anything about the little wire legs coming
24 off the edges. Right?

25 A. Not in this table anyway.

1 Q. Okay. You just call it DRAM. Right?

2 A. It looks like DRAM modules, yes.

3 Q. And your expectation is anybody who reads this paper is
4 going to know exactly what you're talking about when you say
5 DRAM. Right?

6 A. In terms of DRAM modules it looks like, yes.

7 Q. All right.

8 MR. CORDELL: Let's go to DDX 120 at page 1.

9 Q. (BY MR. CORDELL) This is another one of your papers.
10 Right?

11 A. That looks right.

12 Q. All right. Let's look at the right column about halfway
13 down.

14 So here we have DRAMs and controllers, SRAMs and
15 controllers. Right?

16 A. That looks correct yes.

17 Q. You don't say anything about DRAMs having the structure
18 of Rajan. Right?

19 A. Not that I see in this paragraph, no, sir.

20 Q. And you would have been shocked if someone had called you
21 after reading this paper and said, Hey, where do I get these
22 Rajan DRAM structures. Right?

23 A. I wouldn't say that. They might do that depending on the
24 context of what they wanted to know about.

25 MR. CORDELL: All right. Let's try DDX 123.

1 Q. (BY MR. CORDELL) This is your Ph.D. thesis. Right?

2 A. That looks right.

3 Q. You wrote this in April of 2007?

4 A. That's right.

5 MR. CORDELL: All right. Now let's go to page 67,
6 near the bottom of the page.

7 Q. (BY MR. CORDELL) And here you're talking about the
8 physical memory is instantiated. That means it's put into
9 place. Right?

10 A. I think -- that could be what this sentence is saying.

11 Q. And you say it's instantiated as a multiported memory
12 module. Right?

13 A. I should add some clarification, because the words
14 'instantiated' and 'multiported' in the context of this work
15 had to do with also simulation of modeling. But I do see
16 those words.

17 Q. Well, so sometimes you actually make the chip; sometimes
18 you just sort of do it mathematically so you can figure out
19 how it would work. Right?

20 A. I would say you can simulate it at the clock cycle
21 accuracy and get power and performance, yeah.

22 Q. There you go. But here you're talking about memory
23 again. Right?

24 A. I'm referring to memory, yes.

25 Q. And you talk about SRAM and DRAM. Right?

1 A. That's correct, yes.

2 Q. You say nothing about little legs coming out the side of
3 the chips. Right?

4 A. Not in this paragraph, no.

5 Q. You say nothing about Rajan and its structures. Right?

6 A. I don't think so, no, sir.

7 MR. CORDELL: Let's go to DDX 132.

8 Q. (BY MR. CORDELL) You're familiar with this patent,
9 aren't you, sir?

10 A. I don't recall if I cited it in my report or not.
11 Perhaps you could help me.

12 Q. All right. Well, let me look at column 1, line 17
13 through 22. There you go. This says that "The subject matter
14 disclosed herein relates generally to integrated circuit IC
15 design and, in particular, to high-speed dynamic random access
16 memory circuits with low energy consumption."

17 Do you see that?

18 A. I do, yes.

19 Q. And here it's talking about DRAM circuits, it says in the
20 background, the very next line. Right?

21 A. I see DRAM circuits, yes.

22 Q. And the author here tells us that DRAM circuits are
23 sometimes called DRAMs or DRAM devices. Right?

24 A. That's what the author is saying here, yes.

25 Q. And you have no reason to dispute that. Right?

1 MR. BURGESS: Your Honor, may we approach?

2 THE COURT: Approach the bench.

3 (The following was had outside the hearing of the
4 jury.)

5 THE COURT: What is it, Mr. Burgess.

6 MR. BURGESS: I object to this entire line of
7 questioning as improper impeachment. None of this is in
8 evidence. It hasn't been cited in his reports, and he hasn't
9 asked him any questions that he has failed to answer. He's
10 not impeaching anything. He's just throwing up random
11 documents and asking questions about them.

12 MR. CORDELL: I'm having him acknowledge that people
13 in the art disagree with his positions, and if he wants to
14 comment about them, Mr. Burgess is certainly able to ask him
15 on redirect why that might be inaccurate. This witness gave
16 us nothing on the plain meaning of 'DRAM circuit', and I'm
17 just establishing that he has used the term -- others have
18 used the term without the qualifications that he has
19 artificially placed on them today.

20 MR. BURGESS: I believe that the only proper use of
21 these documents that aren't in evidence is as impeachment, and
22 that's not how they are being used.

23 MR. CORDELL: They are being used as demonstratives.

24 THE COURT: We had quite a few objections to this
25 witness' testimony about DRAM and DRAM circuits on direct.

1 There were multiple requests by Mr. Cordell for me to exclude
2 that testimony, which I overruled, and I told him he could
3 adequately address it on cross examination. I'm inclined to
4 grant him some latitude on cross examination.

5 MR. BURGESS: I understand, Your Honor.

6 MR. CORDELL: Your Honor, do you have any idea how
7 much longer you want to go tonight.

8 THE COURT: How much longer do you have?

9 MR. CORDELL: Quite a bit.

10 THE COURT: Well, I think we'll go at least another
11 30 minutes and see where we are.

12 MR. CORDELL: Thank you.

13 (The following was had in the presence and hearing
14 of the jury.)

15 THE COURT: All right. Let's proceed.

16 Q. (BY MR. CORDELL) So just to reframe it, Doctor Brogioli,
17 it is a fact that, at least for this author or this inventor,
18 DRAM circuits are sometimes called DRAMs or DRAM devices.
19 Right?

20 A. That looks what the author is saying, yes.

21 Q. So let me show you a circuit.

22 MR. CORDELL: Can I have DDX 129?

23 Q. (BY MR. CORDELL) Are you familiar with this textbook,
24 sir?

25 A. It looks familiar, yes.

1 Q. This is "Wafer Level 3-D ICs Process Technology."

2 First, tell the jury what a wafer is?

3 A. A wafer is that disk that we had been talking about
4 earlier, I believe.

5 Q. And 3-D ICs, is that three-dimensional integrated
6 circuits?

7 A. I think that's what it's referring to.

8 Q. And process technology, that's talking about the factory
9 where these things are actually made. Right?

10 A. Usually, yes.

11 MR. CORDELL: Okay. So let's look at figure 12.8 at
12 page 11.

13 Q. (BY MR. CORDELL) So here, Doctor Brogioli, we have what
14 this textbook calls a simplified DRAM circuit. Do you see
15 that?

16 A. I do see that.

17 Q. You have no reason to disagree that that is an accurate
18 characterization. Correct?

19 A. This component for this author and this reference, yes.

20 Q. And the way this works is a DRAM cell, I think you called
21 it, consists of a transistor and a capacitor in series.
22 Right?

23 A. Yes. DRAM cells are typically made of that.

24 Q. And so when your engineering students are drawing a DRAM
25 circuit, they draw a transistor followed by a capacitor.

1 Right?

2 A. If they were drawing like a DRAM cell at the white board,
3 there would be transistors and capacitors, yeah.

4 Q. There wouldn't just be any; there would be --

5 MR. CORDELL: May I, Your Honor?

6 THE COURT: You may.

7 Q. (BY MR. CORDELL) It would be a very definite structure
8 with a gate, and there would be a transistor in the middle and
9 a capacitor in the series. Right?

10 A. That looks fair, yes.

11 Q. And that's what your students would draw if you said, Go
12 draw me a DRAM circuit. Right?

13 A. Well, I would refer to that as a DRAM cell, but that
14 could be what they draw.

15 Q. Well, your students would call this a circuit. Right?

16 A. Again, I would say it would depend on the context, sir.

17 Q. Okay. But you acknowledge that this has two electronic
18 components--a transistor and a capacitor. Right?

19 A. That's right. Circuit components, yes.

20 Q. And it's hooked up to ground: Is that right?

21 A. That's correct, yes.

22 Q. And typically you have the bit line here and the word
23 line here. Right?

24 A. Something like that, yes.

25 Q. Okay. And the way this works is every -- actually more

1 than once during a second you've got to refresh this to
2 recharge the capacitor. Right?

3 A. That's right.

4 Q. And that's why the DRAM has to have power all the time.
5 Right?

6 A. Right. The information in the cell deteriorates over
7 time.

8 Q. Okay. And then to make it, you know, scalable so we all
9 can use it, you have to put a whole bunch of these on that
10 wafer. Right?

11 A. That's right.

12 Q. And so we end up with what we have here at figure 12.8
13 where you have a whole string of DRAMs that are DRAM circuits
14 that are lined up like this. Right?

15 A. I would say DRAM cells that are lined up like that, yes.

16 Q. Again, you don't like calling them circuits.

17 A. In the context of the patents and what we've talked about
18 today, I think cell is a more appropriate term.

19 Q. Well, you see that there's a little transistor on each of
20 these. Right?

21 A. That's right, yes.

22 Q. And that there's a capacitor here, too. Right?

23 A. That's correct, yes.

24 Q. And they're hooked together with conductors. Right?

25 A. That's right, yes.

1 Q. But you don't call that a circuit.

2 A. In the context of what I talked about today, no; not a
3 DRAM circuit.

4 Q. And when you talked about those memory die in the HBM
5 products, you acknowledged that there are millions and
6 millions of the structures I've drawn on the white board on
7 that die. Right?

8 A. Millions of those cells, yes.

9 Q. So what's a gate count, Doctor Brogioli?

10 A. It is typically a count for the amount of really, really
11 small components in a large microprocessor, for example.

12 Q. So on that little chip that we were talking about
13 earlier, somebody actually counts up how many of these
14 transistors, for example, exist. Right?

15 A. That's right. I don't think a human does it anymore, but
16 that's right.

17 Q. Right. And if we're talking about one of those DRAM die
18 in the Samsung HBM products, the vast majority of all of the
19 gates would look just like what I've drawn up here on the
20 board. Right?

21 A. It would seem the vast majority of the components would
22 be for storing in these cells, yeah.

23 Q. And by 'vast majority', it would be like 98 percent of
24 all the gates on the die?

25 A. Perhaps. I don't recall seeing firm numbers on that.

1 MR. CORDELL: Okay. Just in case there's any doubt,
2 can I have DDX 130 at page -- well, let me have the cover
3 first.

4 Q. (BY MR. CORDELL) Are you familiar with this book, Bruce
5 Jacobs? He writes a lot of textbooks?

6 A. That's right. I've seen this before.

7 Q. Have you ever used this in one of your classes?

8 A. Maybe. I don't recall.

9 Q. I think I might have had one of his. He used to write
10 one that was red?

11 A. Maybe. I think I've got one of the versions of one of
12 these books at home, too.

13 THE COURT: Let's move along.

14 MR. CORDELL: Let's look at figure 8.2.

15 Q. (BY MR. CORDELL) And you keep calling this a DRAM cell,
16 but to Mr. Jacobs, he just shows you the circuit here. Right?

17 A. He shows what we talked about earlier and refers to it as
18 a DRAM cell.

19 Q. And that consists of exactly these circuit elements.
20 Right?

21 A. The same components you have on the easel that you draw.

22 Q. Okay. And when we're talking about high bandwidth
23 memory, or HBM, we are talking about these same circuits.
24 Right?

25 A. The same type of DRAM cells, yes.

1 Q. Well, let's -- there's no dispute in this case that what
2 you were talking about when you pointed to HBM, is you're
3 talking at DRAMs. Right?

4 A. It's a DRAM type of storage, yes.

5 Q. Okay. So let me show you DDX 134.4.

6 MR. CORDELL: Let me have the cover, please.

7 Q. (BY MR. CORDELL) Are you familiar with this book, sir?

8 A. It looks like it. I don't know if I've referenced it in
9 this matter or not. I don't recall.

10 MR. CORDELL: Let's go to chapter 4. Can you blow
11 up the upper right?

12 Q. (BY MR. CORDELL) Do you see, sir, that this chapter was
13 written by three engineers at Samsung?

14 A. It says they're from Samsung. I don't know if they're
15 engineers or not, but Samsung.

16 Q. They kind of wrote the book on --

17 THE COURT: Just a moment.

18 Yes, sir.

19 MR. BURGESS: Your Honor, I object to the use of
20 this document, which's has been excluded from the case, as I
21 understand it. Their expert tried to rely on it and it was
22 not allowed.

23 THE COURT: Are you telling me this was offered for
24 admission and was rejected?

25 MR. BURGESS: It was excluded from Doctor Robins'

1 report. It was an untimely submission.

2 THE COURT: Well, this is not Doctor Robins.

3 Let me hear a response from Defendant.

4 MR. CORDELL: It was excluded having to do with the
5 Court's orders on the issues that would be tried, and so it
6 has nothing to do with Doctor Brogioli.

7 THE COURT: And your purpose in using this is what,
8 Mr. Cordell?

9 MR. CORDELL: Just to have the witness confirm the
10 nature of the device that he has accused of infringement.

11 THE COURT: In light of that context, do you still
12 have an objection, Mr. Burgess?

13 MR. BURGESS: Yes. It seems to me like this is just
14 a back-door attempt to get in a document that their expert was
15 not allowed to rely on. As I said earlier, it's not proper
16 impeach. It's not in evidence.

17 MR. CORDELL: We will not move its admission, Your
18 Honor.

19 THE COURT: I don't find that it's some kind of
20 back-door impeachment. It seems to me that it is a
21 demonstrative to aid the witness, and that's appropriate. And
22 the fact that it may have been offered through some other
23 expert for admission and excluded in the case for a different
24 reason does not seem to me to prohibit its use as a
25 demonstrative with this witness.

1 I'll overrule your objection.

2 MR. BURGESS: Thank you, Your Honor.

3 Q. (BY MR. CORDELL) So do you see this, Doctor Brogioli,
4 the chapter 4.1?

5 A. I do.

6 Q. And what we're talking about here is DRAM. Right?

7 A. It looks like it's referring to DRAM, yes.

8 Q. Okay. And one of the things that the authors talk about
9 is data bandwidth is constantly increasing and all types of
10 DRAM from server DRAM to mobile DRAM and graphics DRAM and the
11 HBM were high bandwidth memory. Do you see that?

12 A. I think I'm following you. The screen moved on me. That
13 looks correct.

14 Q. And so these authors were quite clear what they're making
15 their HBM out of was DRAM. Right?

16 A. That there is a DRAM style element to what they're
17 referring to as HBM here.

18 Q. And they don't say you have to use the Rajan structure.
19 Right?

20 A. I don't see that referenced here.

21 Q. And they don't say that you have to use the little legs
22 coming off the edge of the die. Right?

23 A. I don't see that reference here, no.

24 Q. And, in fact, if we go to figure 4.23, they tell us
25 it doesn't matter how you wire up the DRAM dies. Right?

1 A. It looks like they're just showing different examples
2 here.

3 Q. That they call them both DRAM, and yet one has little
4 legs coming off the die and the other has the TSVs. Right?

5 A. It looks like they're showing two -- one figure A on top
6 is POP, or package on package, and the figure on the bottom is
7 the TSV 3D package.

8 Q. Yet they call them both DRAM. Right?

9 A. They have a high-level illustration of one of the blocks
10 that has DRAM in it, yes; or has written on it, yes.

11 Q. All right. Okay. Let's move to your testimony about the
12 Patent Office.

13 You showed the jury this slide. Right?

14 A. That's correct.

15 Q. And you seem to offer testimony about what happened at
16 the United States Patent Office. Did I get that right?

17 A. I was pointing to a response from Netlist to the Patent
18 Office.

19 Q. Okay. Now, you weren't at the Patent Office then.
20 Right?

21 A. No, sir.

22 Q. You've never been to the Patent Office?

23 A. As an employee, no.

24 Q. And you weren't there when any discussions took place
25 between the inventors and the patent examiner. Right?

1 A. That's correct.

2 Q. So all you're doing is you're reading the words on this
3 page and then you're telling the jury how they should read the
4 words on the page. Fair?

5 A. I would say the words on the page and the surrounding
6 dialogue.

7 Q. Well, not the surrounding dialogue, sir. You weren't
8 there. Right?

9 A. I was referring to the file history; but no, I was not
10 there.

11 Q. So you are telling this jury that their -- you can read
12 the words better than they can. Fair?

13 A. I didn't say that. I just pointed out the language to
14 them.

15 Q. Okay. Now, do you like cars, Doctor Brogioli?

16 A. I do.

17 Q. So you know the difference between a gas car and a diesel
18 car. Right?

19 A. Generally, yes.

20 Q. Okay. So I want you to imagine that you have invented a
21 new piston material. Okay?

22 A. Okay.

23 Q. Are you with me?

24 A. I'm with you.

25 Q. And you file your patent application and you say, I got

1 this great new piston that can -- man, it can withstand all
2 kinds of heat and high pressure. And that's your patent
3 application. Are you with me?

4 A. I'm with you.

5 Q. And the patent examiner says, You know what? Sorry, but
6 other people have come up with pistons before in the diesel
7 context and they all deal with high temperatures and high
8 pressures. Are you with me?

9 A. Okay.

10 Q. So you say, You know what? My piston is only going to
11 apply to gasoline engines.

12 And so the patent examiner says, Okay. You can have a
13 patent as long as we all understand that your patent is going
14 to apply to gasoline engines. Right? Are you with me?

15 A. I'm with you.

16 Q. So it wouldn't be fair for you, then, to turn around
17 and sue somebody who was making a diesel engine. Right?

18 A. That sounds like a question of law in this sort of
19 hypothetical.

20 Q. You just spent the last three hours testifying about
21 infringement, Doctor Brogioli. You don't know what that
22 means?

23 A. Yes, I do. I'm trying to follow your example of pistons.

24 Q. Right. So would it be fair, after you've told the Patent
25 Office, My patent is not going to apply to diesel engines, to

1 turn around and sue somebody who's making a diesel engine.

2 A. I would say -- so I'm not an attorney, but my
3 understanding is that that's not supposed to be allowed.

4 Q. So when Netlist tells the patent examiner, We're not
5 going to cover DRAM circuits, you think they ought to have
6 to live up to what they told the patent examiner. Right?

7 A. To what was being said in this document, yes.

8 Q. Well, more than that. They said Rajan merely stacks DRAM
9 circuits, and then they label them 206A to D. Right?

10 A. Well, Rajan labels them, but yes.

11 Q. And that's what they told the patent examiner--That our
12 patent is not going to apply to DRAM circuits. Right?

13 A. The DRAM circuits of Rajan, yes.

14 Q. Well, they didn't say the DRAM circuits of Rajan; they
15 said DRAM circuits. Right, sir?

16 A. Well, the language says DRAM -- Rajan merely stacks DRAM
17 circuits.

18 Q. And what you showed on the right-hand side of your slide
19 were these stacks of DRAM dies. Right?

20 A. Rajan's DRAM circuits.

21 Q. Well, DRAM die there, 206A, 206B, 206C, 206D. Those are
22 DRAM circuits. Right?

23 A. They're Rajan's DRAM circuits, yes.

24 Q. And then you told this jury that, Somehow we got to
25 include these purple lines on the edge. Right?

1 A. That's how these are interconnected, yes.

2 Q. Well, more than that, sir. You said you're going to
3 redefine the term 'DRAM circuit' to include the legs on the
4 side on the edges of the die. Right?

5 A. Yes. For these types of DRAM circuits, that's how
6 they're connected versus what's in the patent.

7 Q. So that's how they're connected, sir, but what you told
8 the Patent Office is that you weren't going to cover DRAM
9 circuits. Right?

10 A. I think you've lost me a little bit. I think you said I
11 said to the Patent Office.

12 Q. I apologize. So what Netlist told the Patent Office in
13 order to get them to issue this patent, because somebody had
14 done it before them, is they had to say, We're not going to
15 cover DRAM circuits. Right?

16 A. The DRAM circuits of Rajan, yes.

17 Q. Well, you know, we had this argument earlier in the case
18 and we have a rule in the form of a claim construction.
19 You're aware of that. Right?

20 A. Yes, sir.

21 Q. And the first thing that we were taught in law school is
22 that when the judge makes a ruling, we all have to live with
23 it. Right?

24 MR. BURGESS: Your Honor, I'm sorry to object again,
25 but this is improper impeachment. He's conflating what we

1 said to the Patent Office with the Court's claim construction
2 order, trying to make it seem like they're somehow
3 inconsistent in his testimony. It's not fair. It's not
4 accurate.

5 MR. CORDELL: This is in direct rebuttal to what
6 they did on his direct testimony.

7 THE COURT: I'll allow it, although I doubt this
8 witness knows what we learned in law school.

9 Q. (BY MR. CORDELL) Well, Doctor Brogioli, you've been
10 around for a long time in the patent business. Right?

11 A. A number -- a few years, yes.

12 Q. And you understand that when the Judge gives us a rule, a
13 rule of the road, a claim construction, we all have to follow
14 it. Right?

15 A. That's my understanding, yes.

16 Q. And what the Judge said after looking at all of this is
17 that an array die has to be different from a DRAM circuit.
18 Right?

19 A. That's right.

20 Q. Didn't say anything about little legs coming off the
21 edges.

22 A. Specifically in that quote it does not say that.

23 Q. Doesn't say anything about Rajan. Right?

24 A. In the quote on this screen, no.

25 Q. It doesn't matter if it's wired up on the top or the

1 bottom or in the middle. Right?

2 A. I don't see that language in this quote, no, sir.

3 Q. Okay.

4 MR. CORDELL: Could I have figure 3 at page 132,
5 Mr. Sayers?

6 May I, Your Honor?

7 THE COURT: May you what?

8 MR. CORDELL: Just consult with my graphics fellow
9 for a second?

10 THE COURT: You may.

11 MR. CORDELL: Thank you.

12 (Pause in proceedings.)

13 MR. CORDELL: It's page 140. There we go.

14 Q. (BY MR. CORDELL) You're familiar with this diagram.
15 Correct, Doctor Brogioli?

16 A. It looks familiar. I can't quite place it from just the
17 screen capture.

18 Q. It's from your expert report.

19 A. Okay.

20 Q. And you recall that you used this as part of your
21 infringement analysis in this case?

22 A. That sounds right.

23 Q. Can you remind us of what this is?

24 A. It looks like it is related to that DA or direct access
25 region in some of the test block, the PHY.

1 Q. But what you meant to represent here on the right-hand
2 side are the Samsung HBM DRAM cores. Is that correct?

3 A. That's what the figure says, yes.

4 Q. And you agree with me, sir, that there would be millions
5 of DRAM cells, as you call them, in those DRAM cores. Right?

6 A. That would be correct, yes.

7 Q. And you just don't like calling them DRAM circuits
8 because you understand that the rule would be that there could
9 be no infringement if they're DRAM circuits on those DRAM
10 cores. Right?

11 A. I just refer to them as DRAM cells, sir.

12 Q. But you understand if you admit there are DRAM circuits
13 on the DRAM cores, there's no infringement in this case.
14 Right?

15 A. I guess I would phrase it differently that they're not
16 the DRAM circuits of the prior art.

17 Q. I'll ask you one more time. If you admit there are DRAM
18 circuits on these Samsung HBM DRAM cores, you agree that there
19 is no infringement in this case.

20 A. I'm not sure I can answer as how you phrased that.

21 Q. So you think you can say there are DRAM circuits on this
22 DRAM core and still maintain your opinion that there's
23 infringement in this case.

24 A. If I'm understanding you correctly. I don't think I was
25 asked to perform that analysis.

1 Q. And you can't tell the jury today.

2 A. In the context of what you're saying, again, these DRAM
3 cores are comprised of cells.

4 Q. And those cells have these DRAM circuit components that
5 we've drawn up on the board. Yes?

6 A. They have circuit components, yes.

7 Q. And you don't want to call them DRAM circuits because you
8 understand that if you do so this jury is going to conclude
9 that there's no infringement in this case. Right?

10 A. No, sir. I'm saying they're not DRAM circuits; they're
11 cells.

12 Q. So I've asked it both ways, Doctor Brogioli. If you say
13 there are no DRAM circuits, does that mean that you can find
14 infringement -- or that the jury can find infringement?

15 A. I would say if -- I'm trying to answer your question as
16 simply as I can. But if you are using the Rajan DRAM circuit
17 in this case, that would not be infringing, if I'm
18 understanding you correctly.

19 Q. My question is really precise, so I'll try to be as slow
20 as I can, and precise.

21 If you offered this jury the opinion that the Samsung HBM
22 DRAM cores contain DRAM circuits, you would agree that there's
23 no infringement in this case. Right?

24 A. That sounds like an opinion I haven't offered, and it's
25 not the case of how the HBMs work.

1 Q. You just can't say one way or the other. Is that right?

2 A. Well, if I'm understanding you correctly, it's a scenario
3 I haven't analyzed, as to what your DRAM circuit is in this
4 context.

5 Q. Well, so my DRAM circuit is what I drew up on the board,
6 and I'd like you to assume that there are millions on this
7 Samsung HBM DRAM core. Are you with me?

8 A. Again, I don't -- those aren't DRAM circuits.

9 Q. That wasn't my question, sir. I want you to assume that
10 what I've drawn up on the board -- you're with me. Right?
11 You got that in mind?

12 A. Yes, sir.

13 Q. That structure appears millions of times on this Samsung
14 DRAM core that you put in your report. Are you with me?

15 A. Okay.

16 Q. And we agree that that's true. Right? That structure
17 will appear millions of times on those HBM DRAM cores.
18 Correct?

19 A. Cells like that, yes.

20 Q. If that is true, sir, isn't it a fact that there is no
21 infringement in this case?

22 A. I'm not sure I can answer that question, if I'm following
23 you.

24 Q. Let's talk about electrical communication.

25 THE COURT: Counsel, approach the bench, please.

1 (The following was had outside the hearing of the
2 jury.)

3 THE COURT: Do you have any idea much cross you
4 have?

5 MR. CORDELL: I'm about two thirds through.

6 THE COURT: And you're going to have some redirect?

7 MR. BURGESS: Yes, sir.

8 THE COURT: We'll stop for the day and pick up here
9 tomorrow.

10 MR. BURGESS: Thank you.

11 (The following was had in the presence and hearing
12 of the jury.)

13 THE COURT: Ladies and gentlemen, this witness has
14 got some additional time on the witness stand and we're right
15 at 6:00. We're going to stop for the day at this juncture.

16 I'm going to ask you as you leave the courtroom to exit
17 through the jury room, take your notebooks with you, leave
18 them closed on the table there so they'll be there in the
19 morning when you come back.

20 I'm going to ask you to check your driving conditions and
21 be here on time. You did a good job this morning. We'll try
22 to start as close to 8:30 as possible tomorrow morning as
23 well.

24 Please drive safely to and from your homes. Remember all
25 the instructions I've given you, including not to discuss the

1 case or communicate about it in any way with anyone, including
2 the eight of yourselves.

3 And with that, ladies and gentlemen, you're excused for
4 the evening. I'll see you tomorrow morning.

5 (Whereupon, the jury left the courtroom.)

6 THE COURT: Be seated, please.

7 Counsel, for your information, we've used almost 6 hours
8 and 45 minutes of trial time today. The Plaintiff has 7 hours
9 and 58 minutes remaining; the Defendant has 7 hours and 56
10 minutes remaining.

11 Let me encourage you to the fullest extent of my ability
12 to meet and confer efficiently with regard to disputes that
13 may develop overnight. I will be in chambers, as I have been
14 throughout the trial, ready to meet with you in advance of
15 bringing in the jury and trying to streamline and narrow any
16 disputes that survive your stringent meet and confer efforts.

17 Is there anything that needs to be raised with the Court
18 from either Plaintiff or Defendants before we recess for the
19 evening?

20 MR. SHEASBY: Nothing for Plaintiff.

21 MR. CORDELL: Nothing for Defendants, Your Honor.
22 Thank you.

23 THE COURT: All right. We stand in recess until
24 tomorrow morning.

25 (The proceedings were concluded at 6:00 p.m.)

1 I HEREBY CERTIFY THAT THE FOREGOING IS A
2 CORRECT TRANSCRIPT FROM THE RECORD OF
3 PROCEEDINGS IN THE ABOVE-ENTITLED MATTER.
4 I FURTHER CERTIFY THAT THE TRANSCRIPT FEES
5 FORMAT COMPLY WITH THOSE PRESCRIBED BY THE
6 COURT AND THE JUDICIAL CONFERENCE OF THE
7 UNITED STATES.

8
9 S/Shawn McRoberts

04/17/2023

10 _____ DATE _____
11 SHAWN McROBERTS, RMR, CRR
12 FEDERAL OFFICIAL COURT REPORTER
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Shawn M. McRoberts, RMR, CRR
Federal Official Court Reporter

IN THE UNITED STATES DISTRICT COURT
FOR THE EASTERN DISTRICT OF TEXAS
MARSHALL DIVISION

NETLIST, INC., (CAUSE NO. 2:21-CV-463-JRG
)
Plaintiff, ()
vs. ()
SAMSUNG ELECTRONICS CO., LTD., ()
et al., () MARSHALL, TEXAS
() APRIL 19, 2023
Defendants. () 8:30 A.M.

VOLUME 4

TRIAL ON THE MERITS

BEFORE THE HONORABLE RODNEY GILSTRAP
UNITED STATES CHIEF DISTRICT JUDGE
and a jury

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1 THE COURT: Be seated, please.

2 Are the parties prepared to read into the record those
3 items from the list of pre-admitted exhibits used during
4 yesterday's portion of the trial?

5 MS. TRUELOVE: Yes, Your Honor.

6 THE COURT: Please proceed.

7 MS. TRUELOVE: Thank you, Your Honor.

8 Plaintiff Netlist used the following four exhibits during
9 yesterday's presentation: JTX 0022, JTX 0023, JTX 0027, and
10 JTX 0041.

11 THE COURT: Any objection from Defendants?

12 MS. SMITH: No, Your Honor.

13 THE COURT: Do Defendants have a similar rendition
14 to read into the record?

15 MS. SMITH: We do, Your Honor.

16 THE COURT: Please proceed.

17 MS. SMITH: Your Honor, Samsung used DTX 30, JTX 23,
18 JTX 48, and JTX 65.

19 THE COURT: All right. Any objection from Plaintiff
20 Netlist?

21 MS. TRUELOVE: No objection, Your Honor.

22 THE COURT: All right. Thank you, counsel.

23 All right. Samsung, are you prepared to go forward with
24 Mr. McAlexander as your next witness?

25 MR. CORDELL: Yes, we are, Your Honor.

1 THE COURT: All right. Let's bring in the jury,
2 please.

3 (Whereupon, the jury entered the courtroom.)

4 THE COURT: Good morning, folks. It's good to have
5 you back. Please have a seat.

6 We will proceed with the Defendants' case in chief.

7 Defendants, call your next witness.

8 DR. ALBERT: Samsung calls Mr. McAlexander.

9 THE COURT: All right. If you'll come forward,
10 please, Mr. McAlexander, and be sworn.

11 (Whereupon, the oath was administered by the Clerk.)

12 THE COURT: Please have a seat on the witness stand,
13 sir.

14 All right, Mr. Albert. You may proceed with direct
15 examination.

16 DR. ALBERT: Thank you, Your Honor.

17 JOSEPH COLBY McALEXANDER, III, SWORN,
18 testified under oath as follows:

19 DIRECT EXAMINATION

20 BY DR. ALBERT:

21 Q. Good morning. Will you please introduce yourself to the
22 jury?

23 A. Yes. My full name is Joseph Colby McAlexander, the III.

24 Q. And please tell the jury a little bit about yourself.

25 A. I am about 76 years old. I have been married for over 54

1 years now. My wife and I live and operate a ranch up in north
2 of Dallas about 50 miles, Anna, Texas. I have two sons and
3 five grandkids.

4 Q. And what are you here to testify about today?

5 A. I'm here to testify about three patents in particular,
6 '918, the '054, and the '339, specifically as subjects relate
7 to infringement and invalidity as well as to talk about a
8 license agreement with Rambus.

9 Q. And do you have slides to assist your testimony today?

10 A. Yes, sir, I do.

11 Q. What do you do for a living, Mr. McAlexander?

12 A. I spend my time between two different primary businesses.
13 One, I am the president and founder of McAlexander Sound,
14 Incorporated. It is a Texas-based corporation located in
15 Richardson, Texas. It is a registered professional
16 engineering company.

17 And I also am the CEO and founder of Spirit Song Youth
18 Equestrian Academy, which is on my ranch up in Anna, Texas.

19 Q. What do you do for spirit song?

20 A. We are a state-certified organization. I and my staff
21 are each certified in equine therapy, specifically working
22 with at risk use youth and families assisting them in their
23 mental health issues, specifically giving them some additional
24 life skills so that they can put it back into their personal
25 and professional practice.

1 Q. Is Spirit Song a for-profit company?

2 A. No, sir. It's a non-profit.

3 Q. And how much of your time do you spend between your
4 non-profit activities and your work in your other companies?

5 A. It varies month to month, but I'd say about 60 percent is
6 in the engineering profession and probably 40 percent of my
7 time is devoted to the non-profit.

8 Q. Can you tell the jury about your education, sir?

9 A. Yes. I have a Bachelor of Science degree electrical
10 engineering. I received that in 1969 from North Carolina
11 State University.

12 Q. And how long have you been working in the field of
13 computer systems and memory?

14 A. Over 50 years.

15 Q. And can you tell the jury a little bit about your
16 experience in that field, the computer systems and memory,
17 over that 50 years?

18 A. Certainly. Well, immediately after graduating from the
19 university, I spent three years as an officer in the United
20 States military, specifically with Air Defense, looking at the
21 Air Defense systems and coordinating that for the United
22 States.

23 And then for -- as soon as I left the military, I joined
24 Texas Instruments as a design engineer in dynamic random
25 access memory. And I worked for 14 years directly for Texas

1 Instruments in dynamic random access memory, static memory,
2 charge couple devices that are used in video presentations,
3 and also in non-volatile memory, such as flash.

4 Q. And is any of that evidence particularly relevant for
5 this case?

6 A. Yes. That experience and subsequent experience
7 continuing in this field over the past 50 years, it's all
8 relevant.

9 Q. And are you a member of any professional organizations?

10 A. Yes. I am a member of the Texas Board of Professional
11 Engineers. As I mentioned, I am a registered professional
12 engineer, and I'm also a senior member with the Institute for
13 Electrical and Electronic Engineers. And that's an
14 international organization.

15 Q. Do you have any patents, sir?

16 A. Yes. I have 31 U.S. patents and eight foreign patents,
17 and a number of these are directed specifically to the matters
18 at issue in this case here.

19 Q. And what are you showing the jury on this slide, sir?

20 A. What I summarized on this slide is several of the patents
21 that are related to buffer-type circuitry as well as sensing
22 amplifiers to detect data states and specifically with regard
23 to loading.

24 DR. ALBERT: At this time, Your Honor, we'd like to
25 offer Mr. McAlexander as an expert in computer systems and

1 memory.

2 MR. SHEASBY: No objection.

3 THE COURT: Without objection, the Court will
4 recognize this witness as an expert in those designated
5 fields.

6 Please continue.

7 Q. (BY DR. ALBERT) What are you showing the jury here, sir?

8 A. What I'm showing on this slide is a roadmap of the
9 presentation I'll be giving today in my testimony,
10 specifically focusing on, first, two patents that are directed
11 to the flash DRAM hybrid memory module.

12 I'll then turn my attention to the '339 Patent, which is
13 to byte-wise buffers, and then I will end my testimony by
14 discussing the license agreement known as the Rambus-Samsung
15 license.

16 Q. What types of materials did you consider in forming your
17 opinions in this case?

18 A. I'd say the foundational material I considered was the
19 entirety of the written specifications for the three
20 patents-at-issue in this matter, and I also reviewed in detail
21 the prosecution history of each of these patents. And this is
22 the communication between the inventor and the -- the Patent
23 Office.

24 A number of many documents have been produced in this
25 matter by both the Plaintiff and the Defendant specifically

1 about the accused products, data sheets, and technical
2 documents of accused products, emails, many documents
3 produced, and I have reviewed those in preparation for this
4 testimony.

5 Q. And generally what opinions do you arrive at for the
6 issue of infringement in this case?

7 A. My opinion based upon my findings and my evaluation for
8 each of these three patents as applied to the accused product
9 is there is no infringement by any of the accused products for
10 any of the claims for these three patents.

11 Q. And what is your conclusion as to whether any of the
12 claims are invalid for any of these three patents that we're
13 going to talk about today?

14 A. In evaluating the patents, I have also rendered an
15 opinion that each of these patents is invalid.

16 Q. And when you sought to come to that opinion, what lens
17 did you apply your analysis through with regards to the level
18 of -- of experience that you considered?

19 A. Well, this is referred to as the level of the skilled
20 artisans, sometimes referred to as a person of ordinary skill
21 in the art. I am tasked to look at all of this material and
22 render my opinions through that person's lens.

23 That person is, as I see it for this case, is a person
24 that has a Bachelor degree in engineering such as electrical,
25 and three years of experience in memory, memory-related type

1 systems. And there's an allowance that higher academic, such
2 as a Master's, a Ph.D., perhaps less number of years of
3 experience such as two years.

4 Q. Now, what did you do to familiarize yourself with the
5 patents in this case?

6 A. Well, there's a process that I go through when I do an
7 evaluation of the patent. For each patent, I want to look at
8 the patent as a whole because the entirety of the patent tells
9 the story of what the inventor had in mind.

10 So I start with the face page of the patent because this
11 is where I learn right up front exactly where the inventor is
12 directing his invention to, and the title of the patent, Flash
13 DRAM Hybrid Memory Module, gives me that information.

14 Q. And what did you do next?

15 A. I then turned into the body of the -- the specification,
16 and it starts by providing a number of illustrations that the
17 inventor puts out sometimes called the embodiments of the
18 invention. And I familiarize myself with the illustrations,
19 and then I turn to the written specification portion to help
20 elucidate what the illustrations are about.

21 Q. Were there any particular portions of the specification
22 that you found particularly relevant in this case?

23 A. I found several portions, in fact, the entirety of the
24 patent is relevant, but the specific portions related to, for
25 instance, the field of the invention, which sets forth the

1 environment the invention is placed within.

2 I also look at the background. This is where there's
3 some statements as to what the state of the art was at the
4 time of the invention, and then I'll look at the detailed
5 descriptions that illustrate what the invention is about.

6 Q. And what did you do next in reviewing the patents, sir?

7 A. And, finally, I turned to the back of the patent where
8 the claims are found, and the claims are what I am to then
9 apply to the accused product from an infringement standpoint
10 or for comparing to prior art and other criteria for the
11 determination of whether there's a proffer of invalidity.

12 Q. Now, what is the title of the '918 and '054 Patents tell
13 you about what's disclosed in those patents?

14 A. Well, the title of both the '918 and the '054 Patent as
15 read here is flash-DRAM hybrid memory module. That tells me
16 what the patent is directed to.

17 Q. And what are you showing here on this slide, sir?

18 A. What I'm showing here is the history of the patent from
19 the time it was originally filed as a provisional application
20 in 2007 to the time of the issuance. These are called
21 continuations.

22 Q. So when the patents are described in particular problems
23 and technology in the description, what time frame is the
24 patent describing?

25 A. That is the time that the original was filed. It's

1 around 2007.

2 Q. Now, you were talking about a lot about flash and DRAM in
3 this case. Can you tell the jury your understanding of flash?
4 What is flash, sir?

5 A. Certainly. Flash is sometimes referred to as permanent
6 memory. The way in which the memory cells that store the
7 information are configured and designed are such that, once
8 the data is provided to it, it remains there even if you turn
9 the power off.

10 Q. And what is DRAM?

11 A. DRAM is sometimes referred to as temporary or volatile
12 memory because the way in which that -- those cells are
13 designed, when the memory receives information in the form of
14 binary 1s and 0s and stores it, if one loses the power to the
15 system, all of that information is totally lost and
16 irrecoverable.

17 Q. Now, why do systems -- computer systems have both
18 permanent storage and temporary memory?

19 A. Both are required because the -- one of the benefits of
20 the temporary memory, even though it cannot retain the data
21 for a long period of time, it's very quick, fast to access,
22 fast to write to.

23 An example would be if I'm typing on my phone into a
24 message, typing a message to my wife, it would not be very
25 helpful if I type a letter and it takes two minutes for the

1 letter to appear. So dynamic ram is fast on -- on being able
2 to refresh the video -- the screen, if you will. This is true
3 when you're typing on a keyboard, and it's true even on the
4 internet at the server level. So that's the importance of
5 that.

6 But there is a need to also have an ability to when I
7 finish my message, my letter, or take a photograph or a video,
8 to be able to store that permanently so I can retrieve it
9 later. So both are necessary in a system.

10 Q. Now, do the '918 and '054 Patents provide anywhere where
11 it talks about the problem to be solved?

12 A. Yes. So there is several places in which it does, but it
13 relates specifically to figure 1 in the patent as well as some
14 descriptive language in the specification, such as at column
15 2.

16 Q. Now, what is the patent referring to with regards to this
17 data transfer limitation?

18 A. Well, what it's referring to is the fact that the way
19 that the systems around 2007 and before were architected, that
20 the primary computer brain, called the central processing
21 unit, had to sit there and arbitrate the transactions for data
22 between the temporary memory, dynamic ram, and the permanent
23 memory, which is flash.

24 What's shown on the prior art slide on figure 1 on the
25 left is the section called SSD. That stands for solid state

1 drive. That's non-volatile memory such as flash. But since
2 you had the CPU sitting there like a traffic cop organizing
3 the flow of the data, that creates a bottleneck. It's
4 difficult to make the data transfers in the -- in the time
5 that's necessary.

6 Q. Now, what did the '918 and '054 Patents describe as the
7 solution to this bottleneck problem from 2007?

8 A. What it describes as the solution to that bottleneck is
9 three-fold: One, move the flash memory onto the same module
10 as the DRAM. As a result of moving them closer together, you
11 are able to have a more direct communication of data between
12 the two so that you can then, under a controller that controls
13 this data flow operation, you are able to do a back-up of that
14 information in the case of a pending power loss.

15 So before the power is totally lost, it provides a
16 mechanism by which that data can be transferred; thereby, you
17 do not lose the data from a DRAM.

18 Q. And what is this back-up process that the '918 and '054
19 Patents are describing?

20 A. Well, specifically what it is directed to for those
21 back-up process is that there would be some mechanism whereby
22 the structure would be able to detect that you were about to
23 lose power. And before you fully lose power, it will
24 basically divert the -- the DRAM module of the DRAM flash
25 hybrid module, to a state where there was an additional power

1 source that's available to it for a period of time that would
2 permit the on-board transfer of the data between the DRAM and
3 the flash. And since they're very close together on the same
4 module, you could effect that kind of a transfer.

5 Q. And what did the '918 and '054 Patents say about backing
6 up the DRAM to the flash based on the state of the memory
7 system here on this slide, sir?

8 A. Well, it does refer to the different sources based on a
9 state. State means, for instance, if -- if everything is
10 running perfectly correctly, all the power is being provided
11 to the system, all is working fine, the DRAM and the flash are
12 communicating properly, they're powered up, all is working
13 fine, that's called a normal operating state. The patent
14 refers to that as a first state.

15 There are other states in which there is a detection in
16 the patent that there's a pending power loss. You can start
17 to see fluctuations in the power. We experience that, for
18 instance, in a house if a power starts to fluctuate, we see
19 the lights fluctuate, well, we know we're about to head for a
20 problem.

21 So this patent contemplates a second state where it
22 alters the power source to provide just enough power to allow
23 the transfer to take place. That's called a second operating
24 state.

25 Q. Now, do the '918 and '054 Patents describe specific

1 circuits for enabling this back-up system that transfers data
2 between the flash and DRAM?

3 A. Yes, it does.

4 Q. And what are you showing here, sir?

5 A. What I'm showing here at JTX 3 is figure 12 and figure 16
6 from the '918 Patent. This specifically identifies structural
7 circuitry that would be able to provide these three particular
8 aspects of the patent.

9 Q. And where are you showing the -- the flash memory?

10 A. Well, first I would note that the label at the top on the
11 left side, 1010, represents that flash DRAM hybrid module. It
12 includes a PC board, which is the substrate upon which the --
13 these component circuitry is located.

14 And then you see on the module, a section of highlighted
15 in pink, this is the DDR. It's called a double data rate
16 dynamic random access memory. So this is where the temporary
17 memory is located.

18 Q. And what about the flash memory?

19 A. The flash memory is located on this drawing directly
20 adjacent to it. I've highlighted that in purple. It's called
21 NAND. That stands for not and. It's a logical construction
22 for the flash memory device, non-volatile memory device, both
23 located on the same module.

24 Q. And what are you labeling in green on these figures?

25 A. What I'm labeling in green on these figures are the

1 control mechanisms for setting up the voltage that is to be
2 supplied to the DRAM and the flash. You can see, for
3 instance, over on the right side of this figure 16, that the
4 outputs of the green boxes are basically the voltage lines.
5 They empower the DRAM and the flash both.

6 And what I've defined shown here, the voltage circuits
7 that do that. One is labeled a buck converter, a second is
8 labeled a dual buck type converter, and there's a third one
9 that's labeled as a buck-boost converter.

10 And over on the power module, there is also a buck-boost
11 converter. So these are the ways in which the patent
12 specifically identifies structural circuitry to perform this
13 back-up operation.

14 DR. ALBERT: Mr. Sayers, could you pull up JX 3 at
15 column 29, lines 23 to 27.

16 Q. (BY DR. ALBERT) What is the patent saying here
17 about -- to describe these buck and boost converters that you
18 just described in figure 16?

19 A. Well, specifically at this column and the lines that were
20 highlighted, it says, "The sub-blocks may comprise various
21 converter circuits such as buck-converters, boost converters,
22 and buck-boost converter circuits for providing various
23 voltage values to the memory system 1010." This is what I
24 pointed to with the green boxes in the figure 16 and figure
25 12.

1 DR. ALBERT: If we can go back to the slides.

2 Q. (BY DR. ALBERT) Now, for those yellow boxes that you've
3 labeled on figure 12 and figure 16, what do those do?

4 A. There is the second source of power. If you notice in
5 figure 12, it's called a capacitor bank. Figure 16 is
6 identified in block 1142 as the capacitor array. The
7 capacitor is a type of passive electronic device that's able
8 to store power for some period of time.

9 And so what happens is that during the normal operation,
10 the voltage which is available can be used to charge this --
11 this voltage. It's like having a back-up battery. And so
12 during normal operation, that particular capacitor and
13 capacitor array are charged. Therefore, if you start to lose
14 power, this second power source can take over control and
15 continue to provide enough power long enough for the data to
16 be transferred from the DRAM to the flash.

17 Q. Now, were you in here in court the other day when Doctor
18 Mangione-Smith said that the embodiment in figure 15A shown on
19 slide DDX 3-19 was a DRAM only embodiment?

20 A. I was here for that testimony, and I do remember him
21 saying that, yes.

22 Q. Do you agree with that?

23 A. Absolutely not.

24 Q. And when the jury was shown figure 15A, is that the whole
25 story?

1 A. No. You need to look at the specification and look at
2 the context within which this particular sentence is -- is
3 placed in order to understand what it means.

4 Q. Now, when you looked at the specification, the
5 description in the patent for figure 15A, what does that tell
6 you?

7 A. It tells me that it's a volatile memory subsystem 1030.
8 And if I look at figure 12, that's expressly the DRAM
9 component, subsystem component, that is on the hybrid flash
10 memory module. So 15A is a description of one way in which
11 you can instantiate that memory module.

12 Q. Now, were you here in court when Mr. -- excuse me, when
13 Doctor Mangione-Smith said that embodiment 15B was also a
14 DRAM-only embodiment?

15 A. Yes, I was.

16 Q. And do you agree with that testimony?

17 A. No, I disagree with that testimony.

18 Q. And when the jury was shown figure 15B, was the jury
19 shown the description in the patent describing figure 15B?

20 A. Not that I recall, no.

21 Q. Now, when you look at the description in the patent that
22 tells you what figure 15B is about, what does that tell you?

23 A. It tells me that it's an embodiment, it's a schematic
24 illustration specifically addressing the volatile memory
25 subsystem 1030, which is shown as a part of figure 12 on the

1 hybrid flash DRAM memory module. So it is not directed to a
2 DRAM-only module.

3 DR. ALBERT: And, Mr. Sayers, could we pull up PDX
4 4.23? Plaintiff's demonstrative.

5 Q. (BY DR. ALBERT) Were you here in court when Doctor
6 Mangione-Smith showed demonstrative PDX 4.23?

7 A. Yes, I was.

8 Q. Now, you see that on this Plaintiff's demonstrative, we
9 have some highlighting for RDIMM, SODIMM, and UDIMM. Do you
10 see that?

11 A. Yes, sir, I do.

12 Q. Is there any other part of this sentence that you found
13 relevant to determine whether this was a DRAM-only embodiment?

14 A. Yes, there are other parts of this. So if you look at
15 the context within which this is represented, what was not
16 highlighted was the term 'form factor'.

17 Q. And what is a form factor?

18 A. Form factor just refers to the size and shape. And so
19 what this is showing is that that hybrid flash memory module
20 can be sized and shaped, for instance, like the small outline
21 DIMM or the unbuffered DIMM or the registered DIMM.

22 It is not by any means pointing to the fact that this
23 invention can be instantiated just on a DRAM-only module. No.
24 This is the shape and size of the card.

25 Q. Now, when this demonstrative was shown to the jury, sir,

1 could you see what was behind this box?

2 A. No. It certainly is hiding what the rest of the text in
3 the columns are on that patent.

4 Q. Now, in particular, can you see what is hiding right
5 behind here behind this box?

6 A. I do not have Superman insight. I can't see through it,
7 no.

8 Q. Do you recall what is being referred to right behind this
9 point on this box?

10 A. Yes. Yes, I do.

11 Q. What is being referred to?

12 A. Figure 12.

13 DR. ALBERT: Mr. Sayers, can we bring JX 3 back up,
14 the '918 Patent, and go to column 21? And lines starting with
15 13 through 20.

16 Q. (BY DR. ALBERT) Could you see this reference to figure
17 12 behind the box that Doctor Mangione-Smith showed the jury?

18 A. No, I was not able to. I know it was in the
19 specification, but, no, I was not able to see it.

20 Q. Now, does figure 12 refer to a DRAM flash hybrid memory
21 module?

22 A. That is the only thing it refers to.

23 DR. ALBERT: Can we bring the slides back up?

24 Q. (BY DR. ALBERT) All right. What are you showing here on
25 DDX 3-22 on the left side with the large column of text?

1 A. This is an extract from the '918 Patent, specifically at
2 column 21, and I believe showing lines 10 through 55.

3 Q. The text on the left, is this part of the text that was
4 hidden behind that box that was shown to the jury?

5 A. Yes. The box overlaid this. The rest was grayed out.
6 But this is the text, and you can tell immediately that it is
7 directed to figure -- it helps if I draw it right -- figure
8 12.

9 Q. And with these added circles, sir, what are you showing
10 the jury?

11 A. What I'm showing is that the written part of the text in
12 the specification is the -- is describing the illustrations.
13 And so this entire text is really -- is descriptive language
14 describing what is the -- what are the subcomponents on figure
15 12.

16 And what I've highlighted are the different reference
17 numbers that are -- that you find listed throughout that
18 section in that column 21, and I've also circled where they're
19 located in figure 12.

20 So the entirety of the figure 12 description is really
21 directed to different portions of the subcomponents that are
22 found in figure 12, which is the hybrid flash DRAM memory
23 module.

24 Q. So when the '918 and '054 Patent was talking about the
25 form factor or the size and shape of the printed circuit

1 board, was that limiting figure 12 to only particular types of
2 memory?

3 A. It was not limiting it to a type, but it certainly
4 precluded -- the only thing it's directed to is the flash
5 non-volatile and not volatile memory, DRAM and flash hybrid
6 memory module. That's what figure 12 is directed to.

7 Q. Now, according to the '918 Patent, you could take the
8 contents -- could you take the contents of figure 12, the
9 flash and the DRAM, and put them on different size circuit
10 boards?

11 A. Oh, yes. That was comprehended in the text that's
12 highlighted in column 21 on the left side of this
13 demonstrative in yellow where it says the different form
14 factors. You can take the content, the subcomponents, that
15 are the circuits, if you will, and the structure on figure 12,
16 and you can instantiate that in a number of different form
17 factors, such as the SODIMM, FPDIMM, RDIMM. These are just
18 different shapes and sizes upon which you can place these
19 particular components.

20 Q. So this quote that was shown to the jury on Plaintiff's
21 demonstrative, does this limit the '918 and '054 Patents or,
22 excuse me, does this provide a DRAM only-embodiment?

23 A. No, it does not.

24 Q. What are you showing here, sir?

25 A. What I'm showing here is a outline of a DDR5 Samsung

1 product that's been accused of infringing the '918 Patent.

2 Q. And what are you labeling in red there?

3 A. What I'm labeling in red here is the different black
4 components that you see that are populated on this particular
5 module, and the only components that are there for storage of
6 information, data, is the DRAM.

7 Q. Are there any flash chips on the DDR5 modules?

8 A. No, there are no flash chips on the DDR5 module.

9 Q. And what are you showing here on DDX 3-30?

10 A. What I'm showing here is an additional component that's
11 on the module, DRAM-only module, that is called the PMIC, or
12 the power management integrated circuit, and that is a smaller
13 little black rectangular box more located toward the center of
14 the module.

15 Q. Now, you heard some information about information being
16 stored in the PMIC. Do you recall that -- that testimony
17 earlier?

18 A. Yes, I do.

19 Q. And is there any data transferred back and forth between
20 the DRAM and the PMIC?

21 A. Oh, no, there is -- there is no data communication
22 between the two. The PMIC is there to provide power and
23 control to the -- to the DRAM. But there's no data transfer
24 between them.

25 Q. Does the DRAM ever back up its data into the PMIC?

1 A. Oh, the small number of registers that are available
2 there would not even begin to accommodate what data is stored
3 on the DRAMs.

4 Q. Can the PMIC store things like video files?

5 A. Not at all; never.

6 Q. Can it store things like emails?

7 A. No. It can only store control information.

8 Q. And what is that?

9 A. That is the way in which, when the system turns on, for
10 instance, I think everyone recognizes when you turn a computer
11 system on, it goes through a bios or a power-up sequence. And
12 during the power-up sequence, the computer is out there
13 analyzing what's in my system. So it's polling these
14 different modules and to get some feedback that identifies
15 what it is and how it's configured so the computer knows how
16 to operate.

17 So this is control information to establish how the
18 system will operate, but it has nothing to do with data.

19 Q. Now, have you reached an opinion, sir, on whether the
20 asserted claims of the '918 and '054 Patents are practiced by
21 any of the DDR5 products?

22 A. Yes, I have.

23 Q. What is that opinion?

24 A. With regard to the '918 claims, I have determined that --
25 that these claims are not infringed by the Samsung DDR5

1 product, at least for the reason that there is a limitation in
2 each of the independent claims directed to the requirement of
3 a converter circuit structure, and that is not found on any of
4 these Samsung DDR5 products. And what's being pointed to is a
5 linear drop-out regulator, but that's not a converter circuit.

6 As relates to the '054 claims, the DDR5 product has a
7 normal operating state. The other state is it's turned off.
8 So it does not have a second operable state as required by the
9 '054 claims.

10 Q. And what are you showing here on slide 3-33?

11 A. I'm showing on this slide claim 1 from the '918 Patent,
12 and I specifically have drawn a highlighted box bringing out
13 and enlarging the particular claim limitation directed to a
14 converter circuit.

15 Q. And what are you showing on the next slide?

16 A. The next slide, I'm showing the second independent claim
17 which is claim 16 of the '918 Patent. This also has a claim
18 limitation that requires a converter circuit structure.

19 Q. Now, do any of the accused products, the DDR5 products,
20 have a converter circuit structure, sir?

21 A. No. None of the DDR5 products have that.

22 Q. And how did you determine that the converter circuit
23 structure that Doctor Mangione-Smith identified was not found
24 in the products?

25 A. I referred to the technical data sheets that are provided

1 by the manufacturers for the PMIC device, and looked at the
2 data sheets for what the call-outs were, and specifically
3 identified from Doctor Mangione-Smith that he was pointing to
4 circuits in the PMIC called LDO, low drop-out.

5 Q. And are those LDO, low drop-outs, converter circuits?

6 A. No, they are not. Instead, they are a linear voltage
7 regulator.

8 Q. Now, do the documents that you've seen, do they refer to
9 LDOs as converter circuits?

10 A. I've seen documents that further describe and elucidate
11 the LDOs on this, yes.

12 Q. But do they call them converter circuits?

13 A. Oh, no. They are always called linear regulators.

14 DR. ALBERT: Mr. Sayers, can we pull up JDX 11 at
15 69?

16 Q. (BY DR. ALBERT) What are we showing here at the top?

17 A. What is shown here at section 3.4 specifically calling
18 out figure 3-8 is the linear regulator called the LDO, the
19 linear drop-out.

20 Q. And were you here in court when Doctor Mangione-Smith
21 claimed that the LDO had a switch in it?

22 A. Yes, I was.

23 Q. Do you agree with that opinion?

24 A. No, I do not.

25 Q. Why not?

1 A. If you look at the -- it was very difficult to see on
2 that black schematic that Doctor Mangione-Smith was
3 representing, but basically this is the circuitry that you
4 find for the LDO. And what Doctor Mangione-Smith pointed to
5 was this particular transistor and he called that a switch.
6 It is not a switch. It's a transistor.

7 And the reason that I know that is because you can use
8 transistors in many, many different types of ways, but the way
9 it is used here is not as a switch. This transistor is
10 controlled by an amplifier that's a linear amplifier.

11 And so when you cross a threshold condition that says I'm
12 about to overvolt -- it's an overvoltage protection, I'm
13 getting too much voltage and I got to make sure is DRAMs are
14 not damaged, I need to get into gear and change that.

15 And so what happens is, when that threshold is crossed,
16 this transistor is driven and it continues to be driven so
17 that there is current charged, it flows from the power source
18 which is right here, through that transistor and through these
19 resistors, creating thermal heat.

20 And so the extra energy to account for that overvoltage
21 is basically thrown away in terms of thermal energy. It
22 creates a lot of heat. This is a linear regulator.

23 DR. ALBERT: Can you bring the slides back up, Mr.
24 Sayers?

25 Q. (BY DR. ALBERT) Now, how did you arrive at your opinion

1 that a linear drop-out regulator was not a converter circuit?

2 A. Well, my own experience tells me that it's not because I
3 know it's not. I've also looked at the documents here that
4 describe the linear drop-out regulator that's in the PMIC data
5 sheets themselves, and I've also looked at papers that were
6 available through the technical peer reviews.

7 Q. And shown here on DDX 3-37 is a -- what are you showing
8 here to the jury, sir?

9 A. Well, this is one of the documents that I was able to
10 locate, and this is -- it's called a technician's guide,
11 specifically troubleshooting, maintaining, and repairing
12 personal computers. And this is authored by Stephen Bigelow.
13 And specifically if you look at page 4, it describes exactly
14 what I just mentioned earlier: the linear regulator works by
15 throwing away the extra energy, and it does that by basically
16 expending that energy through a resistor -- a set of resistors
17 which creates a thermal radiation, if you will. So the energy
18 is discarded in the form of heat.

19 And this is -- as I showed you, it is understood in the
20 industry as a very simple and it's a reliable type of
21 structure, but it's very inefficient, and certainly it's
22 creating heat that you now have to account for.

23 Q. Now, how do these linear regulators compare to something
24 called a converter circuit?

25 A. The converter circuit is a fundamentally different

1 circuit structure from the linear regulator. Instead of
2 creating this current path that dissipates heat thermally,
3 instead what they do in the converter circuits is they have a
4 switch and the switch is continually switching on and off to
5 provide charge to an inductor. And it's that combination of
6 the switch and inductor that provides a converter-type
7 circuit.

8 Q. Now, does this reference that you provide on slide 38,
9 Chakraborty, does it provide examples of converter circuits?

10 A. I'm glad you pronounced that name. Yes. This particular
11 one at page 1 of this document that is entitled Digital
12 Combination of Buck and Boost Converters to Control a Positive
13 Buck-Boost Converter.

14 So it identifies both the step-up and the step-down type
15 of converter. Step-down means to move the voltage from one
16 level down to another. That's stepping it down. It's called
17 buck. And if you step it up, that's called boost.

18 Q. And just as a reminder, sir, what type of circuit is
19 Doctor Mangione-Smith pointing you to for this claimed
20 converter circuit? Is it the LDO or are they buck converters
21 or boost converters?

22 A. He is pointing to the LDO, which is a linear regulator.

23 Q. Now, did you see any testimony in this case, in this
24 trial, sir, about whether a voltage converter such as a linear
25 regulator is a converter circuit?

1 A. I saw several testimonies.

2 Q. And what are you showing here?

3 A. Specifically at DDX 3-39, what I'm showing here is the
4 trial transcript testimony extract from Mr. Hun-Joo Lee, who
5 is a Samsung engineer. And specifically on pages 86 to 87 of
6 that transcript, he was asked the question, "Which voltage
7 converter on the PMIC provides the supply voltage in DDR5
8 DIMMs?"

9 His answer was, "The power that you're referring to is
10 not provided by the voltage converter. It is provided by PMIC
11 by way of the LDO linear regulator."

12 Q. Did you see any other testimony, sir?

13 A. Yes. In the trial transcript from Mr. Johnny Kim, who is
14 another Samsung engineer, specifically at page 70, Mr. Kim was
15 asked the question, "Well, what's the difference between a
16 regulator and a converter?"

17 His answer was, "They are two different types. Generally
18 when we say switching converter, we're talking about something
19 that's comprised of an inductor and switch. And typically
20 when we say regulator, we are talking about something without
21 that. It's a simplified circuit."

22 Q. In the schematic for the LDO that was provided to the
23 jury, did you see an inductor in that schematic?

24 A. No. It dissipates the energy via heat. It's a thermal
25 dissipation. It's a linear regulator.

1 Q. So what is your opinion then as to whether a converter
2 circuit is the same as an LDO?

3 A. I have found that the linear regulator, LDO, and the
4 converter circuits are fundamentally different structures.

5 Q. How so?

6 A. The structure to -- to actually instantiate an LDO is one
7 in which you have a linear operation through resistors to
8 dissipate heat. The structure that is used for a converter
9 circuit is one in which you use a switching in combination
10 with an inductor to achieve that. So it's an entirely
11 different structure.

12 And in terms of the overall aspects of it, the efficiency
13 on the LDO linear regulator is very low because you're
14 throwing away energy and heat rather than controlling it, and
15 the converter circuits are a much higher efficiency.

16 So you can use these in different types of environments
17 such as low and high noise, but the thrust of this is they are
18 fundamentally different structures that achieve fundamentally
19 different results.

20 Q. Now, we talked about claim 1 and 16. Do you also have an
21 opinion as to the dependent claims, claim 5, 13, 18, and 19 in
22 this case?

23 A. Yes. Since the Samsung DDR5 accused products do not have
24 the claim limitation for converter circuit structure, that's
25 not found in claims 1 and 16, therefore, in my technical

1 opinion, claims 1 and 16 are not infringed by the DDR5 Samsung
2 products.

3 And because that structure is missing from the DDR5
4 products that's required by the independent claims, thereby
5 the dependent claims are also not infringed.

6 Q. Is it okay to go on and talk about the '054 claims, sir?

7 A. Certainly.

8 Q. Now, what are you showing here?

9 A. What I'm showing on this DDX 3-43 drawing is claim 16
10 from the '054 Patent, and the '054 Patent is directed to
11 operable states. And what I've highlighted here is the last
12 'wherein' clause to claim 16, which says, "In response to the
13 voltage monitor detecting an amplitude change in the input
14 voltage, the memory module transitions from a first operable
15 state to a second operable state."

16 Q. Did the Court provide any constructions or instructions
17 regarding the meaning of these terms?

18 A. Yes, the Court did.

19 Q. And did you apply those instructions?

20 A. Yes, I did.

21 Q. And what were the constructions that you were provided
22 and applied in this case?

23 A. The instructions from the Court was that the term
24 'operable state' means state in which the memory module is
25 operated.

1 Number two, that the first operable state means the state
2 in which the memory module is operated before transition. And
3 the second operable state means state in which the memory
4 module is operated after transition.

5 Q. So for the condition that Doctor Mangione-Smith
6 identifies for this claim second operable state, what is your
7 conclusion as to whether that state is a state which in the
8 memory module is operated after transition?

9 A. It is not a state in which the memory module is operated
10 after transition because in that state that was identified by
11 Doctor Mangione-Smith, the DRAM module is turned off. There
12 is no voltage being provided to the memory system at all.

13 Q. Now, can you walk the jury through what's happening on
14 the DRAM system, the DRAM module, in the first state?

15 A. Sure. In the first state, I've shown here on this --
16 this particular slide, on the left side is a diagram of the
17 PMIC, the power management integrated circuit, and the outputs
18 of the buck converter, which is shown here as SWA plus SWB.
19 There's an SWC buck converter and an SWD buck converter. All
20 of these have outputs that provide voltage to power the DRAM
21 that's on the module.

22 And so under this first operating state, there is voltage
23 being applied and supplied to the DRAM so the DRAM can perform
24 its normal function of data rate reading and writing transfer.

25 Q. And what happens to those voltage outputs from the buck

1 converters when there is an overvoltage or undervoltage
2 condition?

3 A. When there's an overvoltage or an undervoltage condition,
4 then the module PMIC turns those particular converter powers
5 off to where there is zero, there is no voltage being provided
6 to the DRAM.

7 Q. And did you see any testimony on that issue in trial?

8 A. Yes, I did.

9 Q. And what was that?

10 A. Bruce Lo in the trial transcript at page 74, lines 3
11 through 10, the question was asked, "So when there's an input
12 overvoltage or undervoltage condition, the buck converters are
13 all turned off as a default. Correct?"

14 His answer, "Yes."

15 Question, "And when the buck converters are -- are turned
16 off as a default, there will be no supply voltages for the
17 DRAMs. Correct?"

18 His answer, "Yes."

19 Q. And did you see that same condition shown in the
20 documents that you reviewed as well?

21 A. Oh, yes. In the data sheet for the PMIC products, such
22 as the S2FPC01 data sheet, it specifically says that under
23 that condition of overvoltage protection to protect the DRAM,
24 they shut the voltage off so there's no voltage being supplied
25 to the DRAM.

1 Q. And on this point, sir, does Doctor Mangione-Smith agree
2 with you?

3 A. Yes. In fact, at trial transcript page 378, lines 25
4 through page 379 line 10, specifically Doctor Mangione-Smith
5 says, "In the second operable state, after an overvoltage
6 condition, for example, the DRAMs are turned off."

7 Q. Now, do you agree with Doctor Mangione-Smith, though,
8 that that is a second operable state?

9 A. No. In fact, it is a definitely turned-off state. The
10 DRAM module is not able to do what it's intended to do. The
11 computer system cannot communicate with it for data flow one
12 way or the other.

13 Q. So when those buck converters are disabled in the
14 overvoltage and undervoltage condition, what is actually
15 happening to the DRAM?

16 A. The DRAM is basically dead. It's shut down. It cannot
17 access the DRAMs at all for memory flow in either direction.
18 And when -- when the DRAMs are turned off, all of the
19 information that was stored and being used at that time is
20 thrown away. It's discarded. It's never to be retrieved.

21 Q. You can't write to the DRAM. Is that -- excuse me?

22 A. No, you cannot write to the DRAM. You cannot read from
23 the DRAM. You cannot refresh the DRAM. You cannot do
24 anything with the DRAM.

25 Q. So when the buck converters are turned off, can a user

1 use the DRAM to watch a movie?

2 A. No, sir, it cannot.

3 Q. Can they access their email?

4 A. No.

5 Q. Is that memory module operating when a user can't access
6 their email?

7 A. It's not operating.

8 Q. So then what is your conclusion as to whether there is a
9 second operable state in the DDR5 products?

10 A. My conclusion is that there is not. There is a first
11 state in which the normal operation occurs, but there is not a
12 second operable state.

13 Q. Now, were you here when Doctor Mangione-Smith talked
14 about some temperature sensors from the PMIC in this second
15 state?

16 A. Yes.

17 Q. And he said, well, those temperature sensors means that
18 the DRAM module is operating. Is that your understanding?

19 A. That's what my understanding is of his testimony, yes.

20 Q. Does it make any sense to you?

21 A. No, it does not.

22 Q. Why not?

23 A. I'll have to admit that when I heard the testimony, I
24 immediately thought six weeks ago when I was driving up the
25 expressway and I got in a rolling tire falling off of a truck,

1 hit my car. I immediately got over to the side of the road,
2 and I was able to turn my radio and calm myself down from my
3 high temperature thoughts at that time, but the car was not
4 movable. It was not operating. So the intended purpose of
5 that car could not operate.

6 The same thing is true here. When you turn the buck
7 converters off, they're no longer applying voltage to this
8 DRAM module. The DRAM module can no longer perform its
9 intended purpose. Regardless of whether something else might
10 have some bits that are still working, the DRAM module is not
11 working.

12 Q. And what is your conclusion regarding the dependent claim
13 in this case?

14 A. The fact that the second operable state is not found in
15 the accused DDR5 products, which is a requirement of the
16 independent claim, then by definition there's no infringement
17 of the dependent claim because it has to include all of what's
18 in the independent claim as well.

19 Q. So looking through all the documents, all the testimony,
20 ultimately what is your conclusion as to whether any of the
21 '918 and '054 claims are practiced by the DDR5 products?

22 A. My conclusion for all of the claims of the '918 and '054
23 Patent claims is that the DDR5 products do not infringe.

24 With regard to the '918, the DDR5 linear drop-out
25 regulators are not a converter circuit structure. They are a

1 linear regulator structure. And so, therefore, that
2 limitation is missing; therefore, the DDR5 products do not
3 infringe.

4 As it relates to the '054 claims, as I have described to
5 you, the jury, that there is no second operable state. So,
6 therefore, the DDR5 products do not practice the '054 claims.

7 Q. Now, do you also have an opinion as to whether the '918
8 and '054 Patent claims are invalid for failing to satisfy
9 written description?

10 A. Yes, I do.

11 Q. What is that opinion?

12 A. I've looked at the entirety of the specification in the
13 claims, and I've come to the conclusion that the written
14 description for the '918 and '054 Patents, which I'm talking
15 about the text and the illustrations, do not support the
16 asserted claims.

17 I've identified three reasons. One is the claims fail to
18 require the flash DRAM hybrid structure as a memory module.
19 Secondly, the claims fail to require data transfer between the
20 flash and the DRAM memory module -- memory devices on that
21 module. And, third, the claims fail to require backing up the
22 data between the flash and the DRAM, which is the focus of the
23 intent of the specification.

24 Q. Going back to the '918 and '054 title of the patents,
25 what does the title tell you about what was described in the

1 '918 and '054 patents?

2 A. Both patents are directed to a hybrid memory module that
3 includes both flash and DRAM on that module.

4 Q. Do any of the claims that are asserted in this claim, do
5 they require flash?

6 A. None of them do.

7 Q. What does the technical field that was described in the
8 '918 and '054 Patents tell you about what these patents were
9 describing?

10 A. Well, for this demonstrative, I'm showing from the '918
11 Patent, column 1, lines 66 through column 2, line 2. It's
12 exactly replicated on the '054 Patent at the same place. The
13 technical field is specifically directed to combinations of
14 flash and random access memories.

15 Q. Earlier we talked about a problem to be solved. Looking
16 at what's on the left of the screen here, do you see anywhere
17 where the patent describes a problem to be solved in that 2007
18 time frame?

19 A. Yes. It was the problem of data transactions. There's
20 an information transfer bottleneck that I talked about before.
21 There's too much data that's required to be transferred for
22 the way in which the computer architectural systems is able to
23 accommodate.

24 Q. And on the left it's talking about server technology
25 applied to websites and email and video on demand. What is it

1 referring to there from that 2007 technology?

2 A. Well, I guess the best example is even shown on the left
3 side of the screen where it says video on demand. It's the
4 bottleneck can be interpreted to -- can be seen as the -- the
5 ever ending circle on the screen, and so this says that
6 there's a bottleneck. You're trying to get the data down, but
7 there's just not a way to do it well. At the server level,
8 this is an incredible complication, so that's what's being
9 directed here. You can't process the information fast enough
10 for the speed in which it's required.

11 Q. And you showed this earlier to the jury, sir, but how
12 does that data transfer between the DRAM and SSD in figure 1
13 relate to that bottleneck that you just described?

14 A. Well, it says, when you have the computer system
15 architecture as of 2007 where the central processing unit is
16 acting as that traffic administrator or cop between the flash
17 and the DRAM, it says the temporary memory and the solid state
18 memory, which is the permanent memory, are located in separate
19 locations on the -- on the computer system, it's very
20 difficult to get that transfer data to take place in the speed
21 in which it's required.

22 Q. Moving forward to today, do we still have those same
23 problems, or have they been addressed by new architectures and
24 faster CPUs?

25 A. Well, the architectures have totally changed today. They

1 are using different bridge techniques, they've isolated the
2 buses. In fact, in the old art, 2007, this bus communication
3 from the computer system to the solid state drive was a serial
4 bus, one bit at a time; whereas, the data path to the DRAM was
5 8, 16, or 32 bits at a time. And so you had a whole different
6 disparity between the data flow just because the amount of
7 bits that you could put across them were different.

8 That has been resolved in the art today by some redundant
9 systems as one way to do it. Secondly, they have developed
10 new caching techniques where they've got different ways of
11 flowing the data, and they changed the entire architecture in
12 terms of the communication. So that's been solved in many
13 different other ways.

14 Q. Now, what was the patent's offered solution to this 2007
15 bottleneck problem?

16 A. The patent-offered solution was take the slower
17 communication path to the flash SSD, move that onto the same
18 module as where your DRAM is located, at least to account for
19 some part of this data flow mismatch.

20 Q. And what is being referred to there in terms of the data
21 transfer between the DRAM and the flash on-module in the
22 patented description?

23 A. Well, providing a more direct connection without having
24 the arbitration from an external control unit, CPU. Then
25 under the condition, for instance, when you have a detected

1 power problem, then you can make the data transfer; whereas,
2 in the old system, there just -- because of the distance
3 between the two and located in different parts of the computer
4 system, there was no way to back-up that data. The speed of
5 the transactions were much too slow.

6 Q. Now, do you see anywhere in the claims asserted in this
7 case that requires a data transfer between DRAM and flash on
8 the module?

9 A. No, none at all.

10 Q. And what are you describing -- what is the patent
11 describing here with regards to this controller?

12 A. What the patent is describing here is, for instance, in
13 the figure 12 and 16 that I showed earlier where you have an
14 ability to create a separate power source like the capacitor
15 bank, so the event -- in the event when you have a power
16 fluctuation, you can draw from this additional power source to
17 provide enough power to the DRAM module or the DRAM flash
18 hybrid module that you could actually invoke and achieve a
19 data transfer between the two.

20 Q. And is that for data back-up between the DRAM and the
21 flash during that power failure?

22 A. That's specifically what that is for, yes.

23 Q. And do you see the requirements of the back-up transfer
24 between the DRAM and the flash anywhere in any of the claims?

25 A. No, none at all.

1 Q. Did you also analyze the other figures and description in
2 the '918 Patent?

3 A. Yes.

4 Q. And what are you showing here with respect to figures 4A
5 through 7, sir?

6 A. What I'm showing here is additional figures in the
7 patent, and you can see in each one of these figures that
8 there is a combination of flash and DRAM on a common memory
9 module. You see I've highlighted DRAM in pink and flash in
10 purple.

11 Q. And what are you showing here with regards to figures 8
12 through 11, sir?

13 A. Same thing. It's always addressing DRAM and flash
14 together.

15 Q. And what is the patent saying, again, about the backing
16 up of the volatile memory using non-volatile memory, the
17 flash?

18 A. Well, what is shown here at column 20 of the '918 Patent,
19 lines 17 through 28, specifically refers to the controller
20 backs up the volatile memory, which is DRAM, using the
21 non-volatile memory, flash, in the event of a trigger
22 condition.

23 Trigger conditions can include, for example, a power
24 failure, power reduction request by the host system, and so
25 forth. So when you have a power glitch, then this invention

1 pulls into place the ability to do the back-up between the
2 volatile memory and the non-volatile memory.

3 Q. And what do the patents show with regard to figure 12 and
4 16 with regards to that problem -- that solution of combining
5 the flash and the DRAM, having that data transfer and causing
6 that back-up?

7 A. Well, it shows in figures 12 and 16 circuit structure
8 that is able to combat that problem that, by having on the
9 common module the flash and the DRAM, you can then control the
10 voltage to the DRAM and the flash and specifically accommodate
11 with a back-up capacitive bank, a second power source, that
12 you can bring into play to continue to power the DRAM and
13 flash for long enough to get the data transferred between the
14 two.

15 Q. Now, earlier you testified about figures 15A, 15B, and
16 the quote that Doctor Mangione-Smith put up on the screen. Do
17 you recall that?

18 A. Yes, I do.

19 Q. And those three items, figure A, 15B, and the quote, what
20 do they relate to as with respect to your slide here today?

21 A. They relate to the memory module that includes DRAM and
22 flash, and the specific quotes on 15A and 15B related to 1030,
23 the DRAM portion.

24 Q. So 15A and B would be a part of this portion of figure
25 12?

1 A. That's correct. That would be a way in which that
2 particular portion could be implemented.

3 Q. And the quote on the form factor, how does that relate to
4 this item 1020 on figure 12?

5 A. Well, reference 1020 is the entire box, which represents
6 the module. It's the PC board. It's sometimes referred to as
7 a printed circuit board. But basically that's the substrate
8 upon which all the components are fixed.

9 So the form factor just says that I can have that form
10 factor for a small outline or registered DIMM or something
11 like that. All that goes to is I can change how big or how
12 small that box is depending upon different form factors, but
13 it doesn't take away from the fact it's a hybrid memory
14 module. It just has different shape and size.

15 Q. Now, we heard that the '918 and '054 Patents are
16 continuation patents?

17 A. Yes, that's correct.

18 Q. Do you have continuation patents yourself?

19 A. Oh, yes.

20 Q. Is that an okay thing to do?

21 A. Yes.

22 Q. But are there limits?

23 A. Yes. Still have to stay true to the invention.

24 Q. And what do you mean by that?

25 A. What I mean is, is that when you originally file in 2007

1 in this case the provisional application and the
2 continuations, you have to maintain the specification on a
3 continuation, and what you change then is the claims that are
4 directed to that. But it still has to fall within the scope
5 of what is -- what the patent is about.

6 Q. And if the claims that no longer match up with the
7 specification, would those claims be valid or invalid?

8 A. Invalid.

9 Q. And what are you showing here with regards to an earlier
10 claim from the chain of continuation?

11 A. Well, in the previous slide where you see a number of
12 different patents that were part of the continuation, I am
13 looking at one of these, particularly. It was a patent that
14 was issued in 2014, and it's a part of this continuation
15 chain. And you can see from the claim itself that it still
16 maintains consistency with the written description, which is a
17 non-volatile memory subsystem has volatile memory, has
18 non-volatile memory. It's still true to the description.

19 Q. And how about the current claims? Do they match up with
20 the specification?

21 A. No. The only thing they point to is the dynamic random
22 access memory.

23 Q. And where are you showing the dynamic random access
24 memory in this claim 16 of the '054 Patent?

25 A. So specifically I'm showing that in the particular

1 limitation that's directed to a plurality of components
2 coupled to the PC board and specifically identifies the
3 plurality of components concluding the plurality of
4 synchronous dynamic random access memory, SDRAM.

5 Q. Looking at each claim, sir, looking at claim 1, 5, and 13
6 of the '918 Patent, what is your conclusion regarding whether
7 those claims are supported by the specification, the '918
8 specification?

9 A. Well, based upon what I've already described, my
10 conclusion based upon my evaluation is that there's no written
11 description support for claims 1, 5, and 13.

12 Q. And what is your conclusion regarding claims 16, 18, and
13 19 in this case of the '918 Patent?

14 A. The same. I do not find for these '918 Patent claims
15 that the written description supports it.

16 Q. And what is your conclusion with regards to the '054
17 claims in this case, 16 and 17?

18 A. Same conclusion, I do not find any written description
19 for the claims as written.

20 Q. So ultimately what is your conclusion then as to why the
21 '918 and '054 Patent claims have no support in the patents and
22 therefore are invalid?

23 A. Well, as I said in my testimony earlier, I said there
24 were three particularly primary fundamental aspects that the
25 written description is directed to. I did my evaluation, and

1 my conclusions confirm that initial thought.

2 And the statements are: the claims are invalid because
3 they fail to require the flash-DRAM hybrid structure, the
4 claims are invalid because they fail to require data transfer
5 between flash and DRAM memory devices, and the claims are
6 invalid because they do not require backing up data between
7 flash and DRAM.

8 Q. Now, when Doctor Mangione-Smith was doing his
9 infringement analysis the other day and he went claim by claim
10 by claim by claim and he looked at what evidence would satisfy
11 those claims, did he point to anything in the products
12 requiring that data transfer between the flash and DRAM or
13 that back-up --

14 MR. SHEASBY: Your Honor, objection. It's
15 irrelevant for the purposes of invalidity what the claims
16 require for the purposes of infringement.

17 THE COURT: Speak up, Mr. Sheasby.

18 MR. SHEASBY: Your Honor, objection. It's
19 irrelevant for the purposes of -- the question of what the
20 claims require for infringement has nothing to do with the
21 written description analysis that Mr. McAlexander is doing
22 right now. Speaking to the jury about what Doctor
23 Mangione-Smith found to satisfy the claim elements, none of
24 which require these elements, is utterly irrelevant to written
25 description.

1 THE COURT: What's your response, Doctor Albert?

2 DR. ALBERT: May I approach, Your Honor?

3 THE COURT: Let me just hear from you.

4 DR. ALBERT: Sure. The scope of the claim as
5 applied, Your Honor, exactly aligns with the case law from the
6 Federal Circuit on this. The -- the scope of the infringement
7 allegations is directly relevant to the written description
8 requirement and how that is found by the jury.

9 MR. SHEASBY: Your Honor, the claims that were
10 issued by the PTO require none of these three things. And to
11 suggest that there was any defect in Doctor Mangione-Smith's
12 analysis because he did not address these three things is an
13 error of the law and it should not be the subject of
14 testimonial.

15 DR. ALBERT: Absolutely not the intent, Your Honor.
16 It's -- the scope of the allegations on infringement about
17 what is necessary to satisfy these claims is directly relevant
18 to whether that scope is found in the specification.

19 THE COURT: Go ahead.

20 MR. SHEASBY: If he wants to say these three
21 elements are not found in the claims, these three elements
22 were not required by the PTO to be put into the claims, that
23 is fine.

24 The concern is when he speaks about Doctor
25 Mangione-Smith's infringement analysis. That is mixing two

1 entirely distinct issues, and that's what my objection is.
2 The examination can proceed without the improper injection of
3 infringement.

4 DR. ALBERT: And, Your Honor, this exact point was
5 already made to Your Honor earlier, and Your Honor has already
6 resolved it in the pretrial stage. This exact point,
7 objection was made and resolved to allow this exact type of
8 testimony.

9 THE COURT: All right. I'm going to overrule the
10 objection at this point and allow the testimony to go forward.
11 You can certainly probe these issues with the witness on
12 cross-examination --

13 MR. SHEASBY: Thank you, Your Honor.

14 THE COURT: -- Mr. Sheasby.

15 Let's proceed.

16 Q. (BY DR. ALBERT) So, Mr. McAlexander, when you saw Doctor
17 Mangione-Smith's analysis of the claims for infringement where
18 he went claim limitation by claim limitation by claim
19 limitation all through the claims, did he show anywhere where
20 there was a requirement for the flash to transfer data to
21 DRAM?

22 A. No, he did not.

23 Q. Did he show where there was anywhere where there was a
24 requirement for flash to back-up data from DRAM?

25 A. No.

1 Q. So, again, what is your ultimate conclusion as to whether
2 the '918 and '054 claims are -- are invalid as failing to
3 satisfy the written description requirement?

4 A. My findings are that the claims do not have written
5 description support in the patents for the claims that are at
6 issue.

7 Q. Did you also look at the supposed technical benefits of
8 the '918 and '054 Patents?

9 A. I did.

10 Q. Now, what are you saying here in that first bullet point
11 about the alleged technical benefits in the '918 and '054
12 Patents?

13 A. I'm just indicating that, with regard to on-board power
14 management, that others invented that before Netlist.

15 Q. And are there other features incorporated in the DDR5
16 modules that address the issue of speed and power reduction?

17 A. Oh, yes, many others.

18 Q. Can you tell the jury about that?

19 A. When you move to a new technology like DDR5, first there
20 has been significant improvements in the DRAM and the flash.
21 These -- the modules are fabricated in multi, multi tens of
22 billion-dollar facilities, and they are continually improving
23 the equipment and the process to get smaller and smaller
24 feature sizes so you can increase the density, get more
25 storage locations per unit area.

1 So when you're moving from DDR3 to 4 to 5 technology from
2 the early 2000s, DDR2 up through 2020-2023 time frame, there's
3 been significant density improvements. You're getting far
4 more units per unit of memory storage. That is a major, major
5 benefit that helps to reduce speed and power because the
6 feature sizes are smaller, the voltage levels are smaller, the
7 amount of power per bit is smaller. So that's one.

8 The second is the architecture that's for the module card
9 itself, the DIMM card that has the DRAM on it. They have
10 taken -- one of the cards is like 288 pins that have
11 information. You can translate how that information flows
12 across the card, and you can do it -- in some cases do it in
13 DDR5. They move to different channels. They've created
14 separate channels as compared to a single channel of the DDR4.
15 That also had a speed power improvement.

16 So the configuration of the module card itself, the pins
17 going to it, the timing, the lower noise environment, lots of
18 different features contribute to improving speed and power.

19 Q. And are there also lots of different features other than
20 putting power management on the module that contributes to
21 speed and power?

22 A. Yes. Invoking cache systems as well as array type
23 systems where you can do immediate back-up. In fact, it's
24 parallel back-up so that you always have power to one or more
25 of the modules. So even if you lose one rack, the other rack

1 is still operative. So, yeah, there are many other techniques
2 that you can use to solve this problem.

3 Q. Can we move on and talk about the '339 Patent, sir?

4 A. Certainly.

5 Q. Now, what does the '339 Patent generally describe?

6 A. Referred to the face page of the patent, it's
7 specifically directed to memory module with controlled
8 byte-wise buffers. So it's a specific type of buffer.

9 Q. And what are you showing here?

10 A. What I'm showing is here that this also has some
11 continuations before the '339 Patent issued. It dates back to
12 the earlier filing of 2009.

13 Q. And so what time frame would the technology be that is
14 referred to in that -- in the '339 Patent?

15 A. 2009.

16 Q. And what are you showing the jury here with respect to
17 figure 3C of the '339 Patent?

18 A. Figure 3C is directed to a specific type of data buffer,
19 which I'll describe in more detail. You can see here that the
20 memory module has DRAMs, which I've highlighted in pink, and
21 it also has data buffers, which I've highlighted in blue. And
22 you can see that there is, I'll call it, distributed, the data
23 buffers are distributed to align with the different memory
24 location, different DRAMs that are located on that module.

25 Q. Now, was the concept of using distributed buffers

1 available as of the time of the '339 Patent?

2 A. Oh, yes. This was well-known in the art by 2009.

3 Q. And what are you showing on this screen?

4 A. I'm showing a patent which is 8,130,560 to Rajan. It was
5 filed in November 2007. That was is two years prior to the
6 2009 time frame of the '339. And specifically I'm showing
7 figure 6 from that document as well as some recitations from
8 column 2, lines 17 to 19.

9 And you can see that it's quite clear on the 600 module,
10 600 being the reference number for the entire module, that it
11 includes the DRAM circuits. I've shown them in pink. It also
12 has the buffer circuits, and you can see clearly that they are
13 also distributed. And so that is well-known.

14 Q. And what are you showing on the next slide, sir?

15 A. This is another patent. This one is to Ellsberry.
16 Specifically, it's a patent publication. This is prior to it
17 issuing, but the state of the technology is now that when a
18 patent is filed, that in a certain period of time it needs to
19 be published, made public. And that's what this is. It's a
20 published document.

21 It's referred to as a Publication No. 2006/027735581.
22 And on this particular document, the publish date of 2006, you
23 can again see DRAM in pink and the distributed buffers in
24 blue.

25 Q. And on the next slide, sir, you have this quad band

1 memory. What was quad band memory?

2 A. Quad band memory was a product in design from a company
3 called Kentron. And this particular slide that I'm showing
4 here comes from a 2002 technology overview for QBM. And at
5 page 1, you can see a picture of the module, and you can see
6 shown here the DRAMs in pink and the data buffers that are
7 distributed in blue.

8 So same thing. This distributed data buffers is
9 well-known.

10 Q. Now, what was the '339 Patent describing about the data
11 buffer that was in the '339 Patent?

12 A. The data buffer that is in the '339 Patent is where the
13 difference occurs or where it supposedly is going to be
14 different, and that is, it requires what is called a dual
15 path. And I'll use the word 'dual', but really it's multiple.
16 You can have two or more paths. But within the data buffer
17 from the input side to the output side for the data path,
18 somewhere within that data buffer you had the ability to make
19 a decision which way to direct that data.

20 And so you could have one set of bits, it's called eight
21 bits, means a byte in the electronic technology. So
22 one -- each data buffer had a byte come in and that byte of
23 eight bits could be directed to one output or another output.
24 So it basically creates two different data paths within the
25 data buffer.

1 And the specifics of the patent is directed to when you
2 enable one of the data paths, the one that you're selecting,
3 you are then to disable or disconnect the unselected path.

4 Q. Now, what are you showing here with regards to figure 3A
5 and the related description of the '339 Patent?

6 A. Well, what I'm going to show on this slide is an
7 animation I created specifically to show the flow of data
8 through the data buffer. And what's shown on the far left
9 side is the controller 420 that is going to be providing the
10 data input to the module. And you can see I've highlighted in
11 yellow one of those lines.

12 So if you look at 450, any one of those 450 lines would
13 be a byte, eight bits, coming in on that data path.

14 Q. What are you showing next, sir?

15 A. I've also highlighted the blue box 416. That's the data
16 buffer. And what I'm showing here is that when the first data
17 path through the data buffer is selected, it sends the data
18 out on a byte-wise path that then shows where there's a green
19 line that basically sends information to the DRAMs that are
20 located -- or the memory devices located 412A and 412C.

21 Q. And now what are you showing here?

22 A. What I'm showing now, if the selection within the data
23 buffer is the second data path, then the data would be routed
24 through the data buffer along this second path to a second
25 output, and that would go to a different set of memory devices

1 specifically labeled as 412B and 412D.

2 Q. Now, down at the bottom, we see a reference to 402. What
3 is that referring to?

4 A. The data buffer and memory devices are all located on one
5 module, and that would be labeled 402. So what you're seeing
6 is stacked below this or behind the picture, you have four
7 such modules.

8 So what is showing here with the blue and the pink and
9 the data lines, all of that would be replicated on those other
10 three modules behind it.

11 Q. Now, what can you tell the jury about whether each buffer
12 on each module is disclosed to have that multiple data paths
13 that you described?

14 A. Well, the data -- the data path is the multiple data
15 paths within the data buffer. So that data path selection,
16 I'll call it a fork in the road, you have -- you're coming in
17 one input, you can select which output you go to, that dual
18 path or multiple path is located within each DRAM on each
19 module.

20 So what you circled here in red for the blue data
21 buffers, all three of those data -- that row of data buffers
22 would be replicated on each of the other modules. So each of
23 the other modules would have the same thing. Every data
24 buffer has to show a multiple path.

25 Q. Now, what are you showing with regards to figure 5 of the

1 '339 Patent?

2 A. You may remember that the number that's assigned to the
3 data buffer is 416 in that previous figure. So what is shown
4 here is a part of that data buffer 416. And what is shown on
5 figure 5 is one bit. It's the path for one of those eight
6 bits. So whatever I'm showing here and describing would have
7 to be replicated eight times to make that byte-wise data path.

8 So if I can go back, please? Thank you.

9 What you're seeing here for that data buffer, there's an
10 input and an output side. So the data path between those two
11 sides, a first side and a second side, is shown. For
12 instance, if I select path A, then that is going to be a path
13 of data coming through the yellow data lines to the buffer A,
14 and then through the buffer A and then out. That's the first
15 data path.

16 The second data path -- the second data path would be if
17 I selected coming from the output -- from the input through
18 the yellow and then following the pink path or the red path.

19 Those would be the two paths between the input side and
20 the output side, the first side and the second side of the
21 data buffer.

22 Q. Now, what types of products are at issue for the '339
23 Patent?

24 A. They are the Samsung DDR4 products.

25 Q. And, in particular, what types of DDR4 products are at

1 issue?

2 A. Well, the LD. It's called the LRDIMM, load-reduced DIMM.

3 Q. And are these the products -- the DDR4 products that have
4 these data buffers?

5 A. Yes, they do.

6 Q. Now, when you looked at the asserted independent claim in
7 this case, what was your conclusion as to whether that claim
8 is satisfied by the DDR4 LRDIMM products?

9 A. My opinion is, after my technical investigation, is that
10 this claim is not satisfied by the DDR4 products. The claim
11 limitations are not found in the DDR4 products.

12 Specifically, I'm focusing on the last two 'wherein'
13 clauses of the claim, and there's a requirement to drive, and
14 this is within the data buffer, it's a byte-wise buffer, and
15 it's required to drive a respective byte-wise section of the
16 in-bit wide write data from the first side to the second side.
17 So it has to drive a byte, eight bits, from the first side of
18 the buffer to the second side of the buffer.

19 Q. Now, is there a construction that's applicable to this
20 term?

21 A. Yes, there is.

22 Q. What is that Court's construction?

23 A. The Court has construed the term 'drive' to mean enabling
24 only one of the data paths while the other possible paths are
25 disabled.

1 Q. And what is the first part of that construction, sir?

2 A. Enabling only one of the data paths.

3 Q. And what is the second part of that construction?

4 A. While the other possible paths are disabled.

5 Q. Do any of the DDR4 LRDIMM products enable only one of the
6 data paths while the other possible data paths are disabled?

7 A. No, because within the DDR4 data buffers, there is only
8 one path.

9 Q. And what are you showing here with regards to the data
10 buffer of the DDR4 LRDIMM product?

11 A. What I'm showing is a Renesas data sheet of the DDR4 data
12 buffer. And from that, I'm looking at page 1, which is a
13 diagram that shows the structural construct of the data
14 buffer.

15 Q. And what are you showing here, sir?

16 A. What I'm showing here specifically is, again, applying
17 the Court's construction, the drive means enabling only one of
18 the data paths while the other possible paths are disabled, I
19 evaluated the data sheet and documents associated with this
20 data buffer.

21 And you can see, for instance, circled at the bottom, it
22 says DQ (3:0). That's a standard way of denoting that there
23 are four lines--0, 1, 2, and 3--for the data path, DQ. DQ
24 represents data. And there's also right underneath that, it
25 says DQ (7:4). That means those are the data bits 4, 5, 6,

1 and 7. So 0 to 3 and 4 through 7 is eight bits. So it's one
2 byte.

3 So I can see immediately for this data buffer there is a
4 single byte coming in. And then when I look at the top, which
5 is the path from the path going through that data buffer, what
6 comes out is a single byte out. So it's one byte in, one byte
7 out, one path.

8 Q. Now, does Doctor Mangione-Smith agree with you that there
9 is a single input to a single output?

10 A. Yes.

11 Q. And what is shown here on this screen?

12 A. What is shown here is a block -- the same block diagram
13 for the -- for the particular data sheet for the data and
14 buffer, and this is Doctor Mangione-Smith's understanding and
15 drawing of the fact that there is a first side to second side
16 and a single path of the bits through that data buffer.

17 DR. ALBERT: Mr. Sayers, could we pull up PDX 4.157?

18 Q. (BY DR. ALBERT) Were you here in court when Doctor
19 Mangione-Smith testified with regards to this figure for the
20 drive term?

21 A. Yes, I was.

22 Q. Now, is this -- what he said, does it make any sense to
23 you when you read the claims that are asserted in this case?

24 A. It really did not. I understand what he was drawing and
25 what he was referring to here, but the drive aspect for a

1 respective byte-wise requires from the Court's construction
2 enabling the first path, disabling the second. And this is
3 all within the data buffer. The two paths, the two or
4 multiple paths is within a data buffer.

5 So it has -- whatever you look for, there has to be a
6 multiple path that you enable within each of these blue data
7 buffers. It has nothing to do with what happens on another
8 module.

9 Q. Now, is there anything about the claims at issue in this
10 case that discusses what happens in other memory modules?

11 A. No, not at all.

12 DR. ALBERT: Can you come back to the slides?

13 Q. (BY DR. ALBERT) Now, what do you -- can you walk the
14 jury through where you saw that language in the claim that the
15 data buffer from the first side to the second side
16 was -- that -- excuse me, that that data path from the first
17 side to the second side was within the data buffer?

18 A. Certainly. If you look at the claim 1 of the '339 in
19 particular, it's directed to byte-wise buffers, and the
20 byte-wise buffer has a -- it's a plurality of byte-wise
21 buffers, and they have a first side and a second side. And it
22 states that the byte-wise data path between the first side and
23 the second side, well, where is this? It's the byte-wise data
24 path within the data buffer between the first side and the
25 second side.

1 And then continuing, to control the byte-wise data path,
2 you drive a respective byte-wise section of the N-bit wide
3 write data associated with the memory operation from the first
4 side to the second side. Again, this is the first side of the
5 buffer to the second side of the buffer.

6 Q. So going to the Court's claim construction, what is
7 required for the data paths? Where is that located?

8 A. Located within the data buffer.

9 Q. And looking at the data paths in the data buffer, does
10 Doctor Mangione-Smith agree with you that there is only a
11 single byte-wise data path in the byte-wise buffer?

12 A. Yes, he does.

13 Q. So ultimately what is your conclusion regarding whether
14 the DDR4 LRDIMM products infringe any of the claims of the
15 '339 Patent?

16 A. Well, it's clear to me when I look at the data sheet that
17 supports what is actually structurally found on the data
18 buffers is that the DDR4 data buffers have a single data path.
19 There are not multiple paths within that data buffer.

20 So it does not, in quotes, fill-the-fork-in-the-road kind
21 of analogy when you have a single byte coming in and you can
22 have a choice of two or more paths going through that data
23 buffer and you select one and deselect the other or enable one
24 and disable the other. It is not found in any of the DDR4
25 products.

1 Q. Now, moving on to your opinions on invalidity, do you
2 have an opinion as to written description for the '339 Patent?

3 A. Yes. The written description, the specification for the
4 '339 Patent does not support a practice by a single data path.
5 Now, you understand that -- that doctor Mangione-Smith is
6 asking the jury to find that buffers with a single data path
7 infringe these claims?

8 MR. SHEASBY: Your Honor, I object to the recitation
9 of what Mr. Mangione-Smith did or did not say, which I think
10 was inaccurate. It's not --

11 THE COURT: Again, Mr. Sheasby, you're going to have
12 to speak up if you're going to speak from counsel table.

13 MR. SHEASBY: Sure, Your Honor.

14 THE COURT: You're too soft.

15 MR. SHEASBY: I object to the question. The
16 characterization of what Mr. Mangione-Smith said or didn't say
17 is not accurate. He can simply ask the question as a
18 hypothetical.

19 THE COURT: Well, the jury heard Doctor
20 Mangione-Smith's testimony. The jury will know whether
21 counsel is mischaracterizing it or not. And if he is, the
22 jury will hold him accountable. And if he's not, the jury
23 won't.

24 MR. SHEASBY: I understand, Your Honor.

25 THE COURT: Again, this is something you can revisit

1 on cross-examination, but I'm not going to tie Defense
2 counsel's hands as they do their direct examination of this
3 witness.

4 MR. SHEASBY: Thank you, Your Honor. Thank you very
5 much.

6 THE COURT: All right. Objection's overruled.
7 Let's proceed.

8 Q. (BY DR. ALBERT) Well, let's go back to Doctor
9 Mangione-Smith's testimony, sir. What does he say regarding
10 whether there's a single byte-wise data path in the data
11 buffers?

12 A. Well, the question was asked, is -- basically is only a
13 single byte-wise data path in the byte-wise buffer. Correct?

14 And he says, yes, that's what it says.

15 Q. And the question was asked, So you believe that in the
16 accused DDR4 LRDIMMs, there is only a single byte-wise data
17 path in the byte-wise buffer.

18 Is that testimony that you relied upon --

19 A. Yes.

20 Q. -- for your testimony here today?

21 A. Excuse me. Yes.

22 DR. ALBERT: Going back to the slides here.

23 Q. (BY DR. ALBERT) Now, what is your opinion as to whether
24 an embodiment disclosing a single data path in a byte-wise
25 buffer was disclosed in the '339 Patent? Was that disclosed?

1 A. No. I did not find any disclosure for a single data path
2 as being what the patent was directed to.

3 Q. And -- but what did you see in the '339 Patent as to what
4 the patent was directed to? What are you showing here?

5 A. What I'm showing here is this two-patent multi-path
6 architecture. So if you look, for instance, at column 15 and
7 16 in the patent, you will see that, talking about the
8 byte-wise data buffer, you're saying, driven onto two data
9 paths labeled path A and path B. It says, The ranks of the
10 memory devices are likewise divided into two groups with one
11 group associated with path A and one group associated with
12 path B.

13 So there's a first one of the two data paths, and there's
14 a second one of the two data paths. So, again, it's talking
15 about multidata paths. And they're aligned in such a way that
16 the data path A would be communicating to a particular set of
17 memory devices, and the data path B through that data and
18 buffer would be communicating then the data to a second set of
19 memory devices.

20 Q. You already showed these figures to the jury. But,
21 again, do these figures, figures 3A and figure 5, show that
22 path A and path B structure from the '339 Patent?

23 A. Well, they do. But I want to make sure it's very clear,
24 what is highlighted in green and red is showing the output
25 from one of those two -- one or the other of the paths going

1 to the memory devices.

2 As I drew earlier, path A is the path from the input
3 side, one side, through the data and buffer to the output side
4 which then goes to the specific memory modules, and then the
5 second path as shown is the one that goes from the input side,
6 one side, again up through the data buffer and then out
7 through a different port that represents the second path.

8 Q. So ultimately what is your conclusion as to invalidity
9 for written description under the assertion that data buffers
10 somehow infringe where they only have a single data path?

11 A. Based upon that belief that there can be an assertion
12 against a single path structure as complying with the claims,
13 then I find that the claims are invalid. There's no support
14 in the patent for a single path.

15 Q. Thank you, sir. But -- and, again, for the record, what
16 is your conclusion ultimately, though, as to whether the
17 products infringe these claims? Do they actually practice the
18 patents?

19 A. No, they do not because the claims require multiple paths
20 and the ability to choose between those paths, enabling one
21 and disabling the other. And since the products that are
22 accused, the DDR4 Samsung products, only have a single path,
23 they cannot infringe.

24 Q. Can we move on to your opinions on obviousness over the
25 prior art?

1 A. Certainly.

2 Q. What was the QBM alliance?

3 A. The QBM alliance was a group of individuals put together
4 from Kentron specifically for the purposes of what he calls
5 positioning the quad-band memory technology as an industry
6 standard. And this was an international alliance. And you
7 can see that a number of different parties were members of
8 that alliance, including Netlist.

9 Q. Now, those parties that were a member of the QBM alliance
10 for that technology that Kentron was developing, what types of
11 materials would they receive from Kentron?

12 A. Well, it shows in this particular document that the
13 alliance members will receive priority resources from Kentron
14 that includes actual QBM modules, simulation models, design
15 guides, documentation, engineering support for technology
16 valuation and product implementation.

17 So they were -- basically it's an open forum where they
18 actually were receiving the products and the documents.

19 Q. And was Netlist one of those members that would have
20 received the resources from Kentron?

21 A. Yes.

22 Q. All right. Looking at the QBM module, can you describe
23 to the jury what was on it and when it was available?

24 A. Yes. What is shown on the left side is one of the QBM
25 modules, and on the right side of this is shown an extract

1 from page 8 of the QBM technology overview. In fact, the
2 photograph that's shown on the left side is from page 1.

3 And so when we look at these in combination, you can see
4 that on this module were, number one, a plurality of memory
5 devices, DRAMs. They're shown here in pink and also in
6 yellow. There are multiple ones of these DRAMs.

7 You can also see that the blue boxes are the -- what's
8 called a QBM switch. This is actually where data is switched
9 in and out, and it's a buffer circuit. And you can also see
10 highlighted in green, it's called advanced buffer. This input
11 circuit there for advanced buffer was for buffering address
12 and command signals. These are the control signals that
13 direct the traffic. And PLL stands for phase lock loop.
14 That's a clocking system that was built on board.

15 Q. And this QBM technology overview, when was that released,
16 sir?

17 A. I believe it was 2004.

18 Q. So when were these quad band memory modules available?

19 A. As of that time.

20 Q. And what are you showing the jury here, sir?

21 A. What I'm showing here is out of that technology overview
22 at page 15. This is a diagram that shows the data flow
23 through one of those switch buffers, specifically in the right
24 direction.

25 And what's circled in black at the bottom is called DQDM.

1 This is basically the input data. You can see it says 9 bit
2 wide. When you see a slash with a 9, that indicates that
3 those are actually 9 physical lines, each one having a bit.
4 That's an eight bit wide or a byte plus 1 which is used for
5 error connection. So it's a byte-wise buffer that has an
6 additional bit for error correction.

7 So you have a byte coming in at the bottom left, bottom
8 center section, and you can see that there are two paths
9 through right latch A and/or right latch B through the data
10 buffer. So it's clearly two paths, and the description in the
11 overview shows that you can select one or the other. So you
12 enable one and you disable the other. That's what the
13 switches are used for--turn on and turn off.

14 Q. And does that Ellsberry reference that you described
15 earlier from 2006, does that also disclose the DRAM with the
16 distributed buffers?

17 A. Yes. You can see the DRAMs in the pink labeled at DDR,
18 which stands for double data rate RAM, double data rate, DRAM.
19 And you can also see the blue highlighted D boxes. Those are
20 the data buffers.

21 Q. And do you also see the controller as well?

22 A. The controller and the PLL, which is the phase lock loop
23 for the clocking.

24 Q. And does Ellsberry also disclose that fork-in-the-road
25 embodiment?

1 A. Yes. I've highlighted this. You have the DQ path coming
2 in it, says byte. So you have a byte in, and you have two
3 paths, either to export through port A or through port B. So
4 you have two paths through the data switch.

5 Q. So what is your conclusion then as to whether the '339
6 claims are obvious over the QBM system and Ellsberry?

7 A. I find that the claims are obvious over the QBM system
8 and Ellsberry and therefore are invalid.

9 Q. So this memory module limitation, the printed circuit
10 board limitation, the DRAM, the module controller, the
11 byte-wise buffers, the driving and the tri-state buffers,
12 those are all obvious over QBM system and Ellsberry?

13 A. Yes. And throughout the demonstration, I've pointed to
14 the module board, which is a PC board; the controller; the
15 DRAM; the byte-wise buffers that have the two paths. And it's
16 clear from these documents that the buffers that drive the
17 outputs would be tri-state.

18 Q. And what is your opinion as to whether the claims are
19 obvious, 8 and 9 are obvious over QBM system and Ellsberry?

20 A. They are -- the claims 8 and 9 are also obvious over
21 these two documents. I've shown, for instance, the output
22 that goes to different modules, different DRAMs on the module
23 itself, so rank is done. That's done by chip select signals
24 to select which rank gets the -- the output.

25 And also with regard to claim 8, the data lines are

1 configured to carry data from the controller, which is
2 outboard, to the corresponding memory device.

3 Q. If we can move on and talk about your opinion on
4 technical benefits for the '339 Patent?

5 A. Sure.

6 Q. You mentioned here in your slide about other improvements
7 incorporated into DDR4. What do you mean? What other
8 improvements besides what you would see in that '339 Patent
9 contributed to the benefits and improvements of the DDR4
10 products?

11 A. Well, in addition to what is required by the claims,
12 which would be the distributed data buffers and the dual path,
13 that has been accomplished with other way -- other means using
14 channel technology.

15 And I've already described where you can change and alter
16 the way in which the pin counts are used to bring the data on
17 board. They've also established a flow where there's a
18 parallel action between the delivery of the data from the
19 controller to the memory devices.

20 And the other aspects with regard to the speed of the
21 transfer, that makes a difference because if you can change
22 the speed of the transfer operation, you get more data
23 transfer per unit time. That also has a great impact and a
24 benefit.

25 Q. Now, did you also see the benefit of load reduction in

1 the prior art references and systems that we've looked at
2 earlier?

3 A. Yes, in each one of them. And I'm showing here on this
4 diagram the QBM module, the PC board, the DRAMs, and the --
5 and the buffers. And here you can see that by having this
6 divided path through the buffer to go to different-sized
7 loads, rather than the memory controller seeing all 16 loads,
8 it's only seeing one load and that's the data buffer.

9 Q. Now, did you see the same idea of load reduction using
10 the distributed buffers in Ellsberry that came before the '339
11 Patent?

12 A. Yes. In fact, it specifically cites at paragraph 27 and
13 at figure 6 showing that the memory module 106 presents a
14 single load, not the load of the individual memory devices.
15 So this is also a reduction in load.

16 Q. Okay. Can we go on and now talk about the Rambus
17 license?

18 A. Certainly.

19 Q. Tell the jury about the scope of the Rambus patent
20 portfolio as shown on the screen.

21 A. I was asked to look specifically at the Rambus patent
22 portfolio. And in my evaluation, I found that there were well
23 over 2,000 patents that were part of that portfolio. And what
24 I did is I looked at the applicability of what productization
25 those patents would be applicable to.

1 And I found that somewhere around 17-, 1800 of them were
2 actually usable in different types of memory type systems or
3 systems including memory. And then there were a number of
4 other patents that -- that actually were applicable to other
5 disciplines, other products, that would not necessarily
6 include or even be directed to memory.

7 Q. Now, with regards to the other types of patents in the
8 portfolio, what are some examples that you found?

9 A. Well, in the second slide, I'm showing that some of the
10 patents were directed to base band. That's telecommunication.
11 Some were directed to error detection. There's even several
12 patents that were directed to light-emitting diodes.

13 Some were directed to packaging, the integrated circuit
14 packaging, and cryptography and phase lock loops, in other
15 words, clocking. Some were even involving electrical
16 properties such as electrical fault.

17 So the patent portfolio was rather robust and -- and
18 covered a lot of different topics. There was a lot of
19 different potential technologies.

20 Q. Now, how did the scope of the Rambus patent portfolio,
21 those licenses -- the Rambus license, compare to the number --
22 scope of the patents in this case?

23 A. Well, there's a great disparity between 2,000-plus
24 patents across a number of different product sectors as
25 compared to the patents-at-issue in this matter from Netlist.

1 So when I looked at the patent portfolio as a whole, my
2 conclusion was it's not comparable to the patents in this
3 case.

4 DR. ALBERT: Pass the witness, Your Honor.

5 THE COURT: All right. Before we proceed with the
6 Plaintiff's cross-examination, ladies and gentlemen, we're
7 going to take a short recess.

8 You can simply close your notebooks and leave them in
9 your chairs if you will. Please follow all my instructions,
10 and we'll have you back here shortly to continue with the
11 Plaintiff's cross-examination of Mr. McAlexander.

12 The jury's excused for recess.

13 (Whereupon, the jury left the courtroom.)

14 THE COURT: The Court stands in recess.

15 (Brief recess.)

16 THE COURT: Be seated, please.

17 Are you prepared to go forward with cross-examination,
18 Mr. Sheasby?

19 MR. SHEASBY: With the Court's permission, Your
20 Honor.

21 THE COURT: Are there any binders or anything that
22 need to be distributed?

23 MR. SHEASBY: They have already been distributed.

24 THE COURT: Then let's bring in the jury.

25 (Whereupon, the jury entered the courtroom.)

1 THE COURT: Please be seated, ladies and gentlemen.

2 We'll proceed with the Plaintiff's cross-examination of
3 Mr. McAlexander.

4 Mr. Sheasby, you may proceed with cross-examination.

5 CROSS EXAMINATION

6 BY MR. SHEASBY:

7 Q. Good morning, Mr. McAlexander.

8 A. Good morning, sir.

9 Q. It is historically nice to see you again, this time not
10 over Zoom.

11 A. Likewise.

12 Q. Mr. McAlexander, you charge \$595 per hour to Samsung for
13 your work. Is that correct?

14 A. That's what my company charges, yes.

15 Q. And how many hours have you billed in this matter?

16 A. I have personally billed 350 hours in this matter.

17 Q. And how many have your associates billed in this matter?

18 A. Around 60. I think it's around 400 total.

19 Q. Now, you've also -- you worked for Samsung in about 10
20 to 12 other cases. Is that correct?

21 A. I have had other cases with Samsung. I believe it's
22 somewhere around 10 or 12, yes.

23 Q. And Samsung is one of your largest return customers for
24 your expert consulting work. Fair?

25 A. Well, it may be fair to say return. I mean, they -- I do

1 service them from time to time. There are some that I do not.

2 Q. You've serviced them 10 to 12 times. Correct?

3 A. 10 to 12 times, yes.

4 Q. Now, the first issue that you discussed was the '918 and
5 '054 Patents.

6 MR. SHEASBY: And if we could bring up slide 5.

7 Q. (BY MR. SHEASBY) These are the two patents that you
8 discussed first with the ladies and gentlemen of the jury.
9 Correct?

10 A. Yes, that is correct.

11 Q. And in opening, Samsung's counsel said, ladies -- there's
12 no flash, ladies and gentlemen. Do you remember him saying
13 that?

14 A. Yes, uh-huh.

15 Q. And, in fact, in the claims as granted by the United
16 States Patent Office, only one of them requires the presence
17 of flash. Correct?

18 A. One addresses non-volatile, yes.

19 Q. And non-volatile memory, flash is non-volatile memory?

20 A. Flash is a type of non-volatile, yes.

21 Q. And so to be very clear in case there was any ambiguity,
22 claim 1, claim 5, claim 13, claim 16, claim 18 of the '918
23 Patent, claim 16, claim 17 of the '054 Patent, the United
24 States Patent and Trademark Office has granted those claims
25 and there is no requirement for the presence of non-volatile

1 memory or flash in them. Correct?

2 A. As you stated, that's what -- that's correct.

3 Q. Now, there are two non-infringement arguments you make,
4 and they are related to different patents in the family. So,
5 for example, you claim that what's known as an LDO is not a
6 converter. Correct?

7 A. It's not a converter circuit structure, yes.

8 Q. And that applies to the '918. Correct?

9 A. Yes, that's correct.

10 Q. But the '054 Patent, that argument, even if it's
11 accurate, Samsung doesn't avoid infringement. Correct?

12 A. Well, that argument with regard to the converter circuit
13 does not apply to the '054. The '054 is for a different
14 reason.

15 Q. So to be very clear, the LDO is not a converter circuit
16 is only applicable to the '918 Patent. Fair?

17 A. I believe that's what I said, yes.

18 Q. And what is a circuit?

19 A. Say that again?

20 Q. What is a circuit?

21 A. It is a structure. It's an implementation of -- of
22 specific components for the purposes of performing a
23 particular function.

24 Q. Is it a set of wires connected together?

25 A. No.

1 Q. It's a structure -- say it to me again. What is a --
2 what is a circuit?

3 A. It's a structure that comprises electronic components for
4 the purposes of performing together a particular function.

5 Q. So, for example, the -- in this case the converters are
6 on a power management integrated circuit. Is that correct?

7 A. Just to be sure, which converters are you talking about?

8 Q. Sure. The LDOs and the buck regulators are both on a
9 power management integrated circuit. Right?

10 A. Well, the LDO regulators and the buck converters are on a
11 PMIC device, yes.

12 Q. And that's power management integrated circuit. Correct?

13 A. That is correct.

14 Q. And it's describing the fact that the features on that
15 chip are integrated circuits. Correct?

16 A. There -- well, you're --

17 DR. ALBERT: Objection, Your Honor.

18 THE COURT: Just a minute.

19 What is it, Counsel?

20 DR. ALBERT: Outside the scope. Now we're
21 getting --

22 THE COURT: Outside the scope of the direct or
23 outside the scope of his report?

24 DR. ALBERT: Outside the scope of the direct and his
25 report, Your Honor. He's talking about what's on chips and

1 how things are arranged. This is not what Mr. McAlexander was
2 testifying about.

3 THE COURT: All right. I think this is fundamental
4 background. I'm going to overrule your objection.

5 Q. (BY MR. SHEASBY) Power management integrated circuits
6 have circuits on them. Correct?

7 A. Yes. That part I agree with, yes.

8 Q. Okay. And in this case you described them as LDO
9 converter -- you described them as buck converters and LDO
10 regulators as portions of that. Correct?

11 A. That is a part of it. That's what was described in the
12 data sheets, yes.

13 Q. And you said -- you made a distinction. You said, oh,
14 well, it's a buck converter, but it's an LDO regulator.
15 Correct?

16 A. I used the words that were in the description of the
17 spec. So, yes, correct.

18 Q. And by description of the spec, you mean the description
19 of the data sheets. Is that correct?

20 A. Yes, that's correct.

21 MR. SHEASBY: So why don't we go to PX 60, page 1.
22 JX 60, page 1. JX 30, page 1. Excuse me, Mr. Huynh. And
23 let's blow that up.

24 Q. (BY MR. SHEASBY) So this is the data sheet for the
25 Renesas power management integrated circuit. Correct?

1 A. Yes, it appears to be so.

2 Q. And just to orient the ladies and gentlemen of the jury,
3 Samsung purchases these integrated circuits and places them on
4 the DDR5 memory modules. Correct?

5 A. Correct.

6 Q. And these power management integrated circuits are one of
7 the components that allows for what we talked about as
8 on-module power management to occur. Fair?

9 A. That is its function, yes.

10 Q. And in this data sheet, it refers to three step-down
11 switching regulators. Those are what you refer to as buck.
12 Correct?

13 A. That is correct.

14 Q. And it calls them regulators. Correct?

15 A. That is correct.

16 Q. And it refers to the two LDO regulators. Correct?

17 A. That's correct.

18 Q. And it refers to them as regulators. Correct?

19 A. That is correct.

20 Q. It doesn't make a distinction calling one of them a
21 converter and one of them a regulator. Correct?

22 A. Right here it shows them both as regulators. That's
23 correct, they are.

24 Q. And just to put as precise of a point on it, if the LDO
25 is a converter circuit, then your non-infringement argument

1 fails. And if LDO is not a converter circuit, then your
2 infringement argument prevails. Correct?

3 A. I think that's a correct representation. The LDO is not
4 a converter, but your representation is correct.

5 Q. If the LDO is a converter, Samsung is incorrect and
6 Netlist prevails on infringement, at least as to this patent
7 claim.

8 A. With one stipulation--not a converter circuit.

9 Q. That's correct. And we agree that this is a power
10 management integrated circuit. Correct?

11 A. This is a data sheet for it.

12 Q. And on that power management integrated circuits, there
13 are two LDO regulators. Correct?

14 A. That is correct.

15 THE COURT: Mr. Sheasby, would you slow down a
16 little bit?

17 MR. SHEASBY: Sure.

18 Q. (BY MR. SHEASBY) And three switching regulators.
19 Correct?

20 A. That is also correct.

21 Q. And you concede -- well, let's break it down. The LDO
22 regulators are used in a particular situation. Correct?

23 A. Yes, they are.

24 Q. And, in particular, when the power undervoltage or low
25 voltage situation occurs, the LDO will continue to supply

1 power so the PMIC can operate and access its registers.

2 Correct?

3 A. That is also correct, yes.

4 Q. And, in fact, that LDO is a converter. Correct?

5 A. No.

6 Q. Okay. Can you turn to page 85, lines 4 through 10 of
7 your deposition. To orient you, your deposition is behind
8 you, Mr. McAlexander. It's in one of your white binders.

9 A. Okay.

10 Q. It says, Joseph McAlexander, Cross Examination Deposition
11 Transcripts. Do you see that?

12 A. Yes, I have that.

13 Q. And if you turn to the first tab, that's your deposition.
14 Correct?

15 A. Yes, sir.

16 Q. And if you turn to page 85. And just to orient you, it's
17 mini 85. There's four pages per page. So you want to go and
18 look for the 85 in the bottom right-hand corner and tell me
19 when you're there.

20 A. I'm there.

21 Q. So the LDO -- so when there is power -- and why don't you
22 turn to lines 4 through 10.

23 A. Okay.

24 Q. Did you give that testimony under oath?

25 A. The testimony at deposition? Yes.

1 MR. SHEASBY: Your Honor, may I publish?

2 THE COURT: Yes.

3 MR. SHEASBY: No. That's not the right page. If
4 you go to page 84 of Mr. McAlexander. 85. Excuse me. Lines
5 5 through 10. Here it is. No. This is -- I'm going to put
6 it on the screen because I think you have the wrong testimony.

7 Q. (BY MR. SHEASBY) Question: "So when there is -- when
8 there is a power undervoltage or a low voltage situation, the
9 LDO converters will continue to supply power so that the PMIC
10 can operate and access its registers. Maybe that's the
11 easiest way of saying it?"

12 Answer: "Yes. It's probably a little bit more than
13 that, but at least it keeps the registers activated."

14 Did I read your testimony correctly, Mr. McAlexander?

15 A. Yes, and that correctly represents my statement.

16 Q. And when I talked about the operation of the LDO
17 converters on the PMICs in the DDR5 products that occurred in
18 this case -- that are used in this case, you answered my
19 question with a yes. Correct?

20 A. Yes. You're the one to use the word 'converter'. I
21 answered, yes, they probably do that, yes. That's correct --
22 correct testimony.

23 Q. You didn't tell --

24 THE COURT: You need to take that off the screen now
25 that you've used it, Counsel.

1 MR. SHEASBY: Thank you, Your Honor.

2 And if we go to JTX 11.69.

3 Q. (BY MR. SHEASBY) This is something that you showed.

4 This is the LDO that you showed the ladies and gentlemen of
5 the jury. Correct?

6 A. Yes. This is the linear regulator, LDO.

7 Q. And this has components on it. Is that correct?

8 A. It does have components on it, yes.

9 Q. What type of components does it have on it?

10 A. It has a linear amplifier or buffer, it has a transistor.
11 Specifically, a -- it has a transistor on the top, and it also
12 has two resistors, R1 and R2, R1 being a variable resistor.
13 There are additional resistors, capacitors, and then ZL is the
14 load that it drives.

15 Q. This is structure. Correct?

16 A. It's a diagram of a structure, yes. It's a schematic.

17 Q. And this is the structure that will exist in the power
18 management functionality on every single DDR5 that's accused
19 of infringement in this case. Correct?

20 A. As far as I know, the answer is yes.

21 Q. And for the ladies and gentlemen of the jury, if they
22 accept that that's a converter circuit, Samsung is infringing
23 this patent. Fair?

24 A. I'll leave that to the jury to decide. My
25 representation, it's a linear voltage regulator and it's not a

1 converter circuit.

2 Q. If the jury concludes that this is a converter, Samsung
3 infringes. Correct?

4 A. If they -- if the jury concludes it is a converter, then
5 they can render an opinion whether it infringes based on
6 everything else in the claim or not.

7 Q. And the claim requires a converter. Correct?

8 A. No. It requires a converter circuit.

9 Q. And your testimony is that this is not a converter
10 circuit. Correct?

11 A. Correct.

12 Q. And one of the references or the witnesses you showed the
13 ladies and gentlemen of the jury was the testimony of a Bruce
14 Lo from Renesas. Do you remember that?

15 A. Yes, I do.

16 Q. And Renesas is the designer of the -- one of the power
17 managements that is on module in the DDR5s. Correct?

18 A. Yes. Renesas is.

19 Q. Yes. And Mr. Lo is a pretty authoritative source on the
20 design of the power management integrated circuits. Correct?

21 A. Sure.

22 Q. He was the engineer who designed them. Correct?

23 A. At least one of them, yes.

24 Q. And I don't mean this as a criticism. You've actually
25 never designed a power management integrated circuit?

1 A. No. I have designed power management circuits on memory
2 devices, but not a power management integrated circuit
3 separate.

4 Q. And so Mr. Lo --

5 MR. SHEASBY: If we can have PDX slide 5 through 14.

6 Q. (BY MR. SHEASBY) He testified under oath and the ladies
7 and gentlemen of the jury heard yesterday that he described
8 and testified that the LDO is a converter. Correct?

9 A. At that particular place in his testimony, yes.

10 Q. And the claim requires a converter circuit that is
11 configured to reduce the pre-regulated input voltage to
12 provide a fourth regulated voltage. Do you see that?

13 A. Yes, I do.

14 Q. And we agree that that LDO does actually reduce the input
15 voltage. Correct?

16 A. As many types of devices do, yes, it does.

17 Q. Sir, the LDO that Mr. Lo called a converter has the
18 capability and performs the function of reducing the input
19 voltage to provide a fourth regulated voltage. Correct?

20 A. Certainly, yes.

21 Q. And we also spoke about the fact that Samsung designed
22 some of its own power management integrated circuits.
23 Correct?

24 A. Correct, it did.

25 Q. And you would agree that within their specific

1 discipline, the Samsung engineers who design the actual chips
2 would know more about them than you. Correct?

3 A. You asked that in my deposition, and I responded they
4 would know certain aspects more than I would, yes.

5 Q. Well, they would know more about the particular aspect
6 that they were focused on. Correct?

7 A. Yes, they would know more about the particular ones that
8 they were focused on, yes.

9 Q. And you showed the ladies and gentlemen of the jury some
10 testimony from Samsung engineers, and you used this to
11 represent to the ladies and gentlemen of the jury that Samsung
12 engineers agreed with you that LDOs are not converter
13 circuits. Correct?

14 A. I believe it was the opposite. I agreed with them that
15 they were not converter circuits.

16 Q. You agreed with Samsung's engineers that LDO are not
17 converter circuits. Correct?

18 A. That's correct.

19 Q. And so the logical conclusion of that is that you're
20 representing to the ladies and gentlemen of the jury that
21 Samsung's engineers testified that LDOs were not converters.
22 Correct?

23 A. At this particular place they did, yes.

24 Q. Okay. Now, did you read the entire transcript of these
25 witnesses?

1 A. Yes, I did.

2 MR. SHEASBY: Let's go to slide 97. The last slide.

3 Q. (BY MR. SHEASBY) So just to level-set the jury, if we go
4 back, this is Hun-Joo Lee saying LDO is a linear regulator.
5 Correct?

6 A. That is correct. That's what he says.

7 Q. But he goes on to say in the portion that was not
8 displayed to the jury that what that linear regulator does is
9 convert the input voltage. Correct?

10 A. Yes, and I agree with it. That's one function it does.

11 Q. So we agree that the LDO regulator converts an input
12 voltage to provide a fourth regulated voltage. Correct?

13 A. There's no question about functioning to convert. It's
14 the question of the structure, and the structure, it's a
15 linear regulator, not a converter.

16 MR. SHEASBY: I move to strike as non-responsive,
17 Your Honor.

18 THE COURT: Sustained. You can ask the question
19 again or you can move on, Counsel.

20 Q. (BY MR. SHEASBY) Sir, Hun-Joo Lee, the principal
21 engineer and corporate representative, testified that the LDO
22 linear regulator converts the input voltage. Correct?

23 A. Yes, specifically says converting the input voltage.

24 Q. And, in fact, he was not the only engineer who testified
25 at Samsung that the LDO converts input voltage from a high

1 voltage to a low voltage. Correct?

2 A. That is also correct.

3 Q. So, for example --

4 MR. SHEASBY: If we go to slide 5.13.

5 Q. (BY MR. SHEASBY) -- this is Kyungsoo Park. He testified
6 that he was the person most knowledgeable about the power
7 management integrated circuits at Samsung. Correct?

8 A. He was the corporate representative, yes.

9 Q. And he testified that the LDO -- that's the linear
10 regulator we're speaking about. Correct?

11 A. Yes.

12 Q. It converts -- it reduces the 5 input voltage to 1 volt.
13 Correct?

14 A. Says reduces the 5-volt input voltage to 1 volt.

15 Q. And that, you agree, satisfies the limitation, reduce the
16 pre-regulated input voltage to provide a fourth regulated
17 voltage. Correct?

18 A. Please restate your question.

19 Q. Sure. Mr. Park is testifying that this LDO is configured
20 to reduce the pre-regulated input voltage to provide a fourth
21 regulated voltage. Correct?

22 A. Well, he specifically responds yes to the question to
23 reduce the 5-volt input voltage to 1 volt.

24 Q. I understand what he said. For the assistance of the
25 ladies and gentlemen of the jury, that would satisfy the

1 requirement of reducing the pre-regulated input voltage to
2 provide a fourth regulated voltage. Correct?

3 A. If there's a showing of pre-regulated on the input
4 voltage, it would satisfy that function.

5 Q. And you didn't dispute for the ladies and gentlemen of
6 the jury that the DDR5 designs receive a pre-regulated input
7 voltage. Correct?

8 A. No, I did not.

9 Q. And he describes that as the LDO converting the
10 regulated -- converting the input voltage into a regulated
11 output voltage. Correct?

12 A. That's correct. And I agree with it. It does.

13 Q. And so when the ladies and gentlemen of the jury consider
14 whether the '918 Patent claim 16, which recites a converter
15 circuit, is infringed by Samsung, they can take into account
16 that the person you describe as the most knowledgeable on
17 Samsung's PMIC describes the LDO as converting. Correct?

18 A. I would hope that they would take all of this information
19 into account, including this.

20 Q. And the claims-at-issue require a converter. Correct?

21 A. No. The claims-at-issue require a converter circuit.

22 Q. And what's a -- what's a circuit again?

23 A. A circuit is a combination of electrical components for
24 the purposes of performing a certain function, but there are
25 different ways which you can actually integrate the

1 circuits -- the components to create that function.

2 Q. A circuit is electrical components that implement a
3 function. Do I have that right?

4 A. Yes, that's correct.

5 Q. Okay. And a converter is something that converts a
6 voltage. Correct?

7 A. A converter, in part, converts a voltage, but there's
8 also a structural aspect to it.

9 Q. Why don't you turn to your deposition at 39, lines 20
10 through 22.

11 A. You say page 39?

12 Q. Yes. Page 39.

13 A. Okay. All right. I have it.

14 Q. Sir, a converter is something that converts a voltage.
15 Correct?

16 A. A converter is a type of circuit that does function to
17 convert a voltage.

18 Q. Sir, a converter is something that converts a voltage.
19 Correct?

20 A. Well, something. It's a -- if you're talking about a
21 converter, yes, a converter circuit is a specific type.

22 Q. Sir, a converter is something that converts a voltage.
23 Correct?

24 A. A converter does convert voltage, yes.

25 Q. And we saw previously that Samsung's engineers describe

1 their LDO as a converter. Correct?

2 A. We saw previously that they describe the function as
3 converting, yes.

4 Q. And you can't deny --

5 MR. SHEASBY: Let's pull that down and go back up to
6 slide 13, 5.12.

7 Q. (BY MR. SHEASBY) You can't deny that an LDO conversion
8 does take place. Correct?

9 A. I'm not sure I understand what you mean by LDO
10 conversion.

11 Q. In an LDO, conversion of voltage does take place.
12 Correct?

13 A. There is a conversion of voltage, yes.

14 Q. And to be clear, the dispute is whether the LDO regulator
15 satisfies that limitation that's underlined. Correct?

16 A. In its entirety, yes.

17 Q. And that's the only limitation from the '918 Patent that
18 you identified as non-infringed. Correct?

19 A. That is correct.

20 Q. And you don't recall and -- and you agree as well that
21 the LDO produces a regulated output voltage. Correct?

22 A. It does, yes.

23 Q. And it produces a regulated output voltage by reducing
24 the input voltage. Correct?

25 A. There is a reduction in the input voltage to create that

1 output voltage. There are other parts, but that is also done.

2 Q. And you don't recall at your deposition -- you didn't
3 recall any document by a person of skill in the art that
4 supports your position that an LDO is not a converter.
5 Correct?

6 A. I don't recall the construct of that. I believe
7 it's -- the construct was within the framework I didn't have
8 any documents to show at that deposition. Right.

9 Q. At the deposition that you took under oath. Correct?

10 A. Right.

11 Q. Spoke for a very long time. Correct?

12 A. We talked for quite a while, yes.

13 Q. And you didn't identify any document by a person of skill
14 in the art that says an LDO is not a converter. Correct?

15 A. I didn't have any documents in front of me to -- to do
16 that, no.

17 Q. Sir, you had access to your entire report and all
18 documents in your report. Correct?

19 A. I did, yes.

20 Q. And the jury heard you talk about it.

21 MR. SHEASBY: Let's pull up JTX 11.69 again.

22 Q. (BY MR. SHEASBY) You talked about the fact that this has
23 components. Correct?

24 A. Yes.

25 Q. And it has a function. Correct?

1 A. It does.

2 Q. And the jury heard you say that it doesn't have a switch.
3 Correct?

4 A. That is correct.

5 Q. But there is no -- but the reality is that there is no
6 description or requirement of a switch for any of the
7 converters described in the '918 Patent. Correct?

8 A. There's no statement about a switch, if that's what
9 you're asking about.

10 Q. In fact, the '918 Patent doesn't say converters have to
11 have switches. Correct?

12 A. I don't think it does that. It does refer to boost and
13 buck converters which do have switches with inductors.

14 Q. It refers to buck --

15 MR. SHEASBY: Well, I move to strike as
16 non-responsive, Your Honor.

17 THE COURT: The portion of the answer "I don't think
18 it does," that is responsive. The remainder of it is
19 non-responsive, and I'll strike that.

20 Q. (BY MR. SHEASBY) But one thing you did say, that you
21 say, well, it talks about buck. Correct?

22 A. Yes.

23 Q. And a buck is a converter. Correct?

24 A. Yes, it is.

25 MR. SHEASBY: And if we go to PDX 5.15.

1 Q. (BY MR. SHEASBY) Renesas makes the --

2 MR. SHEASBY: 15, please.

3 Q. (BY MR. SHEASBY) Renesas makes the PMICs that are at
4 issue in this case. Correct?

5 A. Yes, they do.

6 Q. And this is a Renesas document. Correct?

7 A. It is a Renesas document, yes.

8 Q. And the designers know more about the intricate parts
9 within their PMIC design than you do. Correct?

10 A. I would say the intricate parts, yes.

11 Q. And Renesas describes linear regulators as a buck.
12 Correct?

13 A. They are showing -- that's what they show here.

14 Q. And you concede a buck is a converter. Correct?

15 A. A buck converter is a converter.

16 MR. SHEASBY: So let's pull that down, and let's go
17 back to the claims. If we can go to claim 4, PDX page 8 -- or
18 page 8 of -- PDX 5, page 8. Page 8.

19 Q. (BY MR. SHEASBY) So we just talked about the fact about
20 the question is whether an LDO is a converter circuit.
21 Correct?

22 A. That is correct.

23 Q. And we heard from the testimony of Mr. Lo. Correct?

24 A. We did.

25 Q. We heard from a number of Samsung engineers on that

1 point. Correct?

2 A. We did.

3 Q. We also saw a Renesas document. Correct?

4 A. Just now, yes.

5 Q. Okay. And so the jury, to determine if Samsung is liable
6 for infringement of this patent, will answer one precise
7 question: Is an LDO a converter circuit or an equivalent of a
8 converter circuit? Correct?

9 A. I think that's fair, yes.

10 Q. Okay. Now, let's go to the second non-infringement issue
11 that you spoke about, and this is '054 Patent claim 16.

12 MR. SHEASBY: If we go to slide 10, Mr. Huynh.

13 Q. (BY MR. SHEASBY) So in this issue, claim 16 doesn't
14 require a converter circuit. Correct?

15 A. On the '054, no, it does not.

16 Q. But what it does require is that the memory module
17 transition from a first operable state to a second operable
18 state. Correct?

19 A. Yes, it does.

20 MR. SHEASBY: And if we can pull up slide 17, Mr.
21 Huynh.

22 Q. (BY MR. SHEASBY) So you agree that this state exists.
23 Correct?

24 A. Yes, I do.

25 DR. ALBERT: Your Honor --

1 THE COURT: Yes?

2 DR. ALBERT: -- I believe this slide was not
3 permitted by Your Honor before.

4 MR. SHEASBY: Your Honor, I'm using a demonstrative
5 to cross-examine a witness.

6 DR. ALBERT: Can we approach?

7 THE COURT: Approach the bench, Counsel.

8 (The following was had outside the hearing of the
9 jury.)

10 THE COURT: Can I have the slide so I can see it?

11 DR. ALBERT: This is that ZZZ sleep slide that you
12 excluded before, sir, Your Honor.

13 MR. SHEASBY: So you didn't exclude it. You
14 concluded that the sleep aspect of it was not shown in Doctor
15 Mangione-Smith's report. And I now intend to establish that
16 that operable state does in fact exist. It's not evidence;
17 it's just a demonstrative.

18 THE COURT: But you intend to use it in this form?

19 MR. SHEASBY: Yes, Your Honor. I intend to
20 establish that that's what that second operable state shows.
21 I'm not going to make any reference to Doctor Mangione-Smith
22 at all.

23 THE COURT: What's your objection, Mr. Albert?

24 DR. ALBERT: Your Honor, this is the exact same form
25 that he's been presenting in Mangione-Smith slides, the exact

1 same graphic that he used. This wasn't permitted before, and
2 it's not something that --

3 THE COURT: I ordered changes to this slide for use
4 with Doctor Mangione-Smith. If this is going to be used with
5 Mr. McAlexander by Plaintiff's counsel, he's not going to
6 reference Doctor Mangione-Smith and he's going to use it
7 solely as a demonstrative, I don't see any reason why I should
8 prohibit that just because a similar slide was in a different
9 form used by Doctor Mangione-Smith, if that's the basis of
10 your objection.

11 DR. ALBERT: Understood, Your Honor.

12 THE COURT: Okay?

13 MR. SHEASBY: Thank you, Your Honor.

14 (The following was had in the presence and hearing
15 of the jury.)

16 Q. (BY MR. SHEASBY) So you agree that this state exists.
17 Correct?

18 A. Yes, I do.

19 Q. And you agree that this state exists. Correct?

20 A. Yes, I do.

21 Q. You also agree that there's a state -- let's not debate
22 about that. There's a state that's depicted here. Correct?

23 A. I'm sorry. Can you say that again, please?

24 Q. Can you agree that there's a state depicted in that
25 middle image? Correct?

1 A. There is a state depicted in that middle image, correct.

2 A state that is depicted there.

3 Q. And that's -- in the DDR5, Samsung's DDR5, the memory
4 controller can write data to the volatile memory subsystem.
5 Correct?

6 A. Say that again, please?

7 Q. Sure. In DDR5, the memory controller can write data to a
8 volatile memory subsystem. Correct?

9 A. Yes, it can.

10 Q. And DRAM is an example of a volatile memory subsystem.
11 Correct?

12 A. It is, yes.

13 Q. And although it's not on the module, the DDR5 systems
14 that use the Samsung modules have the ability to write to
15 data -- write data on those modules to non-volatile storage.
16 Correct?

17 A. Well, there's ability in the -- in the operable normal
18 state, there is an ability to write to DRAMs on the module.

19 Q. I know. And there's also ability to write to
20 non-volatile storage that is off module. Correct?

21 A. Yeah. Through the classic computer system, you can write
22 to non-volatile as well.

23 Q. And the patent discloses this off-module writing system.
24 Correct?

25 A. Can you be more specific with where? I'm not exactly

1 sure what you're talking about.

2 Q. Sure.

3 MR. SHEASBY: Why don't we go to slide PDX 5.28.

4 Q. (BY MR. SHEASBY) So this is a passage from the '918
5 Patent. Correct?

6 A. That is correct.

7 Q. And it says, for example, in one embodiment the host
8 system, a disk controller, writes data to the volatile memory
9 subsystem which then writes data to the non-volatile storage
10 which is not part of the memory system such as, for example, a
11 hard drive. Correct?

12 A. Correct. That's the classic way, yes.

13 Q. And that's described as an off-module write. Correct?

14 A. It is off-module, yes.

15 Q. And the patent describes that as an off-module write.
16 Correct?

17 A. That is correct. That's what it says.

18 Q. And the DDR5 modules that Samsung sells when they're
19 integrated into servers have this ability to participate in
20 off-module write. Correct?

21 A. Oh, yes. That's the way you back -- that's the way you
22 take information from your non-volatile and store it with a
23 save command to the non-volatile.

24 Q. And in this passage, there's no reference to the
25 non-volatile storage being on the module. Correct?

1 A. No. This is non-volatile storage that's on the hard
2 disk. So that's not on the module.

3 Q. So in this embodiment there is volatile memory subsystem
4 on the module. Correct?

5 A. Yeah, there's volatile memory on the module, yes.

6 Q. And there is -- the non-volatile memory is on the system
7 itself. Correct?

8 A. Yes. The system still has non-volatile memory.

9 Q. And that's described in the patent. Correct? In this
10 passage?

11 A. Yes, yes, of course.

12 MR. SHEASBY: Now, let's pull that down and go back
13 to the slide we were looking at, Mr. Huynh. I believe it was
14 17.

15 Q. (BY MR. SHEASBY) So I want to talk about this middle
16 state, if I may. Is that okay?

17 A. You may.

18 Q. And when the -- when a power undervoltage or low voltage
19 situation occurs, the LDO converters will continue to supply
20 power so that the power management integrated circuit can
21 operate and access its registers. Correct?

22 A. The power management linear drop-out regulators will
23 provide power to certain portions of the PMIC so the PMIC can
24 operate.

25 Q. And you've historically described those as LDO

1 converters. Correct?

2 A. LDO regulators, linear regulators.

3 Q. Sir, when I described them to you as LDO converters, you
4 answered my question in the affirmative. Correct?

5 A. On that particular thing, I'll wait to answer that and
6 qualify it later.

7 Q. So why don't we go to 85, lines 4 through 10 again.

8 A. Okay.

9 Q. Why don't you read that to yourself.

10 When there is a power undervoltage or low voltage
11 situation, the LDO converters will continue to supply power so
12 that the PMIC can continue to operate and access registers.
13 Correct?

14 A. Well, that is your question. You didn't read it exactly
15 word for word, but, generally speaking, that was your
16 question.

17 Q. And your answer was yes. Correct?

18 A. Yes, to that function, yes. Correct.

19 Q. And when the PMIC is still receiving power -- and by the
20 way, the PMIC is going to be right there. Is that right?

21 A. On the other side of the board, yes.

22 Q. On the other side. We flip it around.

23 And when the PMIC is still receiving power in that
24 undervoltage situation from the LDOs, the PMIC which is on the
25 module is available to communicate with the host and hold the

1 status registers constant so that it will not lose sight of
2 where it was when the power went down. Correct?

3 A. Yes. That's the control functions. I agree.

4 Q. And those stay awake even when the power to the
5 non-volatile memory is suspended. Correct?

6 A. Just -- what are you referring to in terms of power to
7 the non-volatile memory?

8 Q. Excuse me.

9 A. I think you misspoke.

10 Q. I did misspeak. I'll give it another try, Mr.
11 McAlexander.

12 A. Thank you.

13 Q. Even when the power to the volatile memory has put them
14 to sleep, the on-module power management will continue to
15 receive power and continue to operate that PMIC. Correct?

16 A. The second part of your question, continue to supply
17 power to the PMIC, that part is true.

18 Q. Okay.

19 A. The first part of your question is not wrong -- is not
20 right because it's not putting it to sleep. There's no power;
21 it's turned off.

22 Q. Okay.

23 A. It's not asleep at all.

24 Q. So I'll re-ask it in a more precise way. When the DRAM
25 are turned off, power is still supplied to the on-module power

1 management feature. Correct?

2 A. Yes, that's correct.

3 Q. And in that state, the power management is still
4 operating. Correct?

5 A. For the purposes -- yes, it's still operating.

6 Q. So it talks about a first operable state. Correct?

7 A. For the memory module, yes.

8 Q. And we agree that that's the first operable state for a
9 memory module. Correct?

10 A. Yes. It's normal operation. All is operating.

11 Q. And there is a second operable state required as well.
12 Correct?

13 A. That is correct.

14 Q. And it doesn't say second operable state for the DRAM in
15 the claim. Correct?

16 A. Correct. It doesn't say for the DRAM.

17 Q. It says the second operable state for the memory module
18 as a whole. Correct?

19 A. Memory module as a whole, that is correct.

20 Q. And we agree that the power management is on the memory
21 module as a whole. Correct?

22 A. Well, the power -- the power management integrated
23 circuit, the PMIC, is a component on the module.

24 Q. And it's in this second state, it's still operating.
25 Correct?

1 A. Well, in what you show here is a second state. The PMIC
2 still has power and is operating. The DRAM is the module --
3 the module itself, the memory module is not operating.

4 MR. SHEASBY: Move to strike the second part as
5 non-responsive, Your Honor.

6 THE COURT: Overruled.

7 Q. (BY MR. SHEASBY) To be as precise size as possible, in
8 the second operable state, the PMIC is receiving power and is
9 still operating. Correct?

10 A. That is correct.

11 Q. And there's a very important reason why in this second
12 state the PMIC is still operating. Correct?

13 A. Well, there's an importance for it still to be operating,
14 yes.

15 Q. And that's because the module will need to attempt to
16 restore itself. Correct?

17 A. I'm not sure exactly what you mean to restore itself.
18 Yes, there will be a restoration period. So, in part,
19 correct.

20 Q. And that PMIC, which is operating in this second state,
21 is involved in that restoration. Correct?

22 A. That's correct.

23 Q. And the operation that occurs in the second state is
24 maintaining and reloading the status registers, and that's
25 what allows the module to pick back up again once the voltage

1 interruption issue is resolved. Correct?

2 A. Not exactly.

3 Q. Okay. Why don't you turn to your deposition and look at
4 page 86, line 17 through 87, lines 4.

5 A. Okay.

6 Q. Let me know when you're there. Go ahead and read it to
7 yourself.

8 A. Thank you. Okay. I read it.

9 Q. The operation that is occurring during this second state
10 is maintaining and reloading the status registers, and that is
11 what allows the module to pick back up again once the voltage
12 issue interruption has been resolved. Correct?

13 A. That's correct, yes.

14 Q. So, in other words, this is critically important to the
15 operation of the module when the PMIC is awake and operating
16 and can assist with the bringing the module back online.

17 Correct?

18 A. Yes. Computer needs to know what the identity of the
19 module is.

20 Q. And it can't do that if the PMIC is shut down. Correct?

21 A. Not without rebooting.

22 Q. And so one of the issues -- and for the jury to determine
23 whether Samsung has infringed this patent, what it has to
24 determine is whether this second state in which the PMIC is
25 operating and providing -- maintaining its information so it

1 can assist in -- in restarting the memory module constitutes a
2 second operable state. Correct?

3 A. In accordance with the Court's claim construction, that's
4 correct.

5 Q. And the Court's claim construction says nothing about
6 whether the DRAM is asleep or turned off or on in its
7 construction. Correct?

8 A. I believe the construction is directed to the memory
9 module.

10 Q. And the memory module has more on it than just the DRAM.
11 Correct?

12 A. Yeah, the memory module as a whole has more on it than
13 just the DRAM, but it has the DRAM.

14 Q. And the Court didn't say in its construction that the
15 DRAM has to be on in each state. Correct?

16 A. I believe I already answered that. It's the memory
17 module.

18 Q. The Court did not say that the DRAM has to be on in each
19 state. Correct?

20 A. Did not use the word DRAM. Correct.

21 Q. And you showed the ladies and gentlemen of the jury the
22 testimony of Mr. Bruce Lo regarding this second state.
23 Correct?

24 A. Yes.

25 Q. And you thought that was important for the ladies and

1 gentlemen of the jury to know what Bruce Lo thinks about the
2 second state. Correct?

3 A. I believe it was important, yes.

4 Q. But if we turn to page 5.19, Mr. Lo admits as well in
5 that second operable state that the components that are
6 powered by it are still operating. Correct?

7 A. Yes, I agree with that.

8 Q. And there's more than just the PMIC that is supplied in
9 that second operable state. Correct?

10 A. Yes. Temperature sensor as well.

11 Q. Serial presence detect as well?

12 A. Right.

13 Q. In other words, there are a number of components on the
14 memory module, all of which continue to operate in this second
15 state in which the DRAM has turned off because of a power
16 interruption, but power is still supplied to maintain these
17 critical features. Correct?

18 A. Yeah. The LDO regulators maintain power to these
19 specific small components.

20 Q. And those specific small components are the components
21 that allow the memory to restart again. Correct?

22 A. Allows the CPU to detect what the memory is so that it
23 can restart.

24 Q. And if that doesn't happen, you have to go through an
25 entire new reboot process. Correct?

1 A. That's correct.

2 Q. Now, there is only one claim -- and so just to be as
3 precise as we can for the ladies and gentlemen of the jury, if
4 they believe that the operation of the critical PMIC serial
5 presence detect and temperature sensor constitute an operable
6 state of the memory module as a whole, then Samsung infringes
7 this patent. Correct?

8 A. The -- as I stated before, the jury needs to consider all
9 of the facts and information supported in this case to render
10 that opinion.

11 Q. If the jury concludes that the state in which the PMIC,
12 the SPD, and the temperature sensor are all operating as an
13 operable state, then Samsung infringes this patent. Correct?

14 A. I believe that they can come to that opinion, but that's
15 up to the jury to decide whether they form that opinion or not
16 based upon this information.

17 Q. Totally understand. It's the jury's decision as to
18 whether Mr. Lo's testimony that all the components that are
19 supplied by the LDO still operate in the second state
20 constitute a second operable state. Correct?

21 A. That's correct.

22 Q. And, in fact, the patent in this case, the '918 Patent
23 and the '054 Patent, they both share the same specification.
24 Correct?

25 A. That is correct.

1 Q. And they expressly contemplate the idea that there is
2 going to be a state in which the DRAM, the volatile memory,
3 has been turned off. Correct?

4 A. You'll need to point me to that. Yes, there's always a
5 state where DRAMs are turned off, but in terms of the patent
6 specification, it's very clear the additional power modes, the
7 DRAMs are always operative.

8 Q. I'm asking a different question --

9 A. Okay.

10 Q. -- which is the patent contemplates a mode in which the
11 power to the DRAM is suspended. Correct?

12 A. There's always going to be an off condition to the DRAMs,
13 yes.

14 MR. SHEASBY: Let's go to slide 5.7.

15 Q. (BY MR. SHEASBY) Now, there's only one claim in the
16 patent that are asserted in this case that requires the
17 presence of non-volatile memory, and that's claim 19 of the
18 '918 Patent. Correct?

19 A. That is correct.

20 Q. And if we go to slide 30, just to orient ourselves, this
21 is the specification of the '918 and '054 Patent, and it talks
22 about some type of control chip or logic element. Correct?

23 A. Yes, that's correct.

24 Q. And it says that contemplates that one form that control
25 logic element can take would be an application-specific

1 integrated circuit. Correct?

2 A. Yes, that's correct.

3 Q. And we agree that the PMICs that Samsung uses on every
4 single one of the DDR5 products at issue in this case is an
5 application-specific integrated circuit. Fair?

6 A. It's fair to characterize them as such, yes.

7 Q. And this passage from the specification contemplates that
8 there will be what's described as internal flash on the chip.
9 Correct?

10 A. Yes.

11 Q. And internal flash can also be referred to as embedded
12 flash. Correct?

13 A. It depends upon its location. It can be.

14 Q. One form of internal flash is embedded flash. Correct?

15 A. That's a better characterization, yes.

16 Q. And for this one single limitation, the question is
17 whether there's non-volatile memory on the DDR5 modules.
18 Correct? Just for claim 19.

19 A. All right. Thank you. Yes.

20 Q. It's irrelevant to every other single claim that's
21 asserted in these families. Correct?

22 A. Claim 19 as a dependent claim stands, yes.

23 Q. My point is, none of the other claims require or prohibit
24 the use of flash or non-volatile memory on the module.
25 Correct?

1 A. That was my statement in my testimony, yes.

2 Q. I'm sorry. I misheard you.

3 A. That was the statement in my testimony earlier that there
4 is no flash on the memory module. So, yes, there is nothing
5 that's not comprehended by the claim -- the other claims.

6 Q. To be -- the claims do not require flash on the memory
7 module except for this one dependent claim, claim 19.

8 Correct?

9 A. That's the way the claim's written, yes.

10 Q. And we agree that flash is non-volatile memory. Correct?

11 A. We agree with that.

12 MR. SHEASBY: And if we go to slide 82.

13 Q. (BY MR. SHEASBY) This is JTX 11. This is one of
14 Samsung's documents for their power management system that is
15 put on-module in DDR5. Correct?

16 A. Yes, it is.

17 MR. SHEASBY: And if we go to page 16 of JTX 11.

18 Q. (BY MR. SHEASBY) It describes that that power
19 management, which is on the DDR5, has built in non-volatile
20 memory. Correct?

21 A. Yes, that's correct, in the registers.

22 Q. And, in fact, Samsung's corporate representative concedes
23 that what's here includes non-volatile memory. Correct?

24 A. Yes. There is non-volatile registers in the power
25 management unit.

1 Q. Well, to be precise, he says there's non-volatile memory
2 that has been implemented. Correct?

3 A. That's what he says, yes.

4 Q. So Samsung's corporate representative on the power
5 management modules that are put on its DDR5, he testified that
6 there's non-volatile memory on that power management.
7 Correct?

8 A. Yes, the registers. That's correct.

9 Q. And that non-volatile memory -- he says there's
10 non-volatile memory that has been implemented on the PMIC.
11 Correct?

12 A. His answer at the bottom says non-volatile, NVM, memory
13 circuit has been implemented. Yes, he says that.

14 Q. And he's discussing the figure on the right-hand side,
15 which is JTX 11 at 18. Correct?

16 A. Yes, that's correct.

17 Q. And that structure of JTX 11.18, or some version of it,
18 is going to be on every DDR5 module accused of infringement in
19 this case. Correct?

20 A. That is correct.

21 Q. And some of the other power modules that are used in this
22 case are produced by Renesas. Correct?

23 A. Renesas is one of the suppliers, yes.

24 MR. SHEASBY: And if we go to PDX 93. I'm sorry,
25 5.93.

1 Q. (BY MR. SHEASBY) This is one of the Renesas data sheets.

2 This is JTX 30. Correct?

3 A. That is correct.

4 Q. And we looked at it previously. Correct?

5 A. We have, yes.

6 Q. And it describes that there is non-volatile memory on the
7 power management. Correct?

8 A. Yes, it does.

9 Q. And that non-volatile memory that is on the power
10 management goes into every single DDR5 module that is accused
11 of infringement in this case. Correct?

12 A. The PMIC has that, yes.

13 Q. And the PMIC is on every single DDR5 module that's
14 accused of infringement in this case. Correct?

15 A. Yes, that's correct.

16 Q. May I ask you some questions about validity?

17 A. If you would like.

18 Q. And just for the ladies and gentlemen of the jury, before
19 we do that --

20 MR. SHEASBY: -- let's go back briefly to slide 8.

21 Q. (BY MR. SHEASBY) So we talked about whether there's a
22 converter circuit. Correct?

23 A. We did.

24 Q. We talked about whether there's a second operable state.
25 Correct?

1 A. We did.

2 Q. We talked about whether there is non-volatile memory for
3 that claim 19. Correct?

4 A. Yes, we did.

5 Q. Those are the only three arguments for non-infringement
6 you have presented to the jury for these patents. Correct?

7 A. I would say that characterizes it correctly, yes.

8 Q. Okay. Now let's go to -- let me ask you another
9 question.

10 Did you look at any of the source code or actual
11 technical schematics that describe the operation of the power
12 management integrated circuits at issue in this case?

13 A. I did look at some, yes.

14 Q. You looked at the source code?

15 A. Schematic portions of it, yes.

16 Q. When did you look at those?

17 A. Well, I saw some in the testimony here in the trial.

18 Q. When preparing your report, did Samsung provide you with
19 any source code or the technical schematics that define how
20 the products are actually designed?

21 A. Not as I recall at that time.

22 Q. You've worked for Samsung 10 or 12 times. Correct?

23 A. I've worked for law firms on behalf of Samsung 10 or 12
24 times.

25 Q. And in this time Samsung declined or did not provide you

1 with the source code and technical schematics for the power
2 management integrated circuits at issue in this case. Fair?

3 A. I'm not so sure that they declined. I just did not -- I
4 was not provided with them.

5 Q. But they provided you with information. Correct?

6 A. They?

7 Q. Samsung.

8 A. Documents were produced in this case under Plaintiff and
9 Defendant Bates numbers, so I reviewed those documents, yes.

10 Q. And they were given to you by Samsung's counsel.
11 Correct?

12 A. That is correct.

13 Q. And Samsung's counsel didn't give you the actual
14 schematics and source code for the power management integrated
15 circuits at issue in this case. Correct?

16 A. As I recall from my reports, the answer is, no, they did
17 not.

18 Q. So I'll re-ask the question. May I ask you about
19 validity now?

20 A. Certainly.

21 Q. And this is validity of the '918 and '054 Patents.

22 MR. SHEASBY: And we can go to slide 20.

23 Q. (BY MR. SHEASBY) Validity must be -- the patents are
24 presumed valid and Samsung must prove invalidity by clear and
25 convincing evidence. Correct?

1 A. That is correct.

2 Q. And I want to make sure there is no ambiguity. You have
3 not identified a single reference anywhere in the world or a
4 single design anywhere in the world that anticipates or
5 renders obvious Netlist's designs. Correct?

6 A. You're speaking specifically about the '918 and '054
7 Patents. Correct?

8 Q. I withdraw the question and let me re-ask it.

9 A. Thank you.

10 Q. In your testimony today, you have not identified a single
11 reference or design anywhere in the world that anticipates or
12 renders obvious the '918 and '054 Patents. Correct?

13 A. That is correct.

14 Q. What you do is you claim that they lack what you
15 describe --

16 DR. ALBERT: Sidebar, Your Honor?

17 THE COURT: I'm sorry?

18 DR. ALBERT: Sidebar, Your Honor.

19 THE COURT: Approach the bench.

20 (The following was had outside the hearing of the
21 jury.)

22 DR. ALBERT: Your Honor, he's getting into general
23 aspects of invalidity. I think he's opening the door for what
24 his opening report actually said. He provided his opinion.
25 He just didn't provide it in this court today regarding '918

1 or '054 invalidity aspects. I think this is completely
2 improper.

3 MR. SHEASBY: Your Honor, I precisely asked the
4 question whether he testified about it today. I did that
5 because he discussed historical designs, and I want to make
6 clear the jury is not confused that he's suggesting anything
7 as anticipated or obvious. In fact, I withdrew my question
8 and re-asked it precisely to focus on what was testified
9 today.

10 THE COURT: The sole invalidity defense here is lack
11 of written description. Correct?

12 MR. SHEASBY: That's correct, Your Honor.

13 THE COURT: There is not a 102 or 103 defense, is
14 there, Doctor Albert, to these two patents?

15 DR. ALBERT: As sitting here today in this court,
16 yes, Your Honor. But he has provided those opinions. And so
17 general questions like this are improper.

18 THE COURT: He's provided what opinions.

19 DR. ALBERT: He provided a written 1200-page report
20 providing analysis of -- of anticipation, obviousness. This
21 was narrowed to this case. It's improper for Mr. Sheasby
22 to imply that there hasn't been that kind of analysis.

23 THE COURT: All right.

24 MR. SHEASBY: I didn't imply that analysis.

25 THE COURT: I'm not going to strike anything that's

1 been asked and answered up to this point.

2 I am going to instruct, Mr. Sheasby, that now he's made
3 it clear to the jury that there's not a live 102 or 103
4 defense as to these patents, to move on and not focus on
5 either anticipation or obviousness going forward.

6 MR. SHEASBY: Yes, Your Honor. Thank you.

7 THE COURT: Okay?

8 (The following was had in the presence and hearing
9 of the jury.)

10 THE COURT: Let's proceed.

11 Q. (BY MR. SHEASBY) For written description, you applied
12 the rule that there's no requirement for any particular form
13 of disclosure, and that support may be based on a combination
14 of figures and disclosures throughout the specification.

15 A. Can you restate that, please, or --

16 Q. You did what you described as a written description
17 analysis. Is that correct?

18 A. That is correct.

19 Q. And your idea is you are claiming to the ladies and
20 gentlemen of the jury that what was claimed was not disclosed
21 in the patent. Correct?

22 A. Correct.

23 Q. And the rules that you apply -- the disclosure can be in
24 any form. It can be a combination of figures and texts; it
25 doesn't have to be the exact words, the exact design.

1 Correct?

2 A. As far as within the specification of the patent, yes,
3 that's correct.

4 Q. Yes, sir.

5 And the first issue that you emphasized --

6 MR. SHEASBY: Go to slide 21.

7 Q. (BY MR. SHEASBY) And this was a slide shown in opening
8 about Netlist's, quote, continuation strategy. Correct?

9 A. It was, yes.

10 Q. And what counsel indicated is that there are a large
11 number of applications and patents that have been generated
12 based on this original patent application. Correct?

13 A. Correct.

14 Q. And there is nothing wrong with that. Correct?

15 A. No.

16 Q. In fact --

17 A. Perfectly fine.

18 Q. Samsung does it as well. Correct?

19 A. Yes, they do.

20 Q. So, for example, Samsung over a 16-year period has filed
21 nine separate continuation applications in one of its patents
22 relating to cell phone. Correct?

23 A. I believe that this is a correct representation, yes.

24 Samsung does file continuations as well.

25 Q. And so if there was any suggestion to the ladies and

1 gentlemen of the jury that there is anything improper about
2 continuation practice, there's anything improper about filing
3 an application a number of years after original parent had
4 been filed, that's not accurate. Correct?

5 A. I don't believe that was ever stated, but no, that's not
6 accurate.

7 Q. It certainly wasn't stated by you. Correct, sir?

8 A. No. Continuations are part of the process. You can do
9 it.

10 Q. And in this case it was not a secret that Netlist was
11 seeking a continuation and announcing to the Patent Office
12 that it believed it had the disclosure for these claims in the
13 original patent application. Correct?

14 A. I'm not sure your timing on this -- obviously, the
15 communication of the Patent Office during the time of the
16 communication, it is secret between the Patent Office and the
17 patentee, so I'm not sure your question --

18 Q. Sure. I'm happy to re-ask it.

19 The Patent Office was well aware that Netlist believed
20 that its original 2008 application disclosed all of the claims
21 in the patent. Correct?

22 A. That part is correct, yes.

23 Q. And the Patent Office granted the patent based on this
24 priority claim. Correct?

25 A. They granted a patent.

1 Q. And you have some experience with patent prosecution.

2 Correct?

3 A. I do.

4 Q. And you understand that a patent prosecutor--these are
5 the government officials in the PTO when new or amended claims
6 are presented in an application--they are required to
7 determine if there is written support for those new claims.
8 Correct?

9 A. Yes. That is one of their requirements as I understand.

10 DR. ALBERT: Objection, Your Honor.

11 THE COURT: Yes?

12 DR. ALBERT: We're getting into patent practice,
13 Your Honor, contrary to our standard MILs.

14 THE COURT: Well, the defense here is lack of
15 adequate written description. This is not a time to take up
16 anything such as an inequitable defense under prosecution
17 laches.

18 MR. SHEASBY: I understand that.

19 THE COURT: So, I mean, a bare-bones coverage of the
20 procedural steps before the Patent Office is fine, but the
21 focus needs to be on whether the written description
22 adequately describes the claims.

23 MR. SHEASBY: Right. And what I'm seeking to
24 establish is that the -- Mr. McAlexander will testify that
25 the PTO actually examined this exact question before the

1 patent issued.

2 THE COURT: Well, let's focus on the written
3 description aspect.

4 MR. SHEASBY: Sure, Your Honor.

5 THE COURT: All right.

6 Q. (BY MR. SHEASBY) And when claims are presented, as they
7 were in this case, for the patents that became the '918 and
8 '054 Patent, the patent examiner is supposed to determine
9 whether there has been any omitted limitations that need to
10 be included in the specification. Correct?

11 A. I agree with your premise. They're supposed to.

12 Q. And if the originally-filed disclosure does not provide
13 support for each claim element or limitation that the
14 applicant describes as essential or critical -- let me re-ask
15 the question.

16 If the claims do not include elements that the
17 specification describes as essential or critical, the
18 claims are required to be rejected. Correct?

19 A. I believe that is one of the requirements. It should,
20 yes.

21 Q. And in this situation, the patents at issue in this case,
22 the patent examiner didn't reject them by saying, You're
23 missing a critical element. Correct?

24 A. I don't believe I recall having any statements to that
25 effect in the prosecution history.

1 Q. And to be clear, you're not disputing that the patent
2 teaches a person of ordinary skill in the art how to enable
3 the full scope of these claims. Correct?

4 A. As properly rendered, yes.

5 Q. Sir, you are not disputing that the full scope of the
6 claims have been enabled. Correct?

7 A. I haven't raised that as a dispute, no.

8 Q. Now, one of the things you point to is the concept of
9 whether all the memory has to be on-module, the non-volatile
10 memory. Correct?

11 A. Yes.

12 MR. SHEASBY: And if we turn to slide 28.

13 Q. (BY MR. SHEASBY) This is an embodiment we discussed.
14 Correct?

15 A. Yes.

16 Q. And it's an embodiment in which the non-volatile memory
17 that is written to is off the module. Correct?

18 A. Yes. That's the normal hard disk, yeah.

19 Q. And another passage that you discussed is --

20 MR. SHEASBY: Can I have the elmo, Madam Courtroom
21 Deputy?

22 Q. (BY MR. SHEASBY) It's these figures. Correct?

23 A. Yes.

24 Q. And these figures don't show any non-volatile memory.
25 Correct?

1 A. I -- they don't because they're a different part of the
2 figure 12.

3 MR. SHEASBY: Move to strike the second part as
4 non-responsive, Your Honor.

5 THE COURT: He didn't ask why they didn't, he just
6 asked did they, and you agreed they didn't. So I'll sustain
7 the objection as to the latter part of the response. It goes
8 beyond the question called for.

9 Q. (BY MR. SHEASBY) But they do recite an FPGA. Correct?

10 A. Yes.

11 Q. And that's a control circuit. Correct?

12 A. That is correct.

13 Q. And the patent discloses that that control circuit can
14 have embedded flash memory. Correct?

15 A. It can, sure.

16 Q. And Samsung's design has DRAM. Correct?

17 A. Samsung's design does have DRAM.

18 Q. And it has a controller, this power management integrated
19 circuit. Correct?

20 A. It does.

21 Q. And it has embedded flash. Correct?

22 A. In the PMIC, yes.

23 Q. And a PMIC is a controller. Correct?

24 A. It is.

25 Q. And the FPGA in this design is also a controller.

1 Correct?

2 A. In what design?

3 Q. In the patent.

4 A. Yes.

5 Q. And one of the things that you say is these figures 15A
6 and 15B, they're only used as part of another figure in which
7 there's lots of non-volatile memory present on the design.

8 Correct?

9 A. Yes.

10 Q. But, in fact, those figures are described at column 9,
11 lines 35 to 39, of the patent. Correct?

12 A. Yes, correct.

13 Q. And it says nothing about them needing to be as part of
14 any other module. Correct?

15 A. That is a description of what is on those figures.

16 Q. I understand that.

17 A. Right. Agree with you.

18 Q. Column 9, lines 35 through 39, says nothing about those
19 figures need to be present or added to any other structure.

20 Correct?

21 A. I disagree with your concept.

22 Q. So the words 'figures 15A through 15C have to be embedded
23 with non-volatile memory chips' is present in the passage I
24 boxed, sir?

25 A. I believe I indicated what they were a part of, 1030,

1 which is --

2 MR. SHEASBY: I move to strike as non-responsive,
3 Your Honor.

4 THE COURT: Restate the question.

5 MR. SHEASBY: Sure.

6 Q. (BY MR. SHEASBY) Does the figure -- does column 9, lines
7 35 through 39, say that figure 15A and 15C have to be used
8 with a module that has all the non-volatile memory on the same
9 module?

10 A. No, sir. Those words are not found in that particular
11 paragraph.

12 Q. And another passage that you showed is this passage.
13 Correct?

14 A. I did refer to that, yes.

15 Q. And counsel for Defendant suggested to the ladies and
16 gentlemen of the jury that this critical part of the passage
17 was obscured on the slide. Correct?

18 A. I believe that statement was made, yes.

19 Q. And counsel made a lot of circles. Correct? Do you
20 remember that?

21 A. I constructed a lot of circles on my demonstrative.

22 Q. And in figure 12 above in the previous paragraph, it
23 talks about a block diagram that has both a non-volatile
24 memory subsystem and a volatile memory subsystem on it.
25 Correct?

1 A. That paragraph does discuss that, yes.

2 Q. And we can see that here. Correct?

3 A. Yes, correct.

4 Q. And that non-volatile memory subsystem is 1040. Correct?

5 A. 1040 is the non-volatile. That's correct.

6 Q. And this passage in the paragraph below that I marked in
7 blue talks about SODIMMs, UDIMMs, and RDIMMs. Correct?

8 A. What you've underlined refers to that, yes.

9 Q. And the products are SODIMMs UDIMMs and RDIMMs. Correct?

10 A. They do have that form factor, yes.

11 Q. And that passage makes no reference whatsoever to
12 1040--correct?--in that paragraph.

13 A. Well, the paragraph you're pointing to doesn't reference
14 1040.

15 Q. You spoke about the technical contribution of DDR5 and
16 the patents at issue in this case. Correct?

17 A. I did.

18 Q. And just to set some boundaries, you have some experience
19 in DDR5 -- DDR4 chip design. Correct?

20 A. In chip design, yes.

21 Q. And you don't agree -- you don't believe the DDR4 and
22 DDR5 market is completely different. Correct?

23 DR. ALBERT: Objection, Your Honor.

24 THE COURT: Approach the bench, counsel.

25 (The following was had outside the hearing of the

1 jury.)

2 DR. ALBERT: We're getting into areas of market and
3 marketing. He's not offered as an expert in marketing. He
4 didn't talk about the markets for DDR4 or DDR5 in any way in
5 his expert report or his direct. This is well outside the
6 scope.

7 THE COURT: All right.

8 DR. ALBERT: He's trying to back --

9 THE COURT: What's your response, Mr. Sheasby?

10 MR. SHEASBY: Your Honor, he was talking about other
11 things that contributed to speed and power. He talked about
12 it extensively in his report about the value of these patents
13 and what drives that value. And I am examining him on the
14 fact that speed is a critical aspect of driving value, and he
15 is going to testify that speed is a critical aspect, which is
16 our damages theory in this case.

17 THE COURT: We're not going to go into the
18 conditions of the market and things like that.

19 MR. SHEASBY: I understand, Your Honor. Thank you.

20 THE COURT: Wait a minute. How much longer do you
21 have on this cross? Because it's already 12:15.

22 MR. SHEASBY: I think now would be a natural time to
23 break as soon as I finish this damages model.

24 THE COURT: My question to you is how much time you
25 have.

1 MR. SHEASBY: Thirty minutes at least.

2 THE COURT: And you'll have redirect, I assume.

3 DR. ALBERT: Very short, Your Honor. But the damage
4 model shouldn't be going into this market study anymore. As I
5 understand the Court's order, it is irrelevant and we should
6 be moving on from that. I don't want to have to keep coming
7 up and objecting to every question.

8 THE COURT: I assume you are not going to go further
9 into the market.

10 MR. SHEASBY: I'm going to ask him the question
11 whether speed and density are two of the primary drivers for
12 DRAM, which is a subject that was in his report.

13 DR. ALBERT: Absolutely not, Your Honor. And this
14 is exactly getting into what the customers are going to buy
15 something, why they buy it, what price they paid. He has not
16 gotten into any of that at all. It has nothing to do with his
17 testimony.

18 THE COURT: Tell me how this relates to either one
19 of his invalidity theories or his infringement theories.

20 DR. ALBERT: It doesn't.

21 THE COURT: I'm asking Mr. Sheasby that, Mr. Albert.

22 MR. SHEASBY: Sure. At the end of his presentation,
23 he gave a discussion of market valuation and were there other
24 ways to achieve this valuation, and what's the impact of
25 having the power management on module. He expressly engaged

1 the issues of damages --

2 THE COURT: At the end of his direct?

3 MR. SHEASBY: At the end of his direct, Your Honor.

4 THE COURT: I'll look at it over the lunch break and
5 I'll let you know before we come back. We're going to break
6 now.

7 MR. SHEASBY: Thank you.

8 (The following was had in the presence and hearing
9 of the jury.)

10 THE COURT: Ladies and gentlemen, we're going to
11 break for lunch. It's already a little bit after 12:00 and
12 there's more examination of this witness to go and I'm not
13 going to keep you from your lunch any longer.

14 If you will, take your notebooks with you, follow all my
15 instructions, and we'll try to be back close to or shortly
16 after 1:00.

17 With that, the jury's excused for lunch.

18 (Whereupon, the jury left the courtroom.)

19 THE COURT: The Court stands in recess for lunch.

20 (Lunch recess.)

21 THE COURT: Be seated, please.

22 Counsel, earlier today we met in chambers and reviewed
23 disputed demonstrative slides. We got almost all the way
24 through the slides for Mr. Robins. I carried the last three
25 disputed slides, which are DDX 4-68, 69, and 4-70. I told you

1 I'd give you guidance on that before Mr. Robins is called.

2 Here's what I've decided. I'm going to permit the
3 slides. But at the time they are used or more likely when the
4 slides begin to be presented, that say, as they're titled, the
5 '060 and '160 Patents make no technical contribution. I'm
6 going to tell the jury that whether or not they make a
7 technical contribution or not does not relate to the issue of
8 infringement or validity but can only be considered for the
9 issue of damages.

10 And with that instruction of the Court, those last three
11 slides are permitted as they are.

12 That's the carry-over ruling on Robins.

13 I'm prepared to give you guidance on the outstanding
14 demonstrative disputes regarding Mr. Kline. I'll do that at
15 the next recess. We've certainly got more testimony than we
16 need to get to the next recess.

17 I'm going to strike the testimony of Mr. Halbert
18 completely. He is effectively a JEDEC witness. JEDEC is not
19 a part of the testimony before this jury, and I see no
20 probative reason where the prejudice is outweighed by the
21 probative effect of any of his testimony. He's not going to
22 testify.

23 I'll also give you guidance on Mr. Meyer over the next
24 recess.

25 Also, with regard to Mr. McAlexander, we had a conference

1 at the bench just before we broke for lunch about Plaintiff's
2 counsel going into market conditions and analysis with regard
3 to Mr. McAlexander. The argument was from the Plaintiff that
4 that had been covered earlier and the door had been opened.

5 The only thing I can find earlier is with regard to the
6 issue of speed, and I'll allow Mr. Sheasby to examine this
7 witness with regard to the concept of speed, but not
8 marketability, not density, not any of the other matters that
9 we discussed at the bench.

10 And I believe that's all the guidance that's needed
11 before we continue with this witness.

12 Are there any questions from counsel before I bring the
13 jury in?

14 MR. SHEASBY: Nothing for Plaintiff, Your Honor.

15 MR. CORDELL: Nothing from the Defendants, Your
16 Honor. Thank you.

17 THE COURT: Let's bring the jury in, please.

18 (Whereupon, the jury entered the courtroom.)

19 THE COURT: Welcome back, ladies and gentlemen.
20 Please have a seat.

21 I know you think I can't tell time. I told you shortly
22 before 1:00, and it's 1:30. I promise you we've been busy and
23 been using that time effectively.

24 We're going to continue with the cross-examination of Mr.
25 Joseph McAlexander by the Plaintiff.

1 Mr. Sheasby, you may continue.

2 MR. SHEASBY: Thank you, Your Honor.

3 Good afternoon, ladies and gentlemen.

4 Q. (BY MR. SHEASBY) Good afternoon, Mr. McAlexander.

5 A. Good afternoon.

6 Q. The next issue I wanted to speak about is the concept of
7 on-module power management. And you agree that the DDR5
8 patents employ -- strike that.

9 You agree that Samsung's DDR5 products employ on-module
10 power management. Correct?

11 A. Correct.

12 Q. And if we go to slide 5.34, you were in the courtroom
13 when Mr. Milton testified about the presentation that Samsung
14 received in 2014 about on-module power distribution. Correct?

15 A. Yes, I was.

16 Q. And this was not an issue that you were asked to analyze
17 by Samsung. Correct?

18 A. Correct.

19 Q. The relationship between the presentations made by
20 Netlist and the technology that's accused of infringement in
21 this case. Correct?

22 A. I have not presented that testimony here today.

23 Q. And you were not asked to analyze it. Correct?

24 A. Correct.

25 THE COURT: Gentlemen, let me ask you both to make

1 sure we hear you. You're both a little soft, and we've all
2 had lunch and I want everybody to be engaged and pay
3 attention. So let's make sure you're both heard.

4 Go ahead.

5 Q. (BY MR. SHEASBY) The issue I wanted to discuss with you,
6 Mr. McAlexander, with your permission is the '339 Patent.

7 MR. SHEASBY: And let's go to 5.52.

8 Q. (BY MR. SHEASBY) So the '339 Patent, there are three
9 claims accused of infringement. There's claim 1, 8, and 9.
10 Is that correct?

11 A. That is correct.

12 Q. And the issue that you think is not satisfied by these
13 claims is the idea of whether there is driving of a byte-wise
14 section. Is that correct?

15 A. Yes, that's correct.

16 Q. And just to orient ourselves, you think every other
17 limitation of the claims are satisfied. Correct?

18 A. I did not address any of the others. That's correct.

19 Q. And to be as precise as possible, you don't dispute that
20 any other limitations of this claim is satisfied except for
21 what I've highlighted. Fair?

22 A. Fair. I focused on that particular highlighted section.

23 Q. And I have some slides that you made, and this is a slide
24 that you showed. And I want to speak to you briefly about the
25 orientation.

1 This is a data buffer, and there are going to be multiple
2 of these data buffers on the LRDIMM modules that are accused
3 of infringement in this case. Is that correct?

4 A. That is correct.

5 Q. And, in particular, you described there as being a data
6 path right there. Correct?

7 A. I did, yes.

8 Q. And you said that was one data path. Correct?

9 A. One byte-wise data path.

10 Q. And the -- you said the patent requires there to be
11 a -- and I think I wrote it down. Let me get it precise. The
12 patent requires there to be multiple byte-wise data paths on
13 each buffer. Correct?

14 A. I believe that's what the claim says, yes.

15 Q. And if we go to this figure, you drew the line on the
16 right-hand side for the ladies and gentlemen of the jury.
17 Correct? You depicted that as a data path where I've drawn
18 the write line. Correct?

19 A. Yes, that's correct.

20 Q. There's another data path on this buffer. Correct?

21 A. Where would you like to point to?

22 Q. Sir, is there another data path on this buffer?

23 A. For the write, I'm not aware of another data path.

24 MR. SHEASBY: Move to strike as non-responsive, Your
25 Honor.

1 THE COURT: Overruled.

2 Q. (BY MR. SHEASBY) The write, there's a data path. Is
3 that correct?

4 A. Yes, that is correct.

5 Q. There is also a read data path in the same buffer.
6 Correct?

7 A. Yes, the opposite direction.

8 THE COURT: Let me just stop you, Counsel, and I
9 meant to do this earlier. There's been a lot of testimony
10 about write. That's W-R-I-T-E, not R-I-G-H-T.

11 MR. SHEASBY: Yes, Your Honor.

12 THE COURT: I'm not sure between what's written and
13 what's been verbalized that there's a lack of confusion. So I
14 just want to make it clear when you say write, we're talking
15 about W-R-I-T-E.

16 MR. SHEASBY: And that's correct, Your Honor.

17 Q. (BY MR. SHEASBY) And so just -- here is the memory.
18 Correct?

19 A. Where you've drawn would be memory that the input would
20 drive information to, yes.

21 Q. And there is the -- here is the system or the memory
22 controller down there. Is that correct?

23 A. In a system, yes, that would be correct.

24 Q. And there is two data paths on the buffer, a read data
25 path and a write data path. Is that correct?

1 A. There is a read path and a write path, yes.

2 Q. And they are both data paths. Is that correct?

3 A. Sure.

4 Q. And the -- on each of those read and write data paths,
5 there's tri-state buffers. Correct?

6 A. Yes.

7 Q. And when the tri-state buffers are off, you can't
8 transfer data on the path. Correct?

9 A. Generally speaking, yes, that's correct.

10 Q. And when the TX and RX are enabled, you will be able to
11 transfer data on the path. Is that correct?

12 A. When one of the other is enabled, yes.

13 Q. And the read and write data paths can be turned off and
14 on separately. Correct?

15 A. One would hope so.

16 Q. The transaction control circuitry will determine -- a
17 transaction control circuitry is right over there. Is that
18 correct?

19 A. Yes, that's correct.

20 Q. The transaction control circuitry will determine whether
21 the read path or the write path is enabled. Correct?

22 A. That is correct.

23 Q. And either the read path or the write path will be on.
24 They will not be both on simultaneously. Correct?

25 A. That is correct.

1 Q. And if we look at the claim's construction -- the Court's
2 construction, the Court's instruction doesn't say enabling
3 only one of the write data paths while the other possible
4 write data paths are disabled. Correct?

5 A. No, it doesn't say that.

6 Q. And it doesn't say enabling only one of the read data
7 paths while the other possible paths are disabled. Correct?

8 A. Those words are not there.

9 Q. And we agree that there's at least two data paths on the
10 buffer--a read data path and a write data path. Correct?

11 A. Opposing directions, yes.

12 Q. And the Court's construction says nothing about opposing
13 directions. Correct?

14 A. No.

15 Q. Does the Court's construction use the phrase 'opposing
16 directions'?

17 A. No.

18 Q. And to be clear, this is the sole non-infringement
19 defense that you've raised as to the '339 Patent. Is that
20 correct, sir?

21 A. In terms of the one path, yes.

22 Q. And so if the jury concludes that there are more than one
23 path -- path data paths -- well, I'll leave it at that.

24 The next issue I'd like to discuss is the history of this
25 data path.

1 Did you -- you reviewed the patent specification. Is
2 that correct?

3 A. If you're referring to the '339, the answer is yes.

4 Q. And the -- for both the '598 -- the '918 and '056 [sic],
5 which are the ones we read previously, and the '339, you don't
6 dispute that there is any material difference from the
7 original parent application as it was filed and what's in the
8 application that's going to be in front of the ladies and
9 gentlemen of the jury. Correct?

10 A. That would apply to each one of those, and it's DDR5
11 '054, not the '056.

12 Q. You're absolutely right. I'll re-ask the question.

13 For the '918, the '054, and the '339, you don't dispute
14 that the specification in each of those patents that are in
15 the ladies and gentlemen of the jury's notebook are materially
16 the same as was filed either in 2008 or in 2009.

17 A. As it applies to each one of them, yes.

18 Q. Great. And so this is not a situation in which Netlist
19 slipped in material, additional information, that makes a
20 difference for the purpose of written description. Correct?

21 A. Not for the purposes of written description for the
22 matters in this case.

23 Q. Great. Thank you so much.

24 And one of the -- the product at issue that's accused of
25 infringing this case is LRDIMM. Is that correct?

1 A. Correct.

2 MR. SHEASBY: And if we go to PDX 5.54.

3 Q. (BY MR. SHEASBY) You were here in -- for the examination
4 of Mr. Milton?

5 A. Yes, I was.

6 Q. Mr. Milton is the vice president of research and
7 engineering at Netlist?

8 A. Correct.

9 Q. And he spoke about these materials that were obtained
10 from Samsung. Do you remember that?

11 A. I recall, yes.

12 Q. And this is Samsung saying that Netlist, Inc., is known
13 in the industry as a company that created a system for LRDIMM
14 technology. Correct?

15 A. Yes.

16 Q. And the LRDIMM is what's accused of infringement in this
17 case. Is that correct?

18 A. The Samsung LRDIMM, yes.

19 Q. And did you investigate the relationship between the
20 disclosures that were made by Netlist a number of years ago
21 and the creation of the LRDIMM products that are at issue in
22 this case?

23 A. I don't recall reviewing in detail those documentation,
24 no.

25 Q. Did Samsung ask you to investigate the relationship

1 between the materials that Netlist -- that Samsung accessed
2 leading up to and through the joint development agreement and
3 the products that were ultimately launched?

4 DR. ALBERT: Objection, Your Honor. This is outside
5 the scope of direct. It isn't relevant to any issue
6 whatsoever.

7 THE COURT: Overruled.

8 Q. (BY MR. SHEASBY) Should I ask the question again?

9 A. Please do.

10 Q. Absolutely, Mr. McAlexander.

11 DR. ALBERT: This also addresses the MIL on alleged
12 copying, Your Honor.

13 THE COURT: I don't see that it violates MIL No. 9.
14 Overruled.

15 Re-ask the question, please, Counsel.

16 Q. (BY MR. SHEASBY) Did Samsung, the company you worked 10
17 to 12 times for, ask you to investigate the relationship
18 between the materials that Samsung accessed as part of the
19 JDLA and the products that are accused of infringement in this
20 case?

21 A. As I sit here, I don't recall having gone into that
22 detail.

23 Q. Okay. Now, the next issue that you speak about with
24 regards to the '339 Patent is invalidity. Is that correct?

25 A. Say that again, please?

1 Q. Is invalidity. Is that correct?

2 A. That is correct.

3 Q. And just to level-set for the ladies and gentlemen of the
4 jury, it is Samsung's burden to prove invalidity of the '339
5 Patent by clear and convincing evidence. Correct?

6 A. Correct.

7 Q. And, once again, if we go to slide 6.65 -- sorry 5.65,
8 this is JTX 43, Netlist told the Patent Office that its
9 claims, it believes, were supported by that original
10 application in 2009. Correct?

11 A. That is correct.

12 Q. And the Patent Office in its regular course of dealing
13 would have investigated that. Correct?

14 A. I would say that they were supposed to, yes.

15 Q. Okay. And what you claim is that the combination of
16 something known as QBM and another patent called Ellsberry
17 would render obvious the '339. Is that correct?

18 A. Yes, that's correct.

19 Q. And to set that back, you don't identify any actual
20 product or publication anywhere in the world that anticipates
21 each claim of each element of the '339 Patent. Correct?

22 A. Not as applied to anticipation, correct.

23 Q. Right. In other words, what you do is you're combining
24 things together. Correct?

25 A. Correct.

1 Q. And so this QBM reference that you're using, you're
2 combining it with information from another reference called
3 Ellsberry. Correct?

4 A. Certainly, yes.

5 Q. And you're not saying for the ladies and gentlemen of the
6 jury that what has -- what is in the '339 Patent had been done
7 before by Kentron, the company who presented QBM. Correct?

8 A. I'm representing an obvious combination. No.

9 Q. The answer to my question was no.

10 A. No.

11 Q. And so just so there's no confusion, you are -- it is not
12 the case that the idea of a '339 Patent had been done before
13 by Kentron using QBM. Correct?

14 A. That was not my representation.

15 Q. Were you here in opening, sir?

16 A. I was.

17 Q. Were you here in opening when Samsung's counsel said,
18 quote, and it turns out that the idea of the '339 Patent had
19 been done by a company before called Kentron?

20 A. Yes.

21 Q. And that's not the position you're taking. Correct?

22 A. It's not the position I'm taking based on my -- to my
23 expert report. That's correct.

24 Q. And I want to talk about Kentron. So Kentron had
25 something called a QBM. Correct?

1 A. That is correct.

2 Q. And you showed a slide where you suggested that Kentron
3 had provided QBM information to other companies. Correct?

4 A. Yes, that's correct.

5 Q. But the reality is that you're not aware of any company
6 anywhere in the world that accepted towards commercialization
7 the QBM product. Correct?

8 A. That is correct. It was not -- not for
9 commercialization.

10 Q. And none of the people that QBM attempted to sell its
11 designs to ever accepted them or used them in a commercial
12 product. Correct?

13 A. That is the end result of it, yes.

14 Q. And you are not aware of anyone in history who ever
15 combined this QBM product with the Ellsberry patent to create
16 a product covered by the '339 Patent. Correct?

17 A. I'm not aware of anyone combining them, no.

18 Q. In other words, as far as you know, no one in the world
19 has ever made the combination between Ellsberry and QBM that
20 you've proposed in this case. Correct?

21 A. I'm not aware of someone having combined them.

22 Q. And I want to show you something. We talked about the
23 prosecution history. One of the references that you cite is
24 the Ellsberry reference. Correct?

25 A. That is correct.

1 Q. And the Ellsberry reference was actually considered by
2 the Patent Office. Correct?

3 A. Yes, it sure was.

4 DR. ALBERT: May we approach, Your Honor?

5 THE COURT: Approach the bench.

6 (The following was had outside the hearing of the
7 jury.)

8 DR. ALBERT: We're getting into prosecution history,
9 Your Honor. This is issues that were dealt with in claim
10 construction. If we're going to redelve into that issue, Your
11 Honor, I think it completely would open the door to the
12 reasoning behind the claim construction where Netlist's
13 proposed construction without a fork in the road was rejected
14 by the Court. So I just fear that we were getting really
15 close to the line here.

16 THE COURT: We are not getting into claim
17 construction.

18 MR. SHEASBY: No, Your Honor. I'm simply
19 establishing that both these references were before the PTO.

20 THE COURT: All right. I assume you're ready to
21 move on to something else?

22 MR. SHEASBY: Well, I'm going to show the second
23 reference was before the PTO. I believe it is relevant for
24 obviousness, Your Honor.

25 THE COURT: I don't hear an objection to that and I

1 don't have an objection to that. You're obviously not going
2 to reopen the door to claim construction.

3 But at this point Doctor Albert, I don't find that's what
4 the Plaintiff's counsel is doing.

5 DR. ALBERT: Okay. Your Honor, I do think that this
6 line of questioning is improper. We're getting into
7 individual references, so we object to these individual
8 statements of what was considered and what was not if they're
9 not going to get into the combination --

10 THE COURT: You can certainly return to it on
11 redirect and establish that it is a combination and they're
12 not to be looked at in a vacuum by themselves. That's
13 certainly proper for you to address on redirect. But I'm not
14 going to -- I'm not going to micromanage Plaintiff's attempt
15 to talk about the elements or the references that your
16 invalidity defense combines.

17 DR. ALBERT: Thank you.

18 MR. SHEASBY: Thank you, Your Honor.

19 THE COURT: Let's proceed.

20 (The following was had in the presence and hearing
21 of the jury.)

22 THE COURT: All right. Let's proceed.

23 Q. (BY MR. SHEASBY) This is the prosecution record which is
24 the history of the examination of the patent by the Patent
25 Office. Correct?

1 A. Yes. It appears to be a representation from an
2 interview.

3 Q. And in this interview, the examiner spoke about why
4 Ellsberry did not disclose or teach activating or deactivating
5 data lines. Correct?

6 A. Correct.

7 Q. And the other reference you cite is a reference called
8 QBM. Correct?

9 A. Yes, that's correct.

10 Q. And if you'd turn to your binder, your deposition binder,
11 if you would. And it's the third tab in the binder. It has a
12 date on it of 1/20/2023.

13 Let me know when you're there.

14 A. All right, sir. I have it.

15 Q. And I'm not trying to be pushy so please don't take it
16 this way, but I would like you to read it to yourself and not
17 read it out loud. Can we agree to that?

18 A. Certainly.

19 Q. Okay. And I'd like you to turn to page 45, lines 24 to
20 46, line 2.

21 A. I've got page 45. The lines again, please?

22 Q. Sure. Lines 24 to 46, line 2. So it's just that
23 little -- it's like five or six lines, maybe three lines.
24 It's from 45, lines 24, to 46:2.

25 A. Yes, I've read it.

1 Q. Okay. Does this refresh your recollection that someone
2 named Chris Karabatsos was the employee responsible for
3 designing QBM?

4 A. Yes.

5 Q. And this is another page from the prosecution history for
6 the '339 Patent. Do you see that?

7 A. I see that.

8 Q. And if you look at it, it says that the -- one of the
9 references that was considered was Karabatsos. Do you see
10 that?

11 THE COURT: Just a minute.

12 DR. ALBERT: Objection, Your Honor. What's
13 happening at the Patent Office is not relevant to this.

14 THE COURT: He's entitled to show the reference was
15 before the Patent Office. I assume that's what he's doing and
16 that he's not going to go further than that.

17 MR. SHEASBY: Absolutely, Your Honor. That's all
18 I'm doing.

19 THE COURT: Let's proceed.

20 Q. (BY MR. SHEASBY) So if you look at 6,446,158, you see
21 that the Patent Office -- that was a patent that the Patent
22 Office considered to Karabatsos. Do you see that?

23 A. I see that.

24 Q. And you can actually find that patent in your binder.
25 It's tab 53. Oh, it's a good point. There are two binders.

1 It will be in your second binder.

2 A. Supposedly the one that has tab 53 on it.

3 Q. Yes, sir. One can only hope. And it is called other
4 documents for cross-examination of Joseph McAlexander.

5 A. I have it.

6 Q. Okay. And it's tab 53.

7 A. I have it.

8 Q. And that patent, 6,446,158, the inventor is Chris
9 Karabatsos. Correct?

10 A. Yes, that is correct.

11 Q. And he's the person who identified -- designed QBM.
12 Correct?

13 A. According to the other information, yes.

14 Q. And so both the Karabatsos patent and Ellsberry were
15 before the Patent Office. Correct?

16 A. This Karabatsos patent was before the Patent Office, yes.

17 Q. And did you examine this patent to see it as anything
18 different at all from what is in the QBM references that you
19 provided?

20 A. I recall having looked at this patent. It specifically
21 addresses a part of it but certainly not all of what is in
22 QBM.

23 Q. I understand that. For the purposes of your analysis,
24 does the Karabatsos patent, did you testify on direct to the
25 ladies and gentlemen of the jury that it's a material

1 difference for the purposes that you are using it?

2 A. I didn't testify about the patent.

3 Q. Okay. The -- one of the issues that I'd like to discuss
4 now is damages and the technical evaluation that you performed
5 with damages.

6 May I do so?

7 A. Yes. Are we finished with this binder?

8 Q. We are.

9 A. Thank you.

10 THE COURT: Go ahead, Counsel.

11 Q. (BY MR. SHEASBY) And one of the issues you examined was
12 whether the -- an agreement with Rambus was comparable.
13 Correct?

14 A. Yes, that's correct.

15 Q. And in that Rambus agreement, you actually looked
16 at -- if you go to pages 687 to 706 of your rebuttal report,
17 which is behind you, if you want to look at that.

18 A. The page numbers again, please?

19 Q. It's paragraph Nos. 565 to 566, and I'll find the page
20 numbers, if you give me one moment, sir. Oh, it's paragraph
21 699. Excuse me. And the page number would be --

22 A. 559.

23 Q. 559. And just to level-set, one of the things you did in
24 those paragraphs beginning at 687 and going on into the 700s
25 was you looked at a number of patents that were in the Rambus

1 agreement. Correct?

2 A. Yes.

3 Q. And you -- the patents that you analyzed in those
4 paragraphs, you all concluded were comparable to the patents
5 at issue in this case. Correct? Technologically.

6 A. There were certain ones that I said were technically
7 comparable, yes.

8 Q. In fact, I think you went through eight or ten of them,
9 and each of them you said was technically comparable. Fair?

10 A. I don't remember the exact number. But as a whole, I
11 said they were not, but I did identify some that were.

12 MR. SHEASBY: So I move to strike as non-responsive,
13 Your Honor.

14 THE COURT: Overruled.

15 Q. (BY MR. SHEASBY) In paragraph 690 --

16 MR. SHEASBY: Thank you, Your Honor.

17 Q. (BY MR. SHEASBY) -- you discuss Rambus patents that you
18 think are comparable to the patents-in-suit. Correct?

19 A. Did I hear you say 690?

20 Q. Yes. It's on page 553.

21 A. I'm there. I see that I examine in paragraph 690 several
22 patents, said they relate to DRAM memory module application.

23 Q. And DRAM memory modules are one of the issues in this
24 case. Correct?

25 A. Yes, that's correct.

1 Q. And then in paragraph 692, you identify another three
2 patents that you describe as relating to technologies in this
3 case. Correct?

4 A. That is correct.

5 Q. And then in paragraph 693 -- sorry, in paragraph 694, you
6 point to another three patents that you think are relevant to
7 the technologies at issue in this case. Correct?

8 A. Yes, that's correct.

9 Q. And in 696, you point to another Rambus patent that you
10 think is technologically comparable to the patents in this
11 case?

12 A. I don't see that I said technical comparability here. I
13 said that it was described in paragraph 696 what the patent
14 was related to.

15 Q. And I think in 697, you discuss about how it relates to
16 technologies that are also at issue in this case, voltage
17 regulator module. Correct?

18 A. I just -- well, again, I express what's in figure 1.

19 Q. And then in paragraph 699, you talk about certain Rambus
20 patents relate to the use of stacked dies. Correct?

21 A. In 699, that is correct.

22 Q. And you said you think that's relevant to Netlist's
23 accusations with respect to the '060 and '160 Patents.
24 Correct?

25 A. That is correct.

1 Q. And just to level-set, you actually don't opine that
2 there is a single one of the Rambus patents that has ever been
3 used by Samsung. Correct?

4 A. Say that again, please?

5 Q. Sure. You do not opine that a single one of the Rambus
6 patents has ever been used by Samsung. Correct?

7 A. I don't opine on that, no.

8 Q. You heard yesterday that Mr. Kennedy said that the
9 payment was up to \$1.1 billion. Do you remember that?

10 A. The what was up to?

11 Q. The payment was as much as \$1.1 billion?

12 A. I do not recall, to be honest.

13 Q. But to be clear for the ladies and gentlemen of the jury,
14 no evidence that Samsung is actually using any of the Rambus
15 patents presented by you. Correct?

16 A. I was not asked to evaluate the use. I was asked to
17 evaluate whether or not there was a license agreement.

18 Q. Understood. One of the patents you point to you thought
19 was at least technologically in the same sphere as the '060
20 and '160 Patents were those in -- was that in paragraph 699.
21 Correct?

22 A. I did identify two patents that related to, yes.

23 Q. And you describe it as relating to through-silicon vias.
24 Correct?

25 A. That is correct.

1 Q. And to stacked DRAM. Correct?

2 A. Stacked die packaging with alignment.

3 Q. And just so we can understand that distinction -- one
4 moment.

5 MR. SHEASBY: Why don't we pull up JTX 0031 at 3.
6 And let's blow that up.

7 DR. ALBERT: Your Honor, outside the scope.

8 MR. SHEASBY: Your Honor, I -- I disagree.

9 THE COURT: Just a minute. Restate your objection,
10 Doctor Albert.

11 DR. ALBERT: This is outside the scope. He's
12 getting into particular --

13 THE COURT: Again, you're going to have to clarify.
14 Are you telling me it's outside the scope of this expert's
15 report or are you telling me under Rule 611, it's outside the
16 scope of direct examination?

17 DR. ALBERT: Both, Your Honor.

18 THE COURT: Well, please try to be clear on which or
19 both you are raising in your objections going forward.

20 DR. ALBERT: Thank you, Your Honor.

21 It's outside the scope of his report and his direct
22 examination. We didn't go into anything regarding these
23 stacked die wire bonding that Mr. Sheasby is going into with
24 this document or this document itself.

25 MR. SHEASBY: So I disagree, Your Honor. Mr.

1 McAlexander said that he didn't feel the Rambus agreement was
2 technologically comparable. He discussed the exact subject
3 matter of TSV technology that is in the Rambus patents in his
4 report. I'm going to use it to establish that in fact this
5 agreement was comparable because it relates to similar
6 technology.

7 DR. ALBERT: In his trial testimony, he didn't talk
8 about this at all.

9 MR. SHEASBY: Your Honor, he talked extensively
10 about Rambus in his trial testimony and explained why he felt
11 it wasn't comparable. I am going to use his report to
12 establish that he said something different.

13 DR. ALBERT: His testimony, Your Honor, was that
14 there were lots of memory patents, the Rambus portfolio was
15 vast, extensive. We did not talk about this at all in his
16 direct.

17 MR. SHEASBY: Your Honor, I'm examining him about
18 one of the patents that he expressly discussed in his report.

19 THE COURT: Let me see if I can narrow this dispute.

20 With regard to the objection that's outside the scope of
21 his direct examination, I'm going to overrule that objection.

22 So the question is not what did he testify to earlier;
23 it's what's covered by his report. And so if you believe
24 there's support for this in his report, Mr. Sheasby, show me
25 that.

1 MR. SHEASBY: May I approach, Your Honor?

2 THE COURT: Yes. Both of you approach.

3 (The following was had outside the hearing of the
4 jury.)

5 MR. SHEASBY: Paragraph 699.

6 DR. ALBERT: Infringement analysis, Your Honor, for
7 an expert to provide no infringement analysis for this type of
8 product, it's a different product. There's a whole 'nother
9 expert that's dealing with this, a whole 'nother expert is
10 providing testimony on. This is inherently confusing to the
11 jury to provide cross on a whole infringement issue that he
12 didn't deal with and another expert is going to.

13 THE COURT: What's the probative value of this, Mr.
14 Sheasby?

15 MR. SHEASBY: Yes, Your Honor. Yesterday Mr.
16 Cordell asked Mr. Kennedy and he said that there is no HBM
17 patents covered by the Rambus agreement. I have the quote. I
18 can get it from the examination yesterday.

19 And I intend to establish that there are in fact HBM
20 products covered by the Rambus agreement to show exactly why
21 it's technically comparable.

22 DR. ALBERT: That's not in the questioning so far,
23 Your Honor. He is getting into whether the '060 and '160 are
24 going to cover particular types of technologies, and it's an
25 infringement analysis -- a back-door infringement analysis for

1 an expert that didn't do that and has nothing to do with
2 infringement issues in this case.

3 THE COURT: You're going to need to limit your
4 cross-examination to the comparability issue and you need to
5 tailor it. It's too broad right now.

6 MR. SHEASBY: I understand, Your Honor.

7 THE COURT: All right. I'll allow you to go
8 forward.

9 You can reurge your objection if this gets outside that
10 tailored approach. All right?

11 (The following was had in the presence and hearing
12 of the jury.)

13 THE COURT: As directed at the bench, I'll sustain
14 the objection.

15 You can go forward pursuant to my instructions, Mr.
16 Sheasby.

17 MR. SHEASBY: Yes, Your Honor.

18 Q. (BY MR. SHEASBY) So if you look at paragraph 699 of your
19 report, what you talk is about you talked about a design that
20 has TSVs. Correct?

21 DR. ALBERT: Your Honor, we're getting into the same
22 issues here. We are no longer talking about what you
23 instructed us to talk about. Literally doing --

24 THE COURT: How does this relate to comparability,
25 Mr. Sheasby?

1 MR. SHEASBY: Sure. I am establishing that the
2 design that was paid for by Rambus is a design that uses TSVs
3 does not use wire bonding, can be used in HBM, was not used,
4 which goes to the fact that the reason why they licensed this
5 patent from Rambus and subsequently had to use our patent is
6 because our technology worked, not Rambus'.

7 THE COURT: All right. Do you have anything
8 further, Doctor Albert?

9 DR. ALBERT: Again, we're just getting into the same
10 issues I addressed before, Your Honor. This is not a damages
11 issue. They're trying to get into other issues that Mr.
12 McAlexander didn't address in his direct report.

13 THE COURT: All right. I'm going to overrule the
14 objection. Let's proceed.

15 MR. SHEASBY: Thank you, Your Honor.

16 Q. (BY MR. SHEASBY) So the patent that you referred to in
17 the '060 Patent uses through-silicon vias. Correct?

18 A. Yes, it does.

19 Q. And you've seen as an alternative using through-silicon
20 vias, you can use a DRAM circuit like that on the left-hand
21 side. Correct?

22 A. There are such structures.

23 Q. And you've seen DRAM circuits like those on the left-hand
24 side. Correct?

25 A. Yes, I have.

1 Q. And those are not equivalent to through-silicon vias.

2 Correct?

3 A. They are different ways of connecting.

4 Q. They are not equivalent. Correct?

5 A. I'm not sure what the measure of equivalent is that
6 you're using.

7 Q. Sure. Why don't you go to 163, lines 4 through 8, of
8 your deposition?

9 A. The deposition?

10 Q. Yes. Yes, sir.

11 A. Okay. Okay. Page again?

12 Q. 163, lines 4 through 8.

13 A. Okay. On page 163.

14 Q. And actually go to lines 10 through 14.

15 A. 10 through 14. Okay.

16 Q. Yes. Through-silicon via is not equivalent to DRAM
17 circuits and it is an entirely different process. Correct?

18 DR. ALBERT: Object, Your Honor. These -- this type
19 of technology is not something Mr. McAlexander has testified
20 to today. It's not his opinions. There's another expert
21 that's dealing with these issues in front of the Court who
22 will be called later. Mr. McAlexander is not that person.

23 THE COURT: Well, let me see if I can be clear. I'm
24 persuaded that this witness was examined on direct by the
25 Defendant on the Rambus document and transaction, and I think

1 the door is opened as to Rambus and its comparability or lack
2 of comparability.

3 That doesn't mean, Mr. Sheasby, we can start with the
4 creation of the world and go forward. You do need to tailor
5 this to the Rambus document and what was said on direct about
6 it.

7 MR. SHEASBY: Sure.

8 Q. (BY MR. SHEASBY) And so the Rambus document proposes the
9 use of TSVs for a stacked technology. Correct?

10 A. It does propose that, yes.

11 Q. And that is not equivalent to DRAM circuits like those on
12 the left-hand side. Correct?

13 A. I'm not sure I understand your question about not
14 equivalent to. What's the basis of the equivalence? And I
15 will try to clarify it this way, if I may?

16 Q. Actually let me re-ask the question.

17 A. Okay.

18 Q. If you go to 163, lines 10 through 14?

19 A. Right.

20 Q. Does that refresh your recollection as to what you
21 testified to at deposition?

22 DR. ALBERT: Objection, Your Honor. This is again
23 getting into infringement allegations, infringement issues
24 regarding the '060 with Mr. McAlexander who is not offered an
25 opinion on, hasn't done the analysis. That's a whole 'nother

1 expert.

2 We are getting into the exact claim issues, claim terms,
3 that Mr. McAlexander has not provided an opinion on. It's
4 plainly not relevant to any issue that Mr. McAlexander has
5 offered in front of this jury.

6 MR. SHEASBY: Your Honor, he showed a design that
7 uses HBM -- that can be used for HBM, they used TSVs. He
8 explains why wire bonding cannot be used for that purpose. He
9 explains why a DRAM circuit cannot be used for that purpose.
10 And I intend to establish they licensed this from Rambus, it
11 was not a design that could work, and they subsequently used
12 our design. And I think that's a reasonable area of
13 investigation.

14 DR. ALBERT: Your Honor, this is a back-door
15 infringement analysis for an expert that has nothing to do
16 with any infringement issues in that part of the case.
17 There's a whole 'nother expert that has done the analysis, has
18 done the work, and has applied the Court's constructions to
19 the issues, and is offering his testimony.

20 This is infringement questions for an expert that has
21 nothing to do with that issue.

22 THE COURT: I've heard that objection for about the
23 fourth time, Doctor Albert. I'm not intending to let
24 Plaintiff's counsel pursue a back-door infringement argument
25 through Mr. McAlexander, but I believe the door's opened as to

1 the Rambus document and its comparability here.

2 And I've told Mr. Sheasby about three times he is casting
3 too wide a net and he needs to focus on that.

4 And you've told me about six times the same objection.

5 And I'm telling you both I'm going to allow an
6 examination as to the comparability of Rambus, but not
7 anything further. And I've told you all that and you keep
8 making the same disparate positions or objections and
9 responses to me.

10 I want this tailored to Rambus, and then you're going to
11 move on, Mr. Sheasby.

12 MR. SHEASBY: Sure.

13 THE COURT: Okay?

14 MR. SHEASBY: Sure.

15 Q. (BY MR. SHEASBY) The Rambus design that was licensed by
16 Samsung uses TSVs which you believe are not equivalent to DRAM
17 circuits. Correct?

18 A. In answer to your question, my statement was it's an
19 entirely different process.

20 Q. Thank you. And you don't testify that Rambus has used
21 the -- any of the patents -- that Samsung has used any of the
22 patents in the Rambus agreement. Correct?

23 A. I did not provide any testimony on that, no.

24 Q. And as to the patents that are accused of DDR5, you
25 didn't speak to any Samsung engineers to determine if there

1 were any alternatives that were available if the Netlist
2 technology was not used. Correct?

3 A. I did not communicate with any Samsung engineers.

4 Q. And you did not do an investigation as to whether there
5 was any acceptable non-infringing alternatives. Correct?

6 A. For this particular matter, no.

7 Q. And also as to the '339 Patent, you didn't communicate
8 with a single Samsung engineer. Correct?

9 A. That is correct.

10 Q. And you didn't do any investigation as to whether there
11 was any acceptable commercially-viable alternative to the '339
12 Patent. Correct?

13 A. I did not present that, no.

14 Q. Thank you.

15 MR. SHEASBY: Pass the witness, Your Honor.

16 THE COURT: Redirect by the Defendants?

17 DR. ALBERT: May we approach, Your Honor?

18 THE COURT: Yes.

19 (The following was had at the bench outside the
20 hearing of the jury:)

21 DR. ALBERT: We believe the cross-examination has
22 opened the door on the IPR for the '339 Patent. Questions in
23 cross-examination were about whether these patents -- the '339
24 was allowable at the IPR proceeding at the Patent Office over
25 Ellsberry, the exact same Ellsberry that's being asserted in

1 IPR now where the Patent Office is in fact reviewing it now
2 for patentability.

3 MR. SHEASBY: Your Honor, I did not discuss anything
4 other than the fact that they were previously before the
5 Patent Office.

6 THE COURT: Well, this objection is overruled. I
7 agree with the Plaintiff that it was relevant to show these
8 references were known to them, before the Patent Office. It
9 goes directly to the plausibility of the obviousness
10 combination that Defendants asserted. It does not in my view
11 open the door to the PTAB, IPR proceedings, or anything
12 further.

13 DR. ALBERT: Thank you, Your Honor.

14 MR. SHEASBY: Thank you.

15 THE COURT: Just a minute.

16 Mr. Sheasby, about -- and I'm counting -- about four or
17 five times when the Court's giving you direction from the
18 bench and in the presence of the jury, your response to me has
19 been, "Sure." That's not an appropriate response on the
20 record in front of a jury in a United States district court.
21 I don't want to hear "sure" in response to anything I tell you
22 the rest of the trial. Do you understand?

23 MR. SHEASBY: I understand. I apologize sincerely,
24 Your Honor.

25 THE COURT: All right. Do you have any idea how

1 long your redirect is going to go?

2 DR. ALBERT: It was short. I think it's probably
3 going to be about 15 minutes, I think right now.

4 THE COURT: You're both burning time you don't have
5 to burn.

6 All right. Let's proceed.

7 (The following was had in the presence and hearing
8 of the jury.)

9 THE COURT: All right. You may proceed with
10 redirect examination of the witness, Counsel.

11 REDIRECT EXAMINATION

12 BY DR. ALBERT:

13 Q. Good afternoon, Mr. McAlexander.

14 A. Good afternoon.

15 Q. You were asked a number of questions on cross about the
16 LDO versus the converter circuit. Do you recall those?

17 A. Yes, I do.

18 Q. Now, do the claims at issue for the '918 require a
19 converter or do they require a converter circuit?

20 A. The claims require a converter circuit. It's a
21 structure.

22 Q. So do the claims require the function of a converter or
23 do the claims require a converter circuit structure?

24 A. Well, there's a function that's implied, but it requires
25 a converter circuit structure, an apparatus.

1 Q. And is there anything that was shown to you that changes
2 your opinion about that the LDO is not the converter circuit
3 structure?

4 A. No, nothing changes my opinion. The LDO that's been
5 accused is a regulator, linear regulator. It is not a
6 converter circuit structure.

7 Q. And you were shown some testimony from a Bruce Lo. Do
8 you recall that?

9 A. Yes, I do.

10 Q. And I'll put this on the screen here. Starting with line
11 15: "Is an LDO a linear converter or a linear regulator?"

12 Answer, "Regulator."

13 Is that testimony that you saw presented here in court?

14 A. No.

15 Q. Well, it is testimony from the trial transcript.

16 A. From the trial transcript, yes. That has been presented,
17 that's correct.

18 Q. Is that consistent with your testimony that an LDO is a
19 linear regulator rather than a converter circuit?

20 A. That is consistent with my testimony, yes.

21 Q. And on cross, Mr. Sheasby did not show you this quote
22 from Mr. Lo's testimony. Is that correct?

23 A. No, he did not.

24 Q. Now, one way to slow down your car is to apply the brakes
25 to your car.

1 A. That is correct.

2 Q. Another way to slow down your car is if you're going down
3 the highway and you're in fifth gear, is to pop it in first
4 gear so slow down your car real quick. Right?

5 A. Yes. And it will launch you forward as well.

6 Q. Now, is a brake the same thing as the structure of a gear
7 shift just because both can slow you down?

8 A. No. I think it's rather clear that a braking system and
9 a transmission system are very distinctly different. They're
10 fundamentally different. They both convert, but that doesn't
11 mean, just because it converts energy in a way to stop the
12 car, that all of the sudden a transmission is a converter.
13 No, it's a transmission. It doesn't mean a braking system is
14 a converter. No, it's a braking system.

15 Q. Now, you were also asked a number of questions about
16 operable state. Do you recall those?

17 A. I do.

18 Q. And did you hear anything -- see any testimony that
19 changes your opinion on operable state?

20 A. No, none whatsoever.

21 Q. And what was your conclusion?

22 A. My conclusion is, consistent with the specification, the
23 only operable states that are described in -- in the patent
24 specification are those in which some power source is
25 providing some voltage or voltages to the memory devices, the

1 DRAM, and the flash memory that's on the module.

2 Q. Now, is the function of the memory module to allow access
3 to the DRAM memory on those DDR5 chips?

4 A. It is one aspect of it. But, no, the function of a
5 memory module is for data transactions, being able to
6 communicate with the rest of the system for dynamically
7 reviewing the -- getting information, displaying it to the
8 screen, saving the information to a hard drive later, but also
9 in accordance with the specifications, '918 and '054, is being
10 able to provide that back-up in case of some power glitch.

11 Q. And is the function of providing that access to data
12 available when the DRAMs are turned off?

13 A. No, not at all. When the DRAMs are turned off, the
14 module has no ability to communicate to the CPU. And the only
15 thing the PMIC does, the ones that were talked about before
16 this morning and afternoon, was the ability to have some
17 information there so when the computer is able to communicate
18 later with the module, it has a place -- it knows what the
19 configuration is. It's control configuration information.
20 But there's no data that's stored in the PMIC.

21 Q. And so what was your conclusion then about whether the
22 DRAM is being in an off state or the memory module was in an
23 operable state?

24 A. It's totally inoperable. There is no way for the memory
25 module to communicate according to its intended purpose.

1 Q. Now, you were asked a number of questions about the
2 particular claims of the '918 and '054 Patents and what they
3 covered. Do you recall those questions?

4 A. Yes, I do.

5 Q. Now, is there anything about any of those claims that
6 address the problem of the bottleneck between the DRAM and
7 this flash memory circuits?

8 A. No, none whatsoever.

9 Q. Anything about those claims that require a data transfer
10 between DRAM and flash on that flash DRAM hybrid module?

11 A. No, none.

12 Q. Those requirements are no longer in the claims?

13 A. No. Those requirements are not present in the claim, not
14 required.

15 Q. They used to be in the claims years ago, but not in the
16 claims now.

17 A. Well, they were in claims from prior predecessor patents
18 in that continuation chain, but with specifically the '918 and
19 the '054, no, none of those claims have those features
20 required.

21 Q. Any elements in the claims then that deal with that
22 problem to be solved of accessing the -- the data in the DRAM
23 and backing it up into non-volatile memory?

24 A. No, none whatsoever.

25 DR. ALBERT: Now, Mr. Sayers, can we bring JTX 3 at

1 27, starting at lines 14? The column 27. Sorry.

2 Q. (BY DR. ALBERT) And you were asked a number of questions
3 here about a cache back-up?

4 A. Yes, I see that.

5 Q. And where do you see that? Can you highlight that?

6 A. You're specifically referring to a cache back-up in your
7 question?

8 Q. I'll come back to this.

9 For the --

10 DR. ALBERT: We can take that down, Mr. Sayers.

11 Q. (BY DR. ALBERT) When you were asked the questions about
12 the cache back-up that -- that can take that data and put it
13 off the module onto a non-volatile storage unit off the
14 module, was there still a requirement that on the module you
15 would have that flash and DRAM components that transfer data
16 together?

17 A. According to the specification, yes. When the module is
18 created or used, it doesn't say it's going to sit in a vacuum
19 and not be able to communicate with the rest of the system,
20 because if you want to make a file, a large file back-up or
21 save, you're going to definitely have information data
22 transferred from the module DRAM to a larger data storage such
23 as an SSD or flash.

24 So it doesn't preclude that because that's the way
25 systems work. But that doesn't mean that you don't still have

1 to have, according to the written description, an on-board
2 flash and DRAM in that memory hybrid module.

3 Q. In fact, that ability to take data and move it off
4 through a CPU to some memory storage, that was what the patent
5 was talking about, that bottleneck problem that they wanted to
6 do something different?

7 A. That's correct. And the thing is if you have a memory
8 module that doesn't have data -- ability to store data on the
9 flash, on the module, then when you have a disruption in
10 power, you're going to lose that data, so you will not have
11 the ability to even transfer to an off-module storage
12 facility.

13 Q. And was there anything about the '918 Patent that you saw
14 that dealt with a system that did not have that combination of
15 flash and DRAM and transferring the data and backing up the
16 two?

17 A. I did not see any embodiment that said anything other
18 than that.

19 Q. Anything about the invention that was described in the
20 '918 and '054 Patent that didn't have that flash and that DRAM
21 and with the data transfer and the backing up?

22 A. No. In fact, I went through illustration after
23 illustration to show that in every case there is a requirement
24 for the purpose of the module having both DRAM and flash on
25 it.

1 Q. You were also asked specifically about claim 19 that
2 recites a non-volatile memory.

3 A. Correct.

4 Q. Do you recall that?

5 A. Yes.

6 Q. Now, for that type of memory in claim 19, was there
7 anything in claim 19 that required that flash chips and the
8 DRAM chips being on the module with the data transferring back
9 and forth and the back-up?

10 A. It's required -- it should be required, but, no, it
11 wasn't built into the claim as written.

12 Q. In other words, nothing in Doctor Mangione-Smith's
13 testimony when he did that claim-by-claim analysis on claim 19
14 to show that there was a flash and DRAM chip combination, the
15 data transferring back and forth with the back-up?

16 A. No, none of that.

17 Q. Now, just having a bit of memory on a PMIC, does that
18 address the problem to be solved that the '918 and '054
19 Patents disclosed?

20 A. No, not at all. The FPGA in the patent as well as the
21 ASIC, these have non-volatile memory bits in them. But that's
22 the storage of control information so that -- so that the
23 memory system, when it is brought up and powered up, the
24 computer's system through the CPU external to the memory
25 module can poll that module and determine what are you, who

1 are you, how are you configured. It's control configuration
2 information.

3 But the PMIC non-volatile memory cells that register has
4 zero ability to store any data that's associated with the
5 DRAM. In fact, there's no data line communication between the
6 two.

7 Q. So does that -- does little bits in the PMIC, can that
8 store those video files that the patent was talking about
9 switching back and forth between the DRAM and non-volatile
10 memory?

11 A. No. Impossible.

12 Q. And that few bits in the PMIC, can it store those web
13 files that the patent was talking about as the source of the
14 reason why they went out and got this invention?

15 A. No. Again, it's impossible to do that.

16 Q. Now, you were also asked some questions about the '339
17 and specifically data paths.

18 Now, claim -- the claim that's at issue in the '339
19 Patent we're talking about today is a write path. What does
20 that mean?

21 A. Yes. The claim is directed to a write path, and the
22 write path, and it's W-R-I-T-E, is specifically where the
23 computer system external to the module memory module, is to
24 place data at the pins of that module for the purposes of
25 buffering that data and then putting that data into the DRAM.

1 When you put data into a DRAM, that's called write. It's
2 a write mode. So the write path is and singularly that
3 direction from the outside pins through the data buffer to the
4 DRAM memory. That's the only path that's described and
5 claimed.

6 Q. And so the data that comes in on that write input and
7 then goes into the buffer, does -- will it just go all the way
8 through the buffer and go into the DRAMs?

9 A. Eventually it would. It will follow the path that has
10 been opened up, and it's the only path there is. So that
11 singular right path then permits the data that appears at the
12 input to the data buffer to be transferred through that single
13 path to the output of the data buffer and then to the memory
14 devices.

15 Q. The data on the write path doesn't come in, make a
16 U-turn, and then go into the read path?

17 A. No. When you have a write operation, that's a command
18 operation that everything from the CPU to the data buffer to
19 the DRAMs understand that this is information flow from the
20 external system through the data buffer to be stored on the
21 DRAMs. That's the only way it can go.

22 Q. So that data coming in on that write input, the read
23 direction is not a possible path.

24 A. No, it's not a possible path at all.

25 Q. And then the Court's claim construction refers to

1 multiple data paths. Is that right?

2 A. That is correct, yes.

3 Q. And enabling the selected data path.

4 A. That is correct.

5 Q. And disabling the other possible data paths.

6 A. That is correct.

7 Q. For that write data that comes in, is the read path a
8 possible data path?

9 A. Absolutely not.

10 Q. And so what is your opinion then on whether the DDR4
11 LRDIMM products practice the '339 claims?

12 A. The LRDIMM products that are accused, Samsung products,
13 do not practice this claim of the '339 Patent.

14 Q. And you were just asked some questions on some HBM
15 products.

16 A. Yes, I was.

17 Q. And are you -- have you offered testimony taking the '060
18 and '160 Patents, applying the Court's construction, and
19 determining whether any product actually practices the '060
20 and '160 Patents?

21 A. No, I was not asked to do that. There's another expert
22 that's going to handle that.

23 DR. ALBERT: Pass the witness.

24 THE COURT: Further cross-examination?

25 MR. SHEASBY: No, Your Honor.

1 THE COURT: You may step down, Mr. McAlexander.

2 THE WITNESS: Thank you, Your Honor.

3 THE COURT: You're welcome.

4 Counsel, approach the bench, please.

5 (The following was had outside the hearing of the
6 jury.)

7 THE COURT: According to my records, the Plaintiff
8 has 2 hours and 56 minutes remaining trial time, the Defendant
9 has 1 hour and 58 minutes remaining trial time. I would
10 suggest you both take control of your co-counsel and make sure
11 everything is done efficiently.

12 MR. CORDELL: We are. We are trying, Your Honor.
13 Thank you.

14 THE COURT: All right. You know, Mr. McAlexander
15 started at 8:30 this morning, and he just got off the bench at
16 2:30 this afternoon. And he was listed for three hours. So
17 you-all know what I know.

18 MR. SHEASBY: Yes, Your Honor.

19 THE COURT: All right. Who's your next witness?

20 MR. CORDELL: Doctor Robins, yes.

21 THE COURT: Let's proceed.

22 (The following was had in the presence and hearing
23 of the jury.)

24 THE COURT: Defendants, call your next witness,
25 please. Call your next witness, please.

1 MR. CORDELL: Thank you, Your Honor.

2 Samsung calls Doctor Robins, and his testimony will be
3 presented by my partner, Mr. Colvin.

4 THE COURT: All right. If you'll come forward,
5 Doctor Robins, and be sworn by the Courtroom Deputy.

6 (Whereupon, the oath was administered by the Clerk.)

7 THE COURT: You may go to the podium and prepare,
8 Mr. Colvin.

9 You can leave your suitcase at the table.

10 (Whereupon, the oath was administered by the Clerk.)

11 THE COURT: Please have a seat on the witness stand,
12 Doctor Robins.

13 THE WITNESS: Thank you, Your Honor.

14 THE COURT: If you'd like to pour yourself some
15 water, please feel free to do so.

16 Let's distribute binders.

17 All right, Counsel. You may proceed with direct
18 examination of the witness.

19 MR. COLVIN: Thank you, Your Honor.

20 GABRIEL ROBINS, Ph.D., SWORN,
21 testified under oath as follows:

22 DIRECT EXAMINATION

23 BY MR. COLVIN:

24 Q. Good afternoon, Doctor Robins. Would you please
25 introduce yourself to the jury?

1 A. Good afternoon. My name is Gabriel Robins.

2 Q. Where do you live, sir?

3 A. I live in Charlottesville, Virginia. I've been there 30
4 years.

5 Q. And what have you come to court to discuss?

6 A. I've come here to discuss Netlist's allegations of
7 infringement against the Samsung products.

8 Q. And can you summarize your opinions for the jury?

9 A. Yes. My opinions are that the Samsung HBM products do
10 not infringe on any of the asserted patents.

11 Q. And, briefly, what are the bases for your opinions?

12 A. There's two bases. One is that the claims require no
13 DRAM circuits and Samsung's products do have DRAM circuits.
14 So for that reason alone, they don't infringe.

15 Q. Is there a second reason, Doctor Robins?

16 A. Yeah, there's an additional reason. The claims also
17 require that there is no electrical communication to all dies,
18 but Samsung HBM products are in electrical communications with
19 all DRAM dies. So for that additional reason, they also don't
20 infringe.

21 Q. Do you also have an opinion about the technical benefits
22 of the '060 and the '160 Patents?

23 A. Yes. I will show that there is no technical benefits for
24 either of these patents.

25 Q. Where did you go to school?

1 THE COURT: Doctor Robins, pull the microphone a
2 little closer to you, please. I want to make sure we hear
3 you.

4 THE WITNESS: Okay.

5 THE COURT: Go ahead, Counsel.

6 Q. (BY MR. COLVIN) Where did you go to school, Doctor
7 Robins?

8 A. After high school, I went to a community college, not
9 being able to afford a big university. And so I got my AA
10 degree there while holding two jobs and supporting my mom and
11 brothers. Long story short, I ended up getting my Ph.D. at
12 UCLA in 1992, and I got a Master's along the way.

13 Q. What did you do after your Ph.D.?

14 A. After my Ph.D., I joined the University of Virginia in
15 Charlottesville, Virginia.

16 Q. And what do you do as a professor at the University of
17 Virginia?

18 A. My professorial duties involved research, teaching, and
19 service. So I taught various courses, covering topics
20 including computer chip design, and I also consulted with the
21 U.S. Department of Defense on various technological matters.

22 Q. Have you published any books or articles over your
23 career?

24 A. Yeah. I published a book on chip design, I published
25 over a hundred peer-reviewed papers on topics including

1 semiconductor chip design, and I invented several granted
2 patents.

3 Q. Have you been recognized for your work?

4 A. Yes. I've been a busy boy. I got various awards from my
5 peers as recognition.

6 Q. What other experiences have you gained in this field,
7 Doctor Robins?

8 A. When I advised the U.S. Department of Defense, we
9 explained to them how they can use vast technology in
10 computers to be more effective and more efficient.

11 And some of the site visits were actually right here in
12 Texas, Fort Hood, for example. On the right here, we see a
13 picture of me in a meeting with U.S. Secretary of Defense in
14 his office in the Pentagon.

15 Q. Did that work require a security clearance?

16 A. Some of it did require a security clearance, yes.

17 Q. What other experiences do you have that you consider
18 relevant to this field?

19 A. I published some papers in the mid '90s on the stacking
20 chips using through-silicon vias, or TSVs, and that's a core
21 technology in this case.

22 MR. COLVIN: Your Honor, at this time we offer
23 Doctor Robins as an expert in memory modules, including the
24 subject matter of the '060 and '160 Patent?

25 THE COURT: Is there objection?

1 MR. SHEASBY: No objection, Your Honor.

2 THE COURT: Without objection, the Court will
3 recognize this witness as an expert in the designated field.

4 Let's continue.

5 Q. (BY MR. COLVIN) Doctor Robins, let's talk about some
6 background technology. What is a DRAM?

7 A. A DRAM is a type of volatile computer memory that can
8 store information.

9 Q. And what characterizes DRAM as opposed to other types of
10 memory?

11 A. DRAM has a need to be refreshed occasionally so it
12 doesn't lose its data. It was invented in 1966 by Robert
13 Dennard from IBM.

14 Q. And how many memory cells are there in a device?

15 A. There's millions of these DRAM circuits in a typical DRAM
16 device, and it could be up to billions even nowadays.

17 Q. Does a DRAM memory cell contain a DRAM circuit?

18 A. Yeah, of course. Here we see it. It consists of a
19 transistor, a capacitor, and some wiring.

20 Q. How are DRAM circuits organized?

21 A. They are organized in rows and columns. And once you put
22 enough of them together, you can store a lot of different
23 types of information like emails and photographs and songs and
24 whatever.

25 Q. And what are the DRAMs circuits placed on?

1 A. They are placed on a silicon die.

2 Q. How is it possible to fit so many of these structures on
3 a silicon die?

4 A. Because over the years engineers have figured out ways to
5 miniaturize these circuit components. And today these circuit
6 components are smaller than computer viruses. You need a
7 microscope to actually see them. And that's how you can fit
8 billions of them on a silicon die the size of a fingernail.

9 Q. Can you stack memory die?

10 A. Yeah. You can sure stack them.

11 Q. And are there different ways to connect memory die when
12 they're stacked?

13 A. Yes. You can connect them using so-called TSVs,
14 through-silicon vias, or you can use wire bonding or other
15 methods to stack them.

16 Q. When were TSVs invented?

17 A. TSVs were invented in 1958 Bill Shockley out of AT&T Bell
18 Labs. And this is the same guy, by the way, that invented the
19 actual transistor about a decade earlier for which he won a
20 Nobel prize.

21 Q. Are there different ways to connect stacked die using
22 TSVs?

23 A. Yes. You can connect TSVs using different methods. For
24 example, the so-called multi-drop scheme where every TSV
25 connects to each and every die in a stack. And also you can

1 have a point-to-point scheme where certain TSVs like the one
2 in yellow on the right connects to certain dies. And the one,
3 for example, like the red on the right, connects to other dies
4 while passing through intermediate dies in between.

5 Q. Are there also different types of TSV arrangements?

6 A. Yes. You can design TSVs to have sort of a twist pattern
7 or spiral pattern. In the twist, you have two TSVs kind of
8 twisting around each other. And in the spiral, you can have
9 more.

10 Q. So let's take a look at the figure on the right. Can you
11 explain what's going on there to the jury?

12 A. Sure. You have four TSVs starting at the bottom, and
13 they twist around each other in kind of a spiraling or helix
14 kind of configuration as they go up the stack.

15 And here we see the green one spiraling by going over
16 across the die, then up through the die, then across the next
17 die, and up through that die, and so forth until it reaches
18 the top of the stack.

19 And the red one does the same thing, a little bit offset.
20 And the green one and the blue one, they're all spiraling up
21 in a similar symmetric fashion until they reach the top of the
22 stack.

23 Q. How are TSVs shown visually in this field?

24 A. There's different ways of depicting them. So, for
25 example, on the left here we see a geometric way of depicting

1 them where you can see the actual physicality of them. The
2 yellow pillars there on the left show the TSVs going up the
3 stack and connecting the dies.

4 Or you can do it more schematically or logically like we
5 see on the right. And on the right we see these arrows
6 showing the logical connectivity in a more schematic manner of
7 these TSVs.

8 Q. Doctor Robins, have you analyzed the '060 and '160
9 Patents?

10 A. Yes, I have.

11 Q. What do the '060 and '160 patents describe?

12 A. They describe, first of all, the prior art. They admit
13 that die stacking, DRAM die stacking, was well-known at the
14 time of the patent. And it shows an example of die stacking
15 in this figure on the right. And they explain how die
16 interconnect can be TSVs where it can stack dies like the
17 yellow dies that we see here.

18 And so that's the first thing they cover is the prior
19 art, what was known before.

20 Q. What else do they describe?

21 A. They also describe how TSVs can go up the stack and
22 connect to only selected groups of array dies. So, for
23 example, the green TSV here on the right shows how it's
24 connected to just the two yellow dies, while the blue TSV
25 there is connected to the two upper pink dies and goes through

1 the two yellow dies without connecting to them.

2 Q. Do the patents give an example of how a TSV can go
3 through a die without being connected to it?

4 A. Yes. The patents explain that you can have so-called
5 through-holes through which the TSVs pass without electrically
6 connecting to the dies through which they pass and instead go
7 further and connect to other dies instead.

8 Q. Have you analyzed the asserted claims of the '060 and
9 '160 Patent?

10 A. Yes, I have.

11 Q. There's a lot of text on this slide, Doctor Robins. Will
12 you explain to the jury what we see here?

13 A. Sure. Here we've broken down the claim into smaller
14 chunks called claim elements, and we list them separately so
15 that we can refer to them and discuss them more easily.

16 Q. And what do we see on the left-hand side of the screen?

17 A. So on the left in the dark box, we see the first
18 independent claim of the '060 Patent. And on the right, we
19 see the analogous claim for the '160 Patent.

20 Q. What's the significance of the dark black line?

21 A. The dark black line represents the fact that these two
22 claims, one, are independent, and the other claims below it
23 are dependent.

24 Q. And if an independent claim is not met, what does that
25 mean for the dependent claims?

1 A. If the independent claim is not infringed, then neither
2 is the dependent claim that depends on it because the
3 dependent claim incorporates all limitations of the
4 independent claim upon which it is founded.

5 Q. Doctor Robins, let's take a look at the '060 Patent,
6 claim 1. What's the significance of the two yellow
7 highlighted elements here?

8 A. The significance here is that the first highlighted
9 yellow claim there talks about array dies, and the second
10 claim element below it talks about the not in electrical
11 communication limitation.

12 Q. And are these the elements you're going to discuss today?

13 A. Yes, those are the two.

14 Q. Let's go to the first claim element. Does your analysis
15 focus on any particular part of this claim?

16 A. Yes. I will discuss the concept of array dies. That's
17 the part that will be most relevant here.

18 Q. Has the Court construed the term 'array die'?

19 A. Yes, it has. The Court construed array die as an array
20 die that is different from a DRAM circuit.

21 Q. Did you apply the Court's construction in your analysis,
22 Doctor Robins?

23 A. Of course.

24 Q. How does your application and understanding of the
25 Court's construction relate to Samsung's HBM product?

1 A. It's pretty simple, actually. If the Samsung HBM
2 products contain DRAM circuits, they can't be array dies and
3 there's no infringement.

4 Q. And why is that, Doctor Robins?

5 A. Because the Court construed that array dies must be
6 different than a DRAM circuit.

7 Q. And if we look on the figure on the right that's JTX 8 at
8 page 9, what do we see with respect to Samsung's HBM product?

9 A. We see there several DRAM dies stacked up, each one
10 contains DRAM circuits, and they're connected vertically using
11 TSVs through-silicon vias.

12 Q. And what does that mean for infringement?

13 A. It means, because they're DRAM dies containing DRAM
14 circuitry, they can't be array dies and, therefore, there's no
15 infringement.

16 Q. Is there any agreement between you and Netlist on the
17 nature of the accused HBM products?

18 A. Well, Netlist agrees that the Samsung HBM products
19 contain DRAM dies. They even say so in their complaint shown
20 here on the screen.

21 Q. Doctor Robins, what is a data sheet?

22 A. A data sheet is basically an owner's manual. It
23 describes the product, how it's used, how it operates, and how
24 it interacts with other components.

25 Q. And we're looking at the data sheet that's JTX 68. What

1 does this data sheet show with respect to Samsung's HBM's use
2 of DRAM?

3 A. This page of the data sheet shows the diagram with a
4 bunch of DRAM dies all stacked up, and it makes multiple
5 references to DRAM. So it's very clear that it's talking
6 about DRAM dies.

7 Q. And what do the data sheets for the accused HBM products
8 in this case show?

9 A. All the other data sheets for the other products show
10 very similar diagrams. In fact, in many cases it's identical
11 diagrams that all show DRAM dies. No question about that.

12 Q. And so when the data sheets say DRAM, what do they mean?

13 A. It means a DRAM die containing these DRAM circuits that
14 we saw were invented by Robert Dennard earlier, and there's
15 many millions or even billions of these DRAM circuits on each
16 one of the DRAM dies.

17 Q. And how does that impact your infringement analysis?

18 A. Very simple. Because these DRAM dies contain DRAM
19 circuits, they cannot infringe and they do not infringe.

20 Q. Are you aware of any evidence that supports your
21 understanding of DRAM circuit?

22 A. Yes. There's other references that mention DRAM
23 circuits, and an example is the Rajan reference, which is a
24 patent application that came up during the patent prosecution
25 history of the asserted patents.

1 Q. How does Rajan support your analysis, sir?

2 A. Well, Rajan discusses DRAM circuits. He mentions DRAM
3 circuits 206A through D. These are the orange rectangles that
4 we see here on the right which is a figure from Rajan.

5 Q. How did Rajan become part of this case?

6 A. During the prosecution history, the patent examiner first
7 rejected the claims based on the Rajan reference. And by the
8 way, the prosecution history is simply the set of
9 correspondences between the inventor and the Patent Office
10 itself, all the letters and messages back and forth.

11 And so after the rejection, Netlist decided to try to
12 overcome the rejection by telling the Patent Office that Rajan
13 merely stacks DRAM circuits 206A through D which are different
14 from array dies. In other words, Netlist told the patent
15 examiner that their array dies are not DRAM circuits.

16 Q. Does Rajan limit its description of DRAM?

17 A. No, not at all. Quite to the contrary. Rajan explicitly
18 states that DRAM may take any form whatsoever. In fact, it
19 said so right here in paragraph 18 of the Rajan reference.
20 And, moreover, it says that any other type of DRAM may also be
21 allowable besides this long list of about a dozen different
22 types of DRAM that Rajan explicitly lists. So it's a very
23 broad category of DRAM memories that Rajan refers to.

24 Q. So if we look at this blown-up figure at JTX 63 from
25 Rajan, do the DRAM circuits include the arrows that are shown

1 in this figure that are highlighted in yellow?

2 A. No. The arrows are just interconnects. That's what
3 connects the DRAM circuits to the buffer chip. Those are
4 denoted in yellow there on the right and they're labeled beta;
5 whereas, the DRAM circuits themselves are labeled DRAM. So a
6 DRAM is a DRAM is a DRAM regardless of how it's connected to
7 other things.

8 Q. And what does Rajan have to say about those arrows?

9 A. Rajan says that these electrical connections that we saw
10 in yellow could be done in any desired manner. And that's a
11 direct quote from Rajan.

12 Q. Paragraph 50 that you have highlighted also includes the
13 words DRAM circuit 206A through D. Do you see that?

14 A. Yes.

15 Q. And do the DRAM circuits in Rajan change based on the
16 configuration shown by the arrows?

17 A. No, not at all. Regardless of how you interconnect these
18 DRAM circuits to the buffer chip, and here we see several
19 examples taken right out of Rajan, regardless how you do that,
20 DRAM circuits are still DRAM circuits. Their nature doesn't
21 change by their interconnections that connects them to other
22 things.

23 Q. Do the connection arrows of Rajan depict wire bonding?

24 A. No. What it depicts is logical connections, kind of
25 schematic connections of what connects to what.

1 Q. How would a person in this field illustrate wire bonding?

2 A. If you want to illustrate wire bonding, you can do that
3 like we see here on the right from a patent from 2010. You'd
4 have wires coming in from the bottom, going over, up, and down
5 into the tops of the dies.

6 Q. So if not wire bonding, then what do these arrows of
7 Rajan signify?

8 A. They signify schematic connections, basically topological
9 connections, instructions as to what needs to connect to what
10 but not how it should connect.

11 And we see an example of that in a patent by Matsui from
12 2003 with very similar error connections much like Rajan uses
13 that show topological or schematic connections. And, in fact,
14 the same patent by Matsui also shows an alternative way of
15 connecting them using through-silicon vias. And in this
16 figure 2 of Matsui, he actually depicts the actual geometry of
17 these through-silicon vias.

18 So it's very clear that what Matsui teaches, the
19 connection using either through-silicon vias and also by this
20 schematic way of depicting the exact same thing.

21 Q. How does your analysis of Rajan relate to your
22 understanding of DRAM dies and DRAM circuits?

23 A. So basically it reinforces the understanding that if
24 there is DRAM circuitry and DRAM circuits, there is no
25 infringement.

1 Q. And, finally, let's return back to this claim language in
2 the Court's construction. Summarize one more time for the
3 jury your -- your conclusion here.

4 A. So, again, because these DRAM dies of the Samsung
5 products contain lots of DRAM circuits and DRAM circuitry,
6 according the Court's claim construction order, there's no
7 infringement and there can't be infringement.

8 Q. So looking at the full '060 Patent, claim 1, you've got
9 an X on the right. What does that signify?

10 A. That red X signifies that that claim element is not
11 infringed by the Samsung products.

12 Q. And if we look at the '160 Patent, claim 1, how does your
13 analysis for the '060 apply to this claim?

14 A. The analysis is the same simply because the claim
15 language is so similar across these two asserted patents. So
16 if one is not infringed, neither is the other because it also
17 refers to array dies.

18 Q. Let's move to the next element that you analyze. Any
19 particular part of this element that you're going to focus on,
20 Doctor Robins?

21 A. Yes. In this element, I will focus on the not in
22 electrical communication component of that element.

23 Q. So let's walk through this claim language so that you can
24 explain it to the jury. On the left-hand side of the screen,
25 what do we see, Doctor Robins?

1 A. We see on the left-hand side the claim 1 broken into
2 several elements, and on the right we see an example of -- of
3 this right out of the patent, the '060 and the '160.

4 Q. And what does the first part of this claim element
5 require?

6 A. The first part requires a first die interconnect and a
7 second die interconnect. So in this example figure, we see a
8 green set of TSV and a blue TSV going from the bottom up the
9 stack.

10 Q. And what's required of the first die interconnect in this
11 claim element?

12 A. The first die interconnect connects to a first group of
13 array dies. Here we depict them in yellow. So it's
14 connecting to some array dies. And also it's required to not
15 be in electrical communication with other array dies, and
16 those are shown here in pink.

17 Q. What about the second die interconnect? What does the
18 claim require with respect to that element?

19 A. The second die interconnect is kind of a symmetric
20 opposite. It's got to connect to the second group of array
21 dies, but it must not connect to the first group of array
22 dies. So it must connect to the pink ones but not to the
23 yellow ones. So it's the opposite of what the other one did.

24 Q. So let's take a look at this portion of this claim
25 language that's not in electrical communication with the array

1 die. How would someone in the field understand that claim
2 language, Doctor Robins?

3 A. Well, the notion of electrical communication is a broad
4 concept involving the transmission of electricity. So not in
5 electrical communication with array dies means that there is
6 no transmission of an electrical signal to any circuitry on
7 the die.

8 Q. So what's the take-away from your analysis of the claim
9 language?

10 A. The take-away is the bottom line--that if the die
11 interconnects connect to all the array dies, then there can't
12 be infringement and there's no infringement.

13 Q. So with that understanding of the claim language, what's
14 your opinion about if Samsung's HBM product meet these
15 limitations?

16 A. The Samsung HBM product does not meet this limitations
17 and therefore does not infringe, and I will explain why.

18 Q. What do we see on the right-hand side of the screen,
19 Doctor Robins, in JTX 15?

20 A. We see there the so-called twist pattern where the two
21 TSVs go up the stack in a twisting, meandering manner like we
22 animate here in green and blue. Each one kind of crisscrosses
23 all the different dies as it makes its way up the stack.

24 Q. And you put red Xs on the left-hand side of the screen.
25 Explain to the jury what those signify.

1 A. The red Xs signify that the Samsung products do not
2 infringe on this claim because the die interconnect is TSVs
3 meandering up the stack are in electrical communication with
4 each and every die on the stack. And that does not meet the
5 requirement of the claim elements that says they are not in
6 electrical communication with some group or another.

7 Q. Let's take a closer look at some of these structures.
8 What are you showing the jury on this slide, Doctor Robins?

9 A. On this slide, we kind of do a zoomed-out or blown-up
10 version of the bottom few dies of the stack showing the
11 crisscrossing twist pattern.

12 Q. So in this blown-up version, we see red and blue diagonal
13 lines now that I'm circling on the screen. Explain to the
14 jury what those are.

15 A. Those are TSV interconnects that go across the die, and
16 they connect each die to the next TSV that goes up to the next
17 die and so forth in a left-right kind of meandering,
18 zigzagging pattern as we saw earlier.

19 Q. And the diagonal red and blue lines, where are those
20 lines located with respect to the die?

21 A. They're on the die. They're part of the die.

22 Q. At the bottom of this blown-up image, we see a channel A.
23 Do you see that, Doctor Robins?

24 A. I see it.

25 Q. Explain to the jury what that is.

1 A. That's where the TSV starts at the buffer die, which is
2 the blue die at the bottom, and this channel goes up and over
3 and then zigzags back and forth across the stack of dies until
4 it reaches the top.

5 Q. Is the channel A die interconnect in electrical
6 communication with both of the core die shown here?

7 A. Yes, it is.

8 Q. Explain why you believe that.

9 A. Because it's connected to various circuitry on the die,
10 including these horizontal paths that we see in blue and red,
11 and also including the pads that we see in blue rectangle and
12 green rectangle and so forth. So it's in connection with the
13 various things on the die and, therefore, in connection with
14 the die.

15 Q. Are you aware of other Samsung internal documentation
16 that shows a different depiction of this twisting structure?

17 A. Yes. You can show this twist structure in several
18 different ways.

19 MR. COLVIN: Your Honor, at this time we need to
20 seal the courtroom for Samsung's confidential information.

21 THE COURT: All right. Based on counsel's request
22 to protect confidential information, I'll order the courtroom
23 sealed.

24 I'll direct all persons who are present and not subject
25 to the protective order that's been entered in this case to

1 excuse themselves and remain outside the courtroom until it is
2 reopened and unsealed.

3 (Courtroom sealed.)
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(Courtroom unsealed.)

Q. (BY MR. COLVIN) Your Honor -- sorry. Doctor Robins, we've been discussing --

THE COURT: Just a minute, counsel. Let's let everybody get back in the courtroom who's on their way in.

All right. Now you may proceed.

MR. COLVIN: Thank you, Your Honor.

Q. (BY MR. COLVIN) Doctor Robins, we've been talking about the twist structures. Which of Samsung's HBM products does the twist structure apply to?

A. The twist structure applies to the HBM2 and HBM2-E products while the spiral corresponds to the HBM3 products.

1 Q. And how does your analysis of the twist structure
2 compare -- relate to the spiral structure?

3 A. The analysis is very much the same, because the
4 difference between a twist and a spiral is that in a twist you
5 have two TSVs going up the stack, whereas in the spiral you
6 have four of them together meandering around, going up and
7 twisting up the stack. But all the other observations remain
8 the same.

9 Q. Doctor Robins, were you here for Doctor Brogioli's
10 testimony a few days ago?

11 A. Yes, I was.

12 Q. And what's your understanding of Doctor Brogioli's
13 opinion on this issue?

14 A. Best I can tell, Doctor Brogioli has an overbroad
15 interpretation of this claim element. He construes it to mean
16 communication with other things like drivers and receivers or
17 perhaps memory locations, and I disagree with him on that.

18 Q. And what's the basis of your disagreement?

19 A. Simply because that's not what the claim language says.
20 The claim language, as we see here on the left, do not mention
21 drivers and receivers in terms of not in electrical
22 communication. They don't mention things like memory
23 locations or other things. They mention array dies and array
24 dies alone.

25 Q. In view of all the evidence you reviewed in this case,

1 Doctor Robins, what's your conclusion about if Samsung's HBM
2 products practice this claim limitation?

3 A. The bottom line conclusion is that Samsung products do
4 not practice this not in electrical communication requirement
5 and, therefore, they do not infringe.

6 Q. So let's return to your claim language chart for the '060
7 Patent, claim 1. You now have a second red X, Doctor Robins.
8 What's the significance of that?

9 A. The red Xs indicate there is no infringement.

10 Q. And how many reasons are there for no infringement of
11 '060 Patent, claim 1?

12 A. There's two independent and separate reasons for
13 non-infringement, each one of which by itself would conclude
14 non-infringement of claim 1. And we have two of them, they're
15 independent, so that goes double for non-infringement.

16 Q. Let's look at the '160 Patent, claim 1. What's your
17 opinion on infringement of the '160 Patent, claim 1, by
18 Samsung's HBM product?

19 A. It's the same as the other because the claim language is
20 very similar across these two claim elements and the two
21 asserted patents both have the array dies and not in
22 electrical communication requirements.

23 So for the exact same reasons, the '160 Patent
24 is -- claim 1 of that patent is not infringed by the Samsung
25 products. That's the bottom line.

1 Q. We'll zoom all the way out and look at your big chart.

2 And how do your opinions on the independent claims of
3 claim 1--excuse me--of the independent claims of the '060
4 Patent and the '160 Patent impact the dependent claims?

5 A. So because the independent claims do not infringe, all
6 the claims that depend on those claims cannot infringe,
7 either, because they also include the limitations of the
8 independent claims upon which they depend. So because there's
9 no infringement of the independent claims, there's no
10 infringement of the dependent claims, either.

11 MR. COLVIN: Your Honor, at this time we'll talk
12 about technical benefits.

13 THE COURT: Ladies and gentlemen of the jury, before
14 we do this, I want to mention something to you.

15 You are about to see several slides and hear some
16 testimony indicating the Defendants' view that the '060 and
17 the '160 Patents do not make any technical contributions.

18 You should understand that whether or not these two
19 patents make or don't make any technical contribution does not
20 relate to the issues of infringement or validity, but relates
21 only, if at all, to the issue of damages, and you should not
22 apply this information to your determinations regarding
23 infringement and validity, only damages.

24 So with that, let's proceed.

25 Q. (BY MR. COLVIN) Doctor Robins, what's your understanding

1 of Doctor Brogioli's opinion with respect to the technical
2 benefits of the '060 and '160 Patents?

3 A. My understanding is that Doctor Brogioli believes that
4 Netlist's patents allow Samsung to go beyond four high stack
5 of DRAM dies or higher, and I disagree with that.

6 Q. Why do you disagree with that, Doctor Robins?

7 A. Basically there's lots of prior art references that
8 practice the teachings of the asserted patents, and I'm about
9 to go through a few.

10 MR. SHEASBY: Your Honor, I object. And may I
11 approach?

12 THE COURT: Approach the bench.

13 (The following was had outside the hearing of the
14 jury.)

15 MR. SHEASBY: He just said there's a lot of prior
16 art references that practice the teachings of the '060 and
17 '160 Patent. That's a direct violation of your instructions
18 this morning and it's a direct violation of their stipulation
19 of validity for not a defense.

20 I request a curative instruction, and I think he should
21 have to move on. I don't think this was unintentional. Your
22 Honor said yesterday that everything is considered
23 intentional. He just injected into this case the exact that
24 issue we were in court to talk and he said it intentionally.

25 THE COURT: Mr. Colvin.

1 MR. COLVIN: Your Honor, I had no intention that he
2 was going to phrase that in the way he did, but it is an
3 accurate characterization that there are prior art references
4 and they come before the patents.

5 MR. SHEASBY: Your Honor, he said that each of those
6 elements --

7 THE COURT: We don't have -- we have an invalidity
8 case.

9 MR. COLVIN: Understood, Your Honor. Understood.

10 MR. SHEASBY: I think it should be stricken, it
11 should be ignored, and I think they should move on at this
12 point. I think the damage has been done. There's no way that
13 was unintentional, not on counsel's part, on the -- the -- the
14 witness' part.

15 THE COURT: All right. Thank you, Mr. Sheasby.

16 I'll give an instruction to the jury. I'm not going to
17 circumscribe counsel's continued direct examination of this
18 witness, although I will say I do not expect to hear any
19 language or conclusions that relate to invalidity with regard
20 to these two patents.

21 MR. COLVIN: Understood, Your Honor.

22 MR. SHEASBY: Your Honor, can we also strike that
23 testimony from the record?

24 THE COURT: I'll handle it.

25 MR. SHEASBY: Thank you, Your Honor.

1 (The following was had in the presence and hearing
2 of the jury.)

3 THE COURT: Ladies and gentlemen, I need to clarify
4 one other thing for you. You just heard some testimony about
5 there were prior art references that practice the concepts of
6 the '060 and the '160 Patents.

7 I want you to be completely clear. There are no
8 invalidity defenses to these patents asserted by the
9 Defendants. They are not asserting that the '060 and the '160
10 Patents are invalid under any theory. Their validity is not
11 in question in this case.

12 So what prior art may have practiced or covered what
13 these two patents cover is not an issue that goes to whether
14 or not those patents are valid. Their validity is not in
15 question in this case.

16 All right. Let's continue.

17 Q. (BY MR. COLVIN) Doctor Robins, are you aware of any
18 teachings from before the '060 and '160 Patents that show
19 memory die being stacked eight or higher?

20 A. Yes. There are many of those.

21 Q. And what's one example?

22 A. One example is a patent by Rio from 2009 showing DRAM
23 dies stacked 32 high using TSVs.

24 Q. Do you have another example?

25 A. Sure. There is another 2009 patent showing TSVs used to

1 stack 16 high DRAM dies.

2 Q. Any others?

3 A. Yes. There is a patent by Kang from 2010 showing how to
4 stack up to 16 DRAM dies using TSVs.

5 Q. How about any textbooks, Doctor Robins?

6 A. And there's examples from textbooks as well. Here from
7 2008, we see a textbook by Garou showing how to stack DRAM
8 dies eight high using TSVs.

9 Q. Are you aware of any references from before the asserted
10 patents that show TSVs only connected to some memory die but
11 not others?

12 A. Yes. There's that, too. For example, this patent by
13 Bertin from 1998 shows that you can stack DRAM dies and can
14 have TSVs connect to some dies but not to other dies.

15 Q. Are you aware of any references that show TSVs passing
16 through holes in die?

17 A. Yes. That, too, appears in the prior art. Here's a
18 patent from Jeddeloh from 2009, and it shows TSVs passing
19 through holes in DRAM dies on their way to connect to other
20 DRAM dies.

21 Q. Are you aware of any references from before the '060 and
22 the '160 Patents that teach twist or spiral designs?

23 A. Yes. There's a bunch of those as well. An example is
24 2005 a patent by Shibata, so six years before the asserted
25 patents were filed, and this shows a spiral pattern, the one

1 that we showed earlier in greater detail, and that's similar
2 to the spiral pattern of the Samsung accused HBM products. We
3 see them here side by side.

4 Q. Any other teachings of twist and spiral structure, Doctor
5 Robins?

6 A. Yes. One more example is the 2010 Nishioka patent that
7 again shows a spiral pattern very similar to the Samsung
8 products that use a similar type of spiral pattern.

9 Q. And so how does all of this fit into your analysis of the
10 technical benefits of the '060 and the '160 Patents?

11 A. Well, given that all of these prior art teachings
12 occurred beginning in the '90s and through the early 2000s and
13 basically years before the asserted patents were even filed,
14 they show that the asserted patents are not bringing any new
15 teachings or concepts to the table in terms of technical
16 contribution.

17 MR. COLVIN: Pass the witness, Your Honor.

18 MR. SHEASBY: Your Honor, may I approach?

19 THE COURT: Do you have something to say?

20 MR. SHEASBY: I do.

21 THE COURT: Then stand up and say it.

22 MR. SHEASBY: Your Honor, I have an objection and a
23 motion to strike.

24 THE COURT: Approach the bench, please.

25 (The following was had outside the hearing of the

1 jury.)

2 MR. SHEASBY: He just did it again where he said
3 there was nothing new in these patents based on what's in the
4 prior art.

5 THE COURT: There's no invalidity case here.
6 There's going to be no invalidity question on the verdict form
7 as to the '060 or the '160. I understand the language. I
8 still think the witness intended it to go to the value that
9 these patents would bring, and I think there's much less harm
10 here than you obviously do, Mr. Sheasby.

11 MR. SHEASBY: I understand, Your Honor.

12 THE COURT: It's not accurate, there is no
13 invalidity defense here, and it's at a minimum confusing. But
14 I don't think it's malicious or intentional, and I don't see
15 how it's going to directly result in harm since the jury's not
16 going to get a chance to answer the question about the
17 validity of these two patents.

18 MR. SHEASBY: I understand, Your Honor.

19 THE COURT: All right. We're going to take a short
20 recess, and then we're going to proceed through cross.

21 MR. SHEASBY: Thank you, Your Honor.

22 (The following was had in the presence and hearing
23 of the jury.)

24 THE COURT: Ladies and gentlemen, it's been almost
25 two hours since we came back from lunch. We're going to take

1 is a short recess before we proceed with the Plaintiff's
2 cross-examination of Doctor Robins.

3 If you'll simply leave your notebooks in your chairs,
4 follow all my instructions, including not to discuss the case
5 with yourselves, and we'll be back here shortly to continue.

6 The jury's excused for recess.

7 (Whereupon, the jury left the courtroom.)

8 THE COURT: All right, Counsel. I need to see lead
9 and local counsel in chambers during this recess. We'll try
10 to cover the remaining disputes on Kline and Meyer.

11 In the meantime, we stand in recess.

12 (Brief recess.)

13 THE COURT: Be seated, please.

14 Are you prepared to proceed with cross-examination,
15 Mr. Sheasby?

16 MR. SHEASBY: I am, Your Honor. I would like to
17 make one request on the record, Your Honor.

18 THE COURT: What is it?

19 MR. SHEASBY: There were two questions in which
20 Doctor Robins talked about things not being new or all of the
21 elements being in the prior art. I'd like to renew the
22 request that those be stricken from the record so that they
23 cannot be used in closing.

24 THE COURT: Well, I'll do this. I won't order them
25 stricken from the record. I will direct opposing counsel not

1 to use them in closing argument.

2 MR. SHEASBY: Thank you, Your Honor.

3 MR. CORDELL: Well, certainly, I won't -- I won't
4 suggest they had anything to do with validity, Your Honor, but
5 may I use them for damages purposes?

6 THE COURT: You may not use them for validity.

7 MR. CORDELL: Thank you.

8 THE COURT: Okay?

9 MR. SHEASBY: Your Honor, we're going to hear those
10 two quotes that in their face talk about invalidity in
11 closing. That's what Mr. Cordell just said.

12 THE COURT: I haven't struck them, I'm not going to
13 strike them, I am not going to permit them to be used for an
14 improper purpose. And any argument related to validity of the
15 '060 or the '160 would be an improper purpose, and I'm putting
16 Defense counsel on notice not to do that. And woe unto
17 Defense counsel who would do that in closing argument in light
18 of this instruction from me.

19 MR. SHEASBY: Thank you, Your Honor.

20 THE COURT: All right. Let's bring in the jury.

21 (Whereupon, the jury entered the courtroom.)

22 THE COURT: Please be seated.

23 We'll continue with the examination of Dr. Gabriel
24 Robins, and we'll proceed with cross-examination by the
25 Plaintiff.

1 Mr. Sheasby, you may proceed.

2 MR. SHEASBY: Good afternoon, ladies and gentlemen
3 of the jury.

4 CROSS EXAMINATION

5 BY MR. SHEASBY:

6 Q. Good afternoon, Doctor Robins.

7 A. Good afternoon.

8 Q. It is nice to see you again, this time not over Zoom.

9 A. Always a pleasure, Mr. Sheasby.

10 Q. Samsung's lawyers called you and asked you if you wanted
11 to help Samsung. Correct?

12 A. That's correct.

13 Q. And although you didn't, I think, reference this in your
14 direct examination, you're paid an hourly fee for your
15 testimony and your previous work. Is that correct?

16 A. That's right.

17 Q. And what is that hourly rate?

18 A. It's 550 an hour.

19 Q. And how many hours have you spent on this matter?

20 A. I don't know exactly, but it's a substantial amount.

21 Q. Give us an estimate.

22 A. A few hundred, fair to say.

23 Q. And to be clear, you view yourself as representing
24 Samsung in this matter. Correct?

25 A. I'm here to opine on technology on behalf of Samsung,

1 yes.

2 Q. To be precise, you view yourself as having --
3 representing Samsung, and in fact you've represented them in
4 the past. Correct?

5 A. In the sense I have opined in cases involving Samsung,
6 yes.

7 Q. Sir, you've represented Samsung in the past. Correct?

8 A. As an expert witness, yes.

9 Q. And you think you're representing Samsung now. Correct?

10 A. As an expert witness, indeed.

11 Q. And, in fact, in your resume you describe yourself as,
12 quote, defending the interests of some of the world's largest
13 semiconductor companies. Correct?

14 A. In the capacity of an expert witness, yes, indeed.

15 Q. You describe yourself as assisting in defending the
16 interests of the world's largest semiconductor companies.
17 Correct?

18 A. As an expert witness, yes.

19 Q. And when we go through your examination, I want something
20 to be very clear for the ladies and gentlemen of the jury.
21 You haven't spoken to a single Samsung engineer in preparation
22 for your report or your testimony. Correct?

23 A. That's correct.

24 MR. SHEASBY: And if we can have the slides, Mr.
25 Huynh, and if we can go to slide 2.6.

1 Q. (BY MR. SHEASBY) 2.6, you agree that element A is met.
2 Correct?

3 A. Yes, I do.

4 Q. You agree that element B is met by Samsung's products.
5 Correct?

6 A. I do.

7 Q. Element E. Correct?

8 A. Yes.

9 Q. Element F. Correct?

10 A. That, too.

11 Q. Now, there are a couple portions that I've
12 underlined -- a couple of portions I've underlined that you
13 think are missing from Samsung's HBM products. Correct?

14 A. That's true.

15 Q. And so you'll see that element C is one of those where
16 you think there's something that's different from Samsung's
17 products than what's recited in the claim. Correct?

18 A. Element C, are you referring to the stacked array dies
19 element?

20 Q. Yes, sir. If you see on the left-hand side, it says A,
21 B, C, D, E, and F?

22 A. I see that, and yeah -- go ahead.

23 Q. Sir, you are concerned, I think you told the ladies and
24 gentlemen of the jury, that Samsung doesn't have stacked array
25 dies with data ports. Correct?

1 A. Samsung doesn't have stacked array dies of any kind, yes.

2 That's correct.

3 Q. Sir, Samsung in your opinion doesn't have stacked array
4 dies with data ports. Correct?

5 A. I agree that it doesn't have array dies. It does have
6 data ports.

7 Q. So let's talk about areas where we can agree. You agree
8 that Samsung's -- I think they call them core dies. Correct?

9 A. That's true.

10 Q. And those core dies have data ports. Correct?

11 A. Yes.

12 Q. And those core dies have interconnects internally that
13 can connect to those data ports. Correct?

14 A. Yes.

15 Q. Those interports are sometimes referred to
16 through-silicon vias. Correct?

17 A. These are some of the interconnects, the TSVs, yes.

18 Q. And what you claim is that the limitation is not met
19 because they're not array dies as the Court has construed
20 them. Is that right?

21 A. Yes. It goes against the Court construction order, as I
22 mentioned in my slides.

23 Q. And what the Court said an array die is that it's
24 something different from a DRAM circuit. Correct?

25 A. Indeed, that's the construction, yes.

1 Q. And the Court didn't say array die is something that
2 doesn't have DRAM circuits. Correct?

3 A. It says it's different from a DRAM circuit.

4 Q. Sir, did the Court construe this to say an array die is
5 something that does not have a DRAM circuit?

6 A. Well, if it has DRAM circuits, it is a DRAM circuit.

7 MR. SHEASBY: I move to strike as non-responsive.

8 THE COURT: Sustained.

9 Q. (BY MR. SHEASBY) Sir, did the Court construe this
10 limitation as saying array die is something that does not have
11 DRAM circuits?

12 A. No. The exact wording is different from a DRAM circuit,
13 as you indicated.

14 Q. And, sir, your counsel for Defendant was using some
15 analogies yesterday about gas. Correct? Do you remember
16 that?

17 A. Vaguely, yes.

18 Q. So, for example, there's gas. Correct? Gasoline, you
19 can go to a station and pump it. Correct?

20 A. True.

21 Q. And gas can be put into a gas can. Correct?

22 A. Yes, it can.

23 Q. A gas can is different from gas. Correct?

24 A. Yes.

25 Q. You can put gas in a tank. Correct?

1 A. You could.

2 Q. A gas tank is different from gas. Correct?

3 A. It is.

4 Q. And we can even be more technological about that. When I
5 was growing up, I had CDs. Did you have CDs?

6 A. I did.

7 Q. And CDs is an optical disc. Correct?

8 A. Yes.

9 Q. And it actually has little pits in which information is
10 stored. Correct?

11 A. It does.

12 Q. And the CD ROM is a form of memory. Correct?

13 A. It is.

14 Q. And the way you can read and access that memory is by
15 putting it in a CD ROM player. Correct?

16 A. You could do that.

17 Q. But we all agree that a CD ROM player is not a CD.
18 Correct?

19 A. Yes, that's true.

20 Q. And I wrote this down. It says, if the Samsung HBM
21 products contain DRAM circuits, there's no infringement. Do
22 you remember saying that?

23 A. I remember saying something like that, yes.

24 Q. And just to be clear, the Court's construction does not
25 say an array die is something that does not contain a DRAM

1 circuit. Correct?

2 A. That's not the exact wording. That's true.

3 Q. Sir, it doesn't say an array die is something that does
4 not contain a DRAM circuit. Correct?

5 A. Correct.

6 Q. And you showed this as a picture of a DRAM circuit.
7 Correct?

8 A. Yes, I did.

9 Q. And an array die has in Samsung's products, these core
10 dies that we talked about, have structures far beyond this
11 DRAM circuit. Correct?

12 A. They contain many DRAM circuits, and the entire array is
13 also a DRAM circuit. One doesn't exclude the other.

14 MR. SHEASBY: I move to strike as non-responsive,
15 Your Honor.

16 THE COURT: Overruled.

17 Q. (BY MR. SHEASBY) Sir, the core dies in Samsung's
18 products contain things in addition to that structure that
19 you've depicted on the right. Correct?

20 A. Yes.

21 Q. So one of the things they contain is data ports.
22 Correct?

23 A. True.

24 Q. They contain test structures. Correct?

25 A. They could, but they don't have to, yes.

1 Q. Sir, the core dies in Samsung's products contain test
2 structures. Correct?

3 A. Yes, they do.

4 Q. They contain bypass and signal ports. Correct?

5 A. They contain bypass and signal TSVs, yes.

6 Q. They contain TX and RX. Correct?

7 A. They do.

8 Q. Transistors and receivers. Correct?

9 A. That is correct.

10 MR. SHEASBY: And if we go to slide 2.8, Mr. Huynh.
11 I'll remove the smiley face.

12 MR. COLVIN: Your Honor, if we're going to show this
13 demonstrative, we need to seal the courtroom.

14 MR. SHEASBY: Take it down.

15 THE COURT: Do you want me to seal the courtroom?

16 MR. SHEASBY: I'm fine sealing the courtroom. I
17 think it's already been used publicly, but I have no problem
18 sealing.

19 THE COURT: Well, to avoid any risk of compromising
20 confidential information, I'll order the courtroom sealed and
21 I'll direct those present who are not subject to the
22 protective order that's been entered in this case to excuse
23 themselves at this time and to remain outside the courtroom
24 until it is reopened and unsealed.

25 (Courtroom sealed.)

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(Courtroom unsealed.)

THE COURT: All right. The courtroom is unsealed,
as is the record.

We can proceed with redirect.

Go ahead, Mr. Colvin.

REDIRECT EXAMINATION

BY MR. COLVIN:

Q. Doctor Robins, you recall Mr. Sheasby asking you some
questions about this slide?

A. Yes, I do.

Q. This structure that we see on the left that says -- here,
how would you characterize that, Doctor Robins?

A. That's a DRAM circuit.

Q. Why do you say that's a DRAM circuit?

A. Because it's made of a bunch of smaller DRAM circuits,
such as the one we see on the right, and together they still
constitute a bigger DRAM circuit.

Q. If you turn your attention to the easel, is this a DRAM
circuit?

A. Yes, it is.

Q. And is this the only thing a DRAM circuit can be?

1 A. No, it's not.

2 Q. Explain why you say that.

3 A. So a DRAM circuit is the generalized concept that it can
4 store in a volatile fashion, a volatile memory, a bit or a
5 large number of bits. So the DRAM circuit that's on the easel
6 is the same DRAM circuit that we see on the slide in front of
7 us. But if you take a bunch of those together and put them in
8 a bunch of rows and columns, this whole conglomeration of
9 these smaller DRAM circuits together is still a DRAM circuit.
10 One doesn't exclude the other, as I tried to explain earlier.

11 Q. If we look at just, say, this, what is this, Doctor
12 Robins?

13 A. That's a column decoder. It's basically a kind of a
14 multiplexer circuit that chooses certain columns of the data
15 that comes down.

16 Q. And above it we have regenerator circuits. What is that?

17 A. Those are, so-called, sense amplifiers. They take the
18 tiny chargers in the capacitors of these DRAM circuits and
19 amplify them to a level that the digital circuitry can better
20 recognize and process these signals.

21 Q. Are those types of circuits required for these memory
22 cells to work?

23 A. In DRAMs, yes, they are.

24 Q. And do DRAM circuitries include receivers and
25 transceivers?

1 A. DRAM --

2 MR. SHEASBY: Your Honor, I object. That's outside
3 of the scope of his report.

4 THE COURT: Do you believe there's support in his
5 report for that inquiry, counsel?

6 MR. COLVIN: Your Honor, I believe that there's
7 support in his report for saying that all of the circuitry
8 that's on a DRAM die is DRAM circuitry, and so that was --
9 that's where we we're going, Your Honor, and I believe there's
10 support for that.

11 THE COURT: Well, I'm not so much concerned with
12 where you are going; I'm concerned with the question you just
13 asked that drew the objection.

14 MR. COLVIN: Understood.

15 THE COURT: Do you believe there's support for that
16 question and answer within this expert's report? And, if so,
17 please point it out to me.

18 MR. COLVIN: Your Honor, I would start with
19 paragraph 89 in Doctor Robins --

20 THE COURT: Which report? There are several.

21 MR. COLVIN: His rebuttal report, Your Honor. In
22 particular there's a caption of a figure that's shown there.

23 THE COURT: I've got paragraph 89 in front of me.

24 MR. COLVIN: And, Your Honor, the figure in
25 paragraph 89 shows a picture of a textbook with similar

1 structures we see on the slide, and then Doctor Robins
2 characterizes all of this as showing a DRAM circuit.

3 THE COURT: All right. Based on that I'll overrule
4 the objection.

5 MR. SHEASBY: Your Honor, may I be clear? The
6 question was whether DRAM circuits have transceivers and
7 receivers. That is not in paragraph 89 of his report, and
8 that was my objection.

9 THE COURT: Well, I have the question in the record
10 in front of me and I've looked at paragraph 89. Let me check
11 again.

12 MR. SHEASBY: And I did a word search. The word
13 'transceiver' appears only once in his report with regard to
14 an iPhone.

15 MR. COLVIN: Your Honor, I would also say paragraph
16 97 talks about drivers, and also that this was asked on cross,
17 and so I think this is a fair rebuttal to that cross
18 examination on this very point.

19 THE COURT: Just a moment.

20 All right. My ruling remains. I'm going to overrule the
21 objection.

22 Either restate the question or move on, counsel.

23 Q. (BY MR. COLVIN) Doctor Robins, a DRAM circuit can
24 include things other than just a capacitor and a transistor.
25 Right?

1 A. Of course.

2 Q. What are some of the other things it can include?

3 A. It can contain the sense amplifiers that we saw on this
4 slide, the address decoders, and it can contain various other
5 data conduits. It can also contain transmitters and
6 transceivers so that other things can put data and read data
7 from the memory among other components. Basically everything
8 that's on the DRAM dies is part of the DRAM circuit.

9 MR. COLVIN: Slide 35, please.

10 Q. (BY MR. COLVIN) Do you recall getting asked questions
11 about the Rajan reference?

12 A. I recall that, yes.

13 Q. And you recall getting asked if certain things were like
14 other things and other figures in Rajan?

15 A. I remember that.

16 MR. COLVIN: If we can go to the Court's claim
17 construction, slide 41, please.

18 Q. (BY MR. COLVIN) Doctor Robins, the Court's claim
19 construction of 'array die' is 'array die that is different
20 from a DRAM circuit'. Right?

21 A. That's correct.

22 Q. Does the Court's construction include a qualifier about
23 Rajan's DRAM circuit?

24 A. Absolutely not.

25 Q. And when you applied the Court's claim construction, what

1 did you focus on?

2 A. That the Court construed that array dies are different
3 from a DRAM circuit and that's it, no Rajan reference.

4 Q. And can a product infringe the claims of these patents if
5 that product has DRAM circuits?

6 A. No. And the reason is very simple--because it would run
7 against the Court's very own claim construction order that we
8 see before us.

9 Q. And do the HBM products in this case have DRAM circuits?

10 A. Yes, they do.

11 Q. Is it really just as simple as that, Doctor Robins?

12 A. I think it is.

13 Q. Do the HBM products infringe the asserted claims for this
14 reason?

15 A. That's one of the reasons they don't infringe, and
16 there's other reasons, as I said earlier.

17 MR. COLVIN: Let's look at slide 44, please.

18 Q. (BY MR. COLVIN) Mr. Sheasby asked you about data ports
19 on array dies. Do you recall that?

20 A. I do.

21 Q. And he asked you about those in respect to die
22 interconnects not being in electrical communication with array
23 dies. Right?

24 A. That's right.

25 MR. COLVIN: Let's look at slide 45.

1 Q. (BY MR. COLVIN) Doctor Robins, does this part of the
2 claim require anything to do with data ports with respect to
3 electrical communication?

4 A. No, it does not.

5 Q. What about memory cells?

6 A. It does not require that either.

7 Q. What are the die interconnects required to not be in
8 electrical communication with in order to meet these claims?

9 A. Just the array die and only the array die.

10 Q. Did anything Mr. Sheasby asked you about change your
11 opinions about infringement in this case?

12 A. Not at all.

13 MR. COLVIN: Pass the witness, Your Honor.

14 THE COURT: Further cross examination?

15 MR. SHEASBY: No, Your Honor.

16 THE COURT: Then you may step down, Doctor Robins.

17 THE WITNESS: Thank you very much.

18 THE COURT: You're welcome, sir.

19 Defendants, call your next witness.

20 MR. McKEON: Your Honor, Defendants now call CEO of
21 Netlist, Mr. Chun Hong, to testify by deposition. The total
22 time of the deposition is 16 minutes and 14 seconds; 8 minutes
23 and 40 seconds is for Defendant Samsung, and 7 minutes and 34
24 seconds to be attributed to Netlist.

25 And Mr. Hong had eye surgery right before the deposition.

1 As a result, he had to wear physician-approved glasses that
2 had a dark tint.

3 THE COURT: So he has a reason; he's not trying to
4 be cool.

5 MR. McKEON: That's right, Your Honor.

6 THE COURT: Will you move the easel back, please,
7 Mr. McKeon?

8 MR. McKEON: Yes, Your Honor.

9 THE COURT: All right. Let's proceed with this
10 witness by deposition.

11 MR. CORDELL: Your Honor, I wonder if I could ask
12 Ms. Brunson to switch the audio off and put it back on. It
13 seems like we're having trouble swapping it from one side to
14 the other.

15 THE COURT: All right. Let's proceed.

16 CHUN KI HONG, BY VIDEOTAPED DEPOSITION

17 Q. Good morning, Mr. Hong.

18 A. Good morning.

19 Q. Okay. And Netlist does not sell today DDR5 RDIMM.
20 Correct?

21 A. No. We do. We do sell DDR5 RDIMM.

22 Q. Who would you say your competitors are, Netlist
23 competitors?

24 A. Netlist competitors -- there are a handful of memory
25 module manufacturers in the world that are in our space, but

1 they do not do direct. They do not create intellectual
2 property. They do not create inventions, innovative products
3 like we do, and they are SMART Modular, Viking Components,
4 perhaps Virtium is a competitor.

5 Q. Anyone else?

6 A. I think that's all I'm aware of.

7 Q. Okay. And so you wouldn't consider Samsung a competitor.
8 Is that right?

9 A. Samsung becomes a competitor when they take our high-end
10 technology and water it down into a commodity product. And
11 when they do that, it makes it difficult for us to sell our
12 proprietary solutions.

13 Q. And when you say they 'water it down into commodity',
14 what commodity products are you talking about?

15 A. I think we just covered that. Amongst the handful of
16 memory modules that we discuss, they all contain on them
17 voltage regulators, which is our invention, and DDR5 memory
18 cannot be made, you know, commercially viable. They cannot be
19 made at all without that.

20 Now, when -- we will apply that -- we applied that on
21 NVDIMM initially. We applied it on -- today as we build a
22 DDR5 VLP RDIMM, a very high-end product that no one else
23 offers. When they take that technology, the voltage
24 on-DIMM -- voltage regulator on-DIMM technology, that was
25 designed in order to make those high-end products perform

1 and work properly. When they take that design and apply it
2 to low-end products, like the PC products, commodity, popcorn
3 products, then they water down the value of the technology
4 that we created.

5 So in that sense, yeah, they're using our technology and
6 kind of neutering it. Then that devalues our technology. So
7 I will consider that competition.

8 Q. Who did you first approach with Samsung?

9 A. Did we first approach Samsung? Well, in regards to the
10 patents licensing -- in regards to the patent license, Samsung
11 approached us first --

12 Q. Okay.

13 A. -- seeking a license.

14 Q. When was that?

15 A. I believe that was in April 2015 when Jung Bae Lee
16 visited Netlist in Irvine to seek a license.

17 Q. For the product development side, you said you reached
18 out in 2014. Who at Netlist reached out?

19 A. It was myself. We visited Samsung and met with the
20 president of the memory business unit, as well as Jung Bae
21 Lee, Hyun-Ki Ji, and a whole bunch of other executives.
22 So -- and we had a group of five or six executives that flew
23 out to Korea for the initial meetings regarding our HybridIMM
24 product and a joint development with Samsung.

25 Q. Now, sir, the fee that's -- there's an \$8 million payment

1 by Samsung to Netlist for the license under this agreement.

2 Isn't that right?

3 A. That is incorrect.

4 Q. And do you understand this email was sent after the
5 meeting we just talked about where J.B. Kim is attaching a
6 term sheet? Do you see that?

7 A. Yes.

8 Q. Yeah. This exhibit now, yeah, 7B.

9 A. Okay. Yes, I see it.

10 Q. All right. And this is the term sheet that was sent by
11 Netlist to Samsung. Correct? Do you recognize this?

12 A. Yes, I recognize that.

13 Q. And it's dated 2014, but you understand this to be a 2015
14 document. Correct?

15 A. Yeah. It states '14, but I -- if this was what was
16 attached to the email, then it must have been '15. That's --
17 I'll -- if it's attached to the email that you're talking
18 about.

19 Q. And one of the things that you're -- that Netlist sent
20 here in the right hand, you see "these terms cloak the
21 license/covenant and allow the payments to be cast as NRE
22 rather than royalty payments." Do you see that?

23 A. "These terms cloak the license... and allowed the
24 payments to be cast..." I see that.

25 Q. And this was your -- Netlist saying, consistent with

1 your discussions with Samsung, that you wanted to have the
2 payment -- the license payment cast as NRE because you didn't
3 want to have anyone down the road calling that \$8 million
4 payment a license payment. Correct?

5 A. This is in response to what Samsung wanted at the time.
6 As you indicated, this document came after the meeting. We
7 are catering to whatever Samsung is asking us to do at any
8 given point through these series of negotiations.

9 Q. Oh, sorry.

10 A. So we're -- yeah. We're responding to Samsung's request.

11 Q. Well, this is the first term sheet that was sent between
12 the parties. This term sheet here. Correct?

13 A. That is, but it is -- it is a result of the meeting that
14 we had.

15 Q. Is it your testimony that it was Samsung that told you
16 they wanted to cloak the 8 million as an NRE even though it
17 was a license fee? Are you saying Samsung said that to you,
18 sir?

19 A. Okay. I think, if you step back and understand how a
20 small company like us will negotiate and deal with Samsung,
21 a company of that size, who has -- who's got a bunch of
22 different people with divergent interests within that company.
23 So these negotiations went on for six months. It went all
24 over the place based on who stepped in at Samsung, whether
25 it's the business unit, whether it was the IP people. They

1 all had differently interests.

2 So they're making requests of us to say, Here's what we
3 need. So go back and put that on paper and forward it to us.
4 And then they'll come back and say, Well, let's change this.
5 Let's change that. This won't work for us. So then we go
6 back and make those changes, and they're presented again.
7 That is in reality how the process works.

8 Q. My question is very simple. Is it your testimony that it
9 was someone at Samsung who told you at this meeting that they
10 wanted to have the license fee cloaked as an NRE instead of
11 being indicated as a license fee? Someone at Samsung told you
12 that? Is that your testimony?

13 A. Samsung requested that they wanted to minimize the patent
14 license. They wanted to maximize -- they wanted to show to
15 the world anything that they do with a small company that's
16 got IP -- they wanted to show that off as a strategic
17 agreement that involves joint development and joint work.

18 They did not want to show that this was a naked patent
19 agreement. They wanted to minimize the amount that they
20 wanted to pay in terms of patent royalties, and they made that
21 very clear to us from the beginning.

22 Q. Well, one more time. Is it someone at Samsung told you
23 they wanted to cloak the license payment as an NRE? Are you
24 saying someone at Samsung told you that? Correct?

25 A. Well, in so many words, yes. If you're asking that they

1 stated this verbatim, perhaps not, but they said in so many
2 words they wanted to minimize the patent license and transfer
3 the license in the form of or make it equivalent in the form
4 of other considerations, primarily technology collaboration.

5 Q. Well, let me ask you this, sir. Why would Samsung want
6 to -- you said they wanted to minimize the license payment.
7 Is that what you just said?

8 A. Yes.

9 Q. So you stated that Samsung wanted to minimize this
10 license fee. Is that what you said?

11 A. In general, Samsung does not want to appear as if they
12 are dependent on technology of small companies.

13 Q. You never sent any kind of notice of infringement or any
14 indication of infringement by Samsung prior to the
15 termination. Correct?

16 A. That is correct.

17 Q. Now, you understand that Samsung's invested in Netlist of
18 the amount of \$15million. Right?

19 A. \$15 million. Are you referring to the SVIC note?

20 Q. Yes. Correct?

21 A. That was a convertible loan. That was a loan which we
22 paid back.

23 Q. Why did --

24 A. They didn't make an investment. Yeah. They did not make
25 an investment. That was a loan which we paid back.

1 Q. But did you -- why did you want a loan?

2 A. \$15 -- \$15 million loan at those rates, you know, that
3 was good, good for us to have additional cash in the bank.
4 And it provided enormous incentive for Samsung. In the end,
5 they made 2 to 3X return on that 15 million on top of getting
6 their principal back.

7 Q. And do you know, sitting here today, how much Samsung
8 made off the warrants?

9 A. Well, I recall it was 2 to 3X. So, you know, that means
10 something above \$30 million.

11 Q. And why do you do continuations?

12 A. I think you do continuations to constantly improve the
13 invention, to take your initial invention and continue to
14 refine it so that it can -- that invention can become faster,
15 being able to handle lower power. In our case, it's speed and
16 power and capacity.

17 Q. And do you believe that the claims in this case is
18 supported by the specification?

19 A. I believe that to be the case.

20 THE COURT: Does that complete this witness by
21 deposition?

22 MR. CORDELL: It does, Your Honor.

23 THE COURT: All right.

24 MR. SHEASBY: May we approach just briefly?

25 THE COURT: Approach the bench, counsel.

1 (The following was had outside the hearing of the
2 jury.)

3 MR. SHEASBY: There was one passage that was excised
4 from the deposition. I don't want to make a big deal about it
5 because it was not intentional, I'm sure. We'll just play it
6 tomorrow morning, with the Court's permission.

7 THE COURT: What are you talking about?

8 MR. SHEASBY: So there was an additional portion
9 from Mr. Hong's deposition. We requested that the phrase
10 '\$600 million' be removed but the rest of the passage could
11 stay in. Someone at Samsung just removed the entire Q and A
12 as opposed to just blocking out the \$600 million. It's an
13 important passage for us.

14 THE COURT: That one sentence out of his answer was
15 what I intended to have redacted or deleted.

16 MR. SHEASBY: Yeah.

17 THE COURT: But not the entirety --

18 MR. SHEASBY: Correct. It's not that big of a deal.
19 It's not going to be able to be resolved right now. We can
20 even play it in rebuttal if we need to. I just wanted to --I
21 don't -- I want the Court to be apprised of the fact that
22 there was something extra. I'm not trying to make a big deal
23 out of it at all.

24 THE COURT: Well, actually what you're telling me,
25 it wasn't something extra; there was something taken out that

1 should have been presented.

2 MR. SHEASBY: Yes.

3 THE COURT: All right. Well, I want to know when
4 and how you intend to address that and we'll coordinate that
5 going forward.

6 MR. SHEASBY: Thank you, Your Honor.

7 THE COURT: You have your witness with the
8 interpreter next?

9 MR. CORDELL: I do. I do.

10 THE COURT: Okay.

11 MR. CORDELL: Take a break or put him on?

12 THE COURT: How long is the witness going to be?

13 MR. CORDELL: I'm going to make it as short as I
14 can. I'm low on time.

15 THE COURT: I know you are.

16 MR. CORDELL: So 15 minutes, but it's translated, so
17 I probably should tell you 20.

18 THE COURT: What's the cross time, Mr. Sheasby?

19 MR. SHEASBY: Fifteen minutes.

20 THE COURT: Here's the problem, counsel. I have to
21 be off the bench at 6:00. I'm afraid if we take a recess,
22 we're going to run out of time to get him on and off.

23 Let's go forward.

24 MR. CORDELL: Thank you.

25 (The following was had in the presence and hearing

1 of the jury.)

2 THE COURT: All right. Defendants, call your next
3 witness, please.

4 Ladies and gentlemen of the jury, this next witness is
5 not an English speaker. He's going to testify in Korean with
6 the aid of an interpreter. The interpreter's already been
7 qualified and sworn by the Court.

8 All right. Call the witness for the record, please.

9 MR. CORDELL: Thank you, Your Honor. Samsung next
10 calls Mr. Hyun-Ki Ji.

11 THE COURT: All right. Mr. Ji, go ahead and be
12 seated with the interpreter and then we'll administer the
13 oath.

14 MR. CORDELL: May I approach, Your Honor?

15 THE COURT: Yes.

16 (Whereupon, the oath was administered by the Clerk.)

17 THE COURT: Please have a seat.

18 All right, counsel. You may proceed with direct
19 examination.

20 MR. CORDELL: Thank you, Your Honor.

21 HYUN KI JI,
22 testified under oath as follows:

23 DIRECT EXAMINATION

24 BY MR. CORDELL:

25 Q. Good afternoon, Mr. Ji. Would you please introduce

1 yourself to the ladies and gentlemen of the jury?

2 A. Good afternoon. My name is Hyun Ki Ji.

3 Q. And where do you live?

4 A. I live in Gangnam Seoul.

5 Q. Can you tell us a little bit about your family?

6 A. Yes. I am married and I'm -- me and my wife have been
7 married for 27 years. I have two sons. My oldest son is 25,
8 and he attended university. And while he was attending
9 university, he went to the Army, and actually he came back to
10 the university. And my second son, my youngest son, is 23
11 years old. He's in college but he is serving in the Army
12 currently.

13 THE COURT: Let me ask the interpreter to pull the
14 microphone a little closer to you, please. Thank you.

15 Q. (BY MR. CORDELL) Mr. Ji, I understand that you somehow
16 served in the United States Army. How did that happen?

17 A. When I was in my third year in college, I volunteered to
18 join Katusa, K-A-T-U-S-A, at Camp Humphreys and this was the
19 17th Airborne Brigade, Eighth U.S. Army, and I was there for
20 two years.

21 Q. And did you go back to college after your Army
22 experience?

23 A. Yes, I did.

24 Q. And did you graduate?

25 A. Yes, I did.

1 Q. And from what university?

2 A. I attended Seoul National University.

3 Q. When did you join Samsung?

4 A. That was in 1994.

5 Q. Why did you join Samsung?

6 A. At the time as well, Samsung was introducing a lot of
7 innovative products in the semiconductor area, in the
8 electronics area back then.

9 Q. And what is your job today at Samsung?

10 A. I serve as the team leader of the strategic planning team
11 within the memory business division at Samsung, and my title
12 is executive vice president.

13 Q. And during your almost 30 years at Samsung, has Samsung
14 from time to time collaborated with other companies?

15 A. Yes, we have been collaborating with various other
16 companies.

17 Q. And back in the 2014 time frame, did you begin to explore
18 collaboration with Netlist?

19 A. No. That wasn't the case, because I became the team
20 leader of the planning team in December of 2014, so I became
21 aware of the company Netlist in 2015.

22 Q. And did you explore collaboration in 2015 with Netlist?

23 A. Yes, I did.

24 Q. And why did Samsung explore collaboration with Netlist?

25 A. At the time my boss told me that, There's going to be a

1 proposal made by Netlist about collaboration with Samsung, so
2 when they do, why don't you review it, is what he told me.

3 Q. Okay. Did you have any desire to do technical
4 collaboration with Netlist?

5 A. What I was aware of at the time about Netlist was that
6 they were involved in patent litigation with various -- with
7 many companies.

8 MR. SHEASBY: Your Honor, I object.

9 THE COURT: Just a moment. State your objection.

10 MR. SHEASBY: There's a discussion it's a violation
11 of a MIL. May I approach?

12 THE COURT: Approach the bench, counsel.

13 (The following was had outside the hearing of the
14 jury.)

15 MR. SHEASBY: He just said, because we're involved
16 in litigation, and that's a clear violation of the MIL. We
17 just had this discussion in chambers that he should not be
18 discussing litigation. It's not even true, and it came out
19 clear as day in front of this jury. It is a complete
20 violation.

21 MR. CORDELL: The distinction, Your Honor, I was
22 concerned about is that, again, the translation for 'disputes'
23 and 'litigation' is difficult. I'm happy to move on.

24 MR. SHEASBY: I want a curative instruction. There
25 was no litigation involving Netlist and any other entity that

1 they were involved with at that time.

2 MR. CORDELL: Well, there was -- the hynix
3 litigation was brewing. It was just a few months away.

4 MR. SHEASBY: Your Honor, that is not correct.
5 Listen, I know that we spent yesterday in ways that got --
6 that were not appropriate. Ms. Glasser made a mistake, and I
7 recognize that. It was not intentional. I'm not attributing
8 any intentionality to this, but there's got to be a fair
9 application of these MILs. That was a clear and outrageous
10 violation.

11 Three times I've had to come up here and talk about
12 questions that were clearly outside of this Court's
13 instruction. At some point it has to end.

14 THE COURT: All right. Let me review the question
15 and answer and let me review the MIL. Why don't you take a
16 seat in the meantime.

17 MR. CORDELL: Thank you.

18 (The following was had in the presence and hearing
19 of the jury.)

20 THE COURT: Just a moment, ladies and gentlemen. I
21 shouldn't say this on the record, but an appellate judge could
22 take six months to decide this, and as a trial court you have
23 to act in the moment, so give me just a moment here.

24 All right, ladies and gentlemen. There was a question
25 asked and an answer given as follows: "Okay. Did you have

1 any desire to do technical collaboration with Netlist?"

2 The answer was: "What I was aware of at the time about
3 Netlist was that they were involved in patent litigation with
4 various -- with many companies."

5 You are to disregard that question and that answer. You
6 are not to consider that it's ever been asked or answered.
7 There is a prohibition in this lawsuit that I imposed about
8 testimony regarding other litigation in other places, and this
9 violates that order of the Court.

10 I'm not going to attribute it as an intentional
11 violation. I'm well aware there are differences in
12 translation that sometimes are unintentional. But the
13 statement made is in violation of my prior order, and I am
14 instructing you to disregard the question and disregard the
15 answer that I just read to you in its entirety.

16 All right. Continue with your examination, counsel.

17 MR. CORDELL: Thank you, Your Honor.

18 Q. (BY MR. CORDELL) Did Samsung have a meeting with Netlist
19 in the 2015 time frame?

20 A. Yes, it did.

21 Q. And who was at that meeting for Samsung?

22 A. There was Hyun Hun Hun (ph), VP, and at this meeting
23 there was a Jung Bae Lee, senior VP.

24 Q. And who attended the meeting on behalf of Netlist?

25 A. As far as I remember, Mr. Jibum Kim, the executive vice

1 president, and the president Chun Ki Hong were there.

2 Q. We sometimes call him Chuck Hong, Mr. Chuck Hong?

3 A. Yes, that's how I understand.

4 Q. Now, do you remember every detail of every meeting you
5 had with Netlist over the years?

6 A. No, I don't remember every detail of those.

7 Q. Have you been able to refresh your memory about some of
8 those meetings?

9 A. Yes, I have.

10 MR. CORDELL: Let's look at PX 464, please.

11 Q. (BY MR. CORDELL) So Mr. Ji, this is a PowerPoint from
12 one of your meetings?

13 A. Yes. As I look through this document, looking how there
14 are some colors that are used here, I remember this, yes.

15 Q. And where did this meeting take place?

16 A. That was within the company that I'm working at, within
17 Samsung in Korea.

18 Q. Was the meeting in English, in Korean, or some other
19 language?

20 A. The meeting was in Korean.

21 Q. About how long did the meeting last?

22 A. Probably it was less than an hour, I think.

23 Q. And was there any follow-up action items from the
24 meeting?

25 A. Yes. Netlist was going to send us their proposal.

1 Q. Okay. Let me show you DTX 4. And is this an email you
2 received from Netlist?

3 A. Yes, it is.

4 MR. CORDELL: And can we go to the second page?

5 Q. (BY MR. CORDELL) Is this a term sheet?

6 A. Yes, it is.

7 Q. And is the date just a typo? It should be April 22nd,
8 2015?

9 A. I believe it is correct to be 2015.

10 Q. And what was your reaction when you got this term sheet?

11 A. Since the amount that was proposed by Netlist was just
12 absurdly high, so I thought it was going to be a long
13 negotiation; and secondly, this was a patent license, but --
14 and Netlist wanted to refer to it as NRE by cloaking, so it
15 made me wonder.

16 Q. Is that the yellow box on the right-hand side of the
17 document you're referring to?

18 A. Yes, it is.

19 Q. So the idea here was to take the patent license fee but
20 cloak it as NRE. Is that right?

21 A. That is right.

22 Q. Mr. Ji, I want to be very clear. Who had the idea to
23 cloak the license fee?

24 A. It was Netlist.

25 Q. Now, ultimately did you sign an agreement with Netlist?

1 A. I myself did not.

2 Q. Thank you.

3 Did Samsung ultimately sign an agreement with Netlist.

4 A. Yes, Samsung did.

5 MR. CORDELL: And can I have JTX 29, please?

6 Q. (BY MR. CORDELL) Is this the agreement Samsung signed
7 with Netlist in 2015?

8 A. It is.

9 Q. And was the cash payment under this agreement from
10 Samsung to Netlist in the amount of \$8 million?

11 A. Yes, it is correct. It was \$8 million.

12 Q. After the agreement was signed, what was your role in the
13 Samsung/Netlist relationship?

14 A. Although the agreement was entered into, I did receive
15 updates occasionally from Netlist.

16 Q. And who did you meet with from Netlist?

17 A. Jibum Kim, EVP.

18 Q. And that's -- Jibum Kim is an EVP at Netlist?

19 A. Yes, that's correct.

20 Q. And did Mr. Jibum Kim live in Korea?

21 A. Yes. He lived in Korea then.

22 MR. CORDELL: Let me have PX 1662.

23 Q. (BY MR. CORDELL) Is this a memo about a dinner you had
24 with Mr. Jibum Kim?

25 MR. CORDELL: If I could have page 2, please.

1 THE WITNESS: Yes, it is.

2 Q. (BY MR. CORDELL) Who paid for dinner--you or Mr. Kim?

3 A. Probably I did.

4 Q. And do you remember the dinner with Mr. Jibum Kim in June
5 of 2015 -- 2016?

6 A. Yes, I do.

7 Q. Did Mr. Jibum Kim at any time during this dinner say that
8 Samsung was infringing Netlist's patents?

9 A. No, I never heard anything like that.

10 Q. At any time in your relationship with Mr. Jibum Kim and
11 Netlist, did they ever tell you that they thought Samsung was
12 infringing their patents?

13 A. I have never heard anything like that.

14 Q. You're not in the habit of taking people to dinner who
15 accuse you of patent infringement?

16 A. I don't think I would have dinner with him. Probably
17 not.

18 MR. CORDELL: Let me have PX 446.

19 Q. (BY MR. CORDELL) Are you familiar with this
20 communication, sir?

21 A. Yes, I am.

22 MR. CORDELL: And can I go to the next page?

23 Q. (BY MR. CORDELL) I know this is blacked out, but do
24 you know that there was a list of patents shown here?

25 A. Yes, I do.

1 Q. When you received this communication with the attached
2 chart, was Samsung fully licensed to all of Netlist's patents?

3 A. Yes, it was.

4 Q. Do you have an understanding of whether Mr. Jibum Kim was
5 sending you these patents because he thought Samsung might be
6 interested, or was he talking about another company being
7 interested?

8 A. For me, apparently I thought they were having some issues
9 with another company.

10 Q. And what was that other company they were issues with?
11 Just the name, please.

12 A. It was SK hinix.

13 Q. One last question, Mr. Ji. Does Samsung develop its own
14 technology?

15 A. Yes, Samsung does develop its own technology.

16 Q. Thank you very much.

17 MR. CORDELL: I pass the witness.

18 MR. SHEASBY: Your Honor, may I approach?

19 THE COURT: Approach the bench.

20 (The following was had outside the hearing of the
21 jury.)

22 MR. SHEASBY: This is the second time they've
23 elicited from a witness a claim that Samsung developed its
24 own technology. This is opening the door to the question
25 about whether he checked whether Samsung engineers copied our

1 technology. I understand that Ms. Glasser did not follow the
2 procedure yesterday, but that last question was gratuitous and
3 it's clearly opened the door to my examination about whether
4 there was -- he's investigated whether there was copying.

5 MR. CORDELL: Mr. Sheasby accused us of taking their
6 technology on six separate occasions during the opening
7 statement, Your Honor. I've not been able to respond with my
8 JEDEC willfulness response. This is -- this clearly doesn't
9 open the door; it's simply a statement by the company that
10 they believe they developed their own technology.

11 In fact, I would argue that it's a fairly meager
12 response. I would like to put on something much more
13 forceful but I'm not able to.

14 And while I have the floor, the reason why he wanted --
15 or not wanted, but was talking about patent litigation at all
16 is that we were going to have him address counsel's opening
17 statement slide where he talked about patent litigation and
18 patent risk, and that's why --

19 THE COURT: Let me just save you some both time
20 because you don't have anymore time to waste. I'm not going
21 to grant leave to go into the copying issue.

22 MR. SHEASBY: Thank you, Your Honor.

23 THE COURT: All right. Let's proceed.

24 (The following was had in the presence and hearing
25 of the jury.)

1 THE COURT: Let's proceed with cross examination by
2 the Plaintiff.

3 MR. SHEASBY: May I approach, Your Honor?

4 THE COURT: You may.

5 CROSS EXAMINATION

6 BY MR. SHEASBY:

7 Q. Good afternoon Mr. Ji. It's nice to meet you.

8 A. Nice to meet you.

9 Q. You understand that your testimony is under oath.
10 Correct?

11 A. Yes, I do.

12 Q. And a substantial part of the purpose of the JDLA was to
13 obtain access to Netlist's patents. Correct?

14 A. A major purpose was to resolve any potential patent
15 litigation issues.

16 THE COURT: Just a minute. What are you doing,
17 Mr. Sheasby? We are in the middle of cross examination and
18 you are walking across the courtroom.

19 MR. SHEASBY: I'm sorry, Your Honor.

20 THE COURT: Are you ready to continue with cross
21 examination?

22 MR. SHEASBY: I need to go get a deposition, Your
23 Honor.

24 THE COURT: Then you need to stop the cross
25 examination and ask for leave to do it.

1 You may go get whatever you're missing.

2 MR. SHEASBY: Thank you, Your Honor.

3 (Pause in proceedings.)

4 THE INTERPRETER: Your Honor, I was not quite done
5 with the interpretation. May I just complete the
6 interpretation?

7 THE COURT: We'll pick back up when we're ready.
8 Are you ready to proceed?

9 You want to approach? Approach the Court Security
10 Officer.

11 All right. Before counsel asks the next question, I'll
12 allow the witness to complete the answer that they were in the
13 process of giving.

14 Go ahead.

15 THE WITNESS: A major purpose was to resolve any
16 potential patent litigation issues that might be filed by
17 Netlist.

18 Q. (BY MR. SHEASBY) If you'll turn to your deposition and
19 look at page 20, lines 5 through 17.

20 THE COURT: Do I have a copy of this deposition,
21 counsel? Please get me one, if you don't mind.

22 MR. SHEASBY: May I approach, Your Honor?

23 THE COURT: You may.

24 Q. (BY MR. SHEASBY) Have you read your testimony at column
25 [sic] 20, lines 5 through 17, Mr. Ji?

1 A. Sorry. I'm not done yet. May I enlist help from the
2 interpreter?

3 Q. Sure.

4 THE COURT: All right, counsel. Please continue
5 with your next question.

6 Q. (BY MR. SHEASBY) After listening to Netlist's
7 suggestions, your company felt that from your perspective it
8 will be beneficial for Samsung to collaborate with Netlist.
9 Correct?

10 A. The issues involving patent litigation risk we thought
11 was related to patent-related matters.

12 MR. SHEASBY: Your Honor, may I publish his answer?

13 THE COURT: You'll need to start with line 2 where
14 the question begins.

15 MR. SHEASBY: Sure.

16 THE COURT: You may publish lines 2 through 17.

17 MR. SHEASBY: Thank you, Your Honor.

18 MR. CORDELL: Your Honor, may we approach?

19 MR. SHEASBY: Page 20 --

20 MR. CORDELL: Before that's published, may we
21 approach, Your Honor?

22 THE COURT: Approach the bench.

23 I'm going to charge you both for new carpet after you
24 wear a hole in it.

25 (The following was had outside the hearing of the

1 jury.)

2 MR. CORDELL: So, Your Honor, here's the problem.
3 This answer talks about Netlist made suggestion in regards to
4 IP-related matter. That's the dispute with hynix.

5 MR. SHEASBY: That is not --

6 THE COURT: It is what it says it is, and the
7 question is, "When you are referring to IP you are referring
8 to patent, so, in your view, a substantial part of the purpose
9 of the JDLA was to obtain access to Netlist's patents?" It's
10 actually lines 14 through 17 that he's trying to get in.

11 MR. SHEASBY: I'm actually trying to get in all of
12 it.

13 MR. CORDELL: The problem is now he is asking what
14 we just went through when the witness talked about the
15 litigation with hynix. That's exactly what he's talking
16 about.

17 THE COURT: I'm going to let him publish this. If
18 on redirect you want to go back to SK hynix and the threat of
19 litigation, you can do that. All right?

20 MR. CORDELL: Thank you.

21 (The following was had in the presence and hearing
22 of the jury.)

23 THE COURT: All right, counsel. Let's proceed.

24 You may publish those designated lines of the deposition
25 to the jury.

1 MR. SHEASBY: Thank you.

2 This is not the right deposition. May I approach the
3 assist?

4 THE COURT: Yes.

5 Let's go.

6 MR. SHEASBY: Yes, Your Honor.

7 Question: "What do you understand the purpose of the
8 JDLA to be?"

9 Answer: "Netlist made suggestion in regards to
10 IP-related matter. After listening to Netlist's suggestions,
11 our company felt that, from our company's perspective, it will
12 be beneficial for us to collaborate with Netlist."

13 Question: "When you referred to IP, are you referring to
14 patents?"

15 Answer: "Yes, I am."

16 Question: "So, in your view, was it a substantial part
17 of the purpose of the JDLA to obtain access to Netlist's
18 patents?"

19 Answer: "Yes, that's correct."

20 Q. (BY MR. SHEASBY) Did I read your testimony correctly,
21 sir?

22 A. Yes, you did.

23 Q. And another purpose of the collaboration was the joint
24 development of an NVDIMM product?

25 A. Yes.

1 Q. And you talked about the fact that -- in fact, Samsung
2 has repeatedly reached out to Netlist for its technology.
3 Correct?

4 A. I'm sorry. The interpreter spoke too fast, so I couldn't
5 quite understand.

6 Q. No problem.

7 A. I'm not sure about that.

8 Q. Let me see if I can refresh your recollection. If you'll
9 pull up PX 1787, which is in your binder at tab 20.

10 And you read English. Isn't that correct, Mr. Ji?

11 A. I can do slowly.

12 Q. And so this is a 2017 email from Yong Hwangbo to net
13 licenses saying Samsung is making a proposal for evaluation
14 and development, and then the technology it's seeking is
15 Netlist's HybriDIMM and related technology. And you see it
16 says -- down below it wants IP, it wants evaluation and
17 development. Do you see that, sir?

18 A. What line are you on when you read that?

19 Q. So Samsung is making a proposal for evaluation and
20 development. Do you see that, sir?

21 A. Yes, I do.

22 Q. And in this email it doesn't make any reference to threat
23 of litigation, does it, sir?

24 A. It doesn't appear to be.

25 Q. And you testified regarding this term sheet that you said

1 you received. Is that correct?

2 A. Yes, I did.

3 Q. You testified under oath to that. Correct?

4 A. Yes, I did.

5 Q. And you were very definitive with the jury about what
6 this term sheet meant. Correct?

7 A. Yes, I was.

8 Q. You are not listed as a recipient on this document, are
9 you, sir?

10 A. I am not listed as a recipient.

11 Q. And Samsung also asked you about this document. Correct?

12 A. Yes, that's right.

13 Q. If you turn the page, it's a list of patents and it says
14 under 'covered products' 'HBM'. Is that correct?

15 A. Yes, that's right.

16 Q. When's the first time you've seen this document, sir?

17 A. Probably the person who received this document would have
18 been one of my co-workers, the people who were working under
19 me, so I think it might have been around the time he or she
20 received it.

21 Q. Do you have any actual recollection of having reviewed
22 this document when preparing for your testimony in this case?

23 A. Yes, I have seen this document.

24 Q. Did you see this document in preparation for the case?

25 A. During the course of preparation for this case I did.

1 Q. And you sat with your lawyers to discuss this document.
2 Correct?

3 MR. CORDELL: Your Honor, I'm going to object.

4 THE COURT: Wait a minute. We're getting awfully
5 closed to privileged communications.

6 MR. SHEASBY: I'll withdraw that.

7 THE COURT: Let's do that.

8 Q. (BY MR. SHEASBY) Did you ask -- you didn't testify under
9 oath in your direct examination that you asked Samsung
10 engineers if they were using Netlist technology. Correct?

11 A. I didn't quite understand the question. Actually I
12 didn't quite understand the questions at the end of the day,
13 so...

14 Q. When you received this document from Netlist, did you ask
15 your engineers if they were using Netlist technology?

16 A. Not to my recollection.

17 MR. SHEASBY: Pass the witness.

18 THE COURT: Redirect?

19 MR. CORDELL: Can we have that same document back?
20 Well, you can start with this one. Go back to the first page,
21 please. So this is DTX 04.

22 REDIRECT EXAMINATION

23 BY MR. CORDELL:

24 Q. Mr. Ji, counsel asked you about whether or not you were
25 on the address line of this document. Do you remember that?

1 A. Yes, I do.

2 Q. What was your relationship with Jung Bae Lee in April of
3 2015?

4 A. I was a team leader of the strategic planning team and
5 Jung Bae Lee was the team leader of the product planning SVP,
6 he was.

7 Q. And can you explain how it was that you came to receive
8 this document?

9 A. He probably would have sent this document over to me.

10 Q. But do you have any doubt in your mind looking at the
11 second page that you did receive this document?

12 A. There's no need for me to have any doubt about this.

13 MR. CORDELL: Can I have PX 446, please? And can we
14 go to the second page?

15 Q. (BY MR. CORDELL) When you received this document,
16 Mr. Ji, was Samsung licensed or not licensed?

17 A. Samsung was licensed.

18 Q. So would there have been any reason for you to go to talk
19 to anyone about whether or not you were using these patents if
20 you were already licensed?

21 A. No, not at all, because from our perspective we were
22 already licensed so there was no need for us to check on that
23 at all.

24 MR. CORDELL: May I have the elmo?

25 Q. (BY MR. CORDELL) Counsel showed you a piece of your

1 deposition from a few years ago. Do you remember that?

2 A. Yes, I do.

3 Q. And he specifically pointed you to this line relating to
4 Netlist making a suggestion in regards to an IP-related
5 matter. Do you remember that?

6 A. I do.

7 Q. And I know counsel objected when you mentioned an
8 IP-related matter earlier. Do you remember that?

9 MR. SHEASBY: Your Honor, can we approach?

10 THE COURT: No.

11 THE WITNESS: Yes, I do.

12 Q. (BY MR. CORDELL) What was the IP-related matter that you
13 had told counsel for Netlist about?

14 A. I'm sorry. I didn't quite understand the question.

15 THE INTERPRETER: May I just re-interpret?

16 MR. CORDELL: Yes. Go ahead.

17 THE WITNESS: It was a matter related to patents.

18 Q. (BY MR. CORDELL) And was that matter related to patents
19 related to hynix?

20 A. I remember not.

21 Q. Well, you had mentioned something about hynix earlier.
22 Did it have anything to do with this IP-related matter?

23 A. No.

24 MR. CORDELL: Your Honor, I pass the witness.

25 THE COURT: Any further cross?

1 MR. SHEASBY: No, Your Honor, no further cross.

2 THE COURT: All right. You may step down, Mr. Ji.
3 Thank you.

4 Ladies and gentlemen of the jury, we're going to recess
5 for the day at this point. I'm going to ask you to take your
6 notebooks with you and leave them in the jury room as you exit
7 the courtroom.

8 I'll remind you to continue to follow all of my
9 instructions, including not to communicate with anybody about
10 the case. We are getting close to finishing the evidence.
11 It's my anticipation we'll probably finish the evidence before
12 lunch tomorrow, so please be sure that you follow all the
13 instructions that I've given you.

14 Travel safely to your homes. Try to be back so that we
15 can start at 8:30 tomorrow morning. I will do my best to be
16 on time. I won't give you any guarantees, but I promise you
17 if I'm not right here at 8:30 it will be because I'm working
18 on other matters related to this case. But I'm going to do my
19 best for us to start at 8:30 in the morning, so please be
20 prepared for that.

21 Please travel safely. I'll see you in the morning.

22 With that, the jury's excused for the evening.

23 (Whereupon, the jury left the courtroom.)

24 THE COURT: Counsel, the Plaintiff has 1 hour and 23
25 minutes remaining of designated trial time, and the Defendant

1 has 31 minutes of designated trial time.

2 I will see you in the morning. I'll be available in
3 chambers. Continue your meet and confer efforts overnight.

4 With that, we stand in recess until tomorrow.

5 (The proceedings were concluded at 6:15 p.m.)
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1 I HEREBY CERTIFY THAT THE FOREGOING IS A
2 CORRECT TRANSCRIPT FROM THE RECORD OF
3 PROCEEDINGS IN THE ABOVE-ENTITLED MATTER.
4 I FURTHER CERTIFY THAT THE TRANSCRIPT FEES
5 FORMAT COMPLY WITH THOSE PRESCRIBED BY THE
6 COURT AND THE JUDICIAL CONFERENCE OF THE
7 UNITED STATES.

8
9 S/Shawn McRoberts 04/19/2023

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